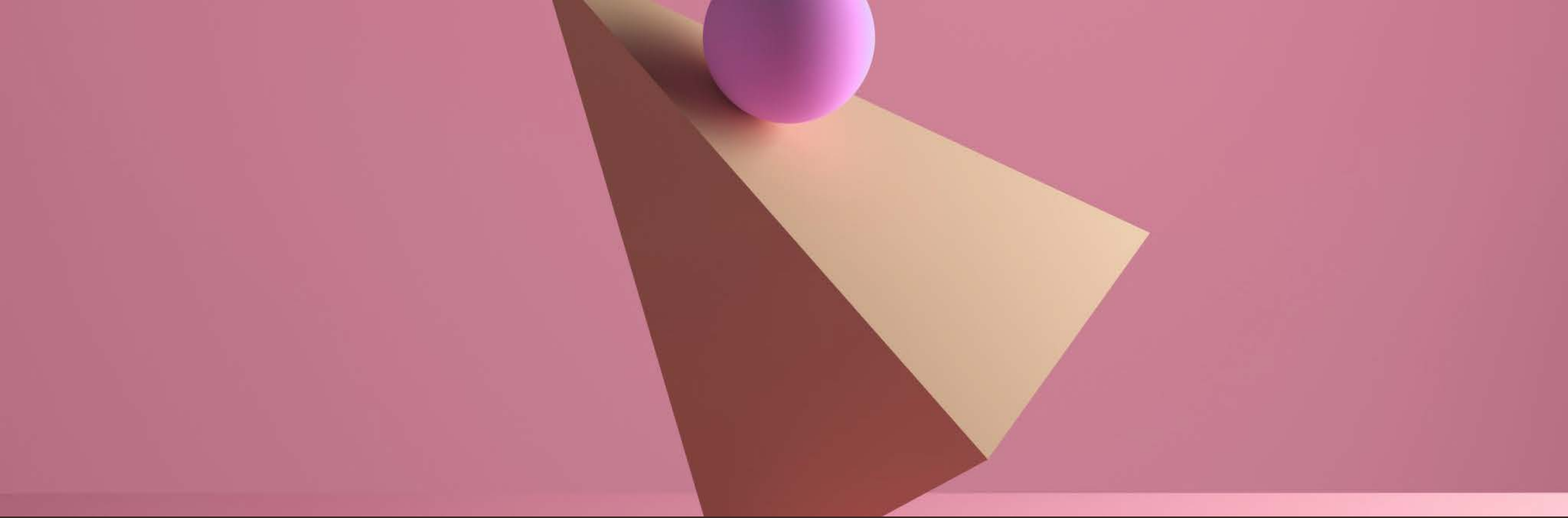


2D POSITION AND ORIENTATION

Lecture 2

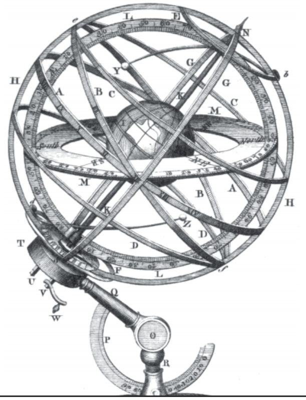
Reading: 2.0-2.1 Representing Pose in 2D

Part I Foundations

Chapter 2 Representing Position and Orientation

Chapter 3 Time and Motion

2 Representing Position and Orientation



A fundamental requirement in robotics and computer vision is to represent the position and orientation of objects in an environment. Such objects include robots, cameras, workpieces, obstacles and paths.

A point in space is a familiar concept from mathematics and can be described by a coordinate vector, also known as a bound vector, as shown in Fig. 2.1a. The vector represents the displacement of the point with respect to some reference coordinate frame. A coordinate frame, or Cartesian coordinate system, is a set of orthogonal axes which intersect at a point known as the origin.

More frequently we need to consider a set of points that comprise some object. We assume that the object is *rigid* and that its constituent points maintain a constant relative position with respect to the object's coordinate frame as shown in Fig. 2.1b. Instead of describing the individual points we describe the position and orientation of the object by the position and orientation of its coordinate frame. A coordinate frame is labelled, $\{B\}$ in this case, and its axis labels x_B and y_B adopt the frame's label as their subscript.

The position and orientation of a coordinate frame is known as its pose and is shown graphically as a set of coordinate axes. The relative pose of a

Section 1 Geometry recap

Euclidean geometry

- After Euclid of Alexandria

Euclid of Alexandria (325-265 BCE)

School of Athens (Raphael)
Marie-Lan Nguyen

Euclidean geometry

- After Euclid of Alexandria

https://www.claymath.org/euclid_index/

Around 300 BCE he wrote "The Elements" (13 books)

Scroll from "The Elements"
Euclid | Public domain, Wikimedia Commons

Euclid of Alexandria (325-265 BCE)

School of Athens (Raphael)
Marie-Lan Nguyen

Euclidean geometry

- The geometry you learnt at school

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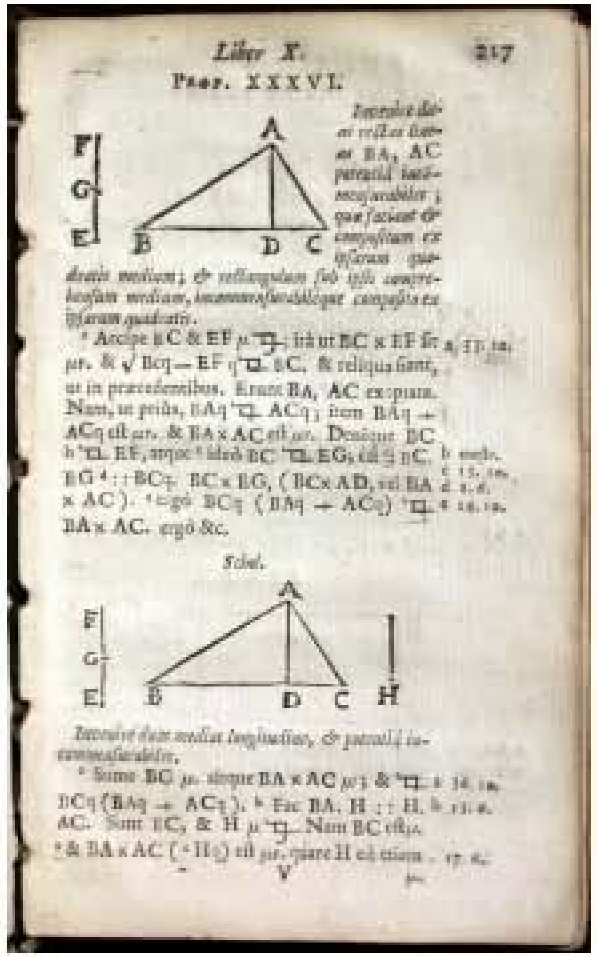

Liber X. 217
Prop. XXXVI.

Euclid, Elementorum Libri XV: Cantabrigiae, 1655 2005

Euclidean **geometry**

- The geometry you learnt at school
- The geometry of a **plane** and **3D space**
  - ➥ lines, triangles, circles etc.

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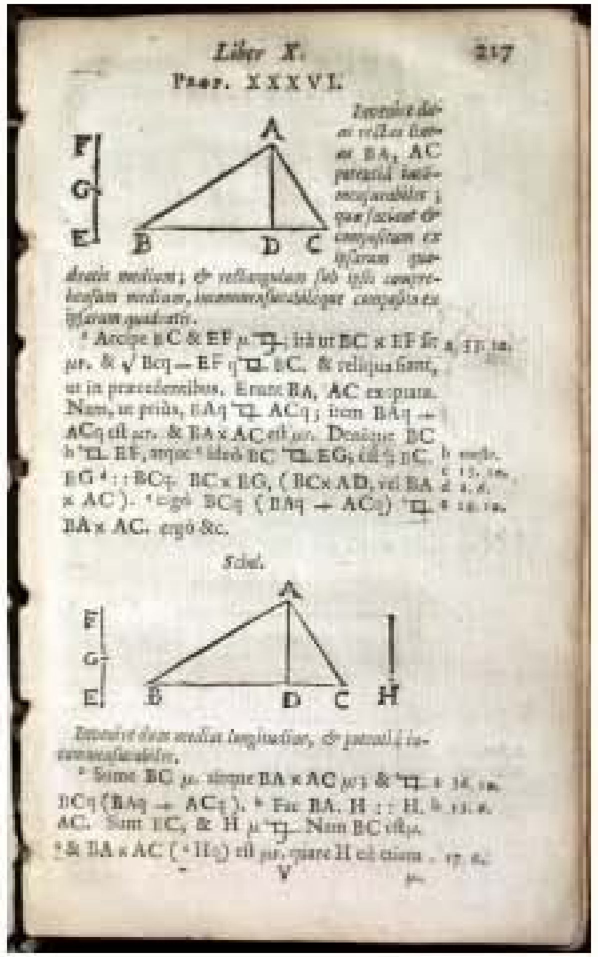

Liber X. 217
Prop. XXXVI.

Euclid, Elementorum Libri XV: Cantabrigiae, 1655 2005
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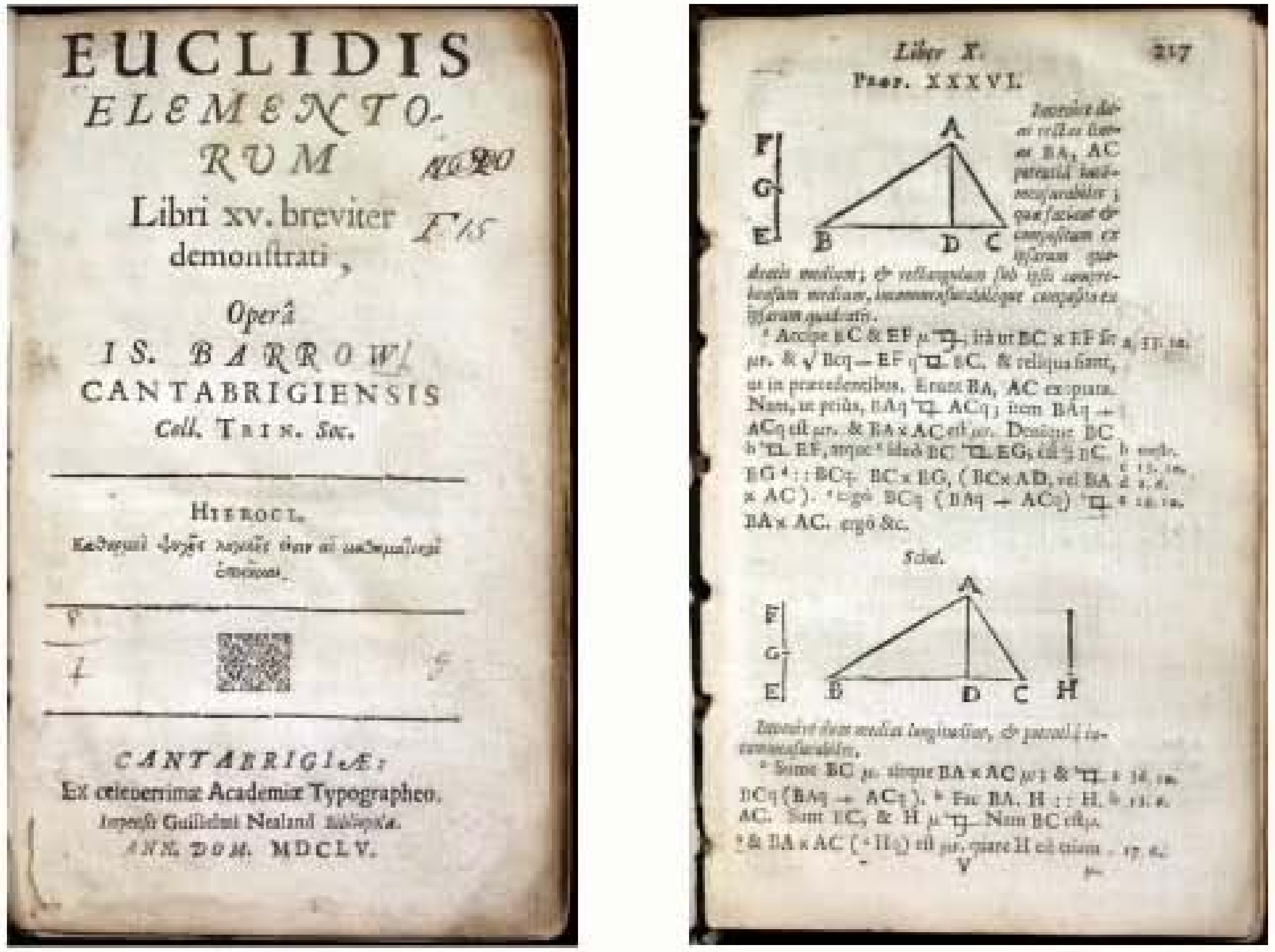

Euclid, Elementorum Libri XV: Cantabrigiae, 1655 2005

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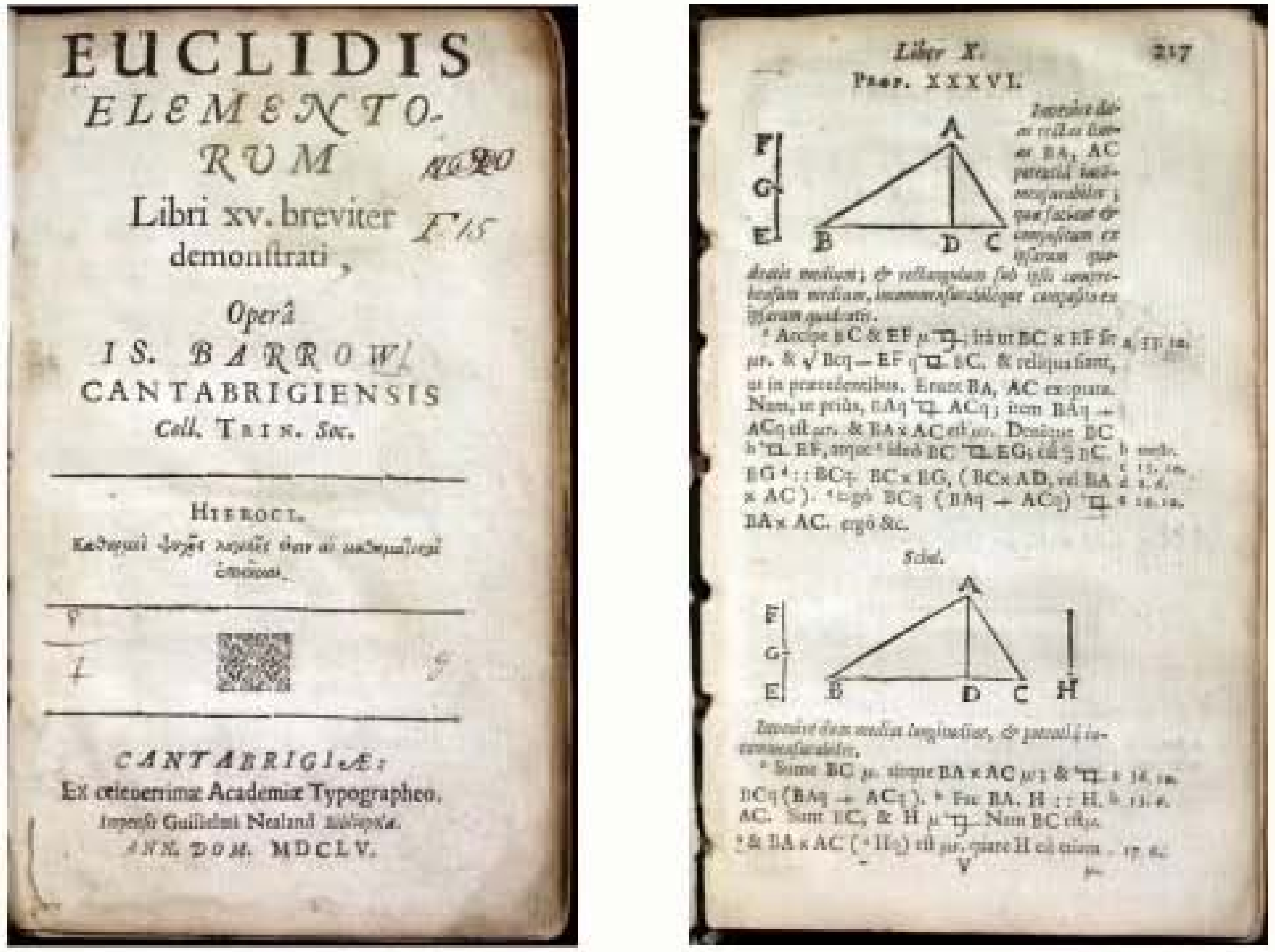

Euclid, Elementorum Libri XV: Cantabrigiae, 1655 2005

Euclidean geometry

- The geometry you learnt at school
- The geometry of a **plane** and **3D space**
  ➥ lines, triangles, circles etc.
- Based on a few axioms
- No numbers
- No coordinate frames

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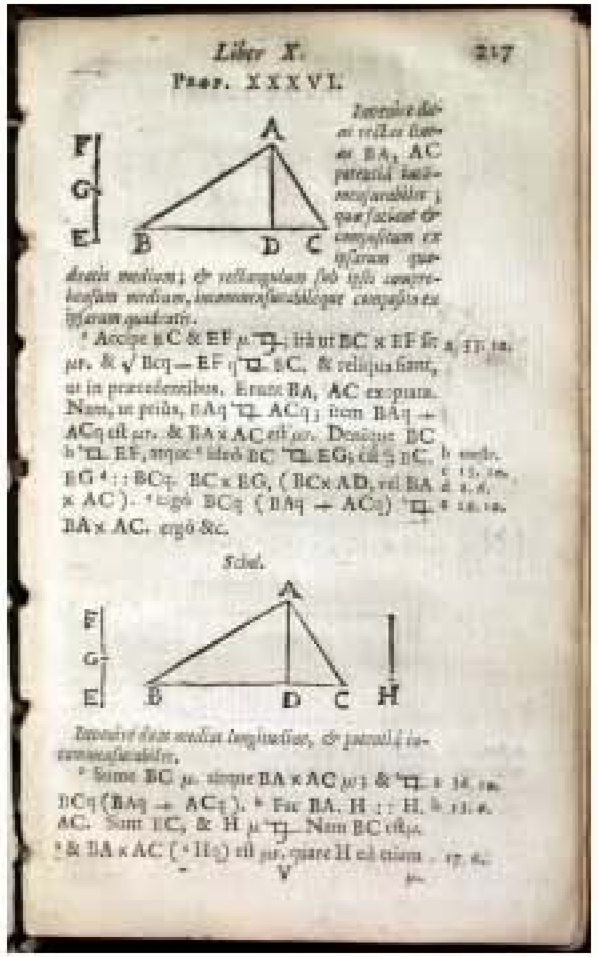

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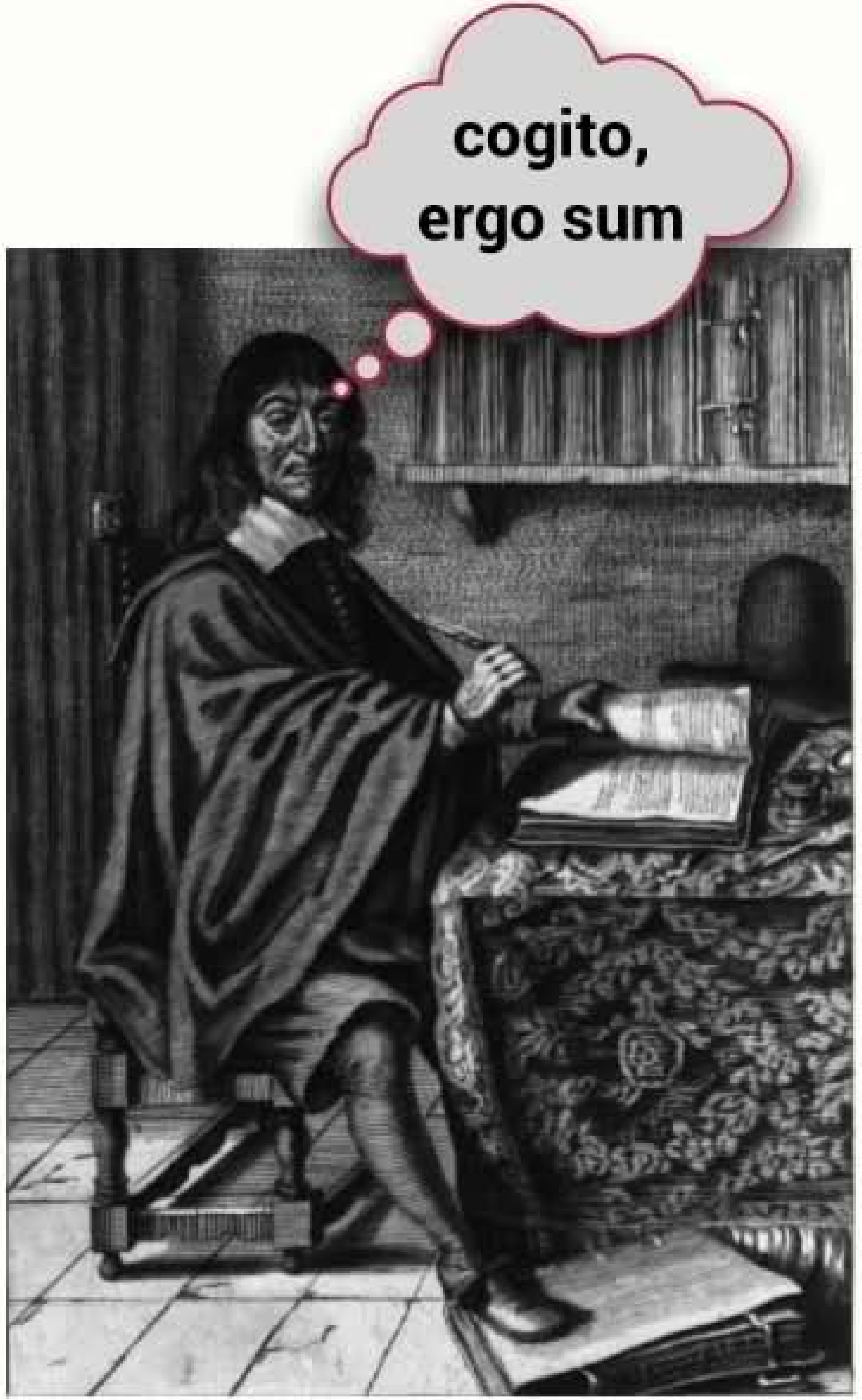

Rene Descartes (1596-1650)

- Philosopher
- Mathematician
- Part-time mercenary

Rene Descartes (1596-1650)

- Philosopher
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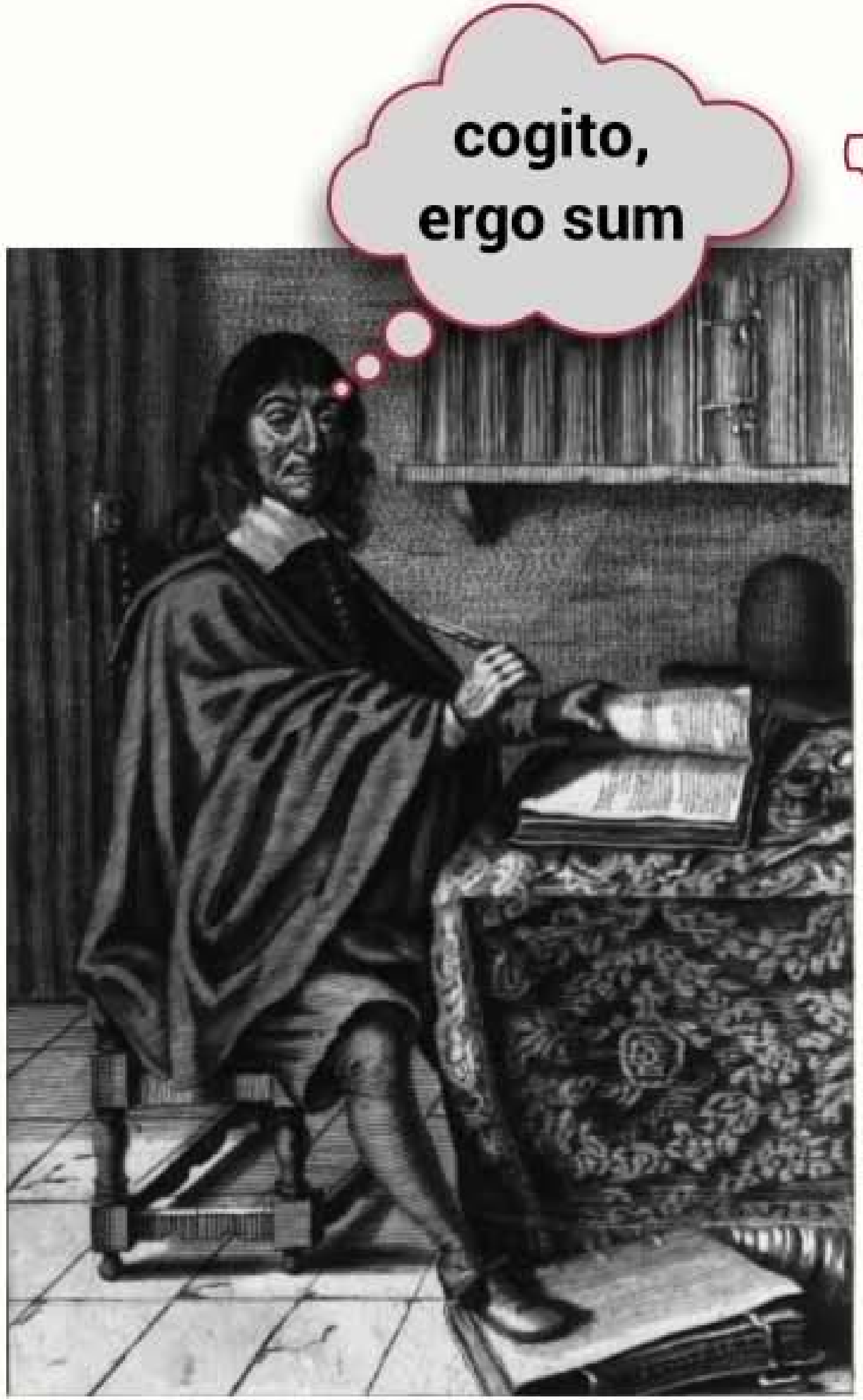
- **The father of modern philosophy**



I think, therefore I am

Rene Descartes (1596-1650)

- Philosopher
- Mathematician
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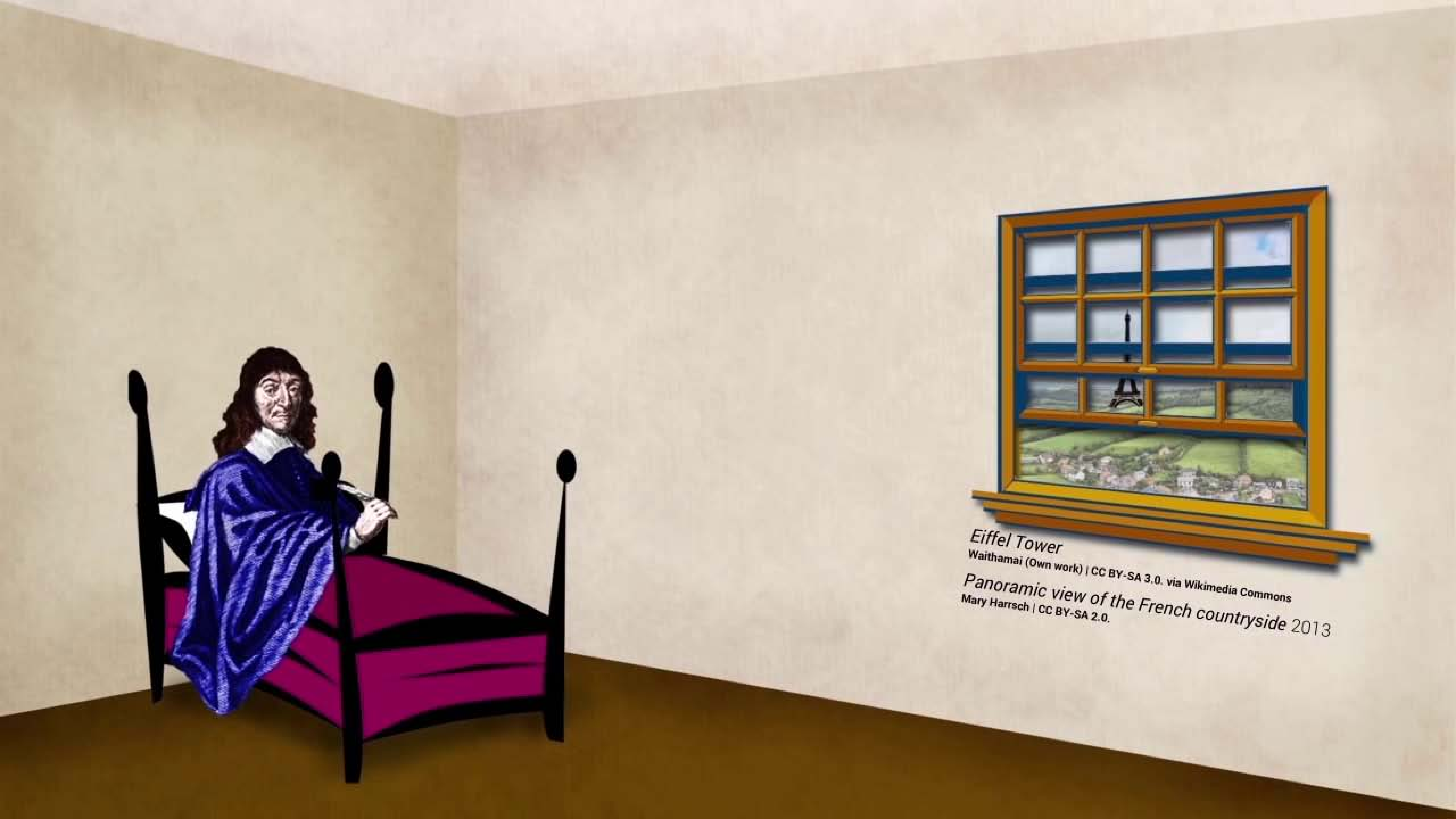
- **The father of modern philosophy**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Panoramic view of the French countryside 2013
Mary Harrsch | CC BY-SA 2.0.


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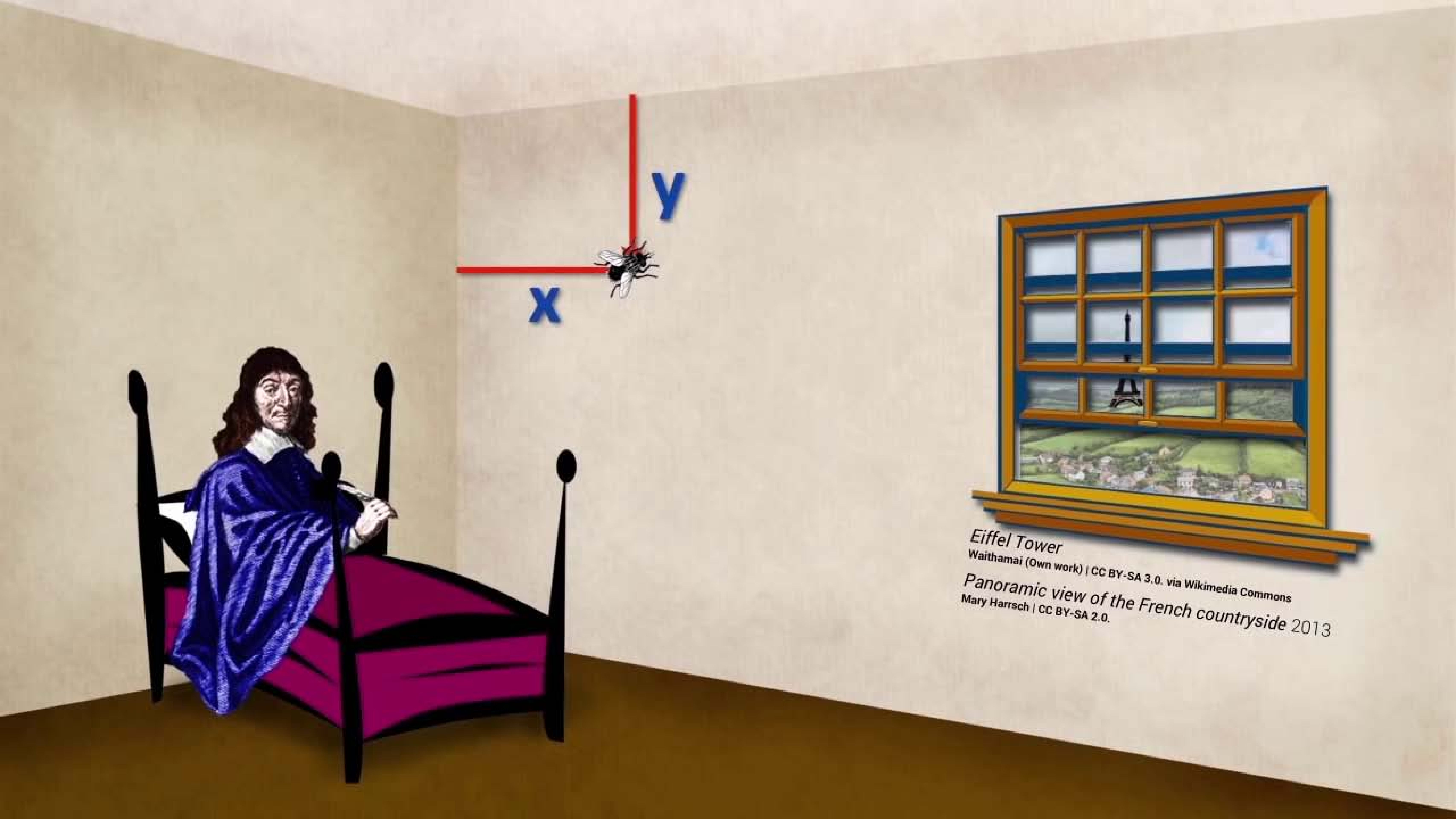
y
x
Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
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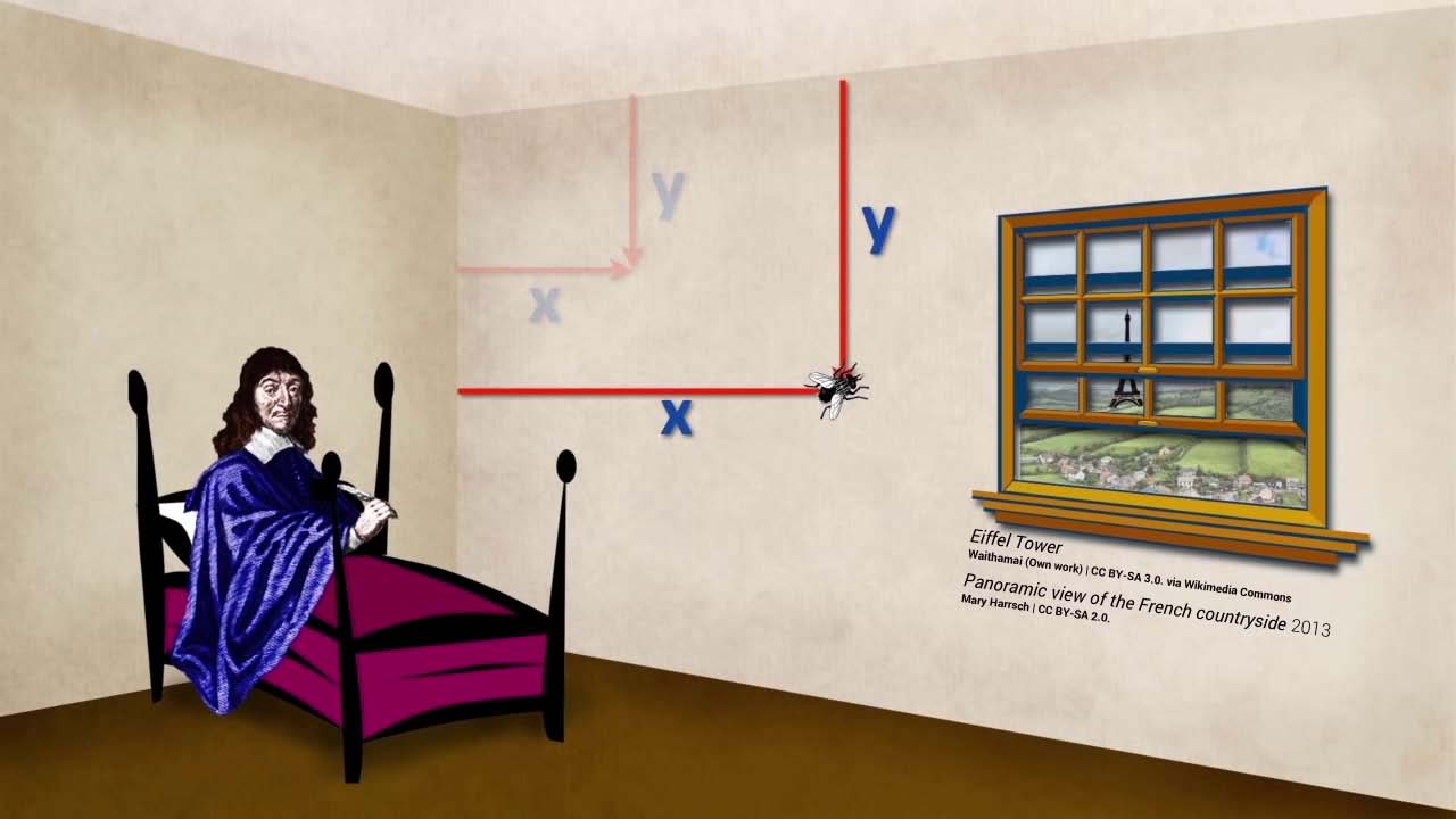

y
x
y
x
Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
Panoramic view of the French countryside 2013
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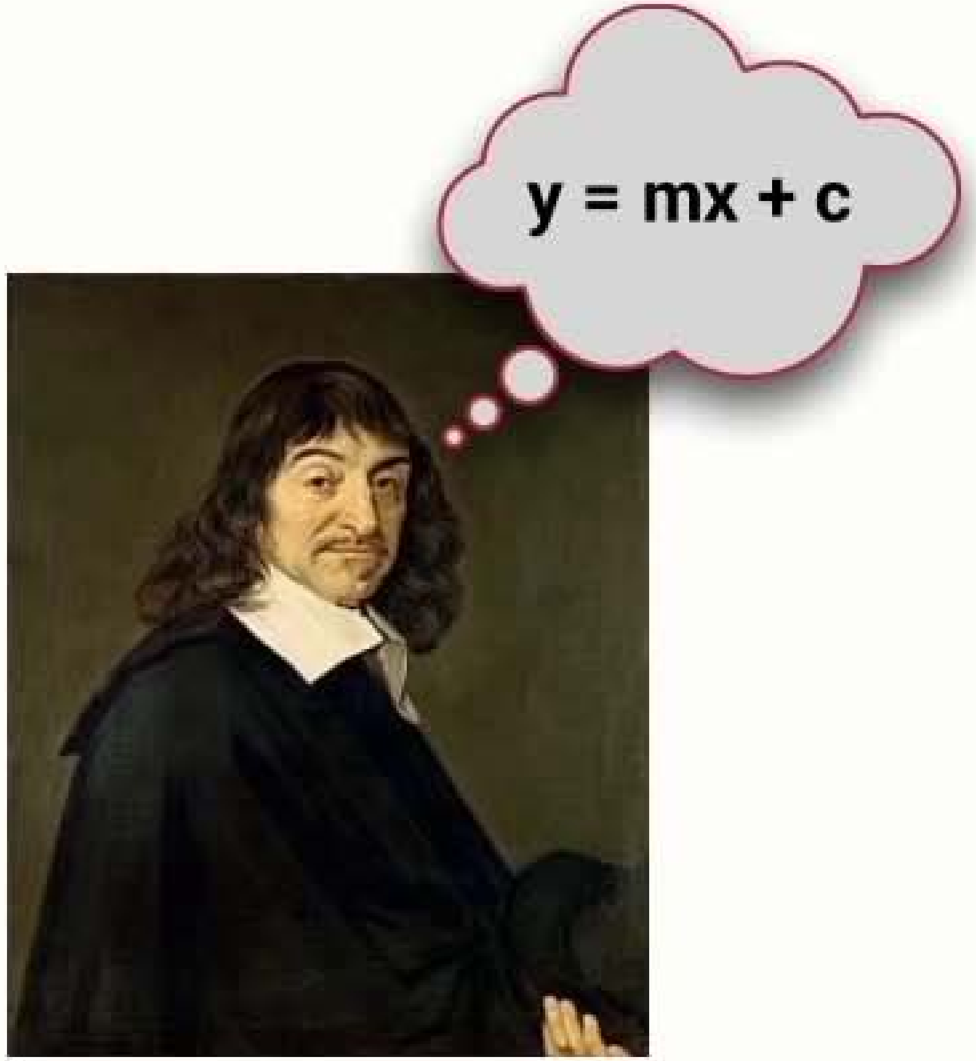
Cartesian **geometry**

- Added algebra to Euclidean geometry

Cartesian **geometry**



- Added algebra to Euclidean geometry
  - Shapes could be expressed by equations

Cartesian **geometry**

$y = mx + c$

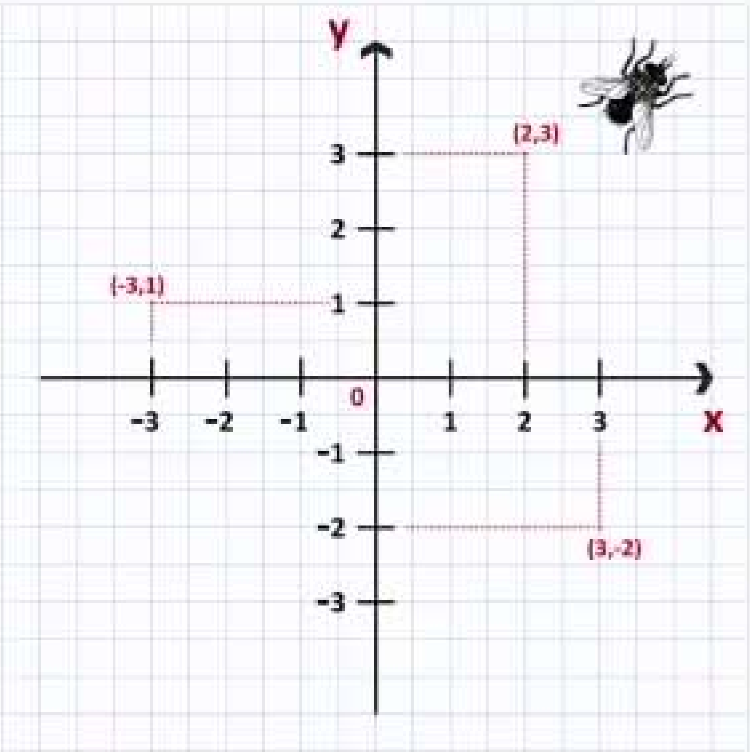
- Added algebra to Euclidean geometry
  - Shapes could be expressed by equations
  - Cartesian (or analytic) geometry

His name in Latin was Renatus Cartesius

Cartesian **geometry**

$y = mx + c$

Descartes
After Frans Hals | Public domain, Wikimedia Commons



- Added algebra to Euclidean geometry
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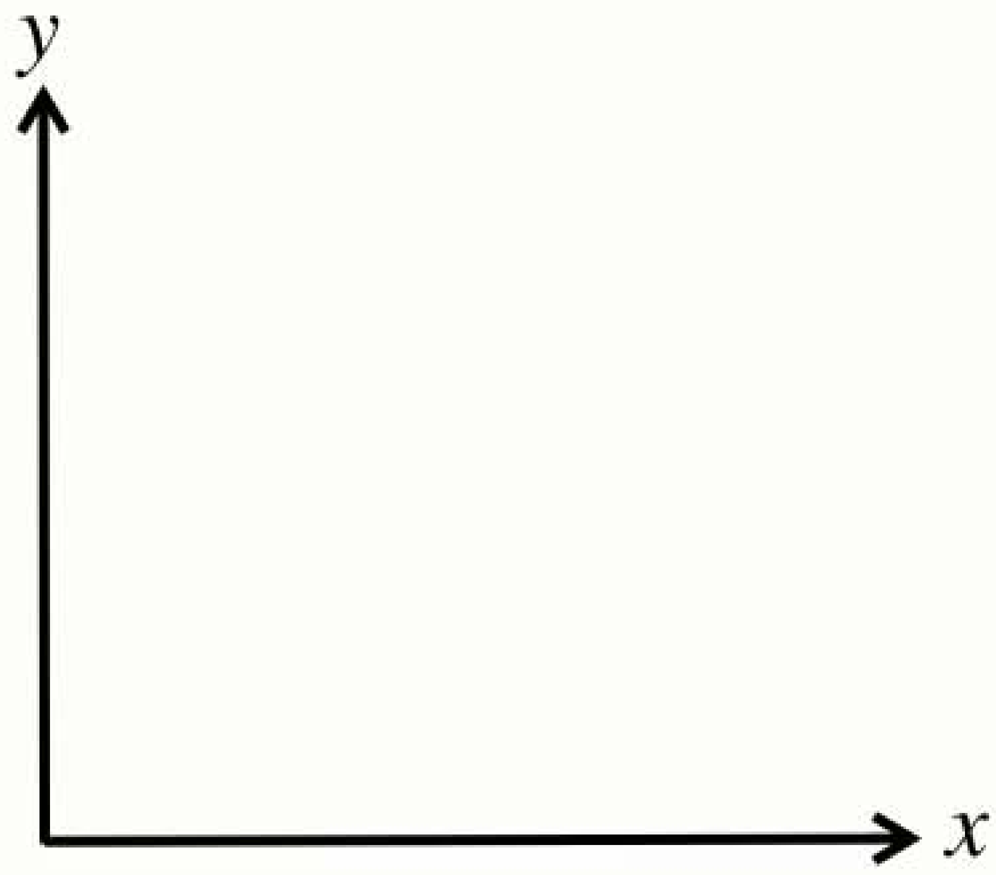
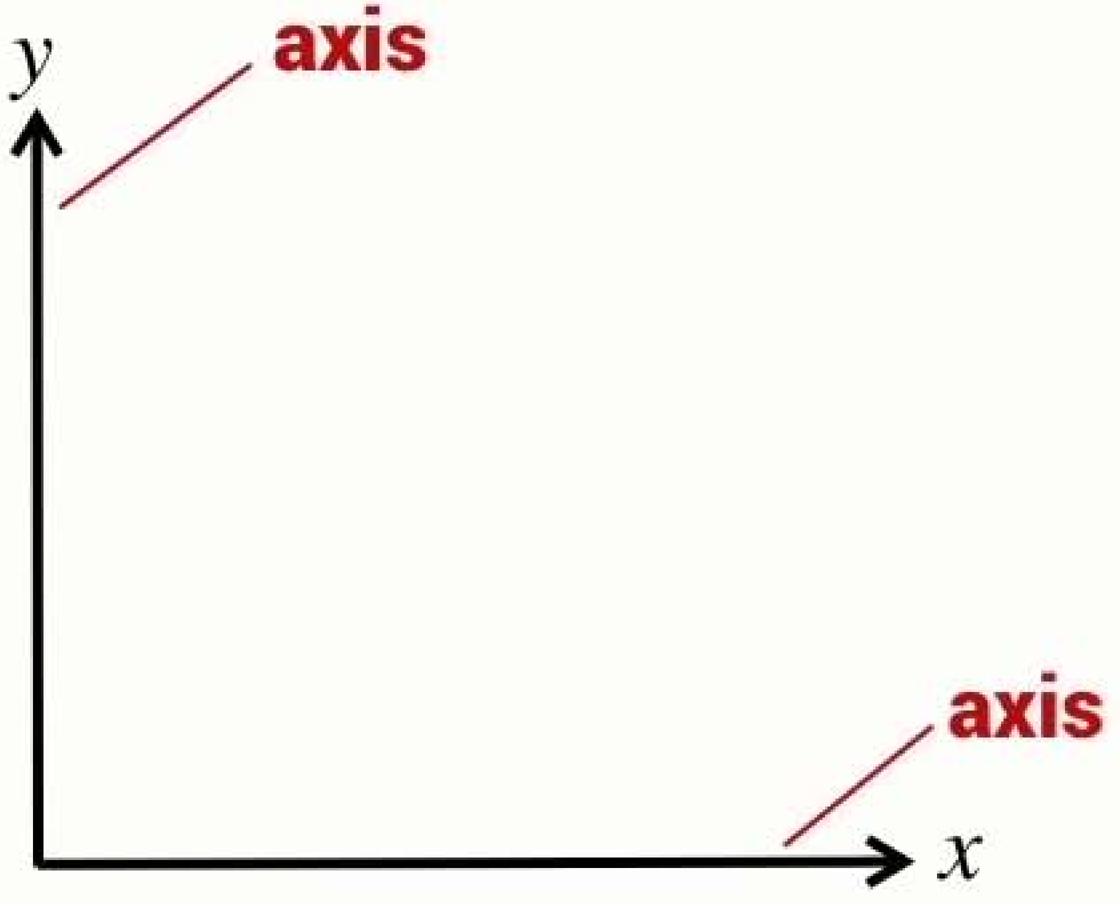
- Developed the coordinate system

2D coordinate frame

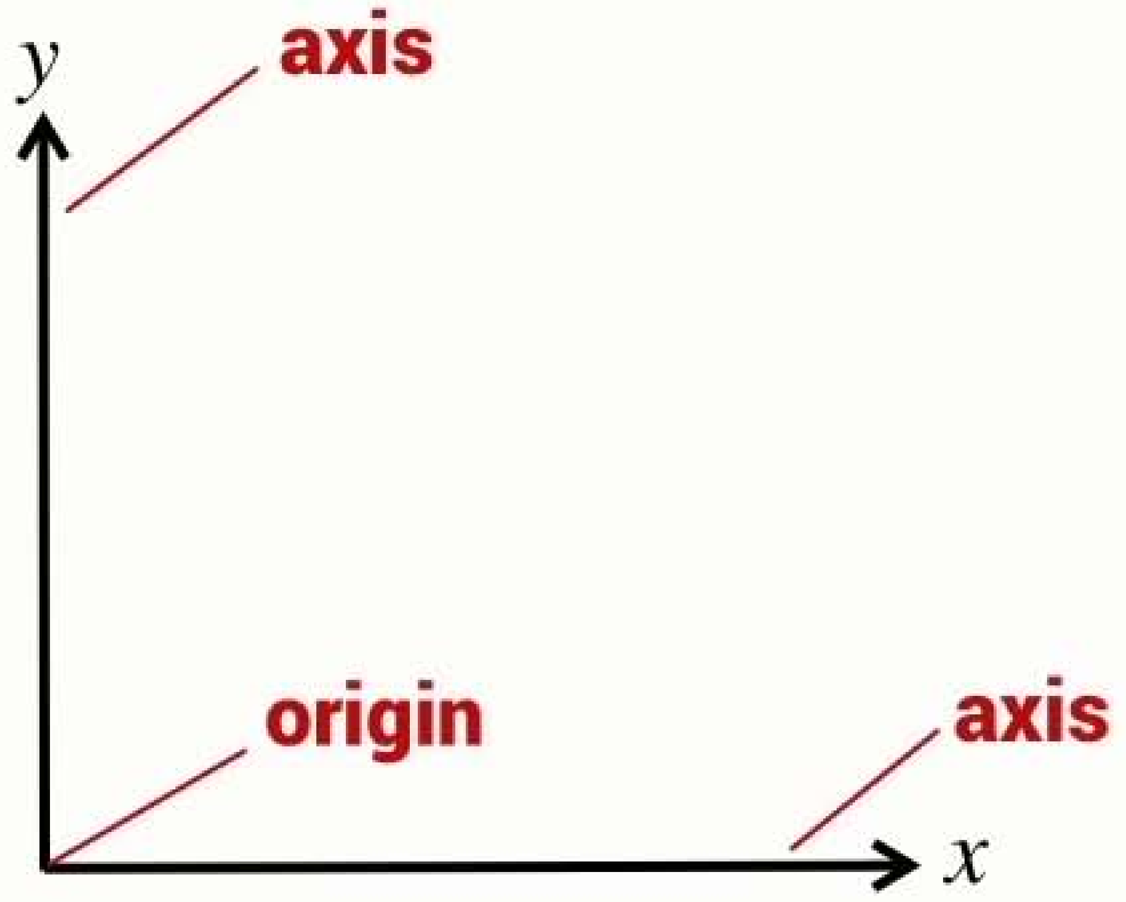


2D **coordinate frame**



- Two axes form a 2D **coordinate frame**

2D coordinate frame



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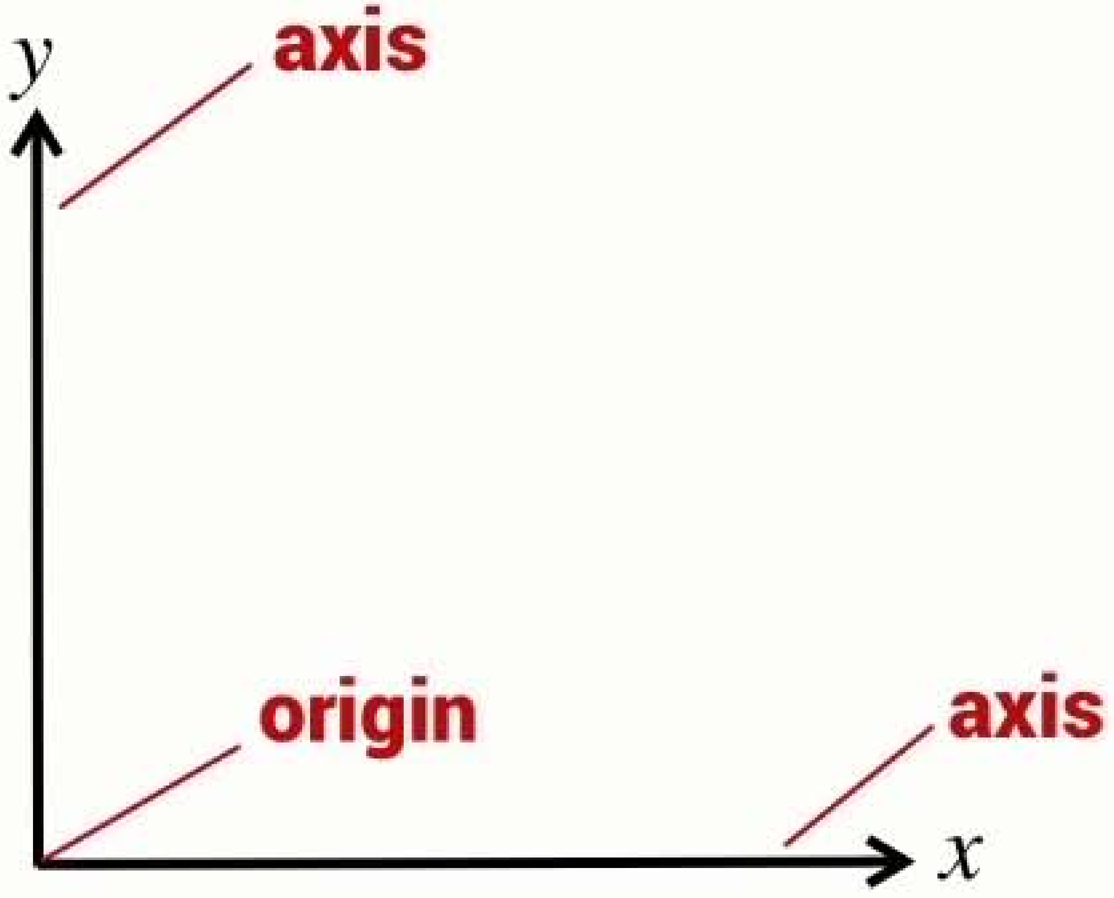
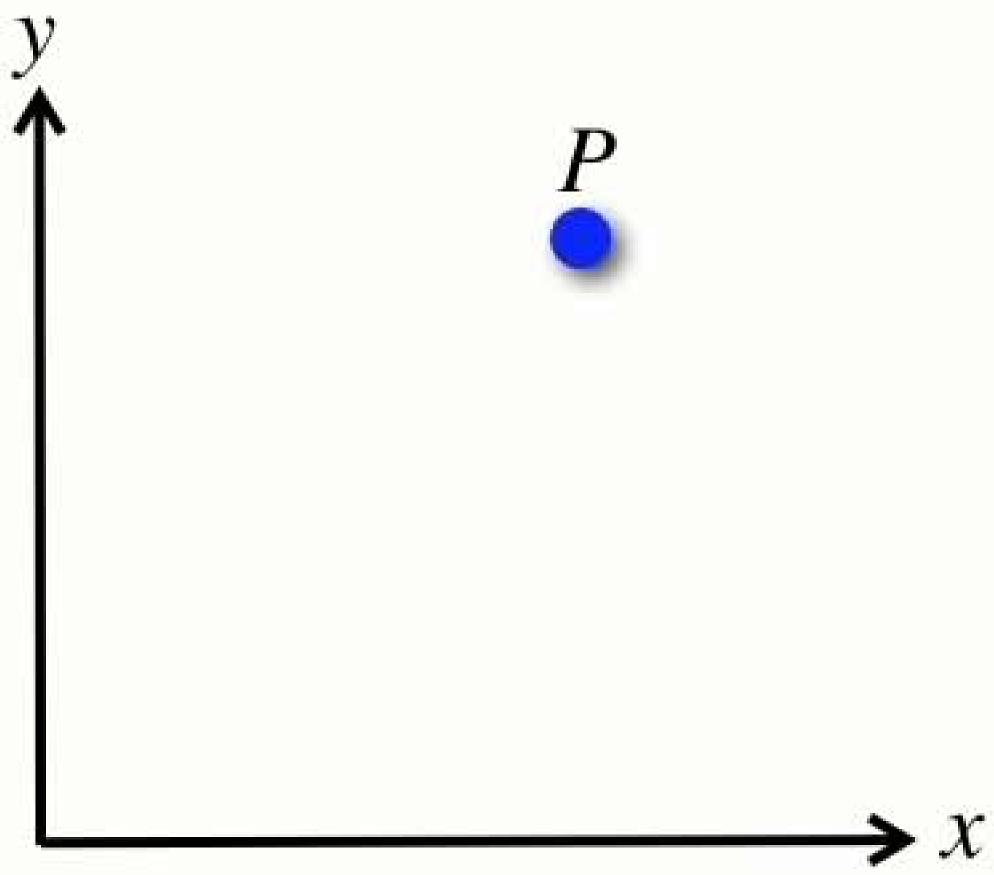
2D coordinate frame

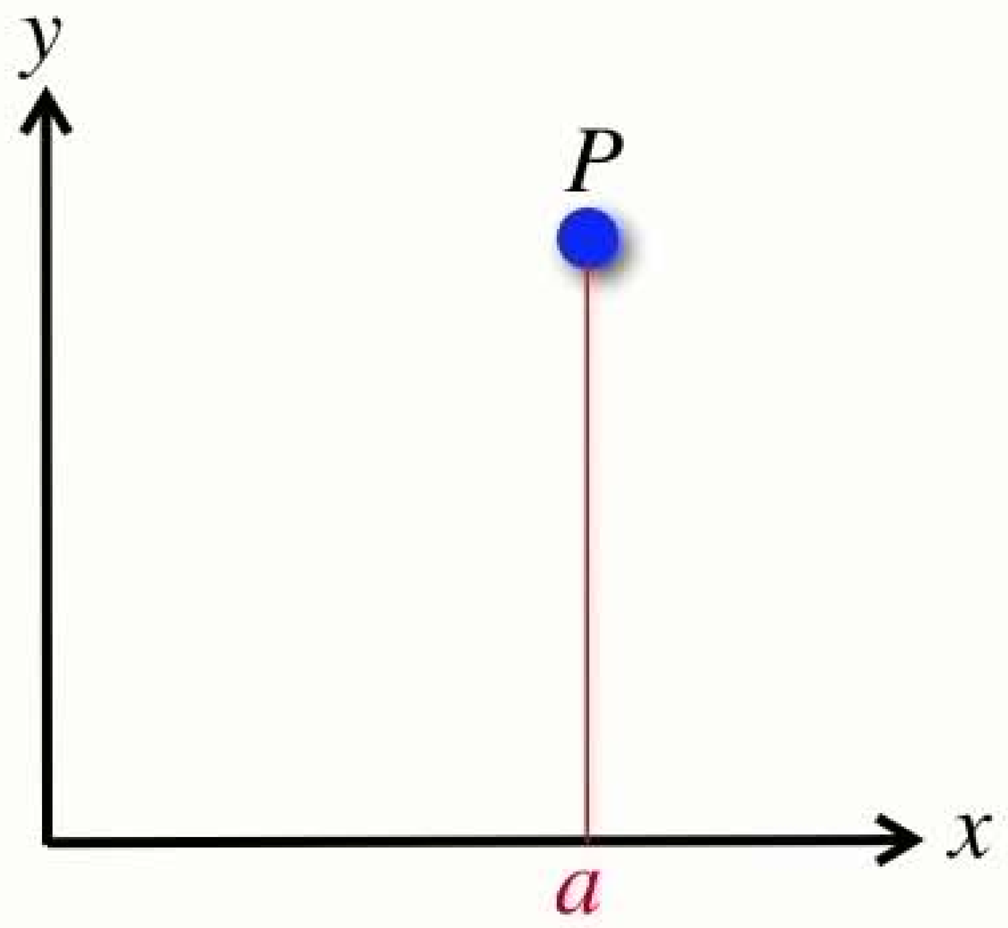


- Two axes form a 2D **coordinate frame**

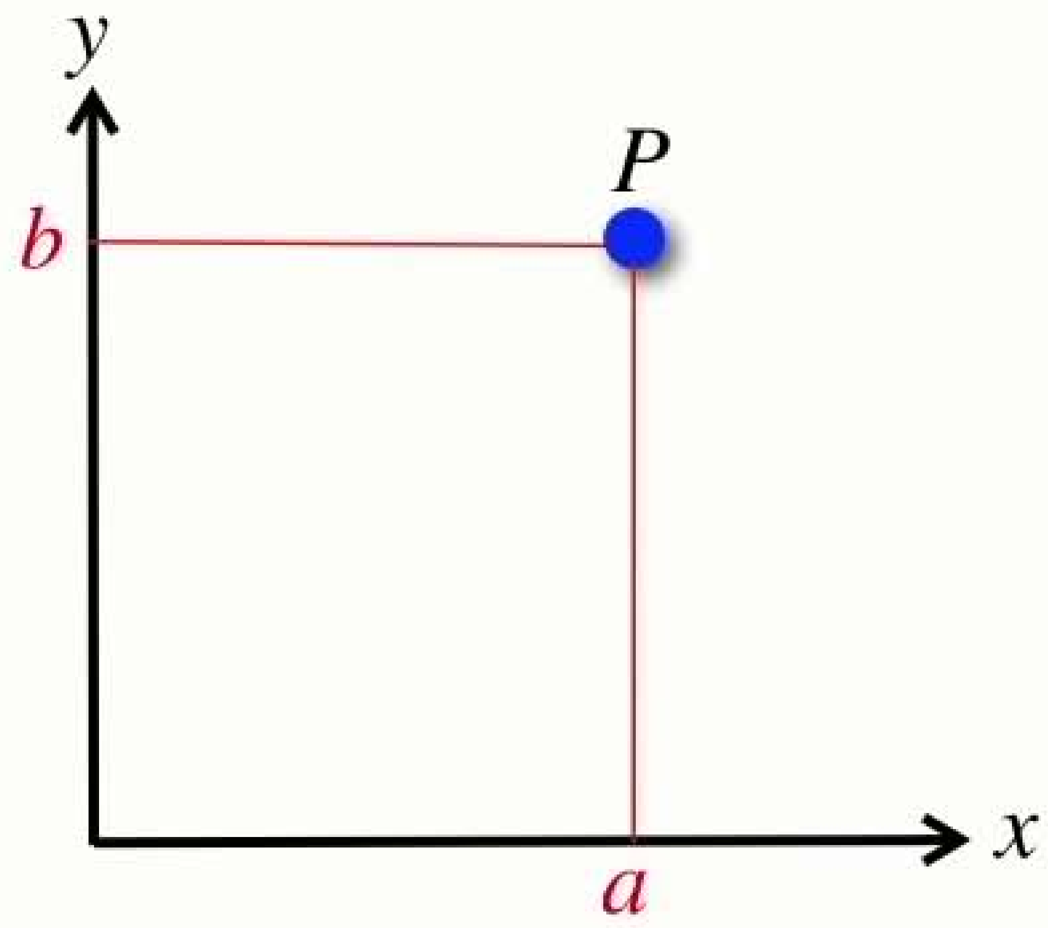
Points



Points

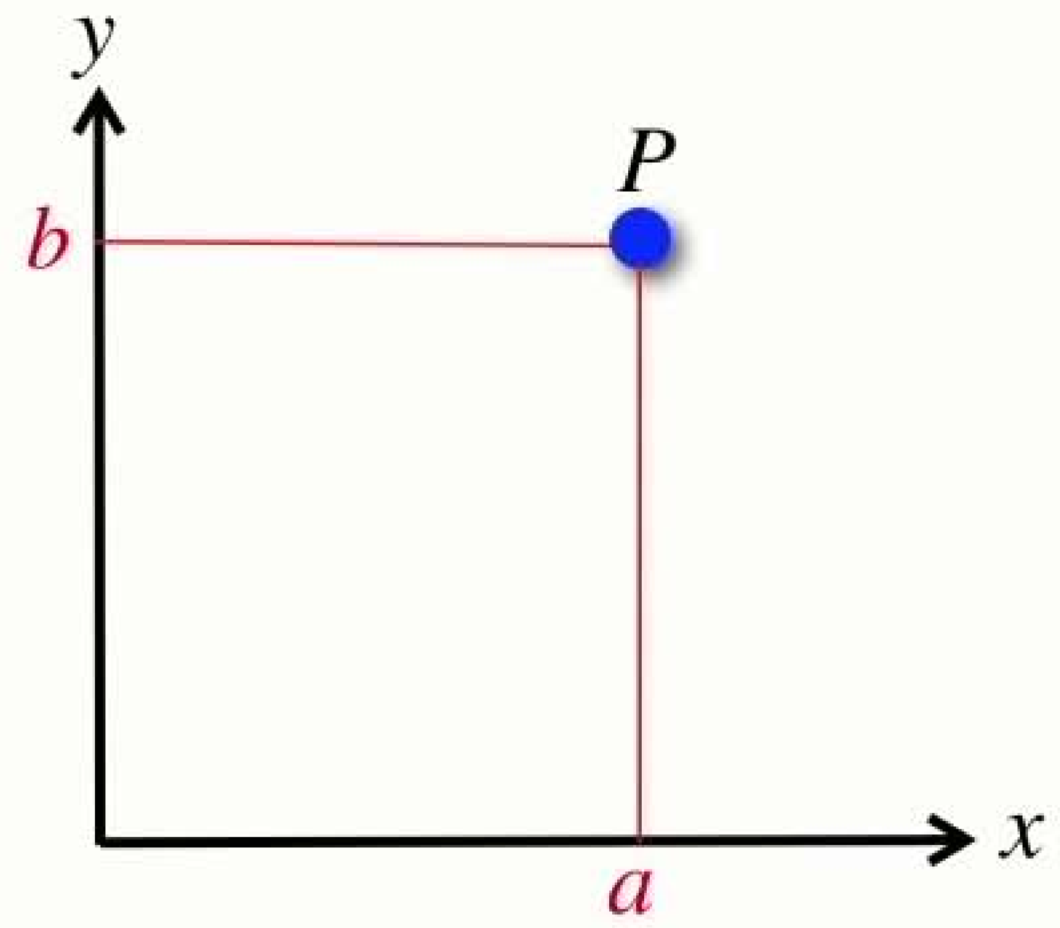


Points



$$P = (a, b)$$

Points



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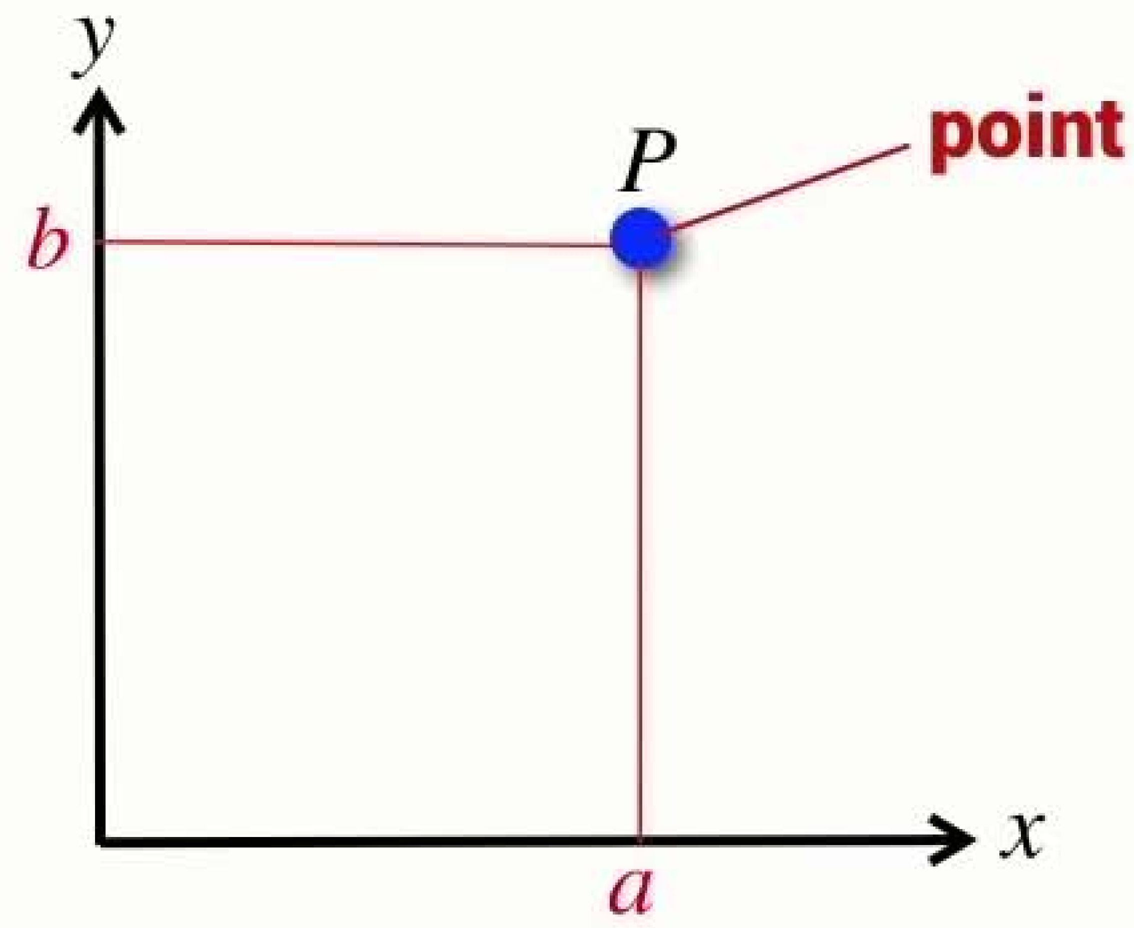
- A point is a location in n-dimensional space $P \in \mathbb{R}^n$

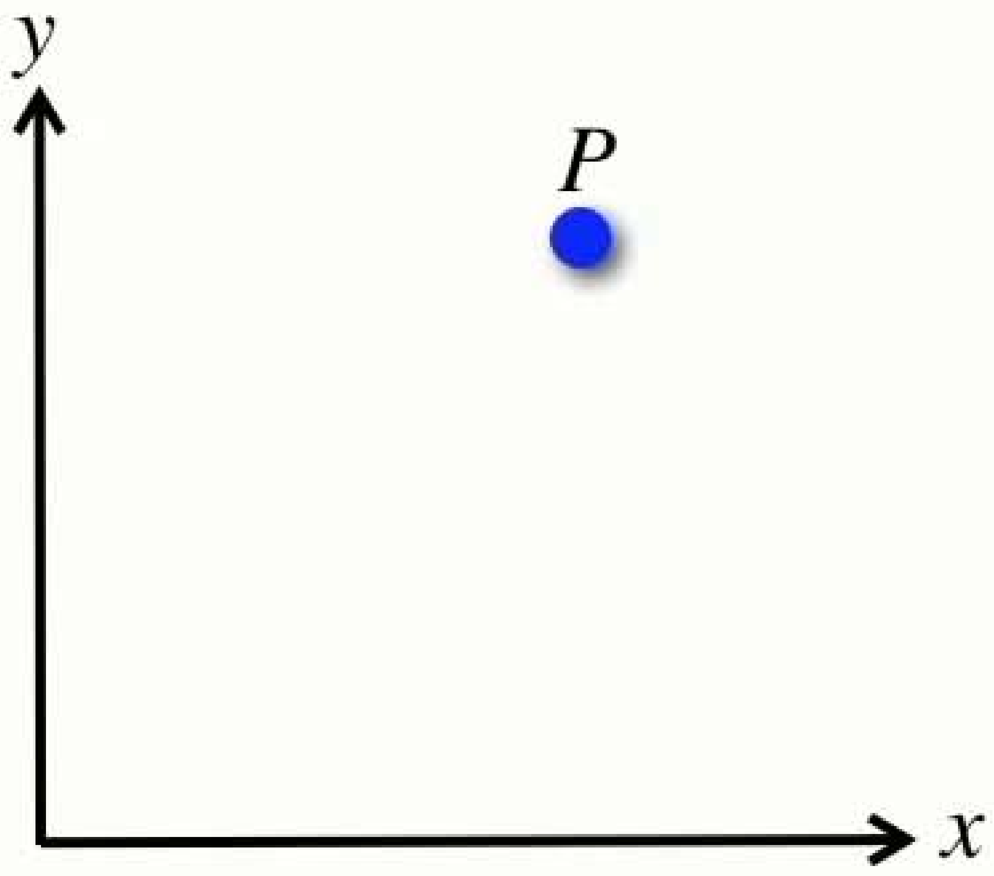
Points



$$P = (a, b)$$

- A point is a location in n-dimensional space $P \in \mathbb{R}^n$
- Represented by a Cartesian coordinate or an n-tuple $(x_1, x_2 \cdots x_n)$

Vectors



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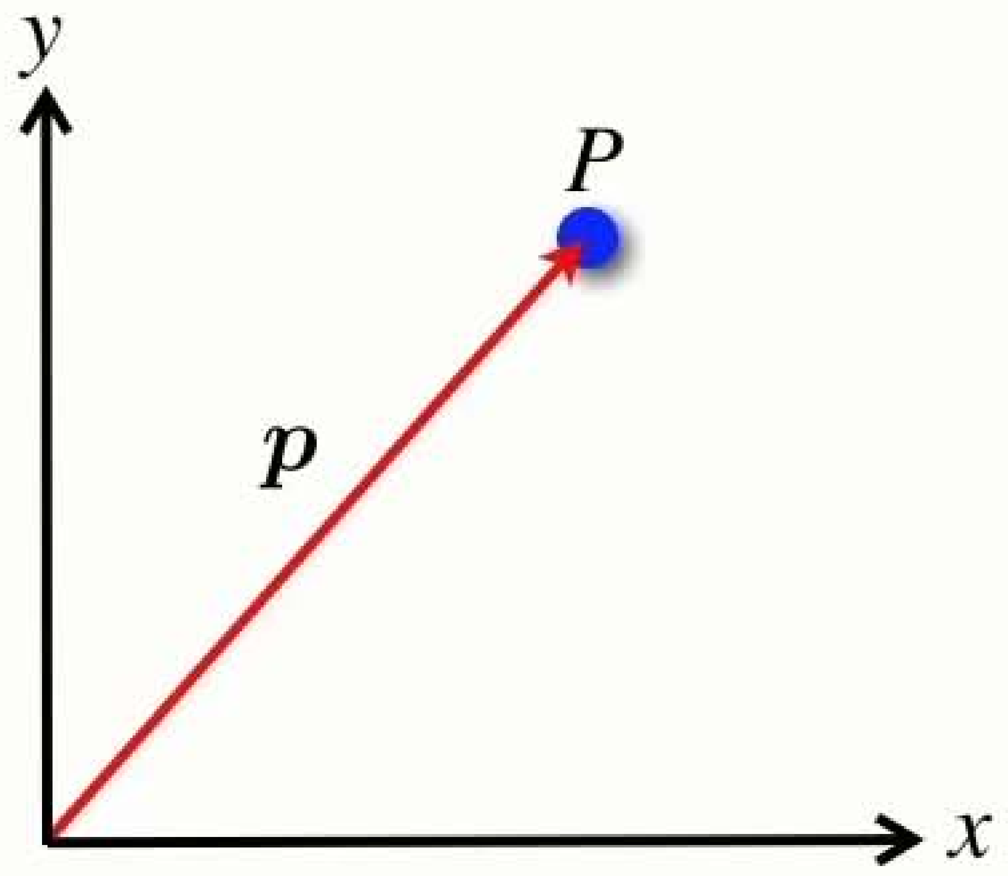

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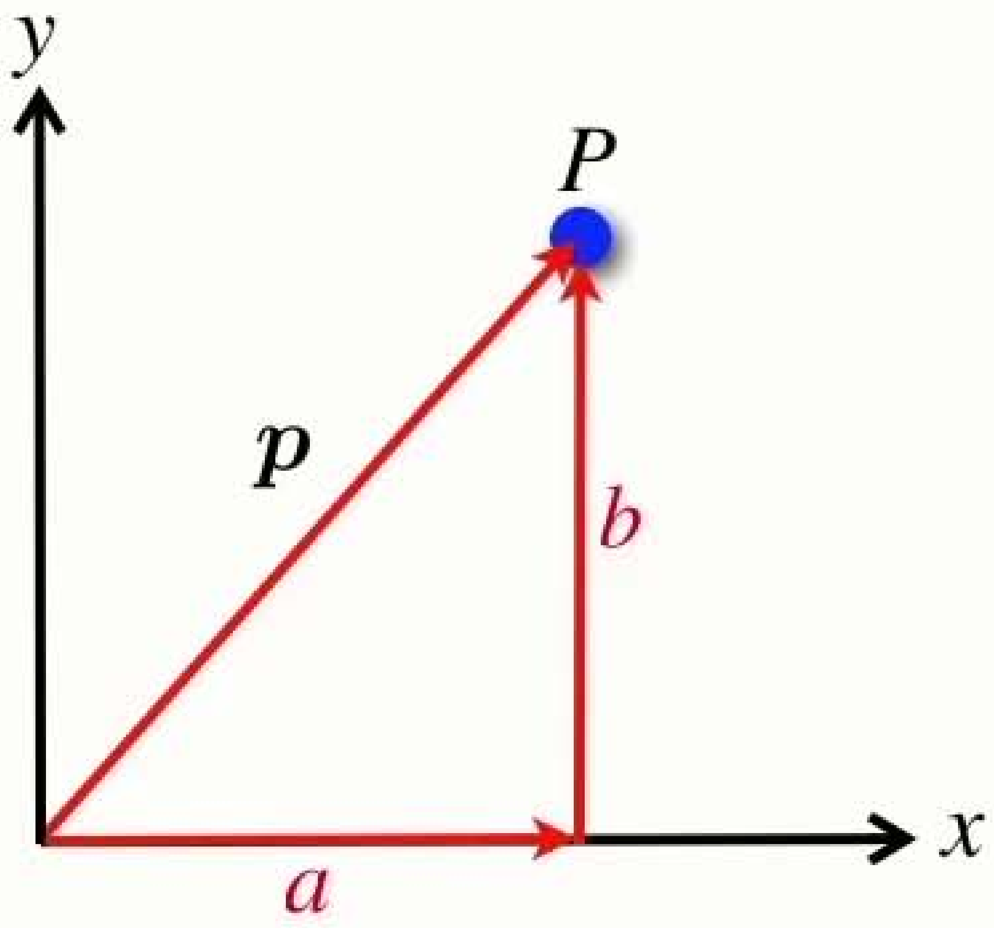
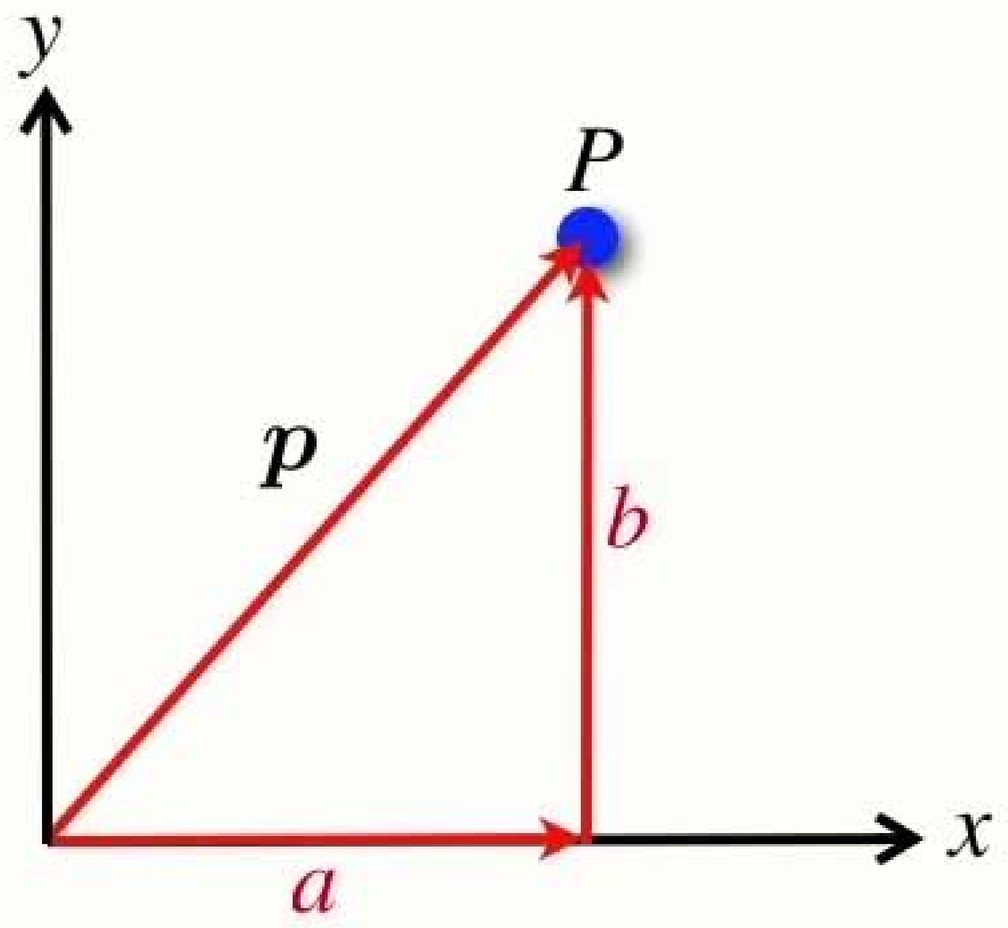


$$\boldsymbol{p} = \begin{bmatrix} a \\ b \end{bmatrix}$$

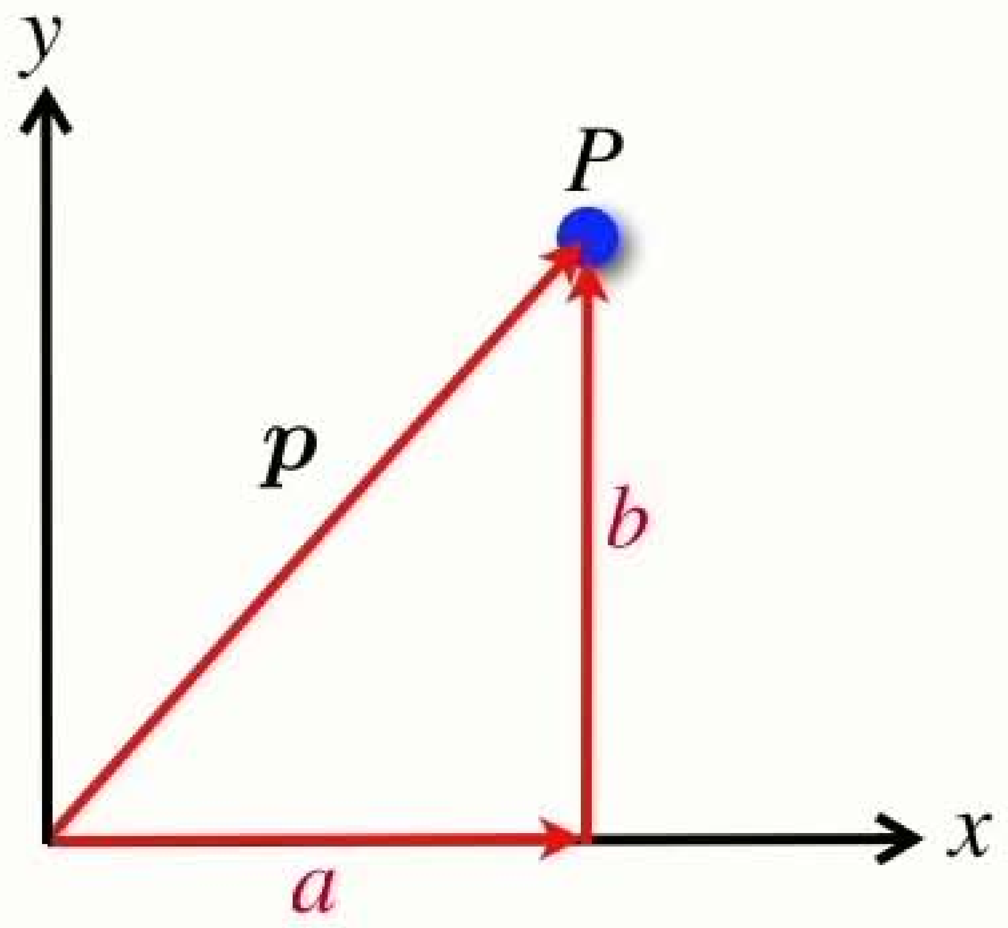
Vectors



$$\boldsymbol{p} = \begin{bmatrix} a \\ b \end{bmatrix}$$

- A vector is an element of an n-dimensional vector space $\boldsymbol{p} \in \mathbb{R}^n$

Vectors



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- A vector is an element of an n-dimensional vector space $\boldsymbol{p} \in \mathbb{R}^n$
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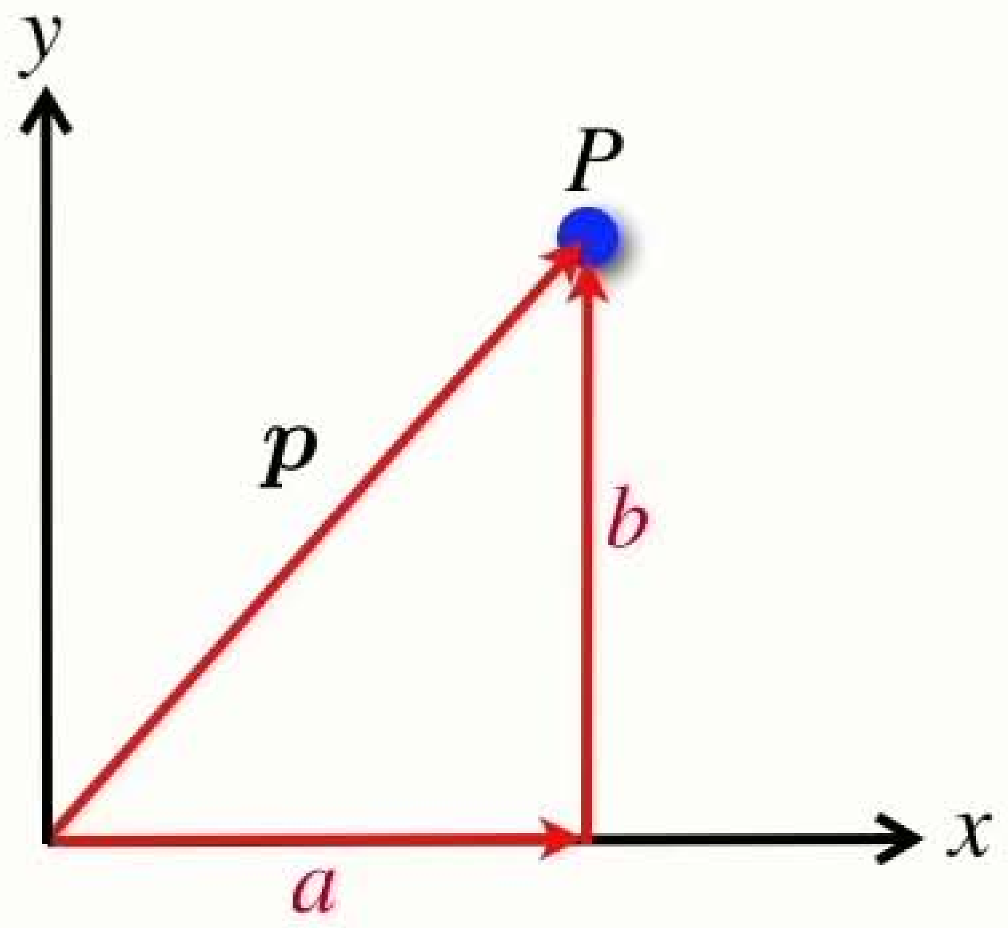
Vectors



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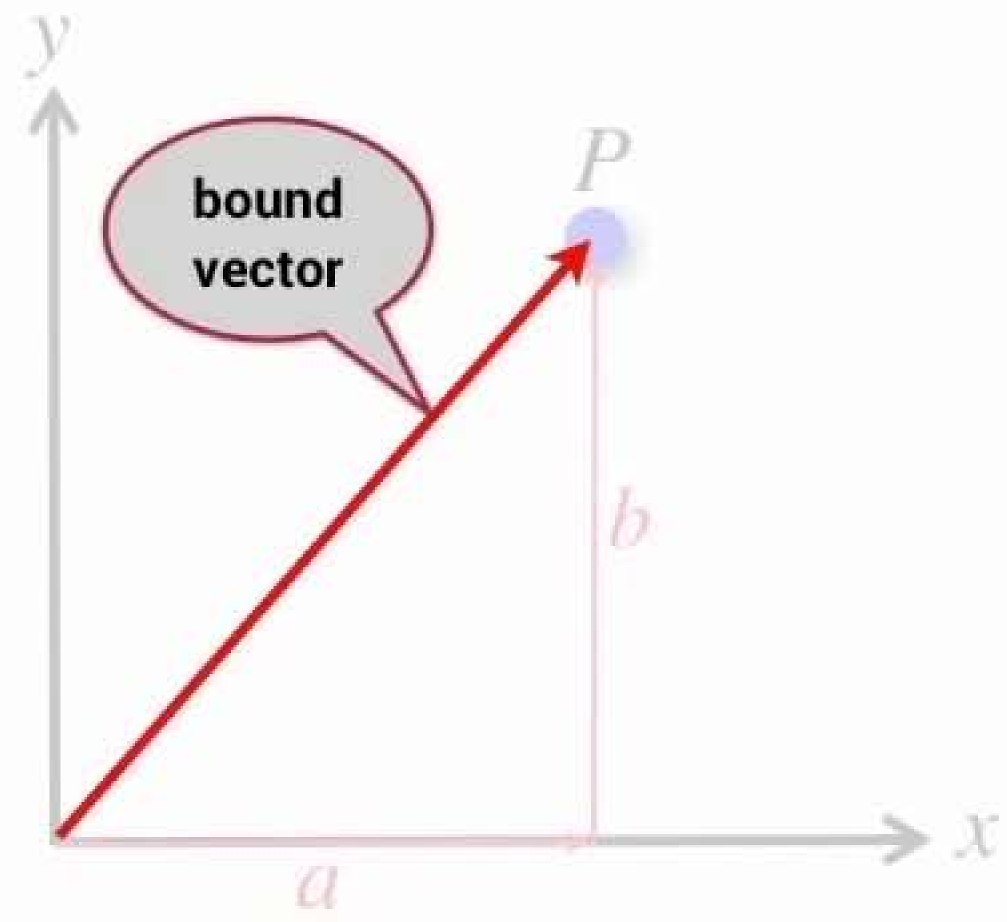
- Can be considered a relative displacement

Vectors



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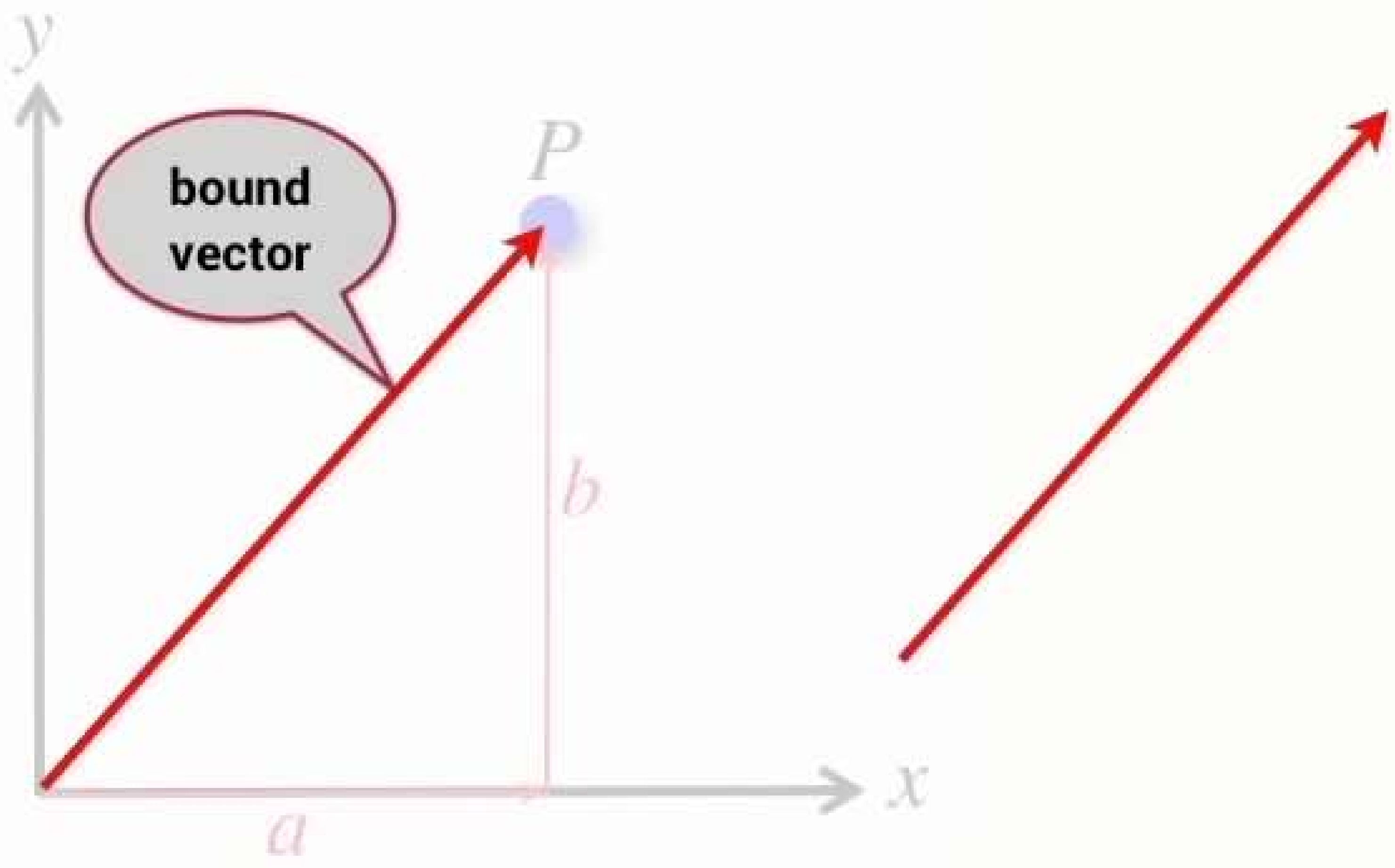
- Can be considered a relative displacement
- The vector has a particular starting point

Vectors



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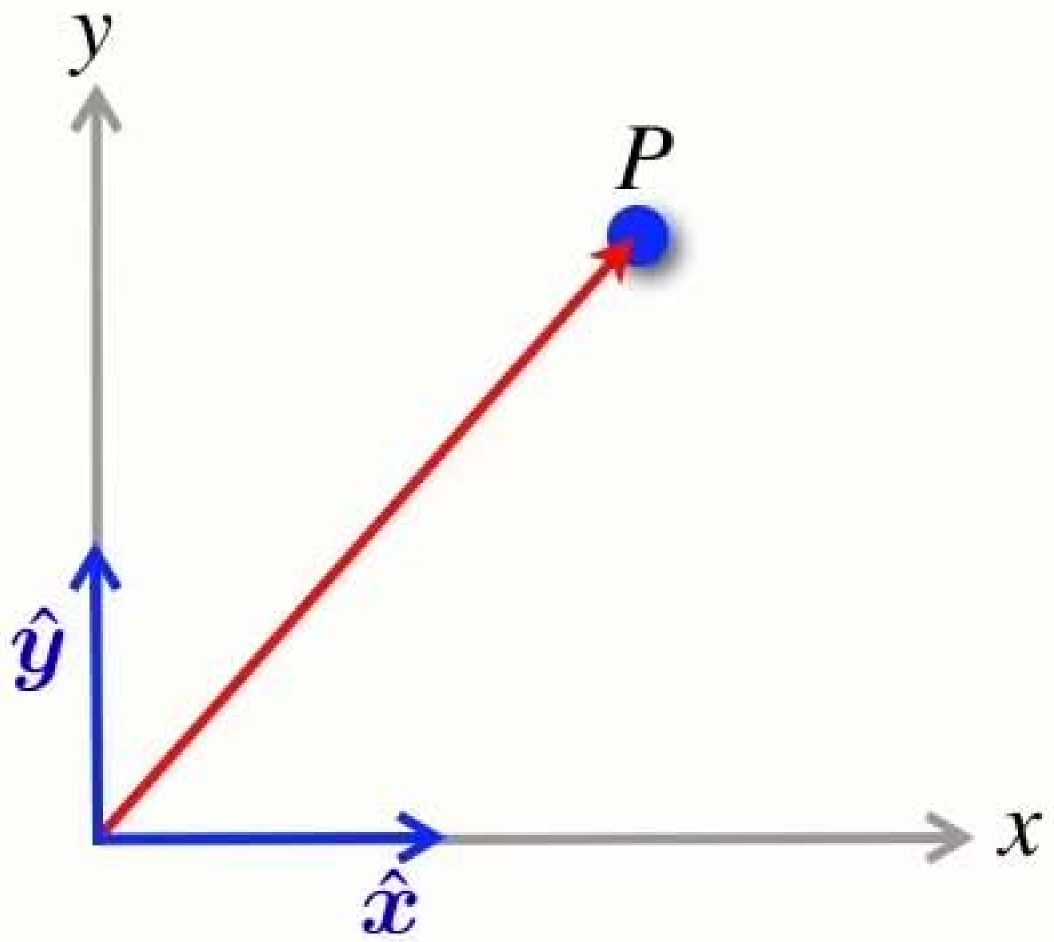

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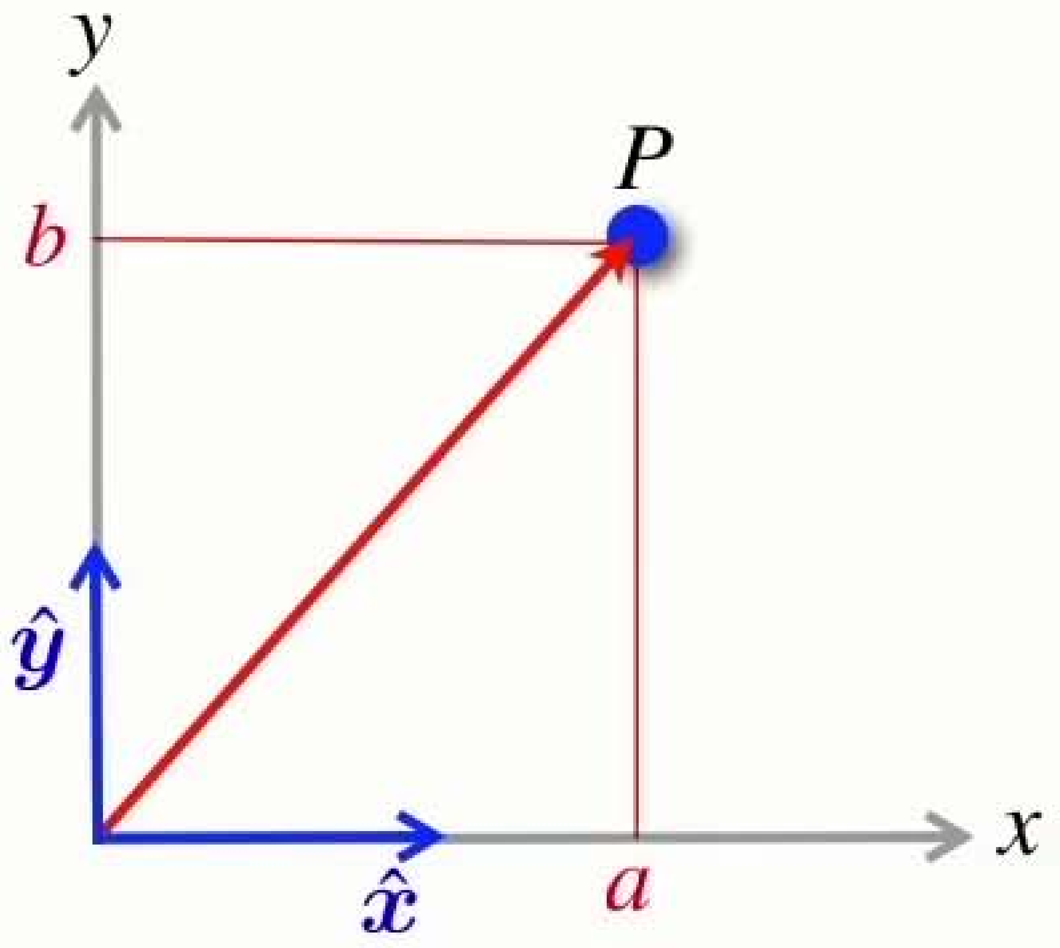

$$\boldsymbol{p} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\boldsymbol{p} = a\hat{\boldsymbol{x}} + b\hat{\boldsymbol{y}}$$

Points and vectors

- Points
  - Define a location
  - Cannot add or multiply points

Points and vectors

- Points
  - Define a location
  - Cannot add or multiply points
- Vectors
  - *Do not* define a location
    - specify how to get from one location to another
  - Can add & subtract vectors
  - Can multiply a vector by a scalar

Points and vectors

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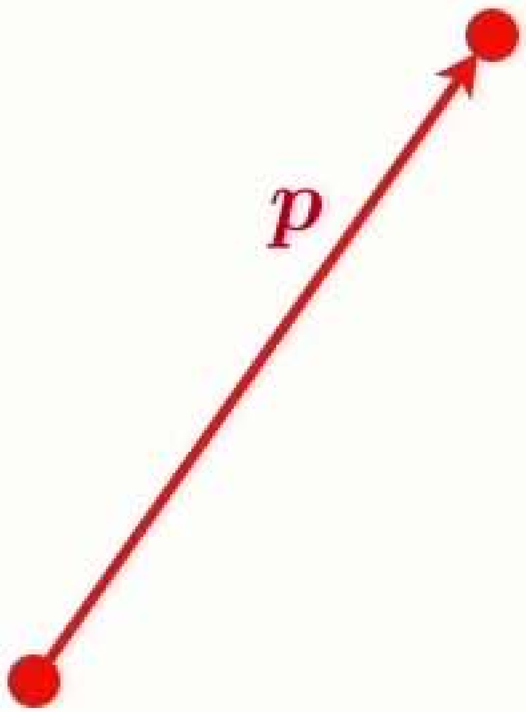
- The difference between two points is a vector



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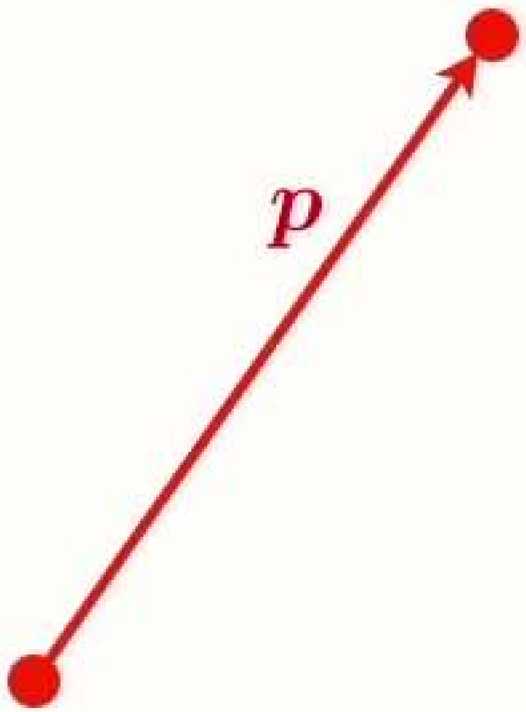

- The difference between two points is a vector
- A point is defined by a vector displacement from another point (typically the origin of a coordinate frame)

Points and vectors

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  - ➥ Cannot add or multiply points
- Vectors
  - ➥ *Do not* define a location
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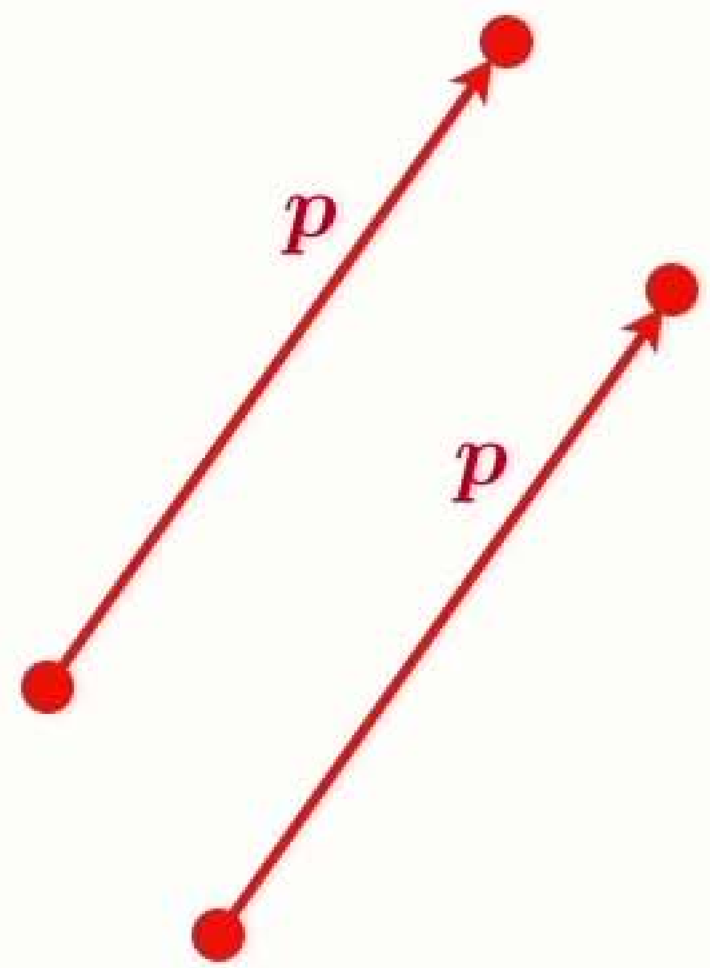
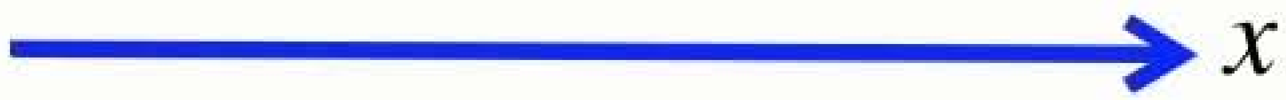
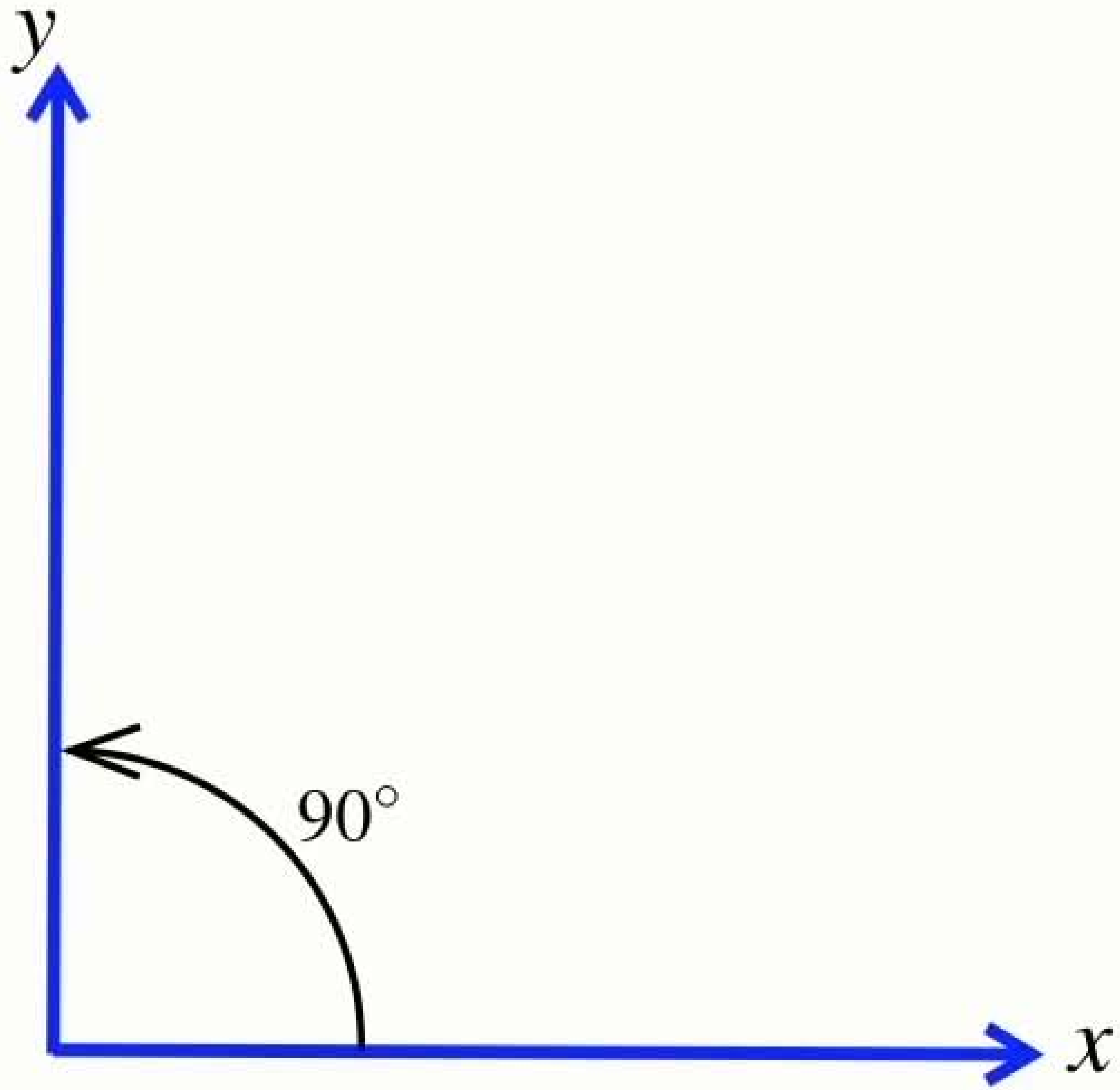


- The difference between two points is a vector
- A point is defined by a vector displacement from another point (typically the origin of a coordinate frame)
- Both can be represented by *n* real numbers

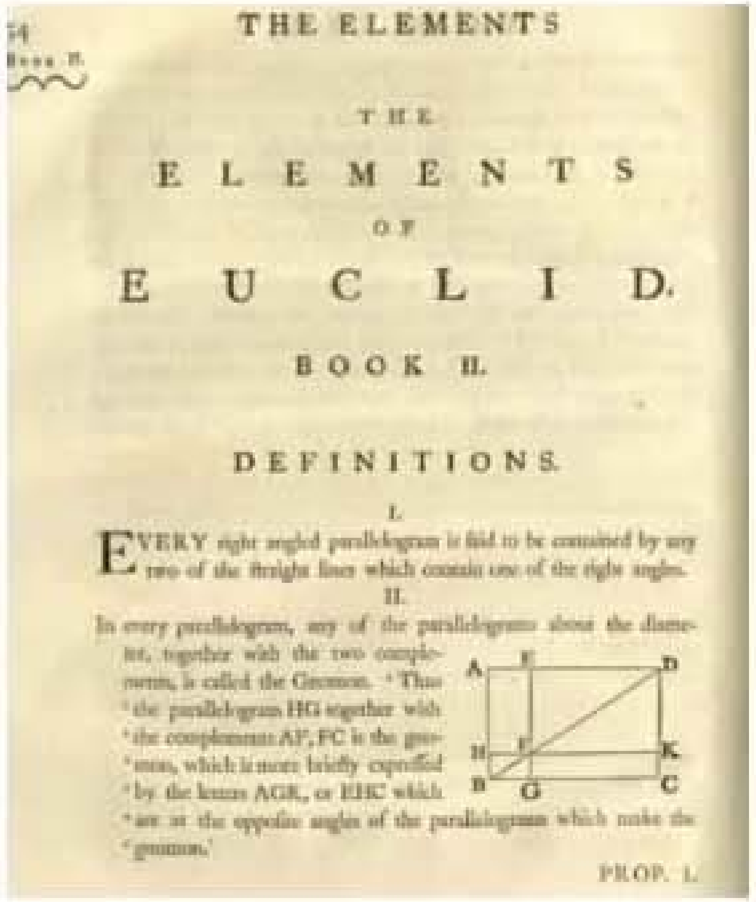
Right-handed **coordinate frame**



Right-handed coordinate frame



Euclidean geometry applies to planes



54
Book II.

THE ELEMENTS

THE
ELEMENTS
OF
EUCLID.

BOOK II.

DEFINITIONS.

I.

EVERY right angled parallelogram is said to be contained by any two of the straight lines which contain one of the right angles.

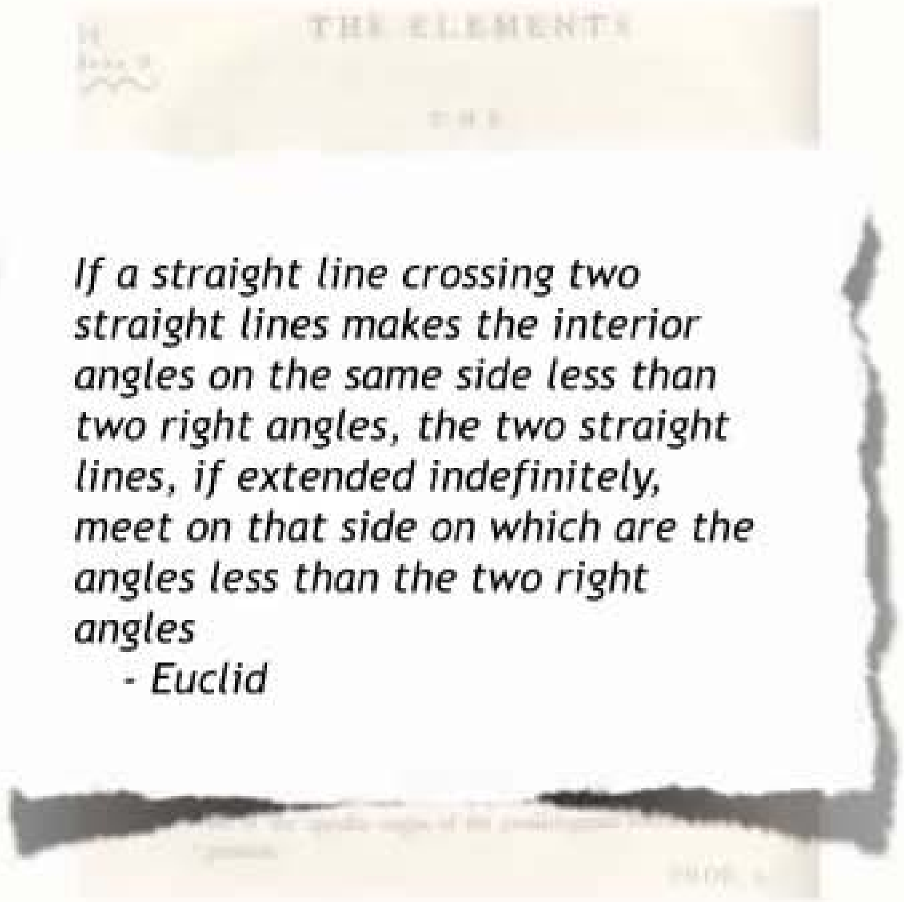
II.

In every parallelogram, any of the parallelograms about the diameter, together with the two complements, is called the Gnomon. 'Thus 'the parallelogram HG together with 'the complements AF, FC is the gnomon, which is more briefly expressed 'by the letters AGK, or EHC which 'are at the opposite angles of the parallelograms which make the 'gnomon.'

PROP. I.

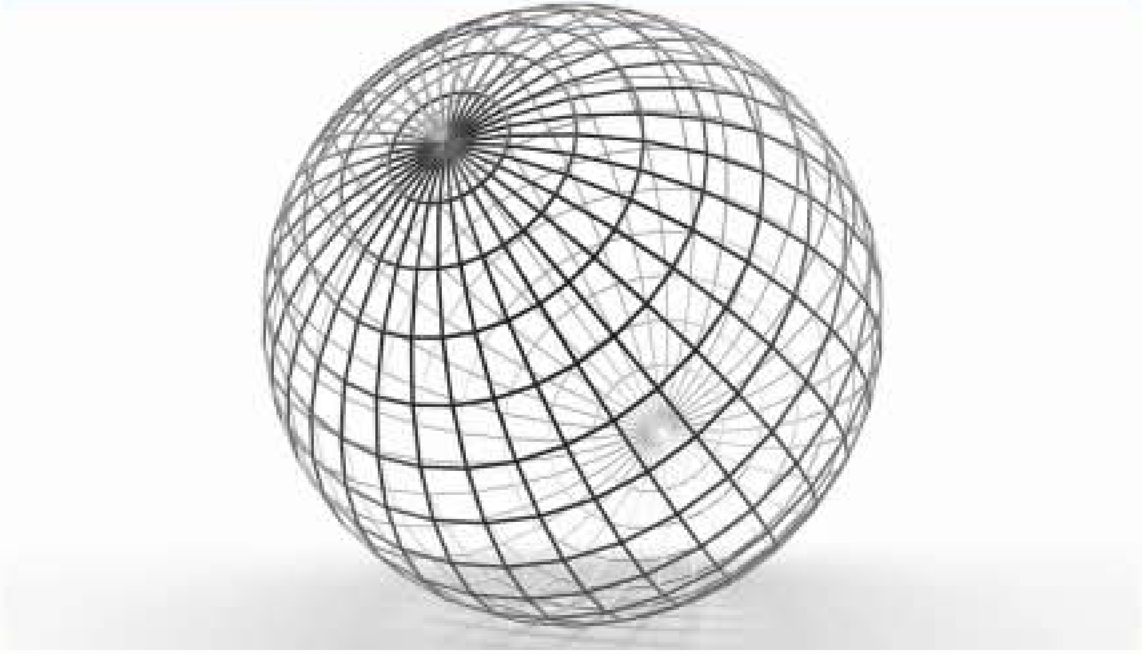
Euclidean geometry applies to planes

- Euclid's geometry (postulate 5)
  - ➥ parallel lines never intersect



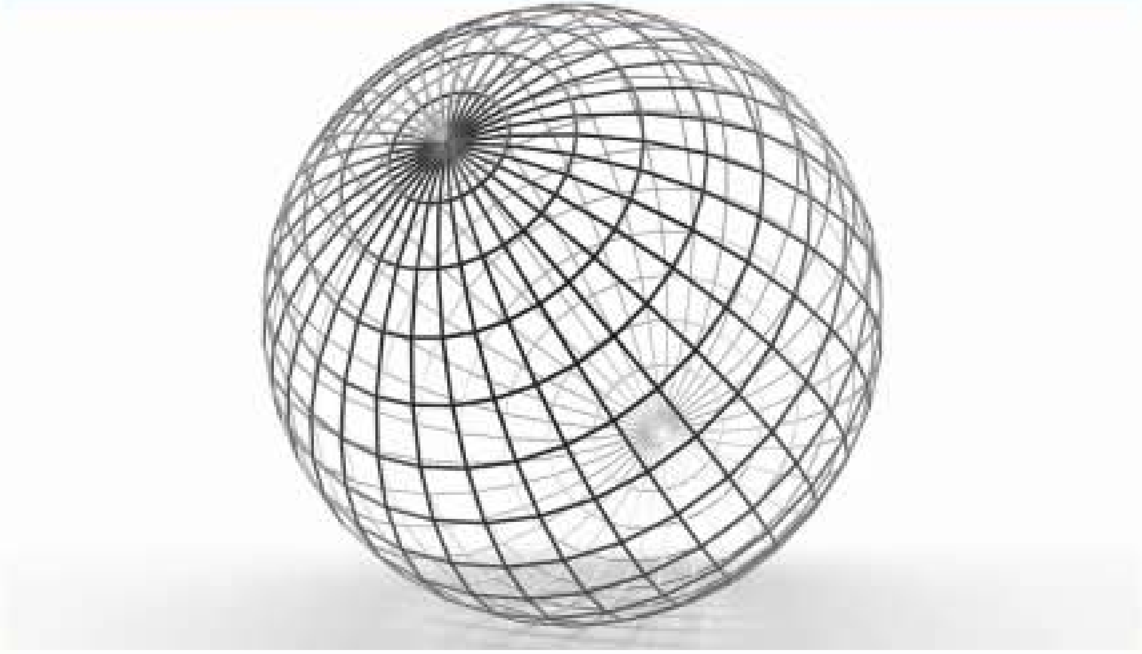
Euclidean geometry **applies to planes**

- Euclid's geometry (postulate 5)
  - ➥ parallel lines never intersect
- On the surface of our planet, parallel lines at the equator will intersect at the poles



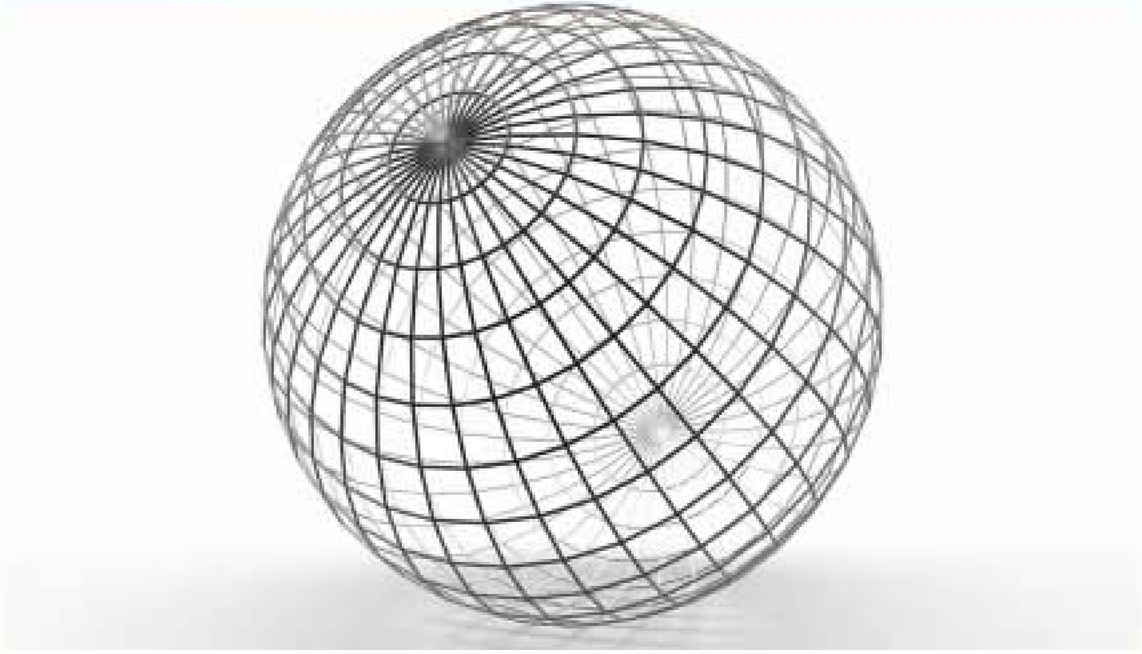
Euclidean geometry applies to planes

- Euclid's geometry (postulate 5)
  - ➥ parallel lines never intersect
- On the surface of our planet, parallel lines at the equator will intersect at the poles
- Euclidean geometry does not apply
  - ➥ it is a non-Euclidean geometry



Euclidean geometry applies to planes

- Euclid's geometry (postulate 5)
  - ➥ parallel lines never intersect
- On the surface of our planet, parallel lines at the equator will intersect at the poles
- Euclidean geometry does not apply
  - ➥ it is a non-Euclidean geometry
- Euclidean geometry is a good approximation over small areas



Section 2 Position and pose

Pose

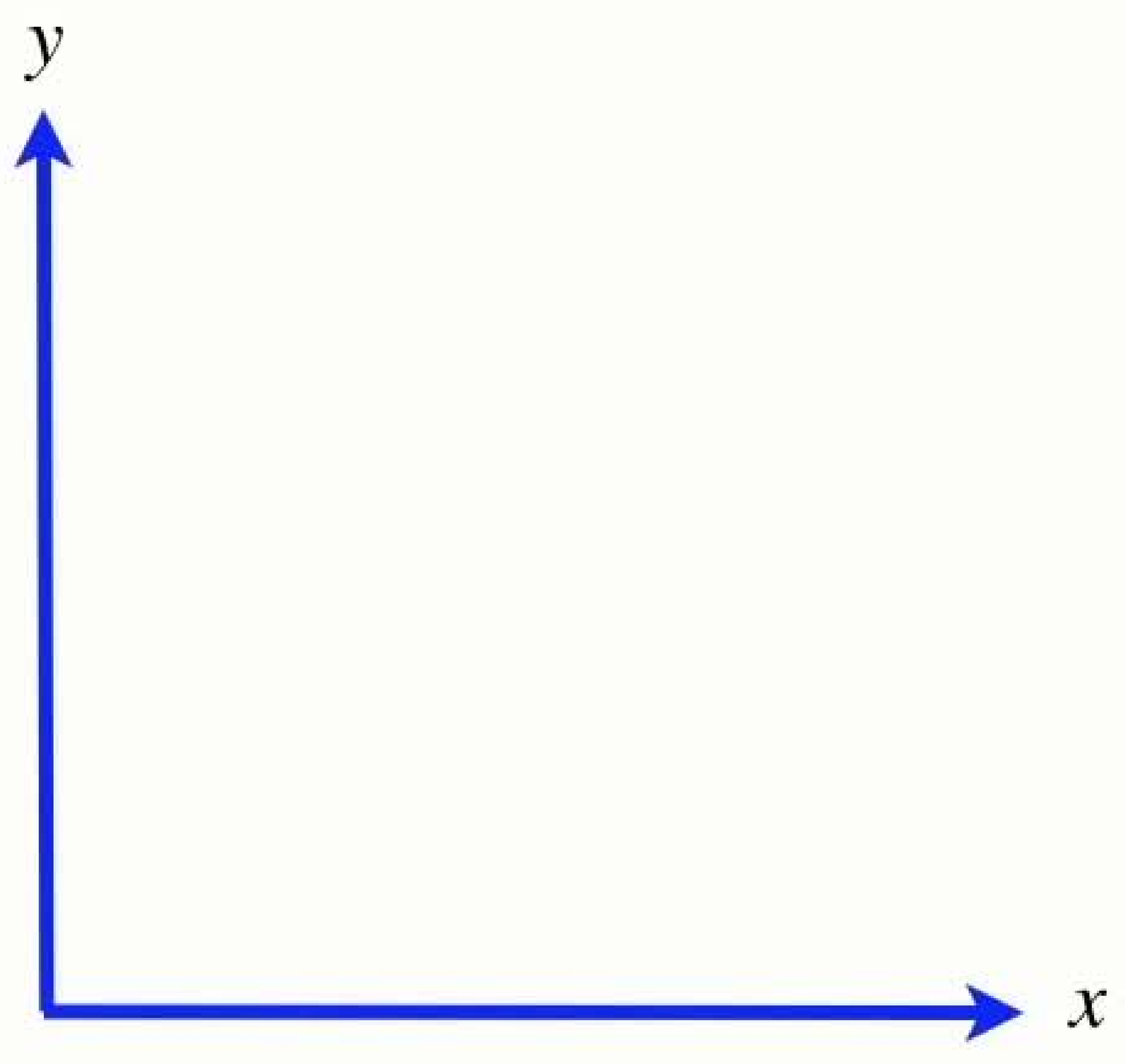
- It has 3 parameters:

$$\xi$$

pronounced ksi

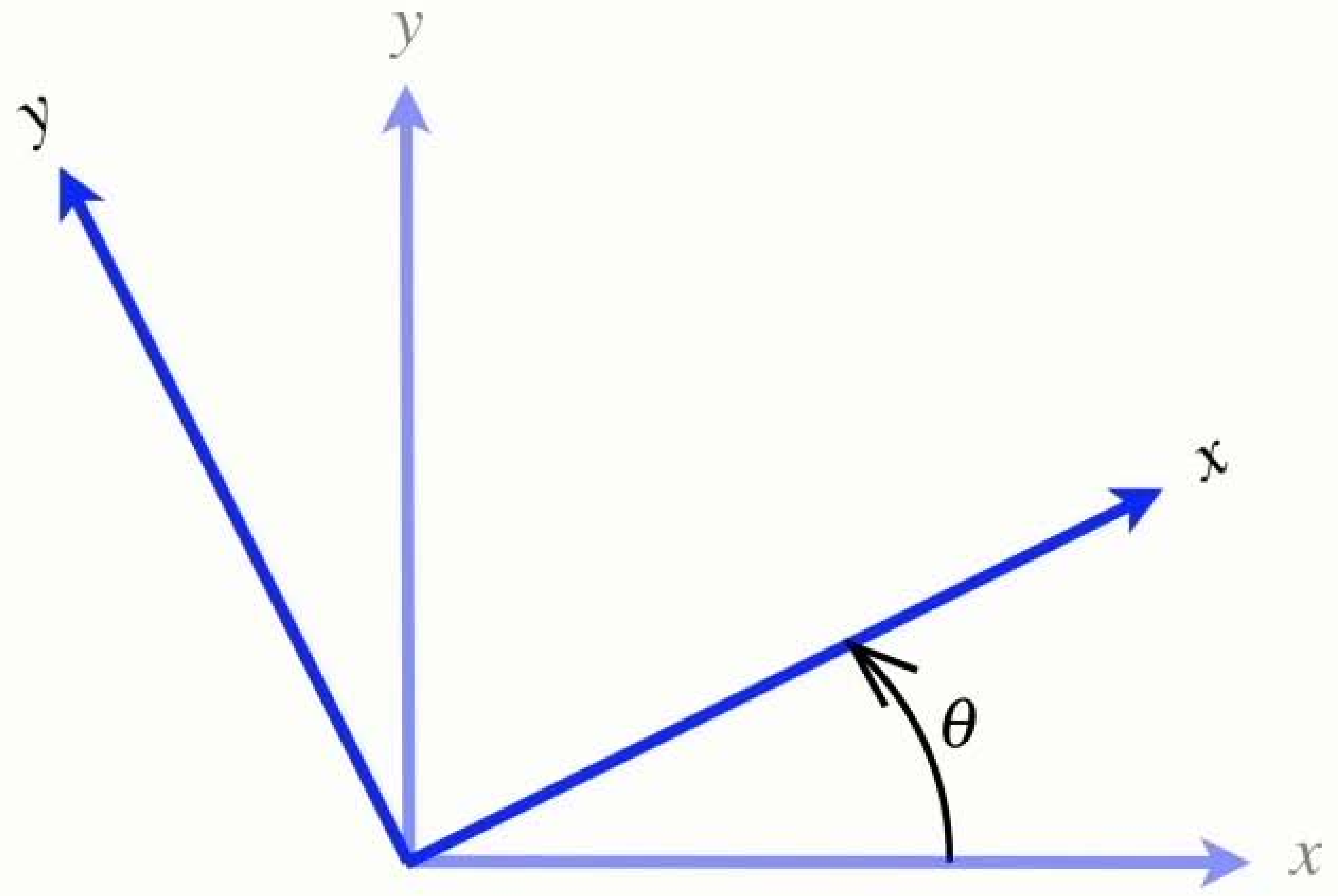
$$(x, y, \theta)$$

Coordinate frame rotation convention



- Angles increase positively in the anti-clockwise direction

Coordinate frame rotation convention



- Angles increase positively in the anti-clockwise direction

Pose

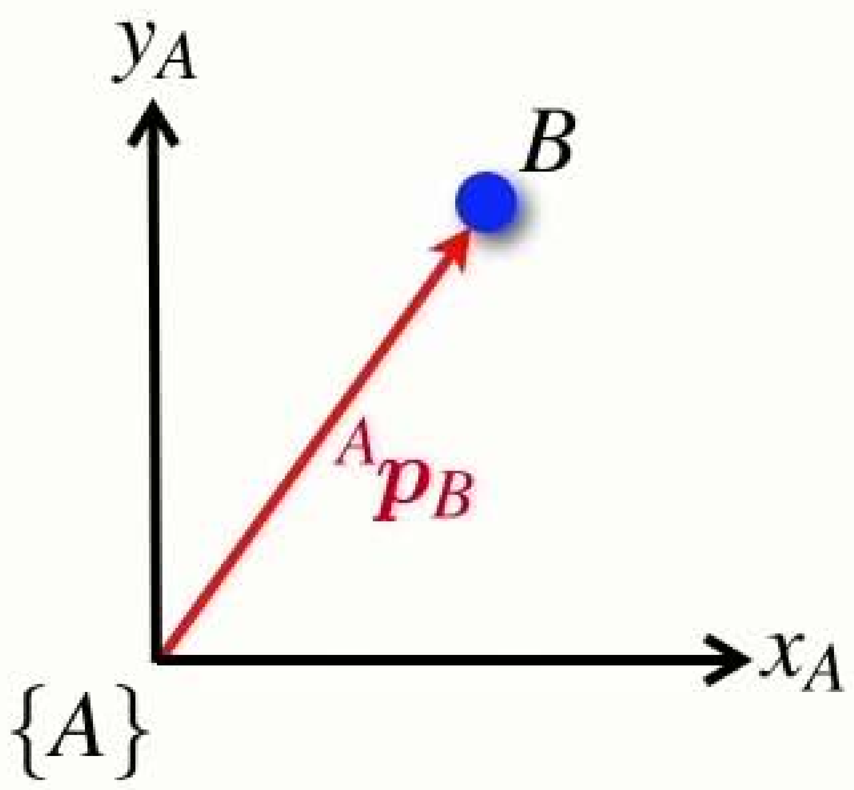
- It has 3 parameters:
- Think of it as just a move (x, y) and a twist (θ)

$$\xi$$

$$(x, y, \theta)$$

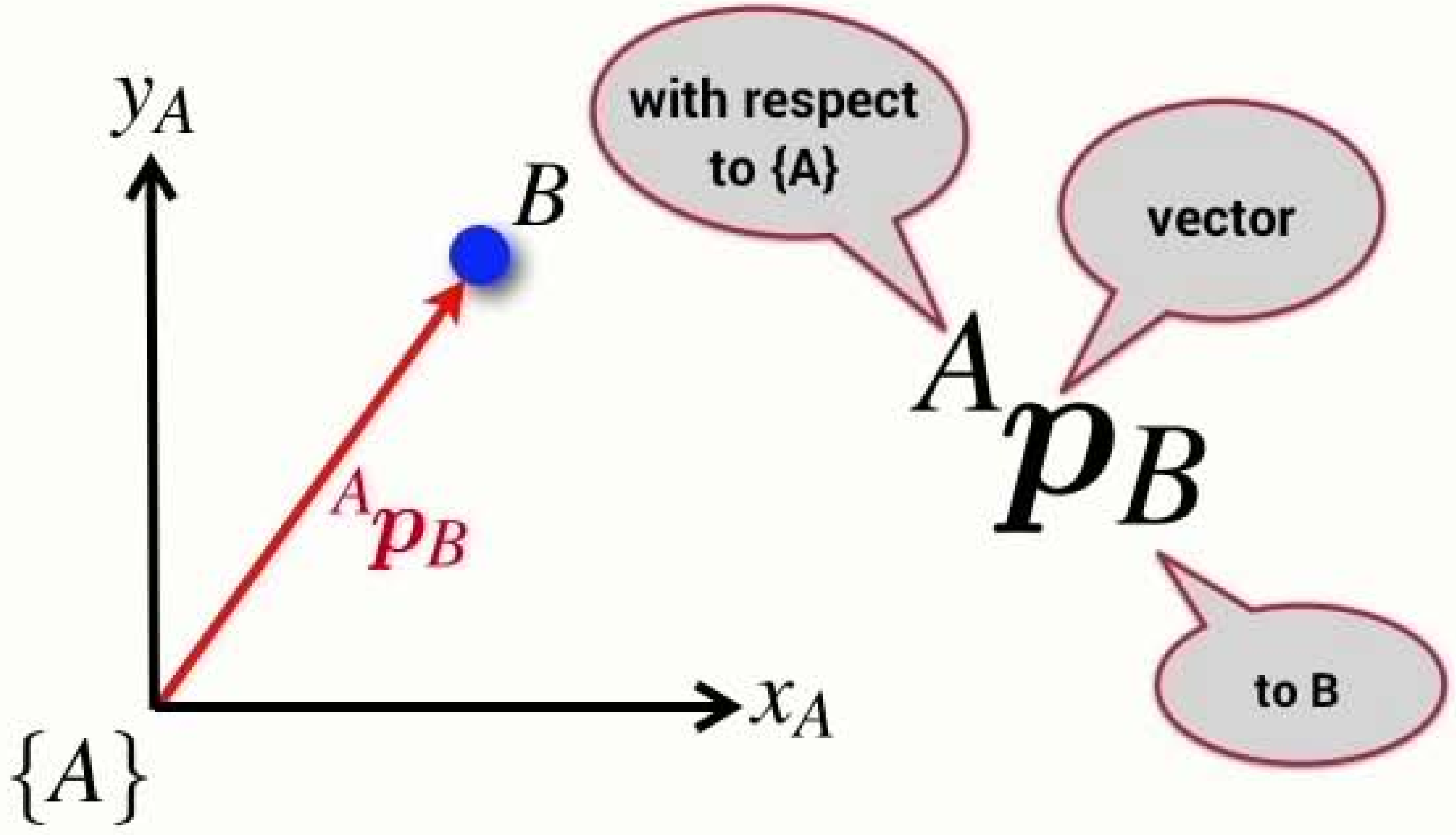
Summary

- A point is described by a vector with respect to a coordinate frame



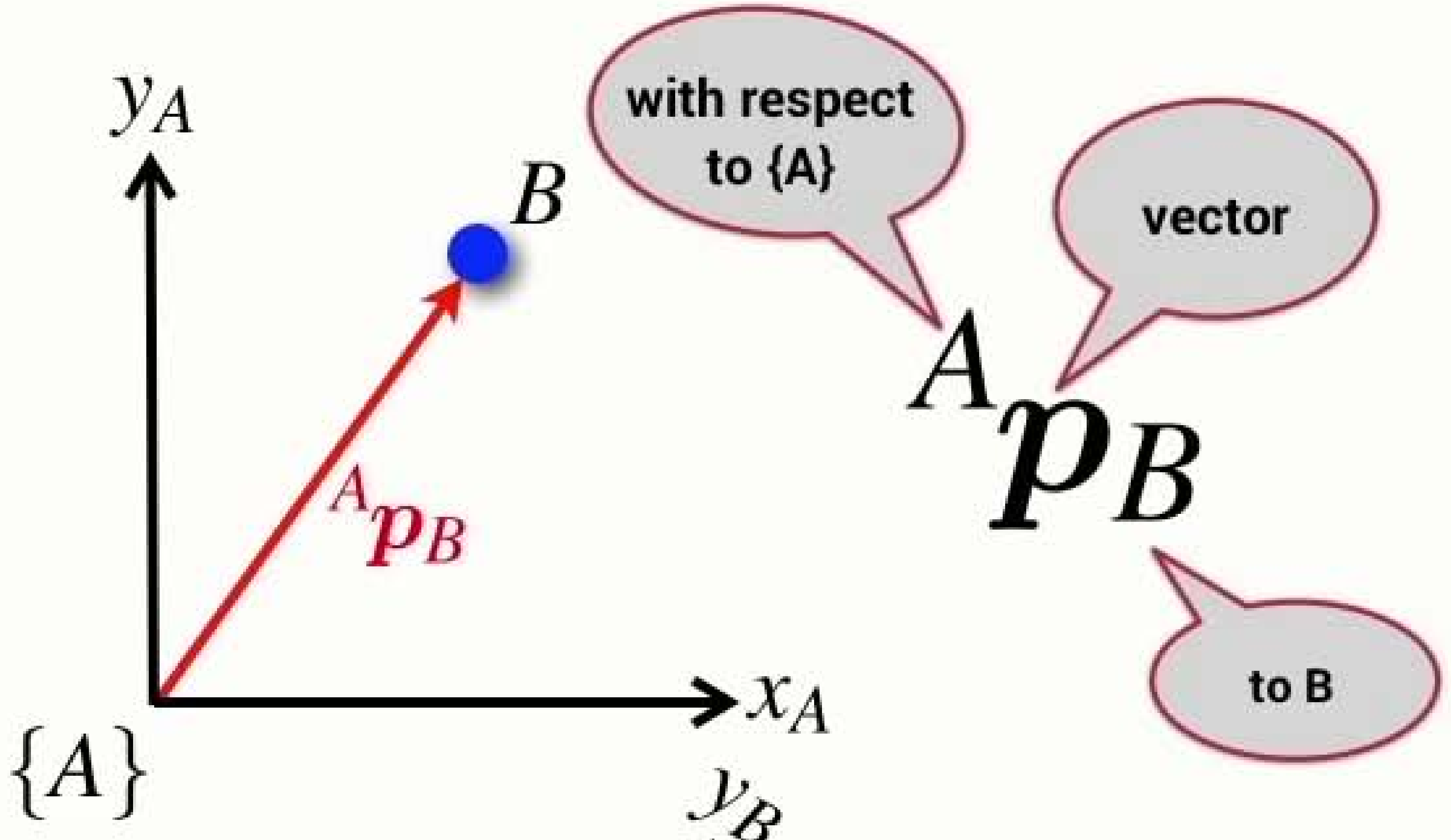
Summary

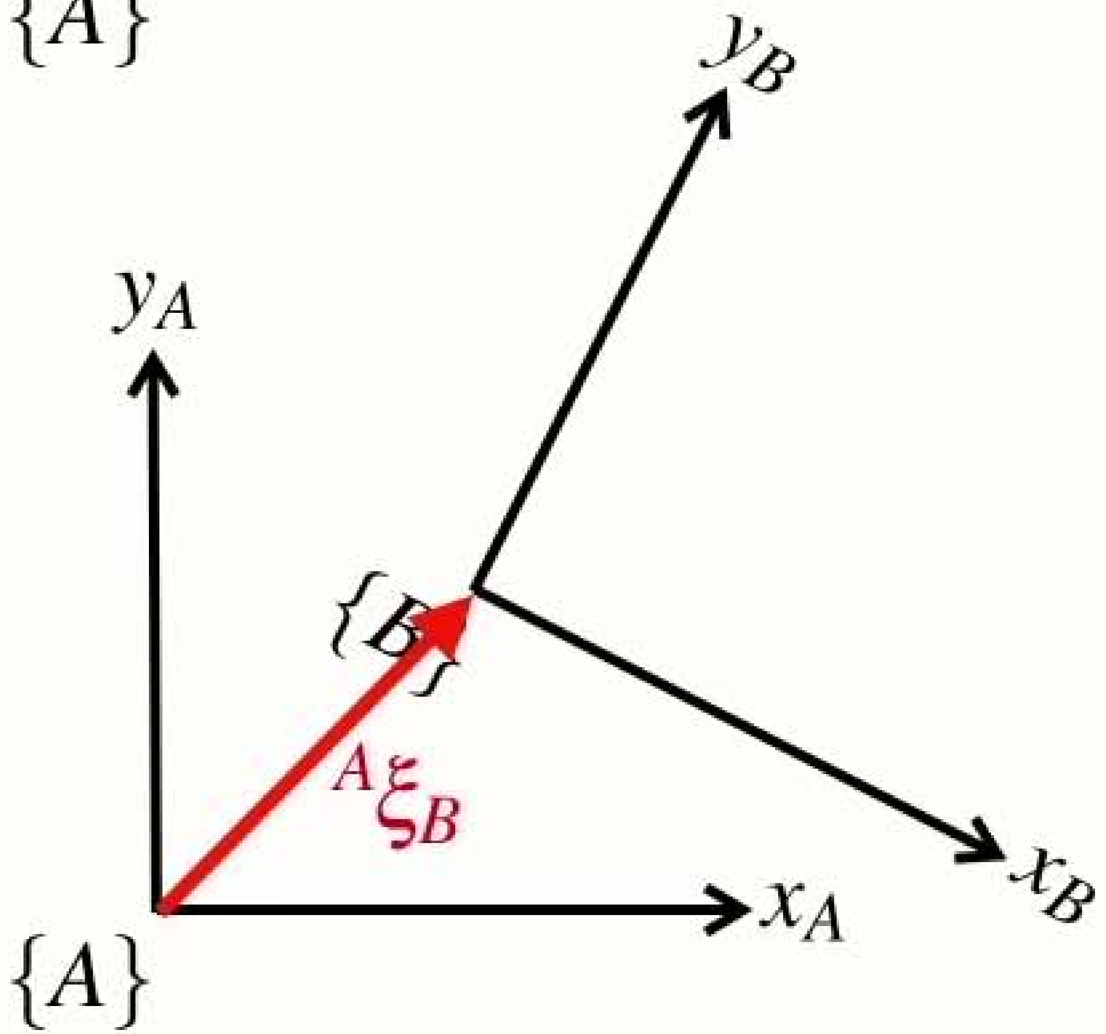
- A point is described by a vector with respect to a coordinate frame



Summary

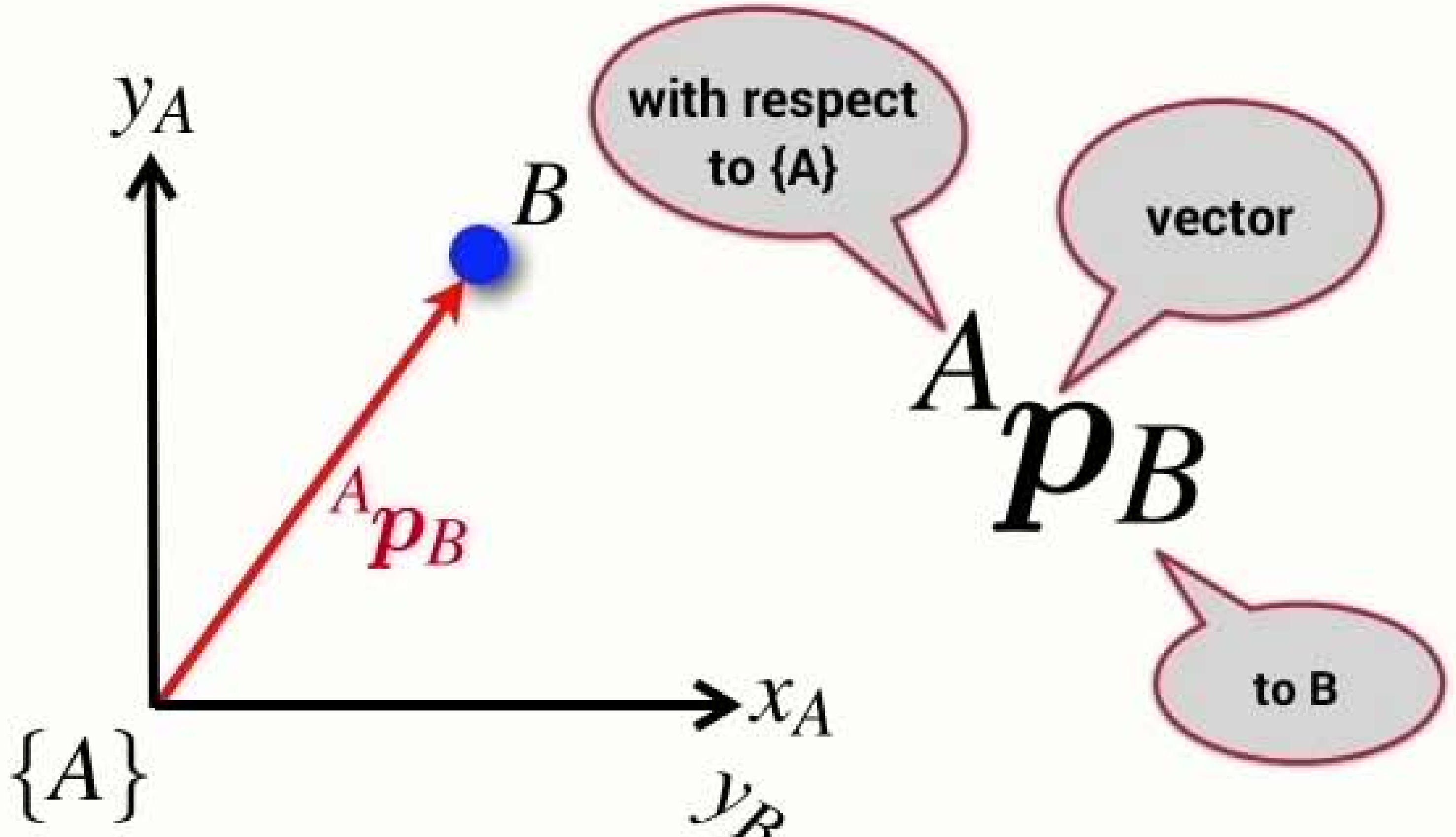
- A point is described by a vector with respect to a coordinate frame
- The pose of a coordinate frame can be described with respect to another coordinate frame

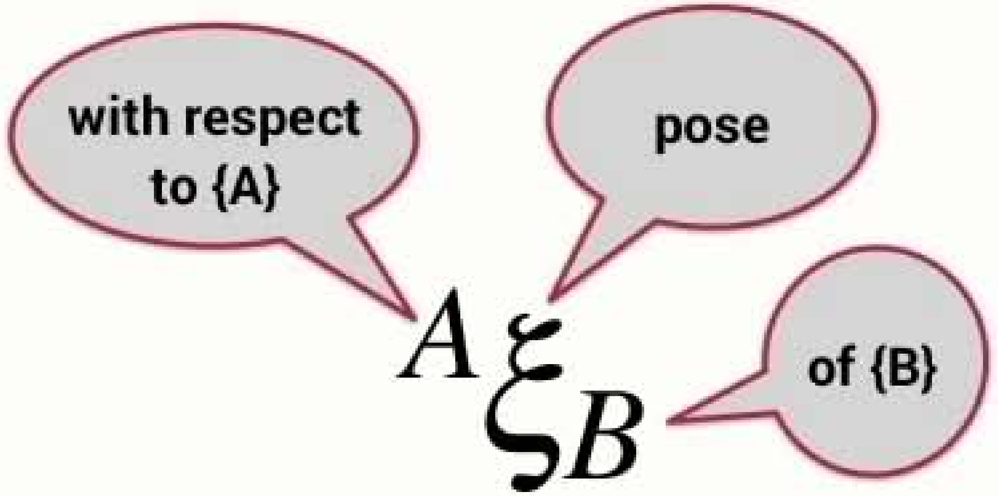


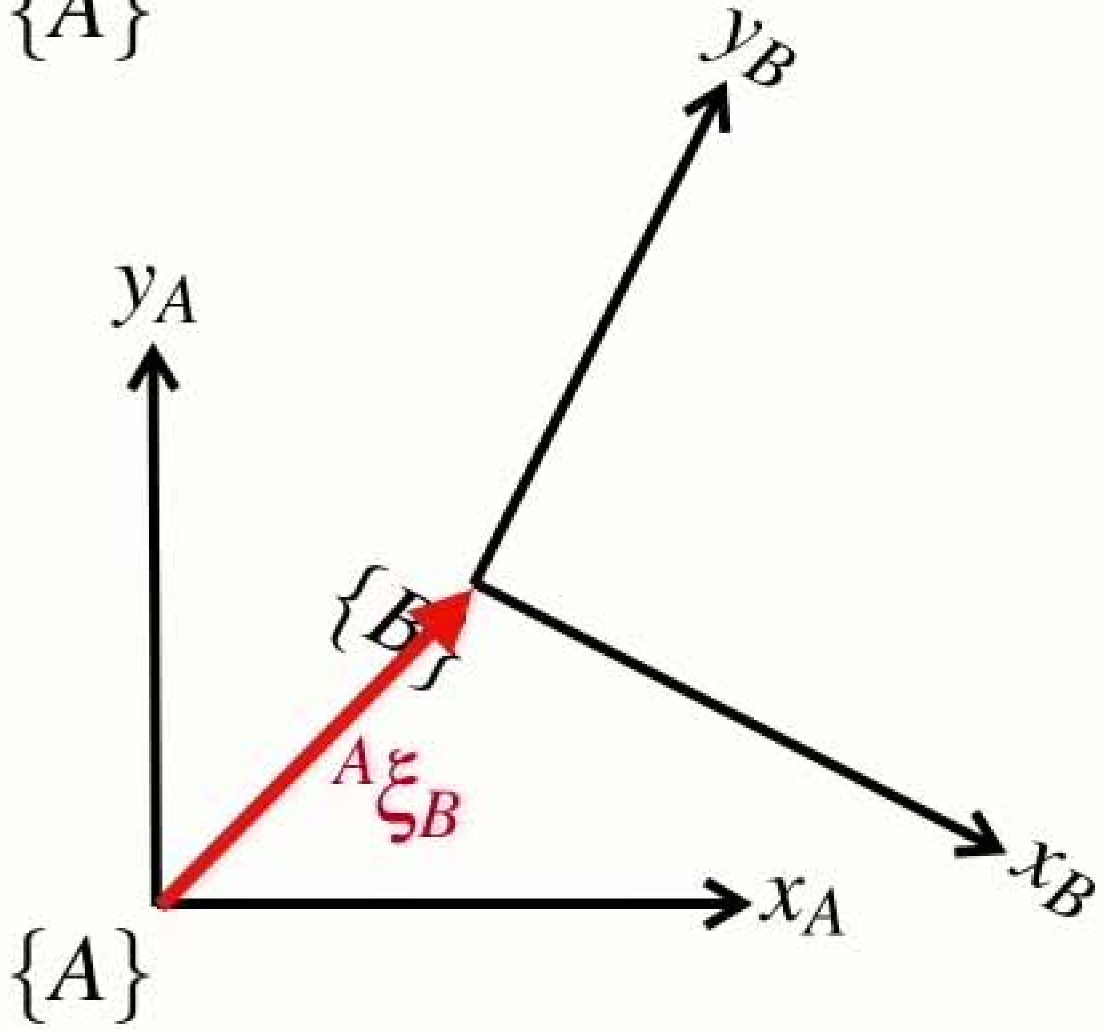


Summary

- A point is described by a vector with respect to a coordinate frame
- The pose of a coordinate frame can be described with respect to another coordinate frame

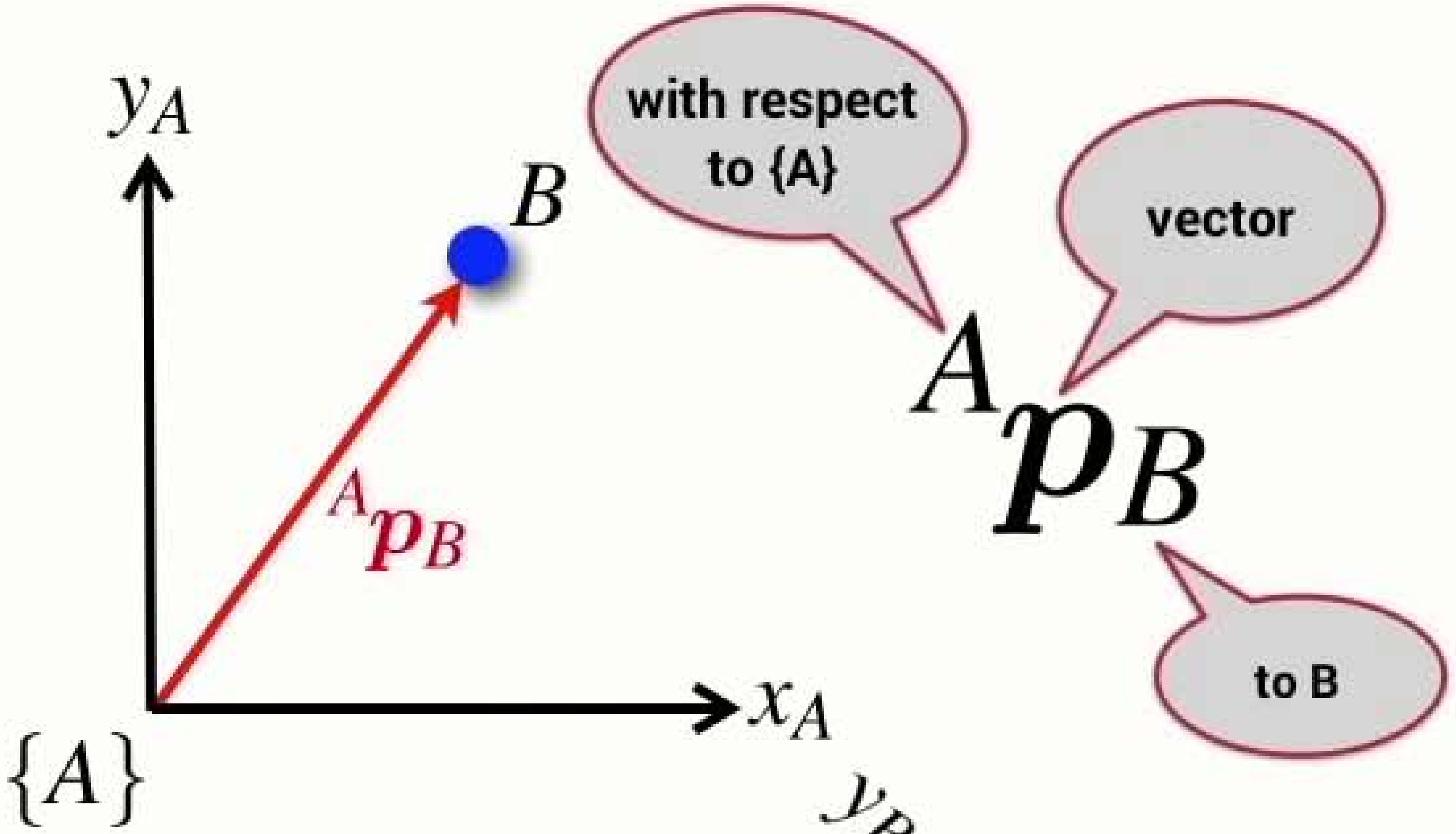


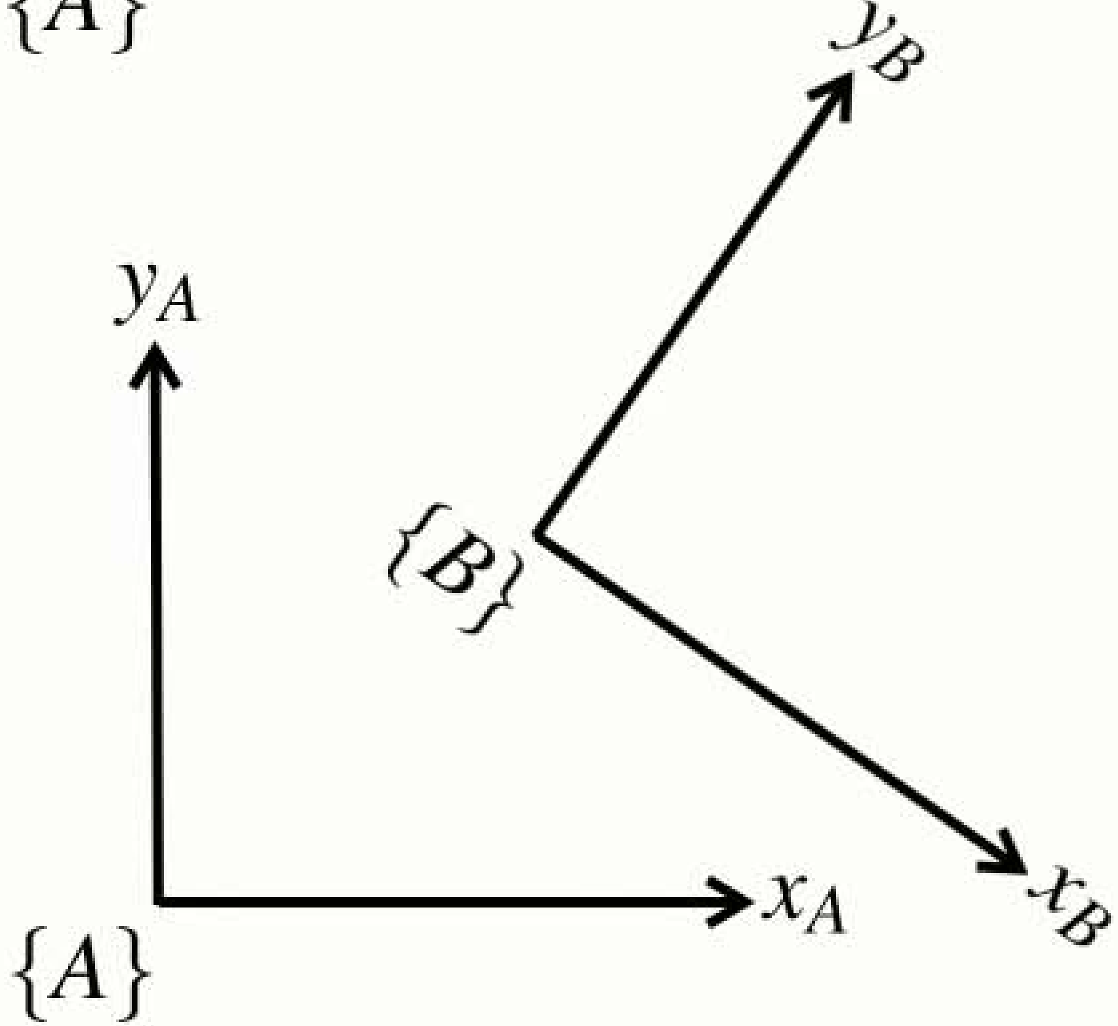


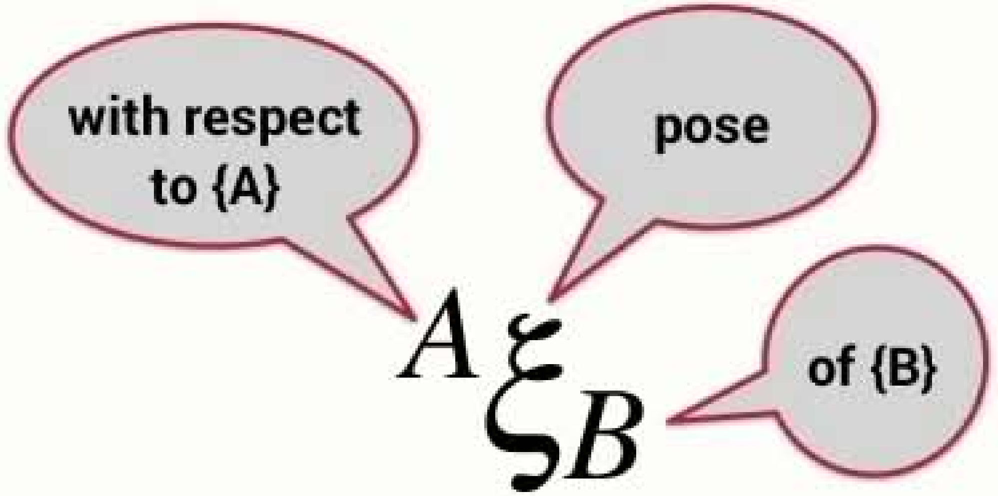


Summary

- A point is described by a vector with respect to a coordinate frame
- The pose of a coordinate frame can be described with respect to another coordinate frame
- Pose can be considered as the motion of a coordinate frame
  - ➥ a translation and a rotation

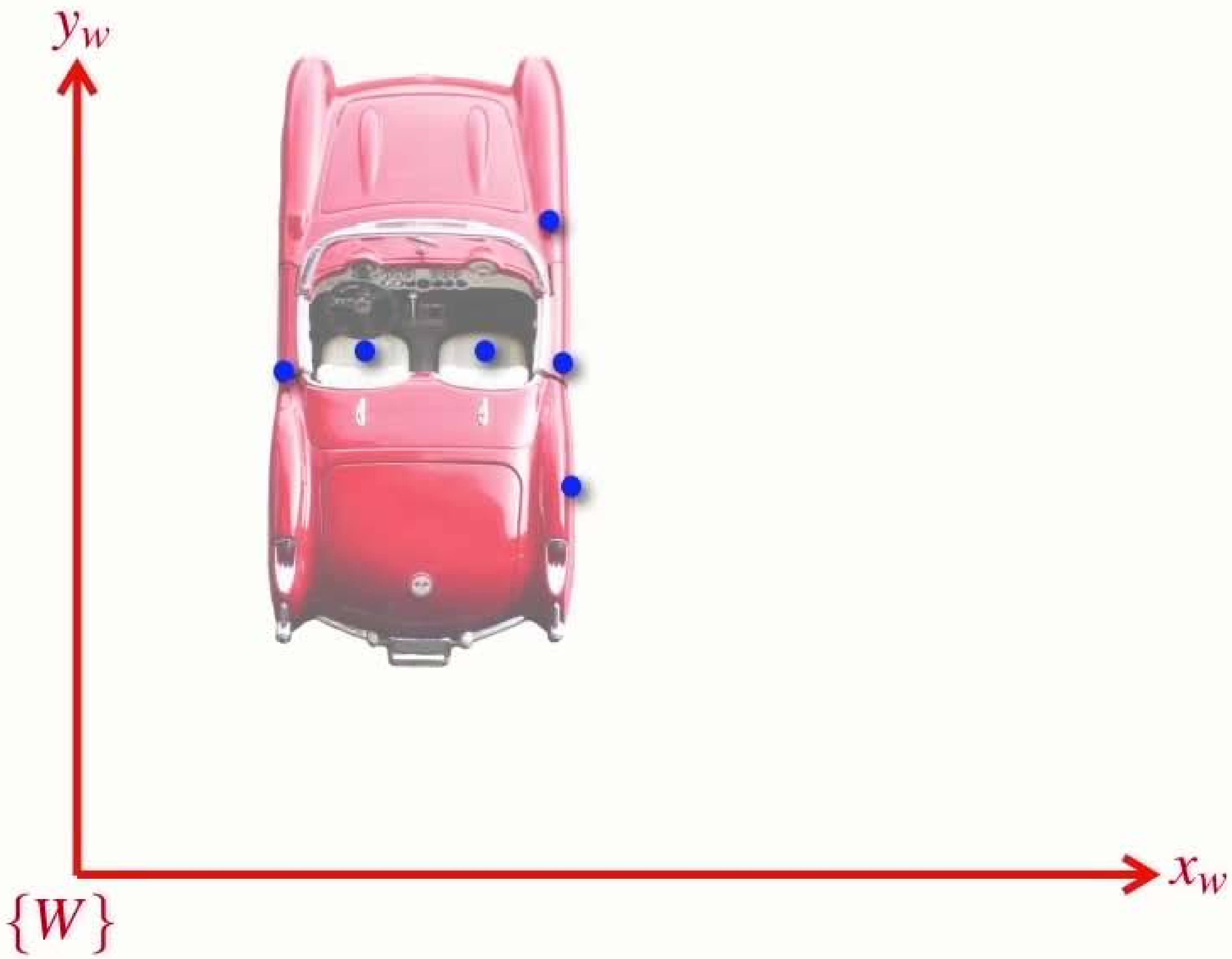


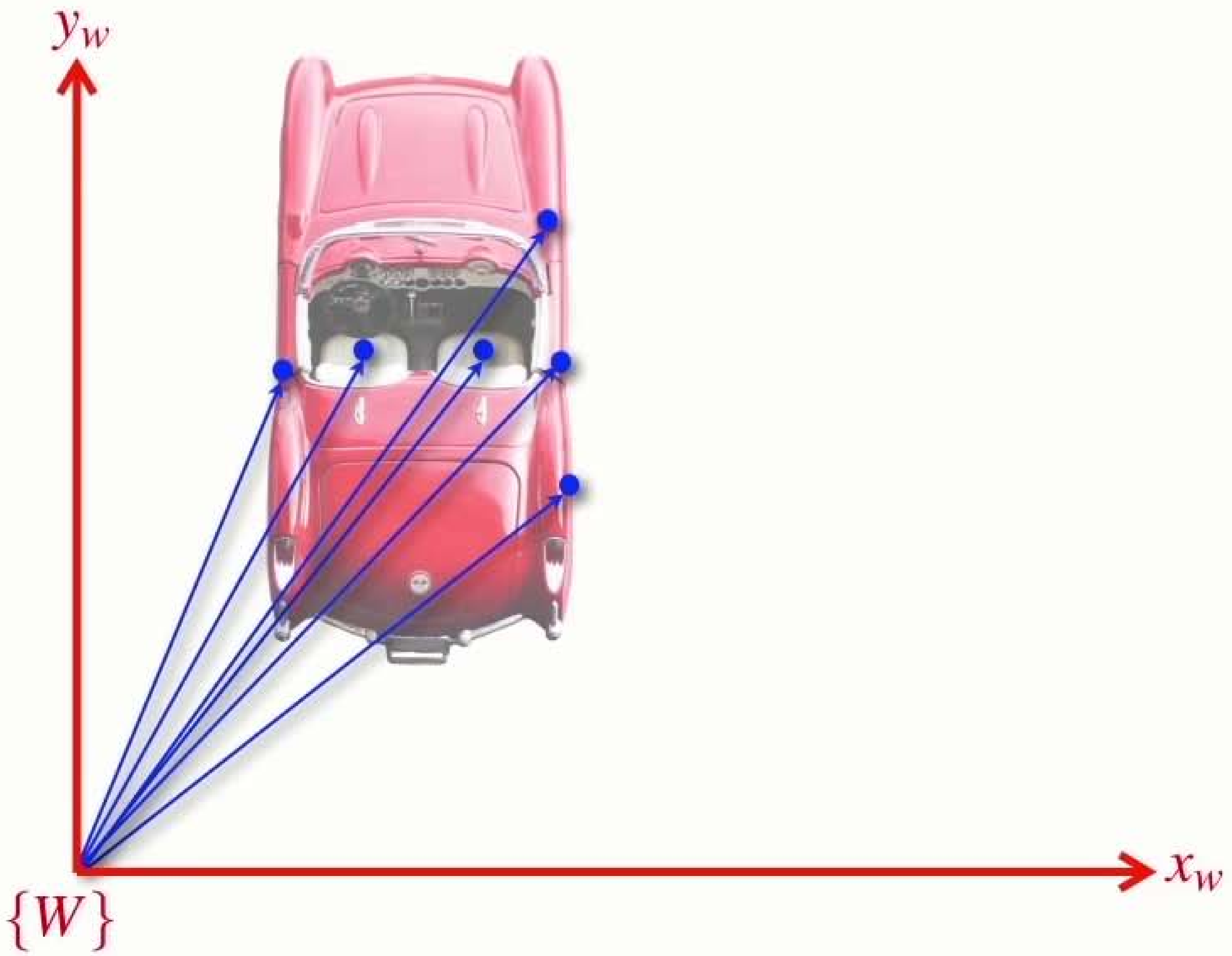


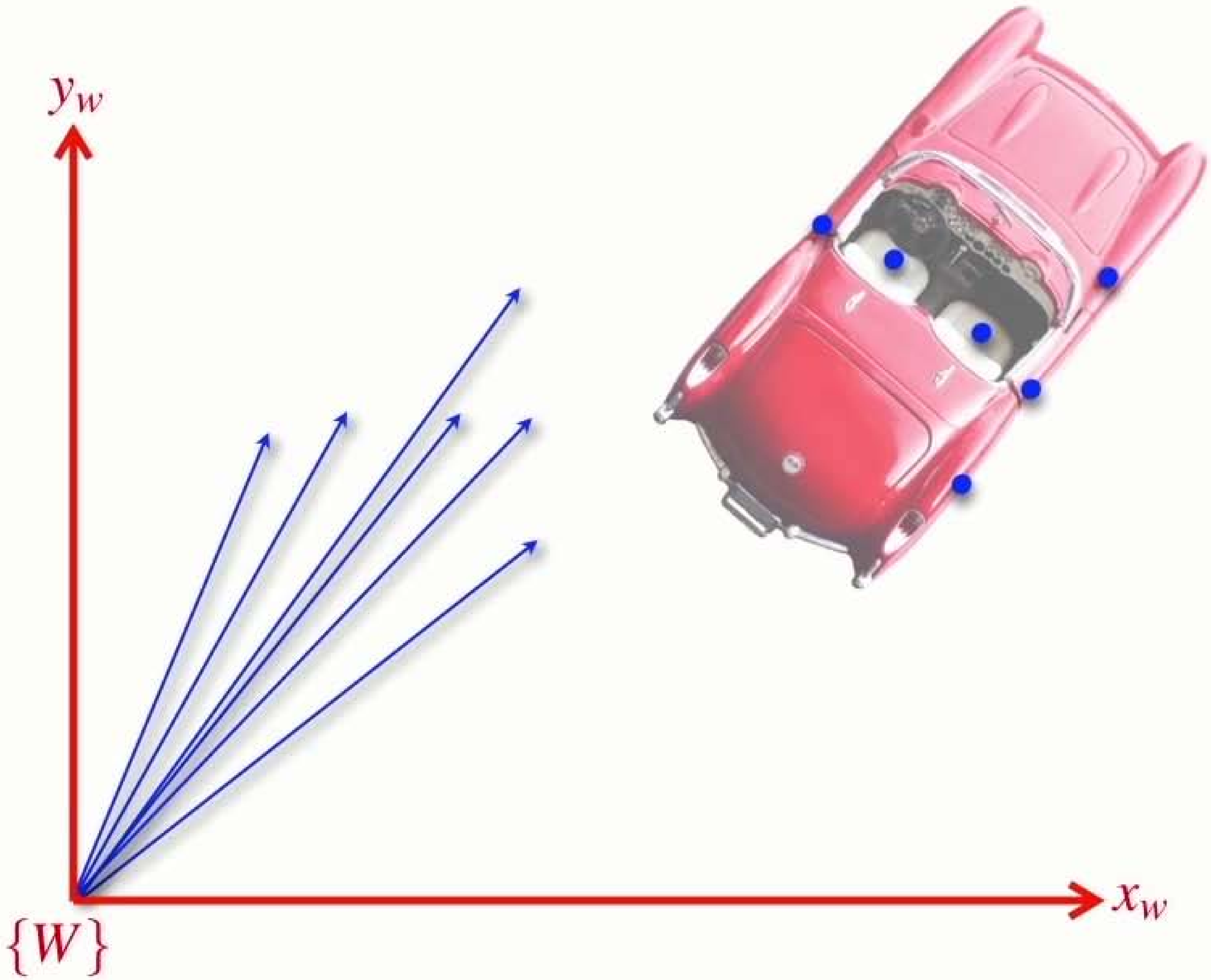


Section 3 Relative positions

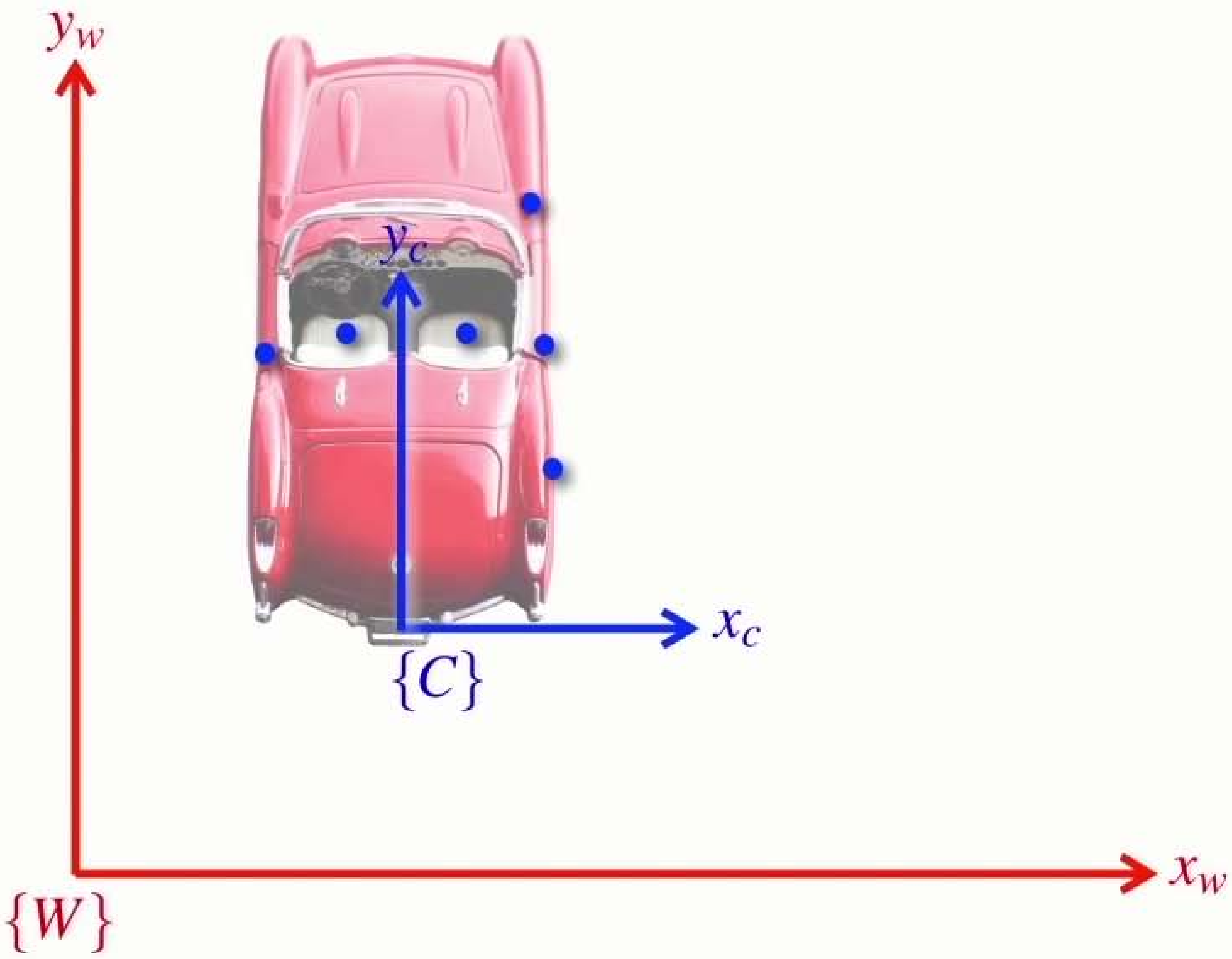

y_w
x_w
$\{W\}$


y_w
x_w
$\{W\}$

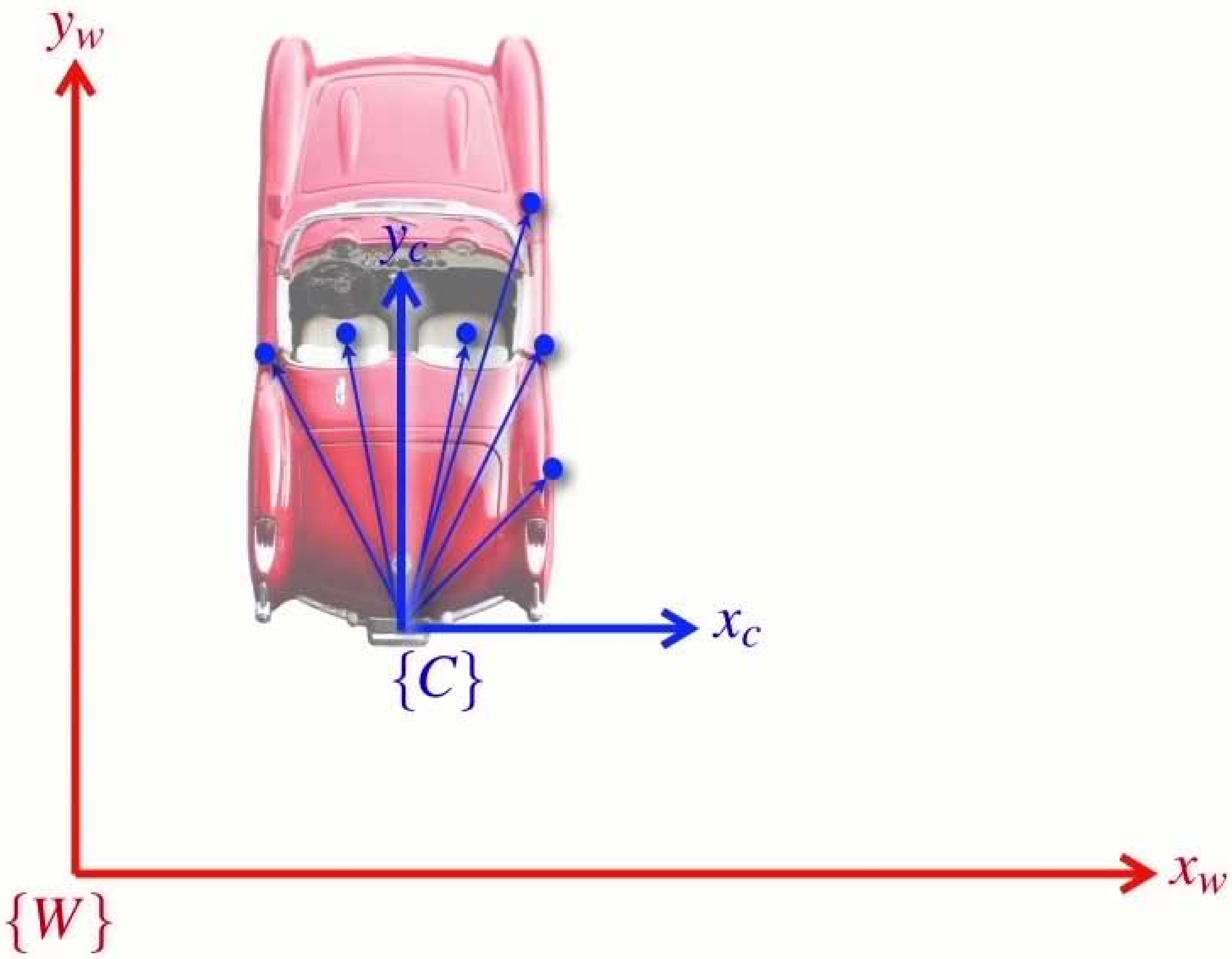

y_w
x_w
$\{W\}$

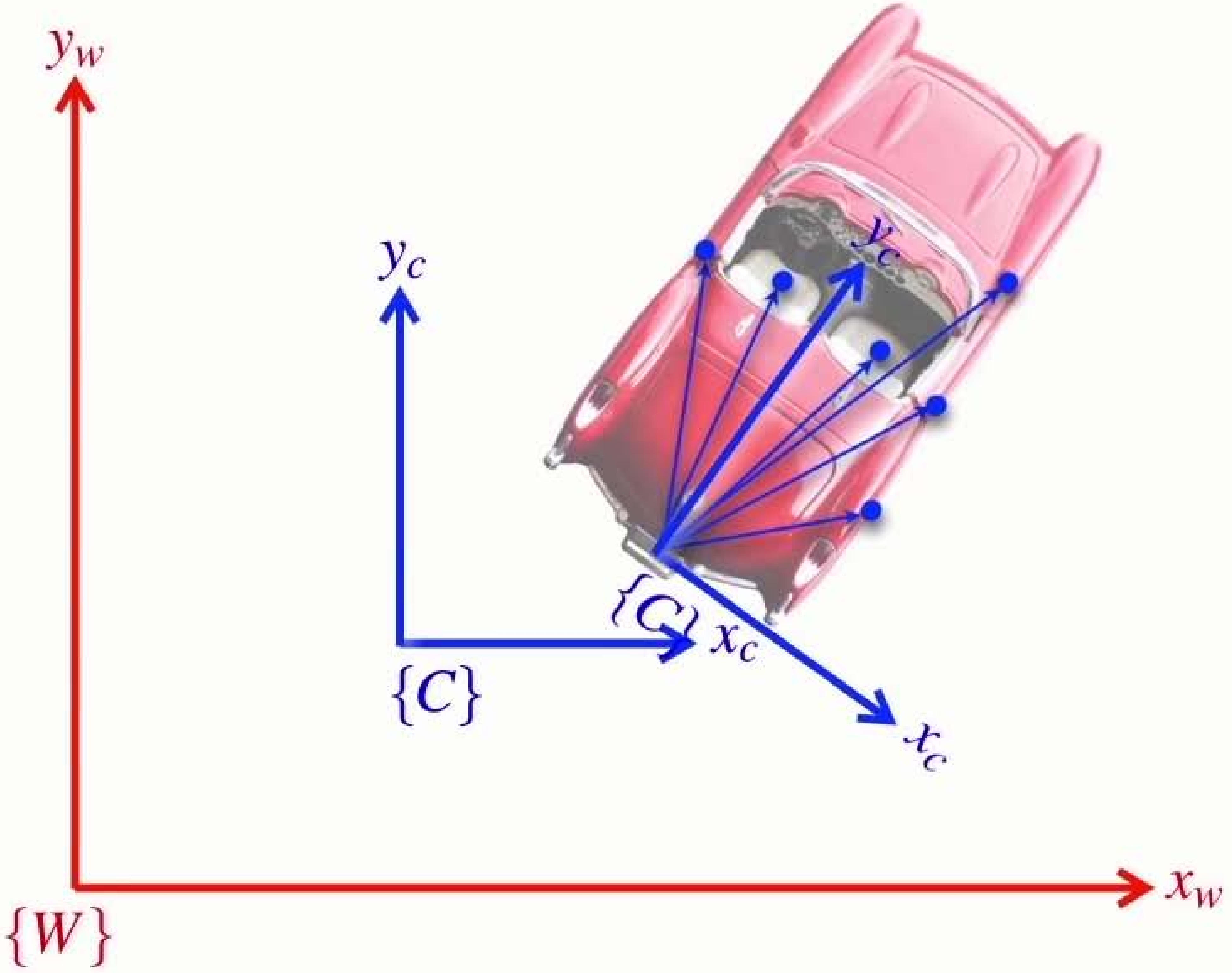


y_w
x_w
$\{W\}$

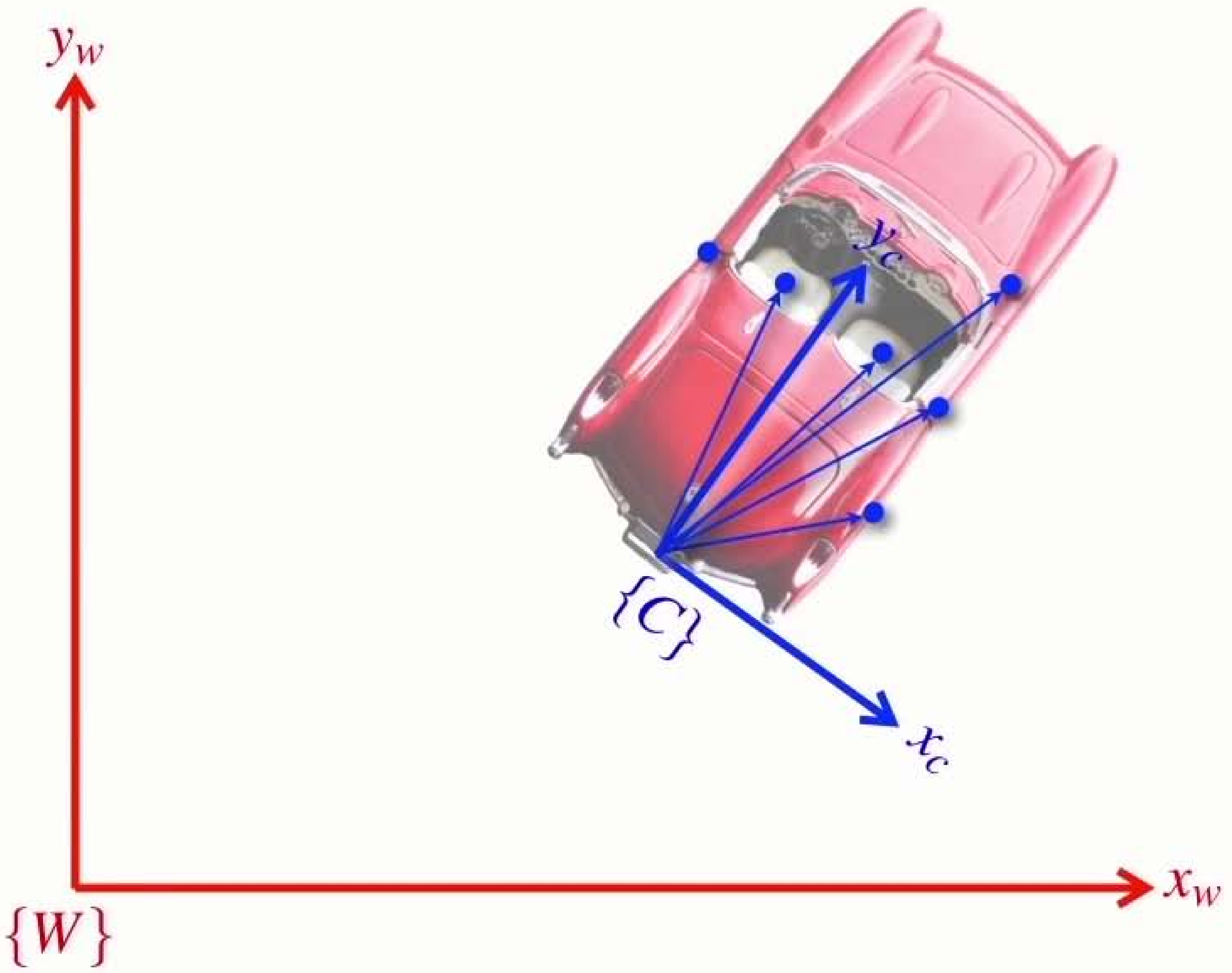


y_w
y_c
x_c
$\{C\}$
x_w
$\{W\}$

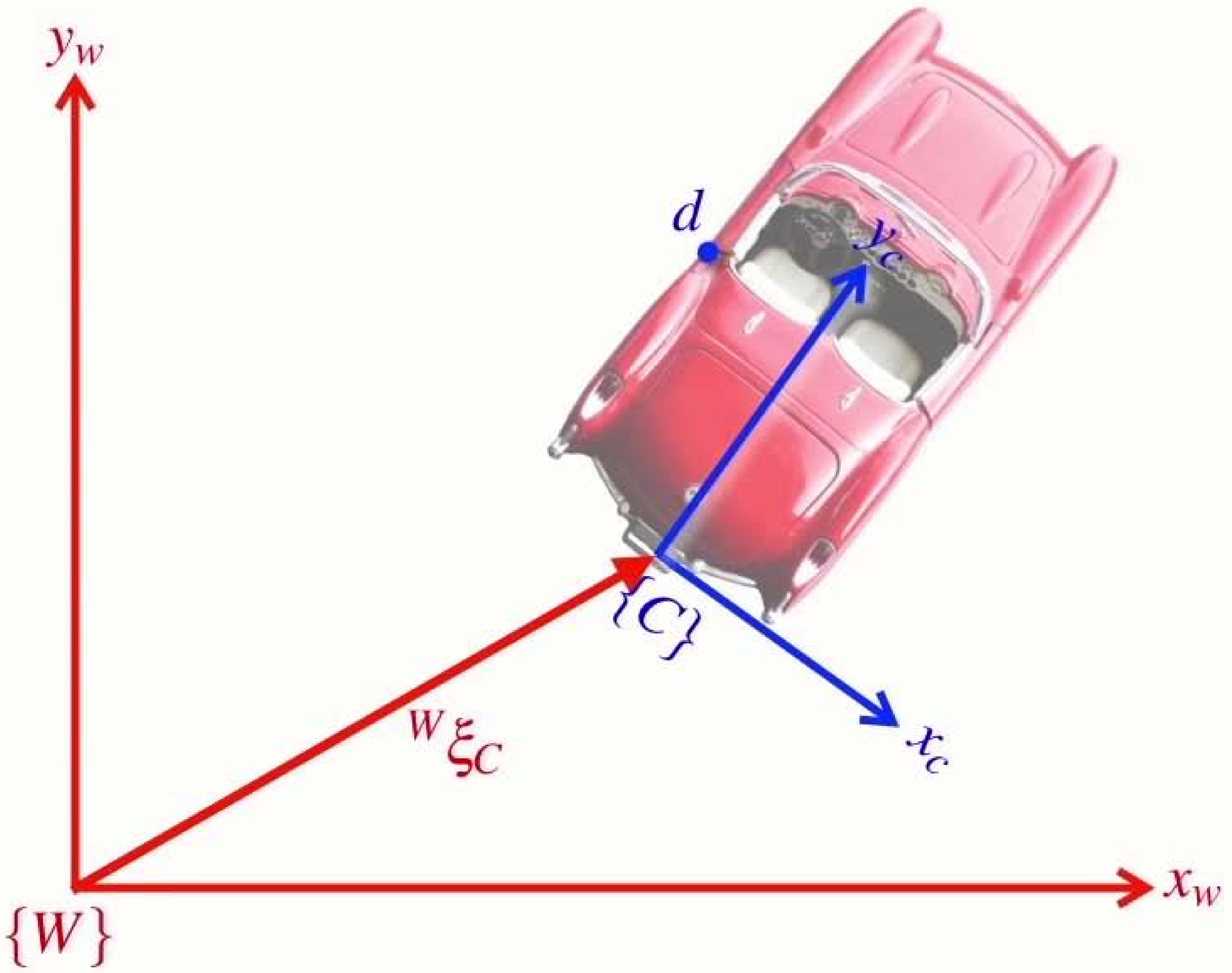

y_w
y_c
x_c
$\{C\}$
x_w
$\{W\}$



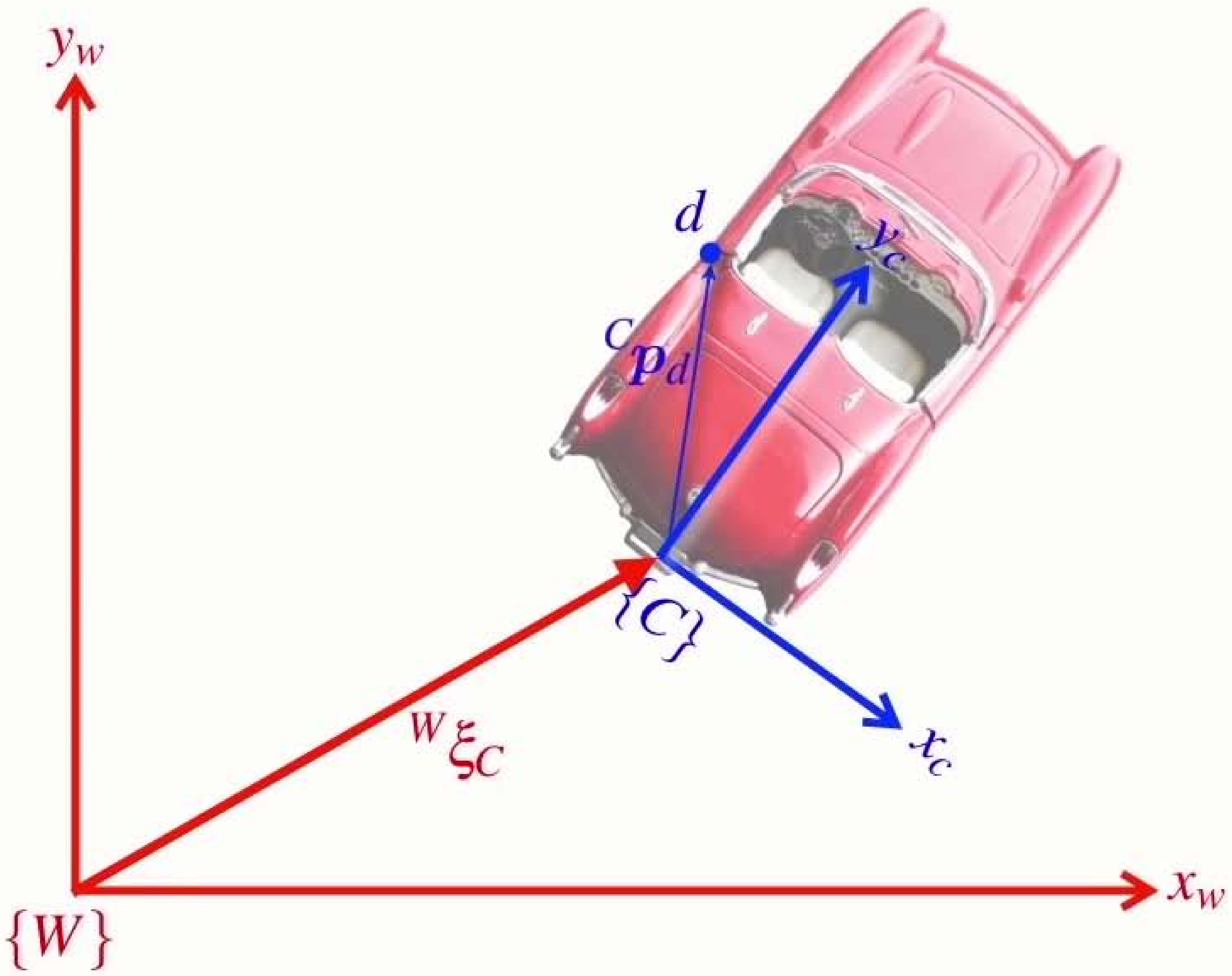
y_w
x_w
$\{W\}$
y_c
x_c
$\{C\}$
y_c
x_c
$\{C\}$



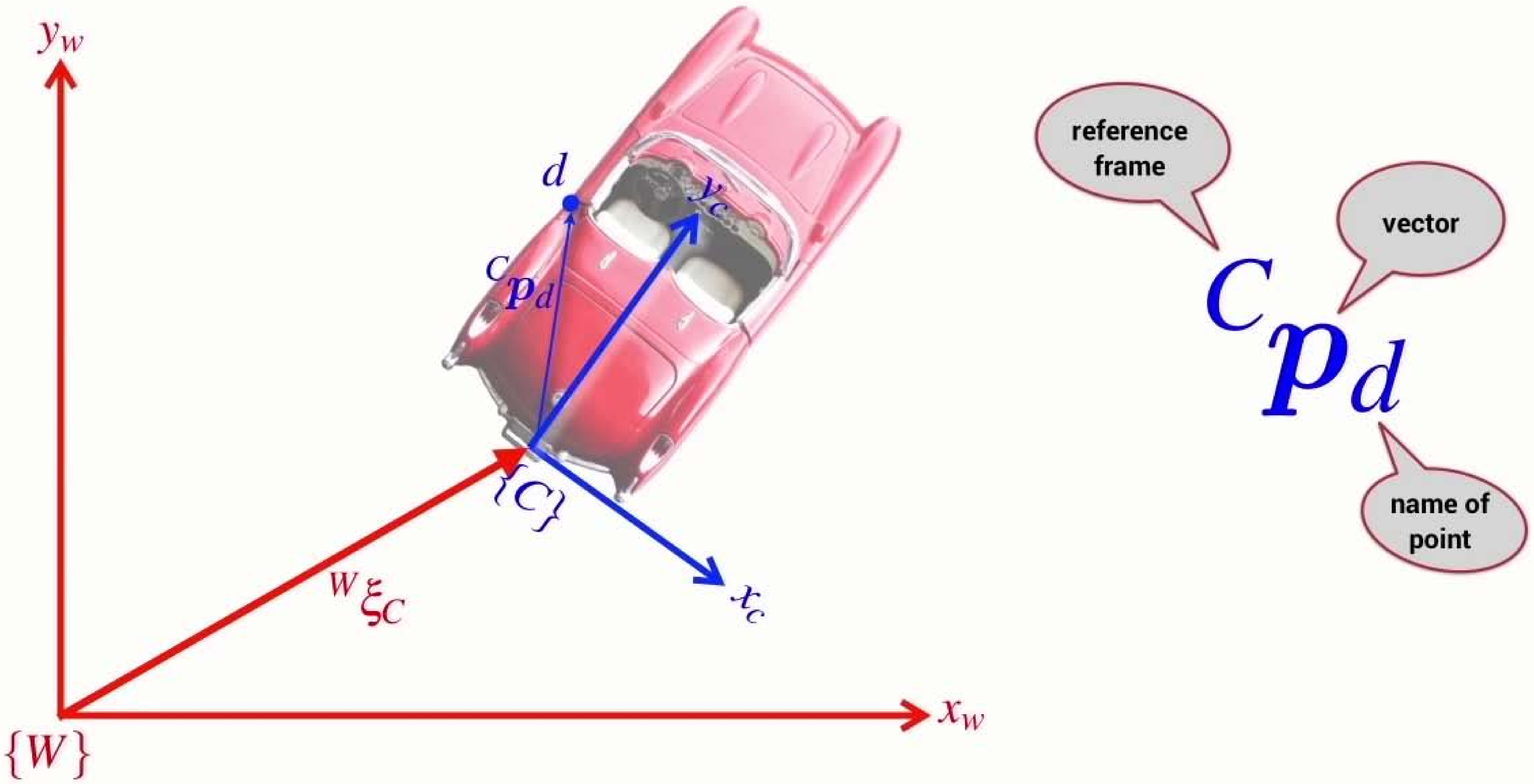
y_w
y_c
$\{C\}$
x_c
x_w
$\{W\}$

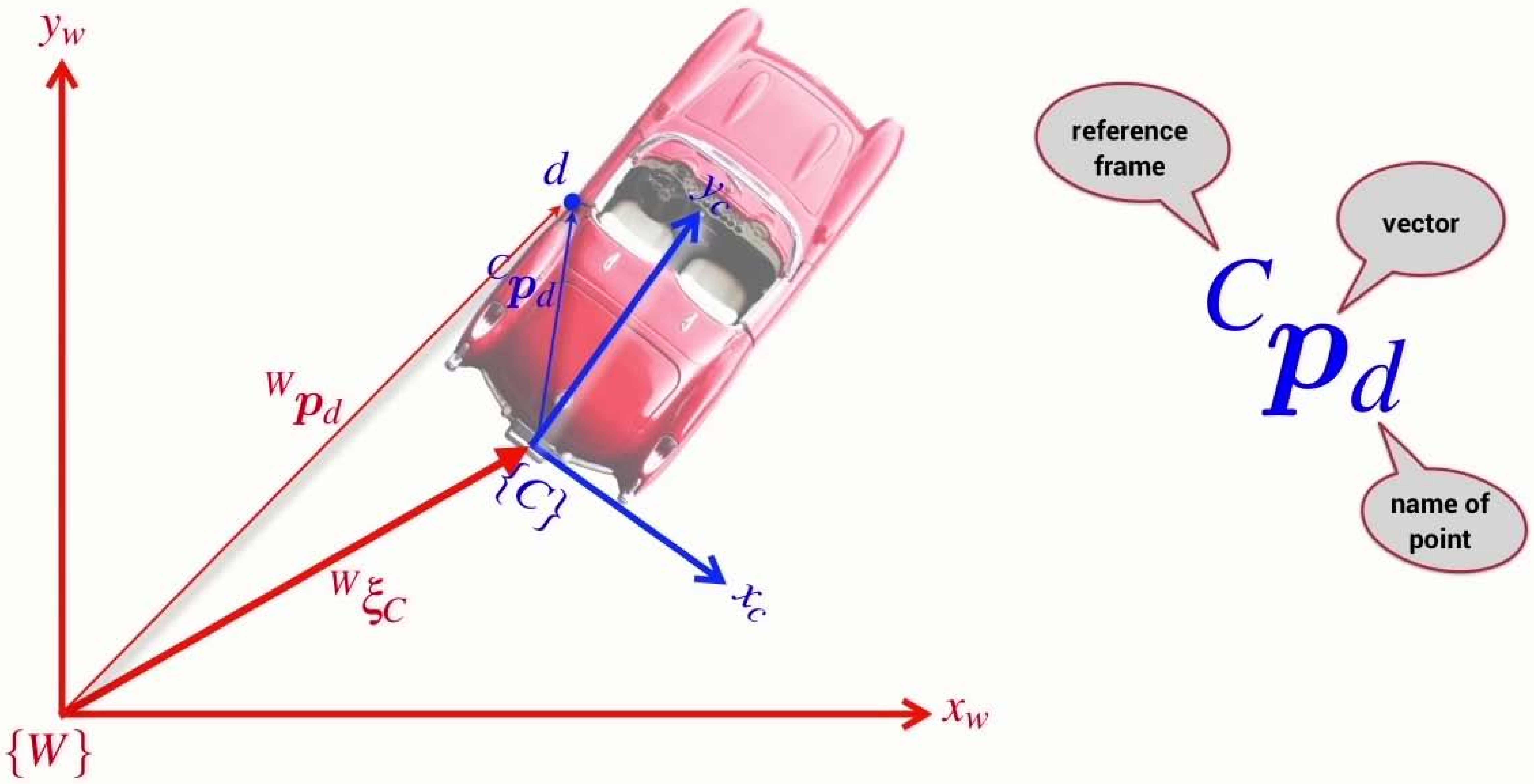


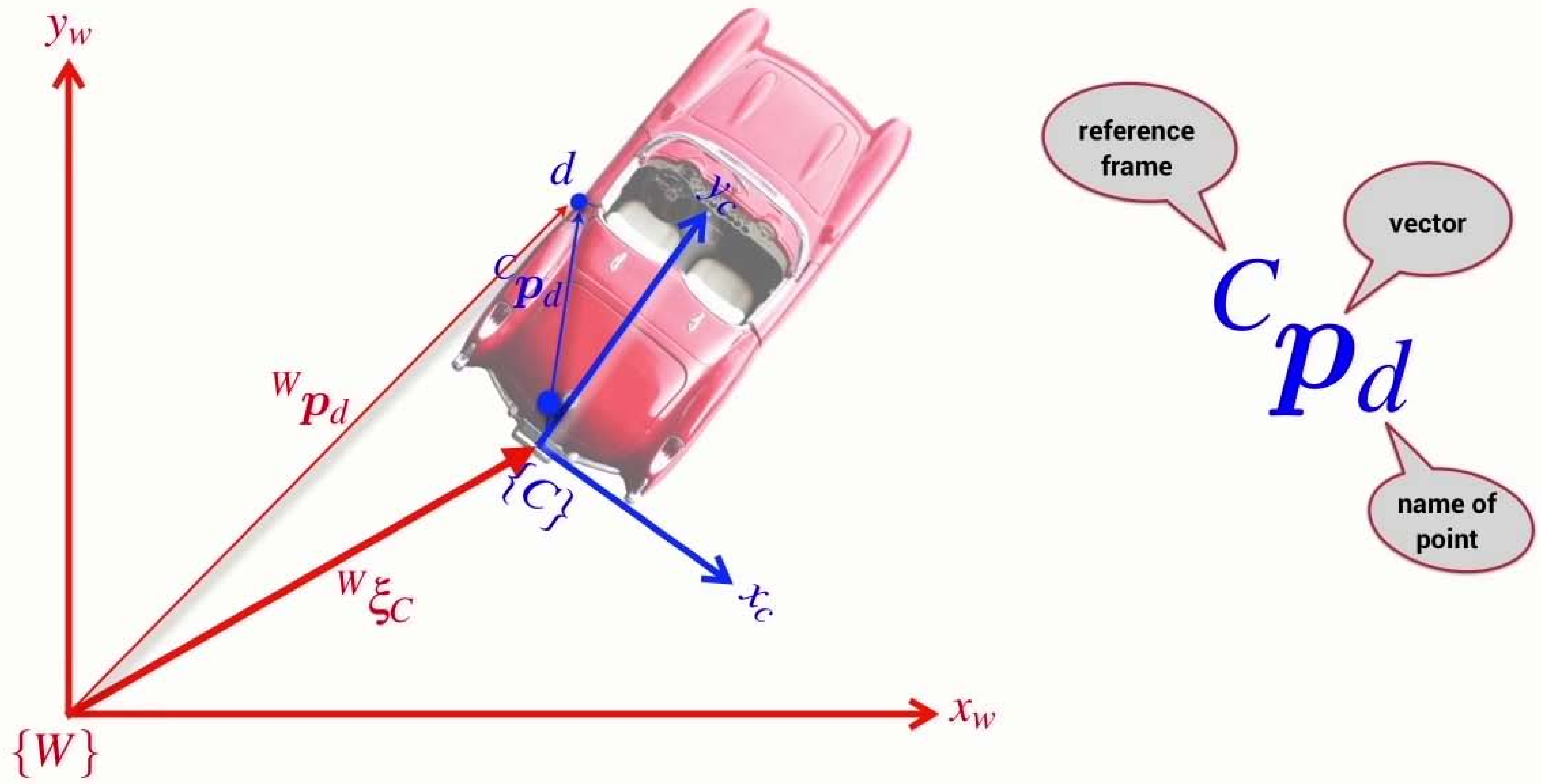
y_w
d
y_c
$\{C\}$
$^{W}\xi_{C}$
x_c
x_w
$\{W\}$



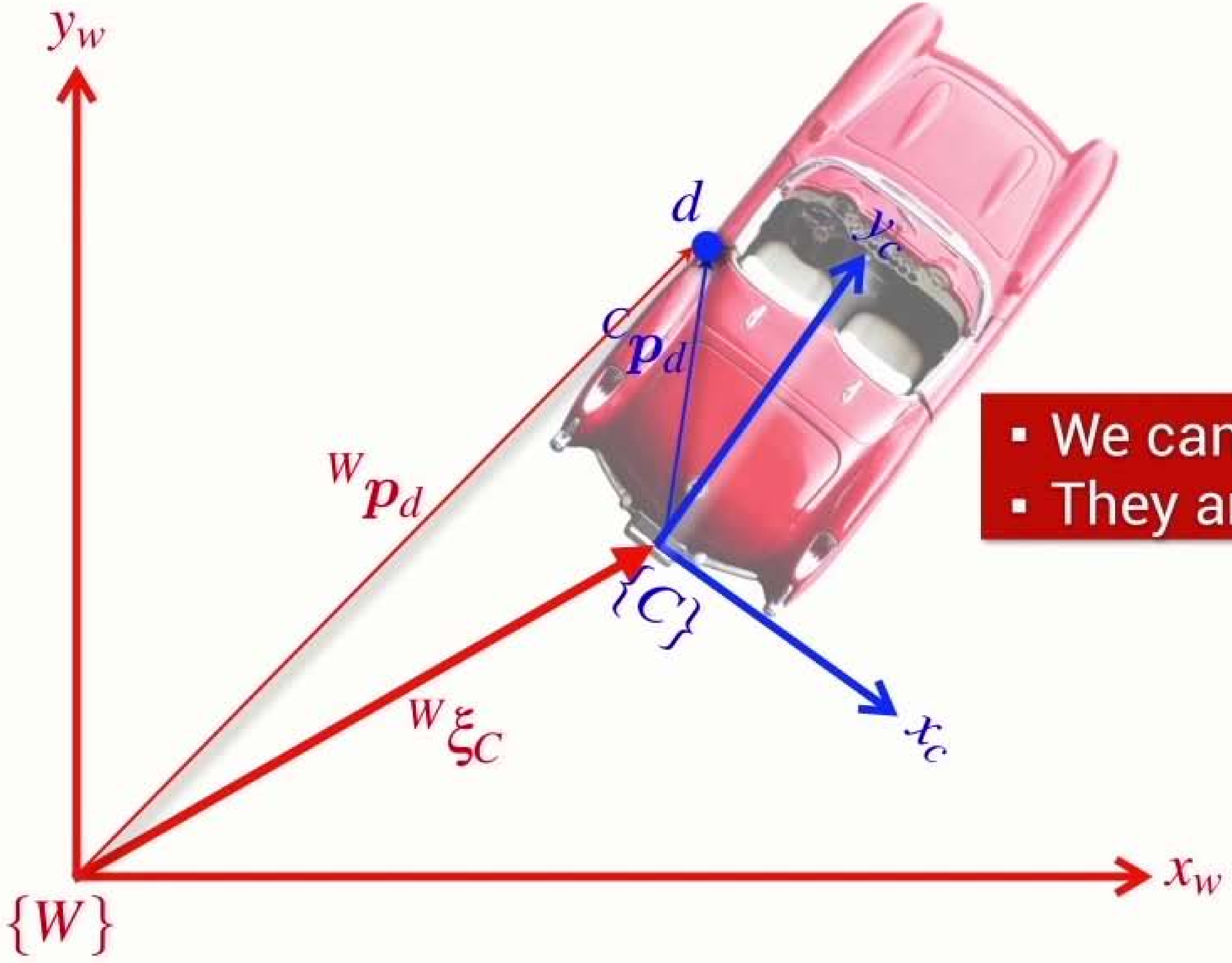
y_w
d
y_c
${}^{C}\mathbf{p}_d$
$\{C\}$
x_c
${}^{W}\xi_C$
x_w
$\{W\}$

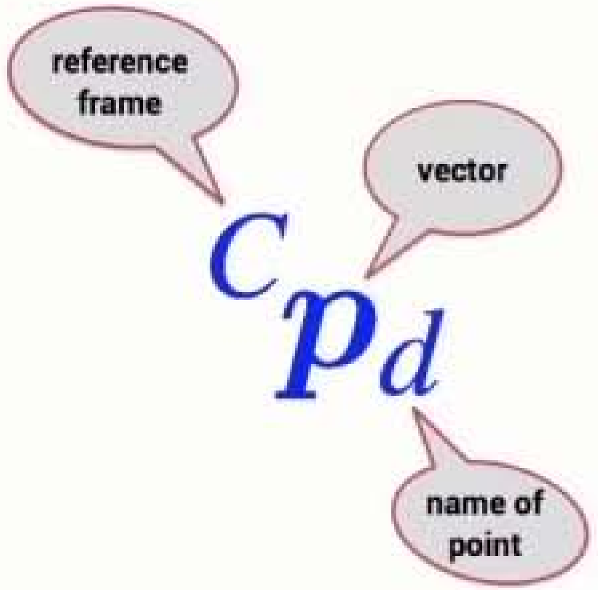

y_w
d
y_c
${}^{C}\mathbf{p}_d$
$\{C\}$
${}^{W}\xi_C$
x_c
x_w
$\{W\}$
reference frame
vector
${}^{C}\mathbf{p}_d$
name of point


y_w
x_w
$\{W\}$
d
y_c
x_c
$\{C\}$
${}^{C}\boldsymbol{p}_d$
${}^{W}\boldsymbol{p}_d$
${}^{W}\xi_C$
reference frame
vector
name of point
${}^{C}\boldsymbol{p}_d$

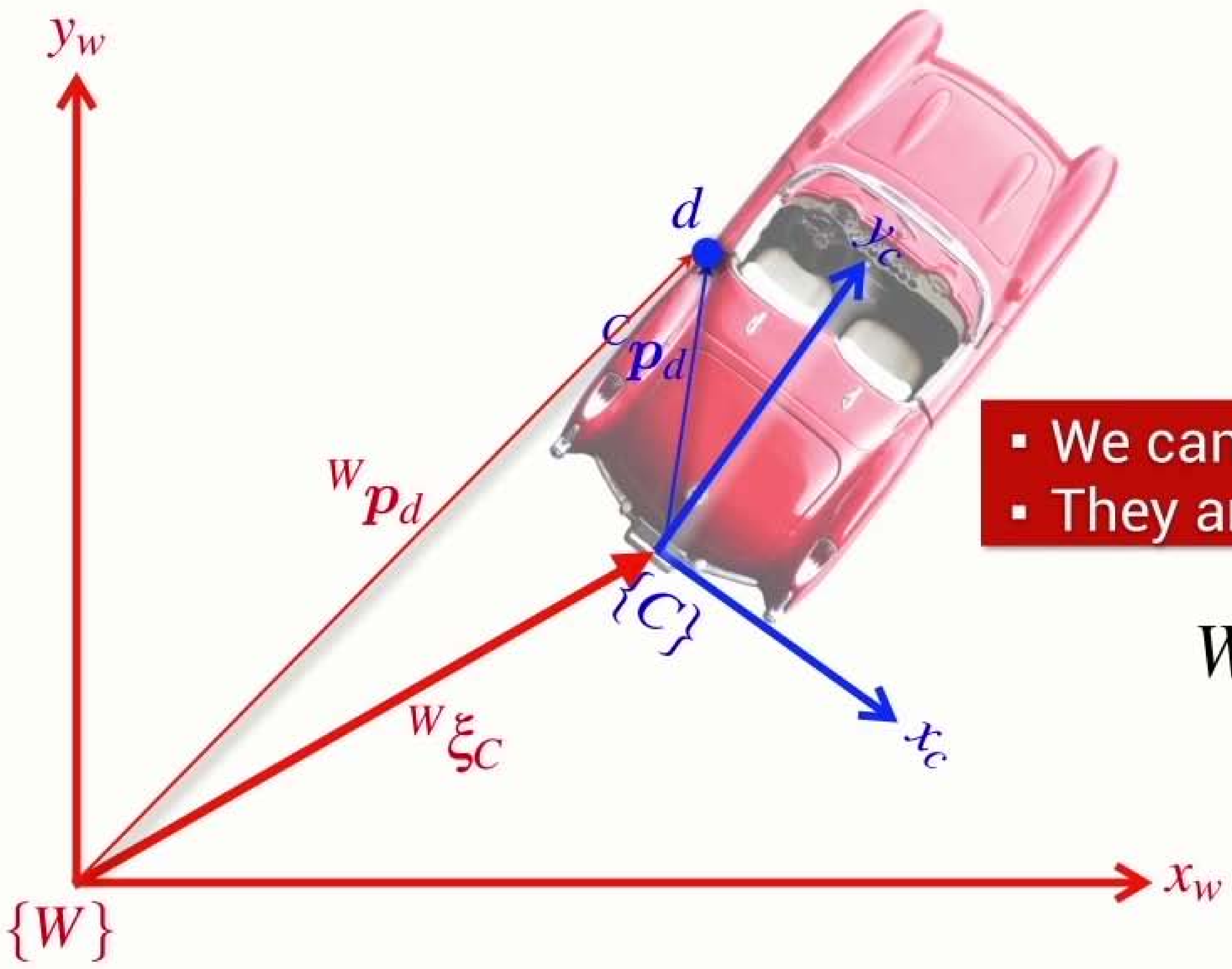


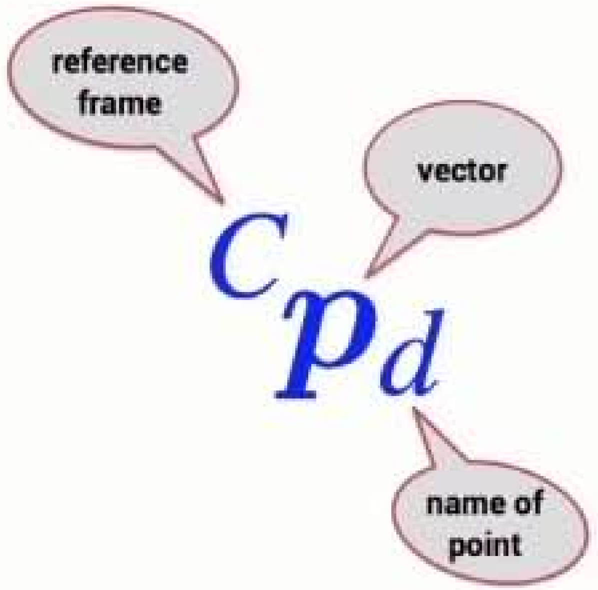
y_w
x_w
$\{W\}$
d
y_c
x_c
${}^C\boldsymbol{p}_d$
${}^W\boldsymbol{p}_d$
$\{C\}$
${}^W\xi_C$
reference frame
vector
${}^C\boldsymbol{p}_d$
name of point





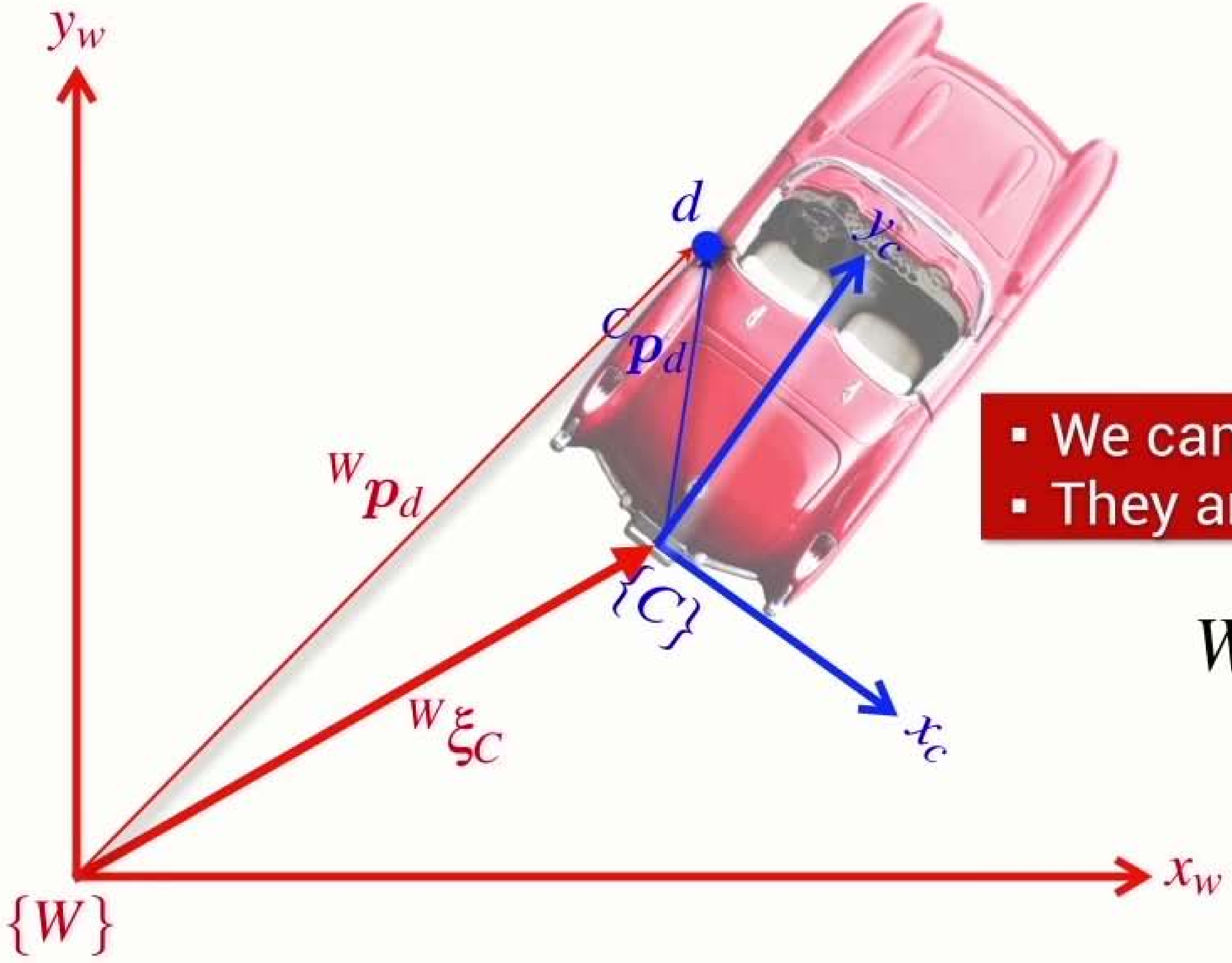
- We can't add **Poses** and **Vectors**
- They are different things





- We can't add **Poses** and **Vectors**
- They are different things

$$^W\boldsymbol{p}_d = {^W\xi_C} \cdot {^C\boldsymbol{p}_d}$$



reference frame

vector

$$^C\boldsymbol{p}_d$$

name of point

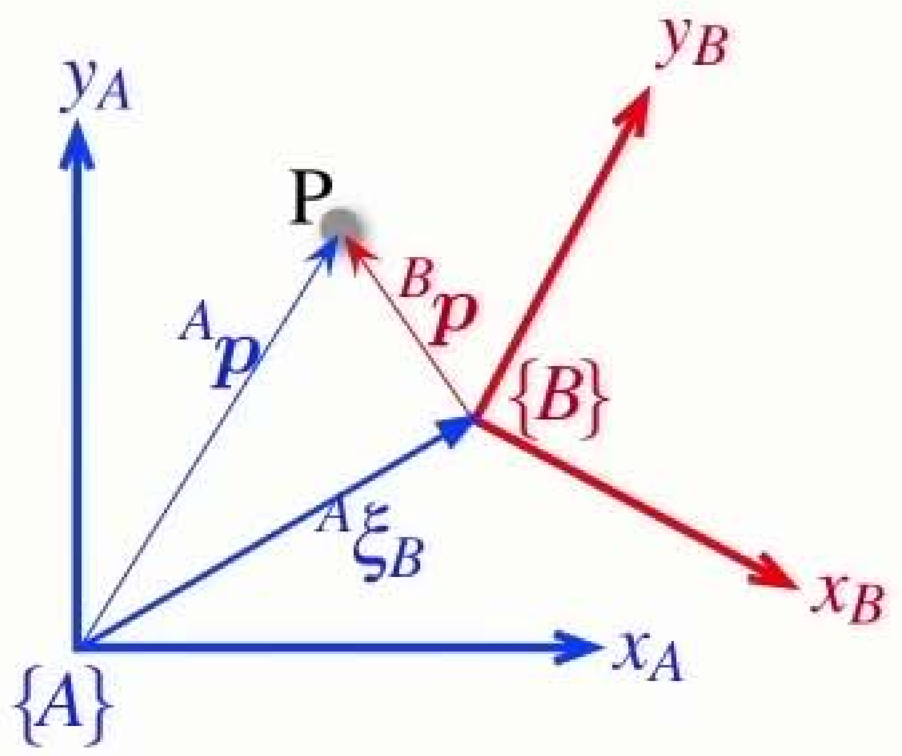
- We can't add **Poses** and **Vectors**
- They are different things

$$^W\boldsymbol{p}_d = {}^W\xi_C \cdot {}^C\boldsymbol{p}_d$$

transform
vector from one frame
to another

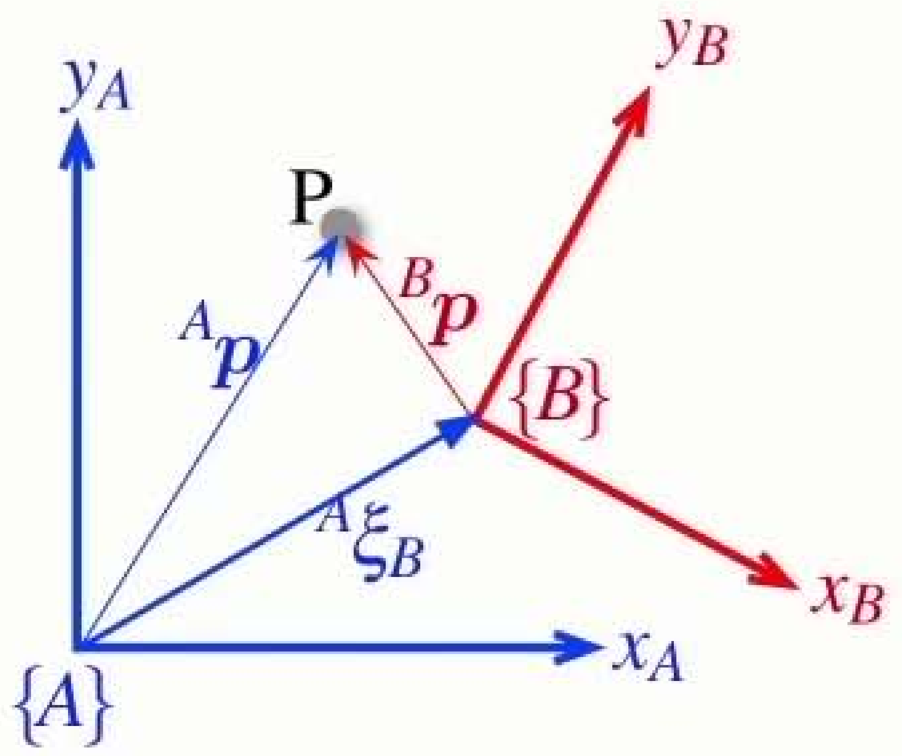
Section 4 Relative poses

Relative **points**



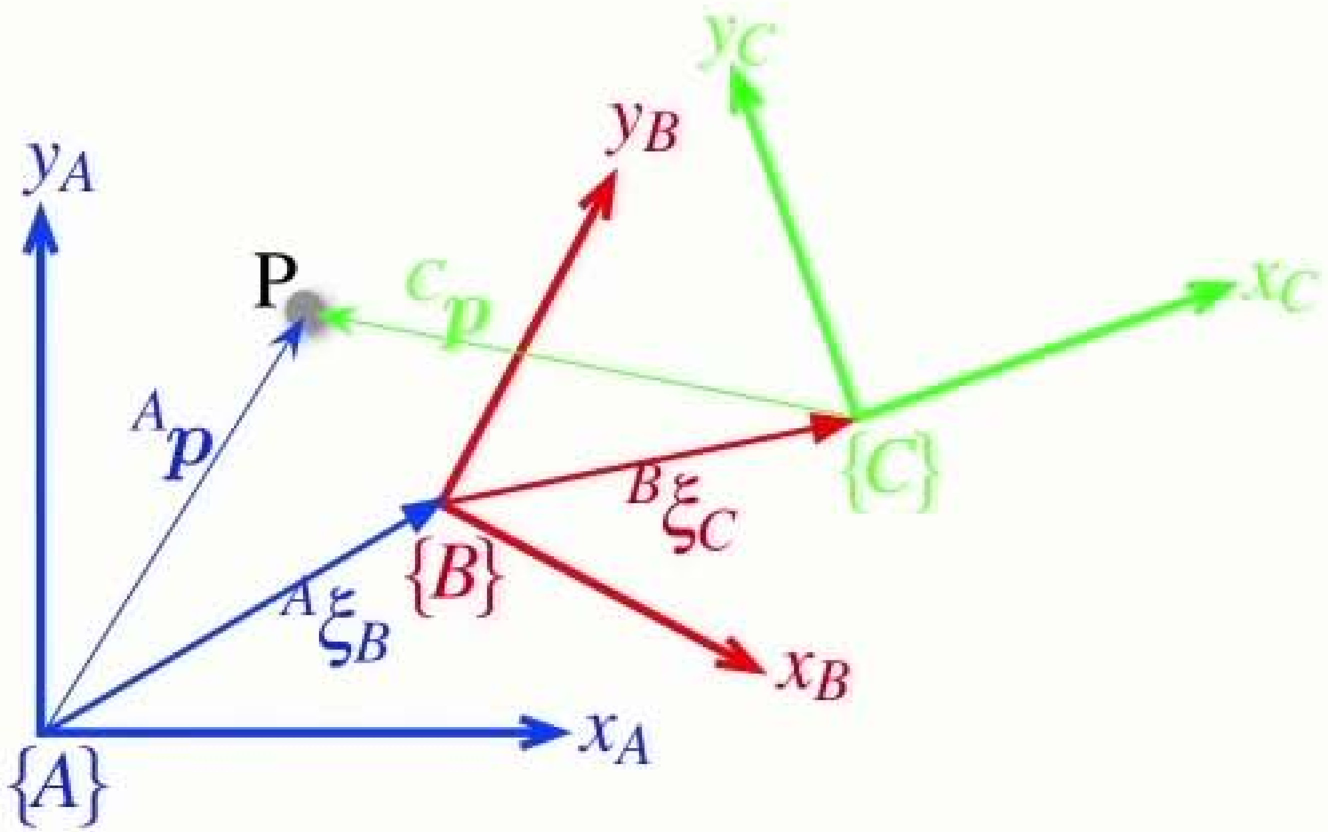
Relative **points**

$$ {}^{A}\boldsymbol{p} = {}^{A}\xi_{B} \cdot {}^{B}\boldsymbol{p} $$



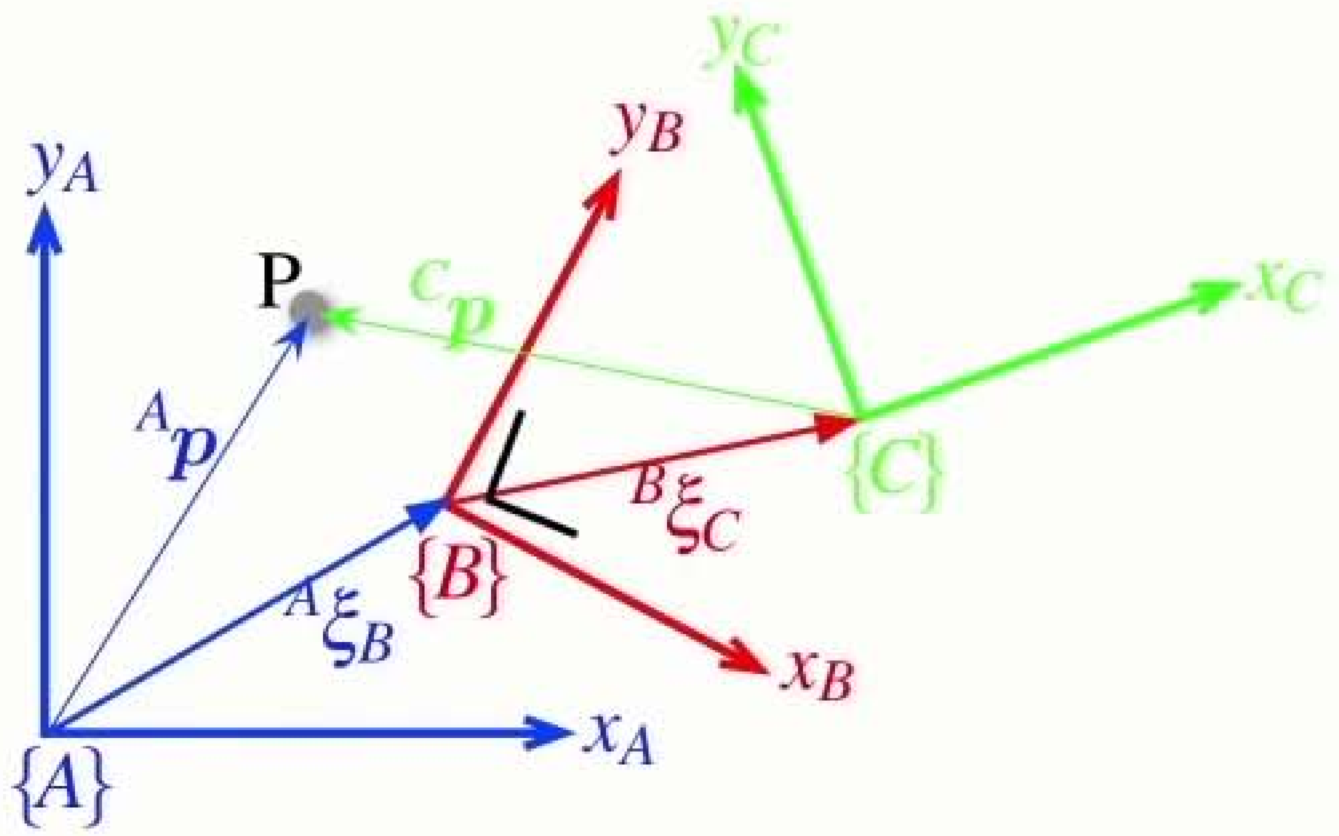
Relative **points**

$$ {}^{A}\boldsymbol{p} = {}^{A}\xi_{B} \cdot {}^{B}\boldsymbol{p} $$



- Relative poses can be **compounded** or **composed**

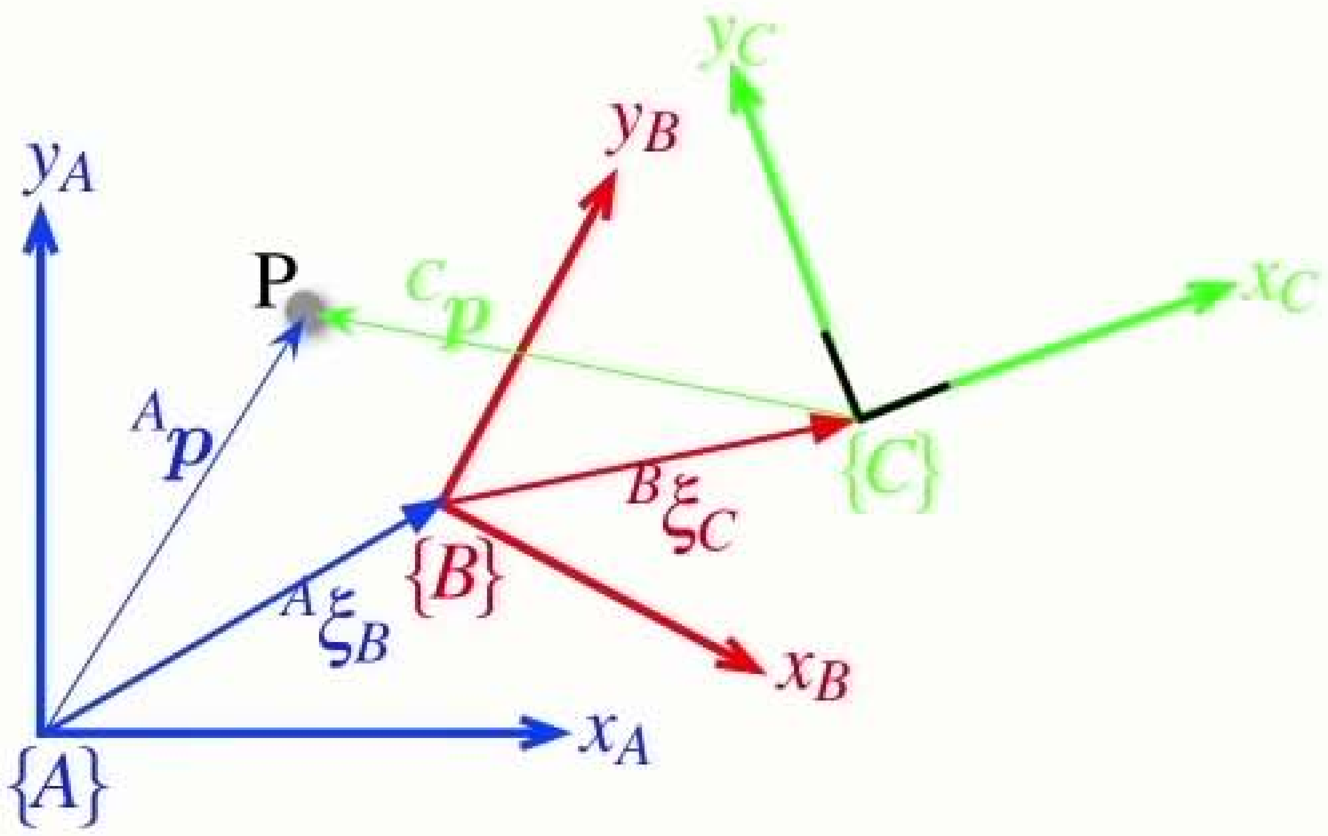
Relative **points**



$$ {}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p} $$

- Relative poses can be **compounded** or **composed**

Relative **points**

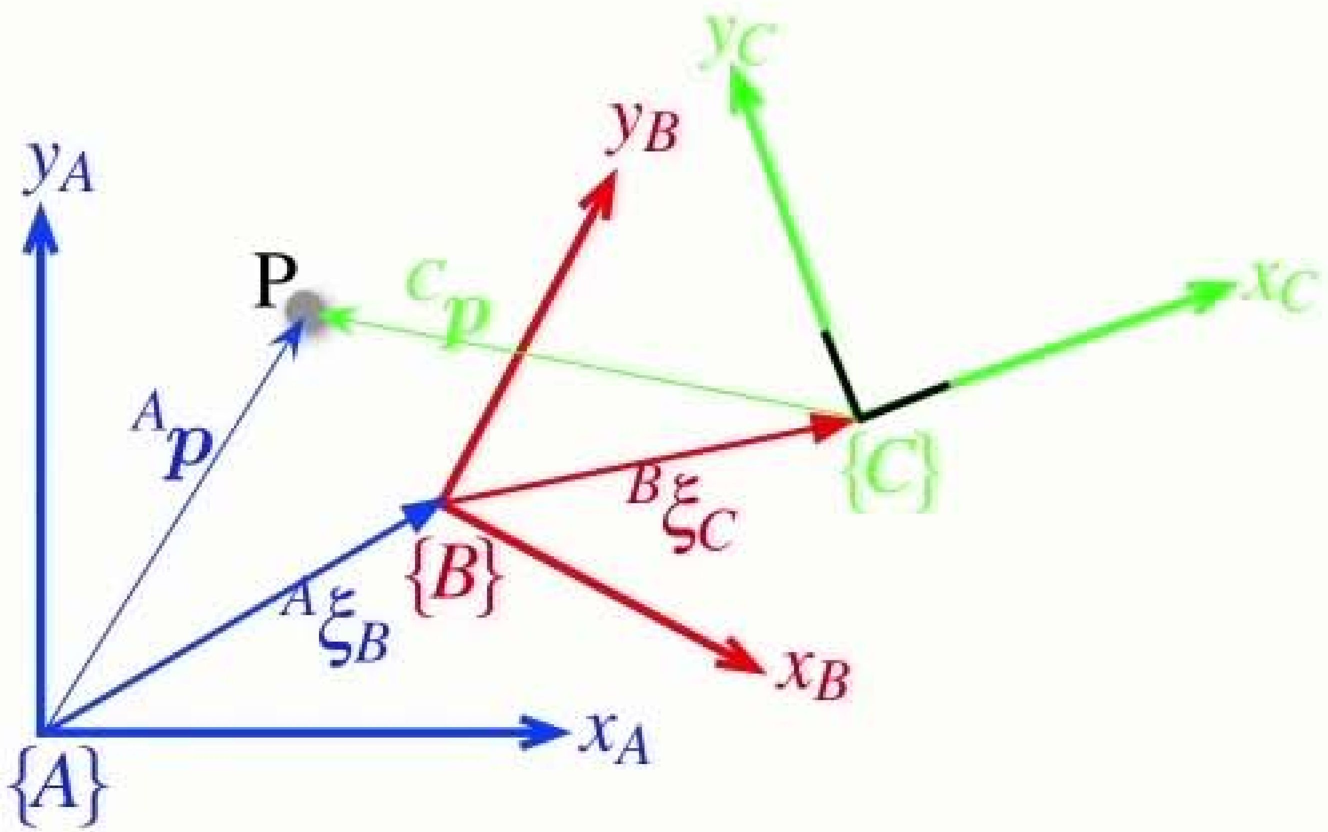


$${}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p}$$

$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

- Relative poses can be **compounded** or **composed**

Relative **points**



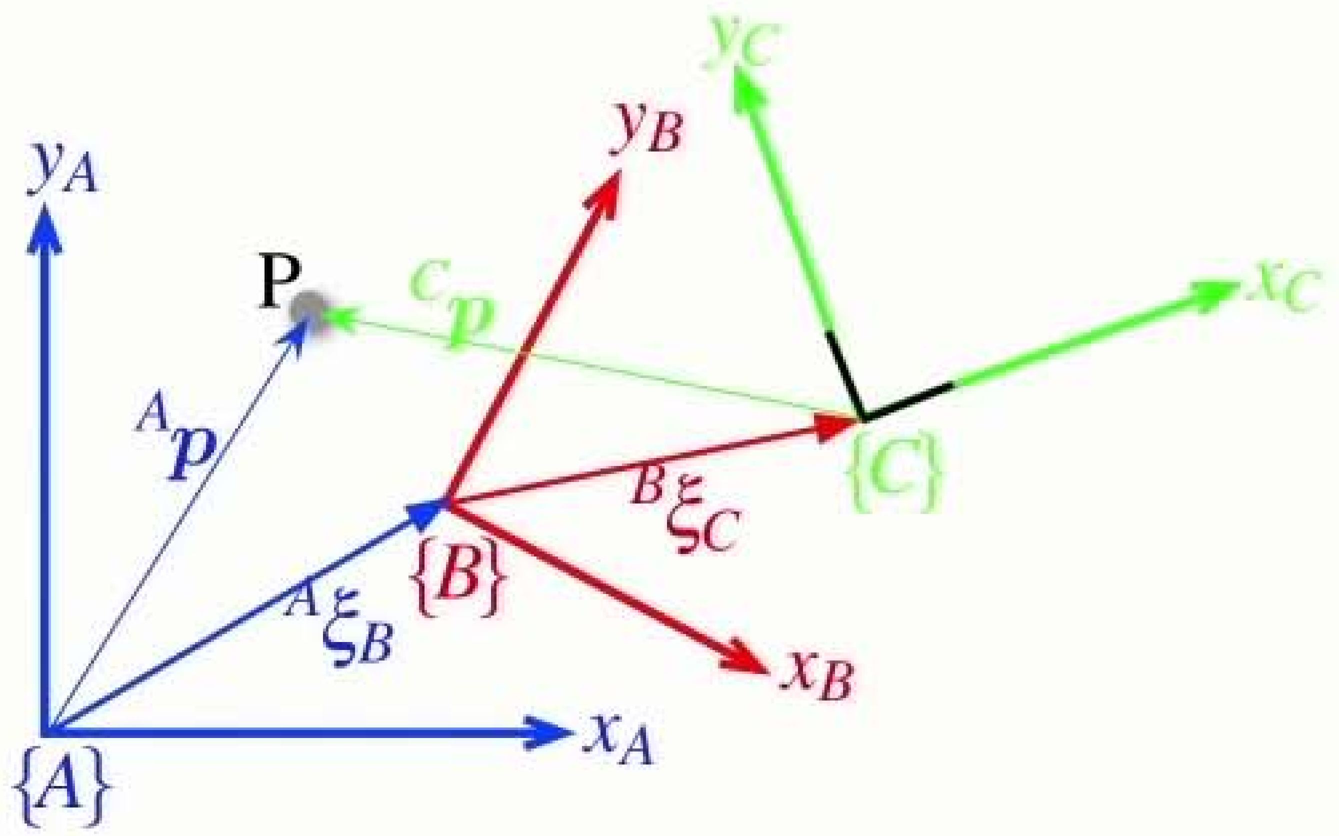
$$ {}^{A}\boldsymbol{p} = {}^{A}\xi_B \cdot {}^{B}\boldsymbol{p} $$

composition

$$ {}^{A}\xi_C = {}^{A}\xi_B \oplus {}^{B}\xi_C $$

- Relative poses can be **compounded** or **composed**

Relative **points**



$${}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p}$$

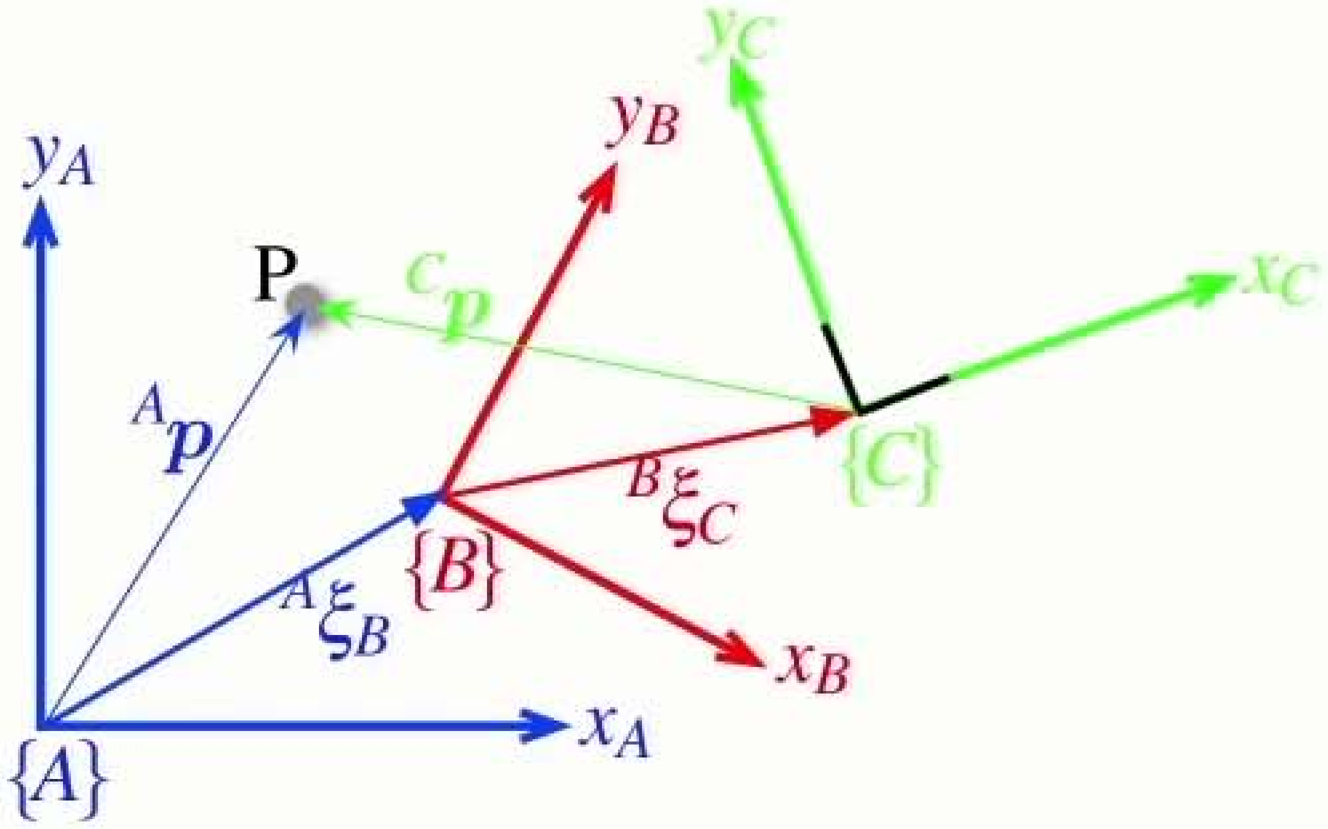
composition

$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

$${}^A\boldsymbol{p} = \left({}^A\xi_B \oplus {}^B\xi_C\right) \cdot {}^C\boldsymbol{p}$$

- Relative poses can be **compounded** or **composed**

Relative **points**



$$ {}^{A}\boldsymbol{p} = {}^{A}\xi_B \cdot {}^{B}\boldsymbol{p} $$

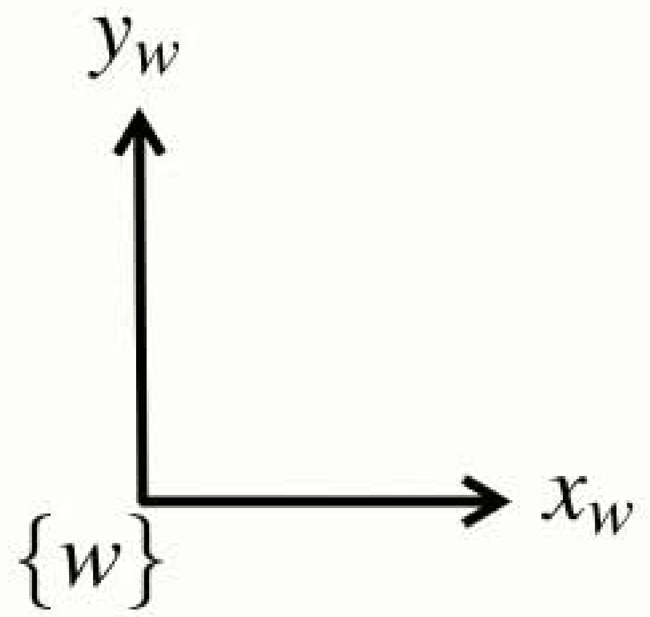
composition

$$ {}^{A}\xi_C = {}^{A}\xi_B \oplus {}^{B}\xi_C $$

$$ {}^{A}\boldsymbol{p} = \left({}^{A}\xi_B \oplus {}^{B}\xi_C \right) \cdot {}^{C}\boldsymbol{p} $$

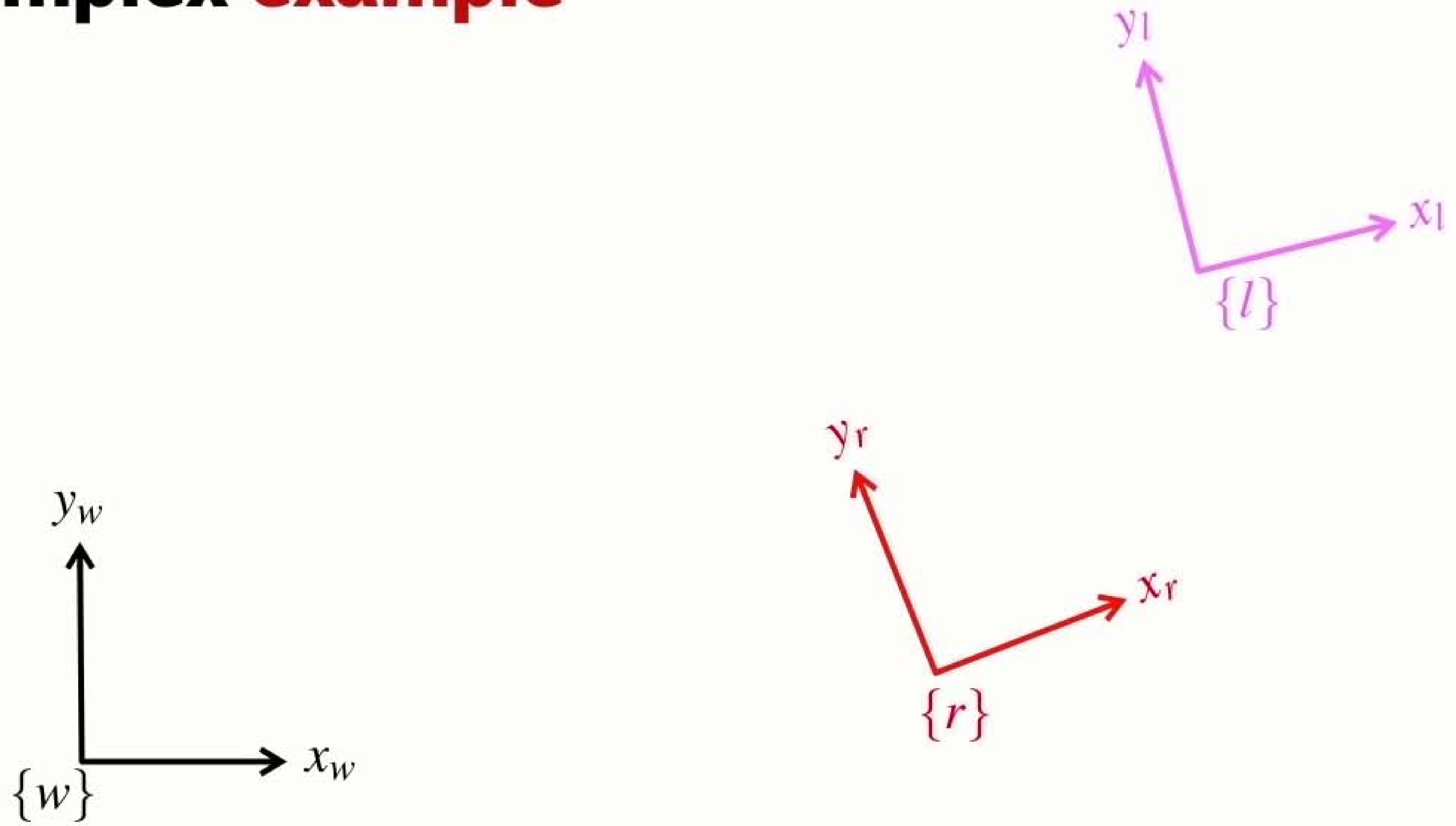
- Relative poses can be **compounded** or **composed**
- We can extend this approach indefinitely...

Complex example

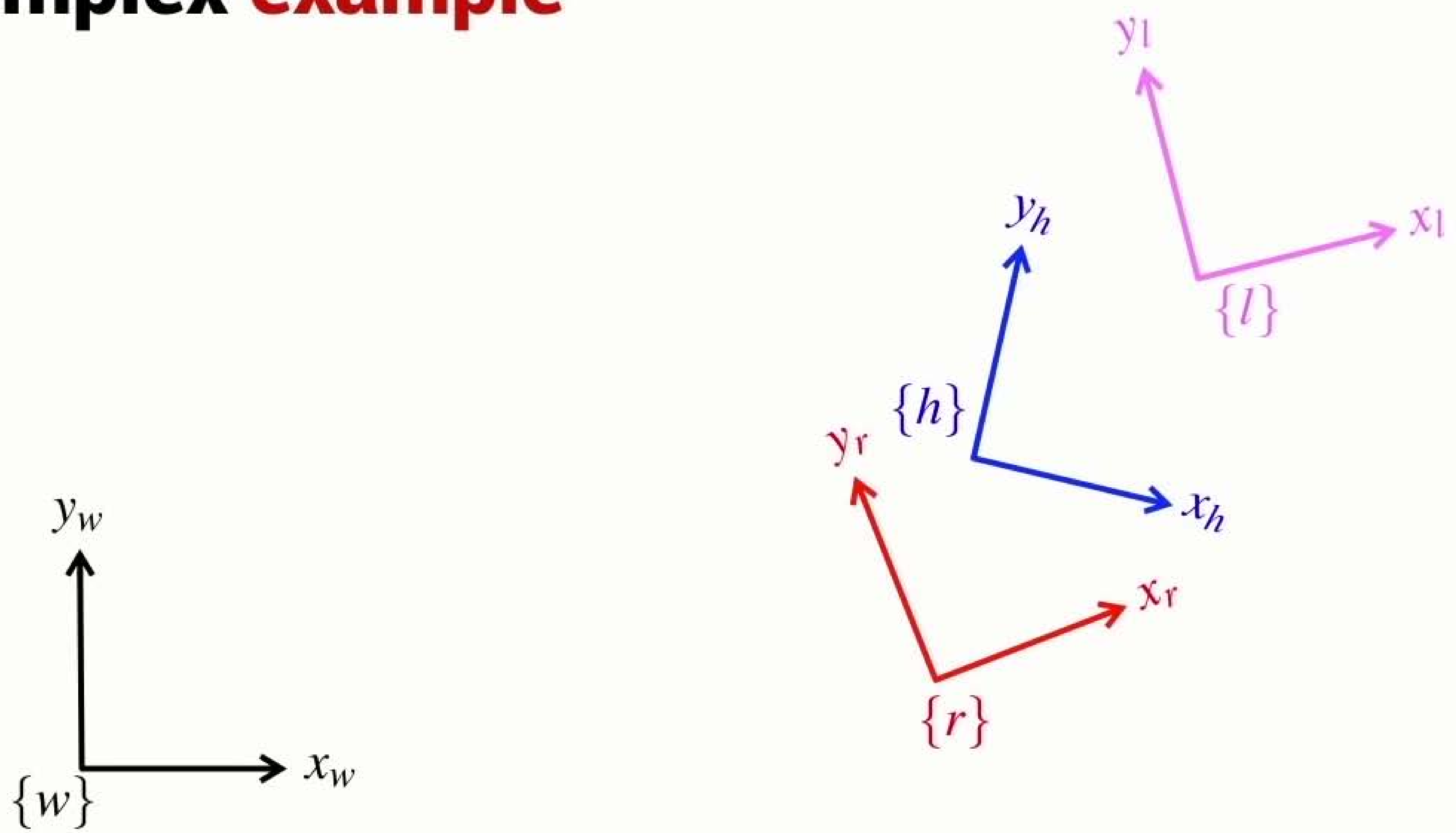


Complex example

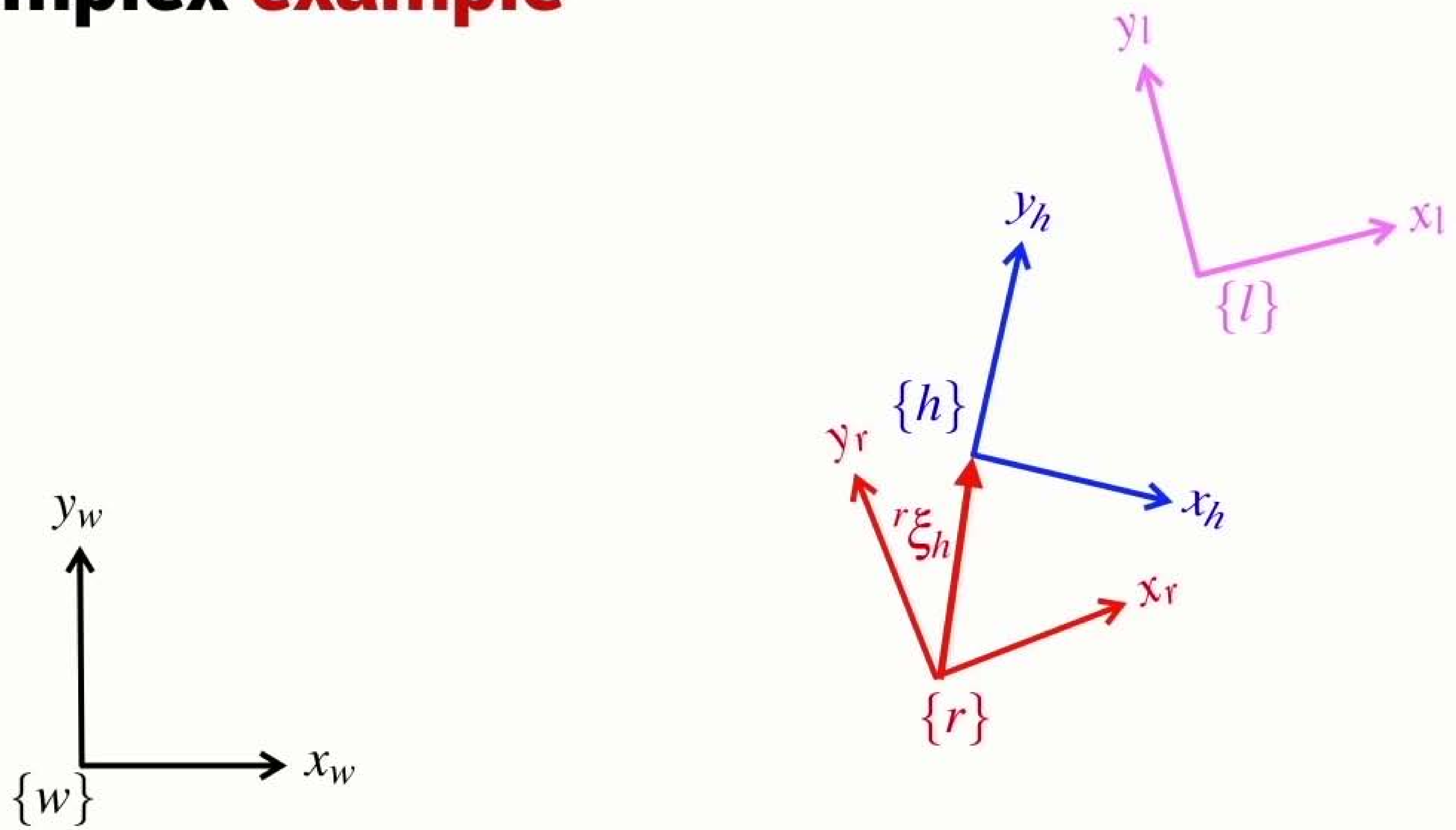
Complex **example**



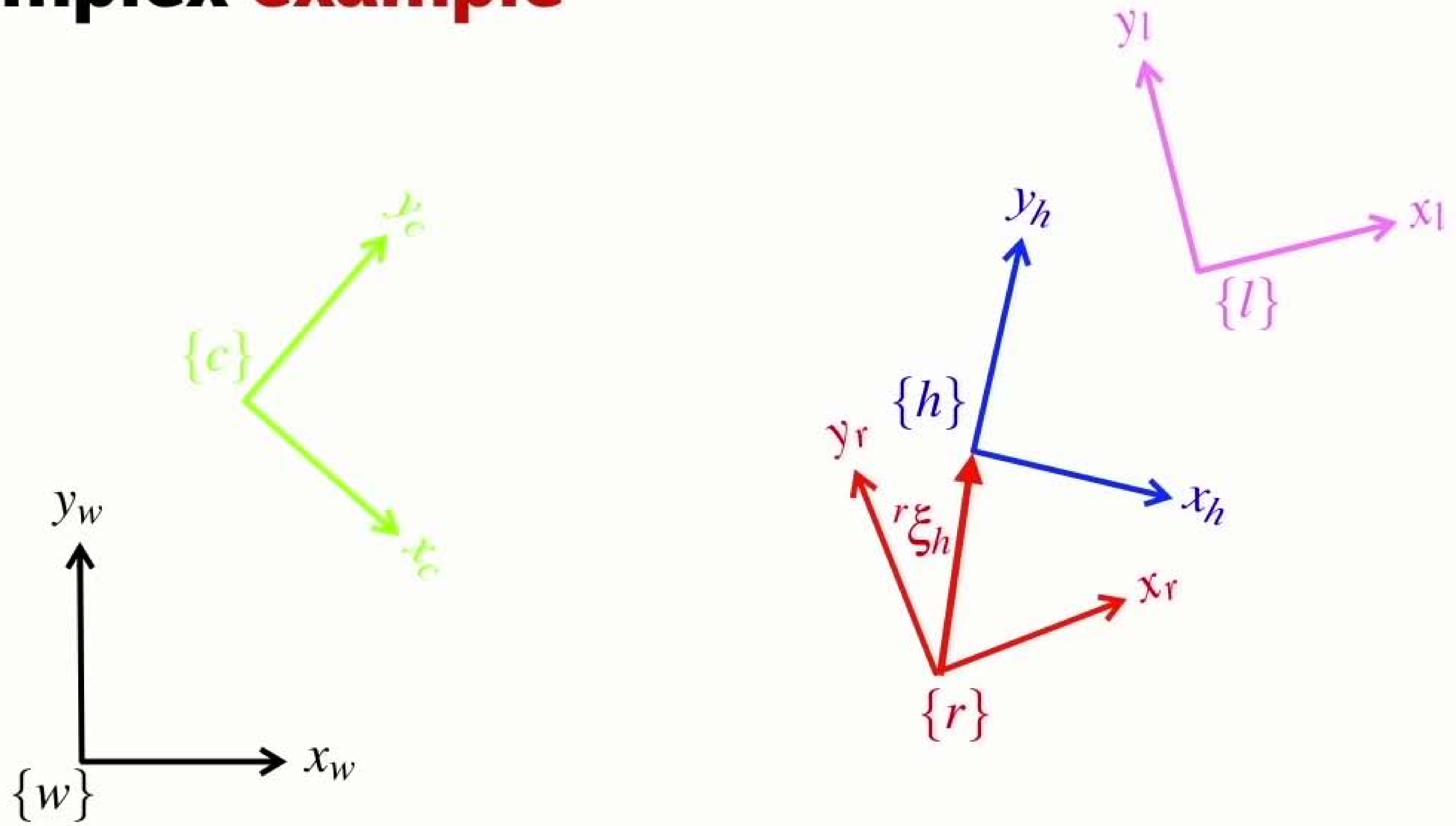
Complex **example**



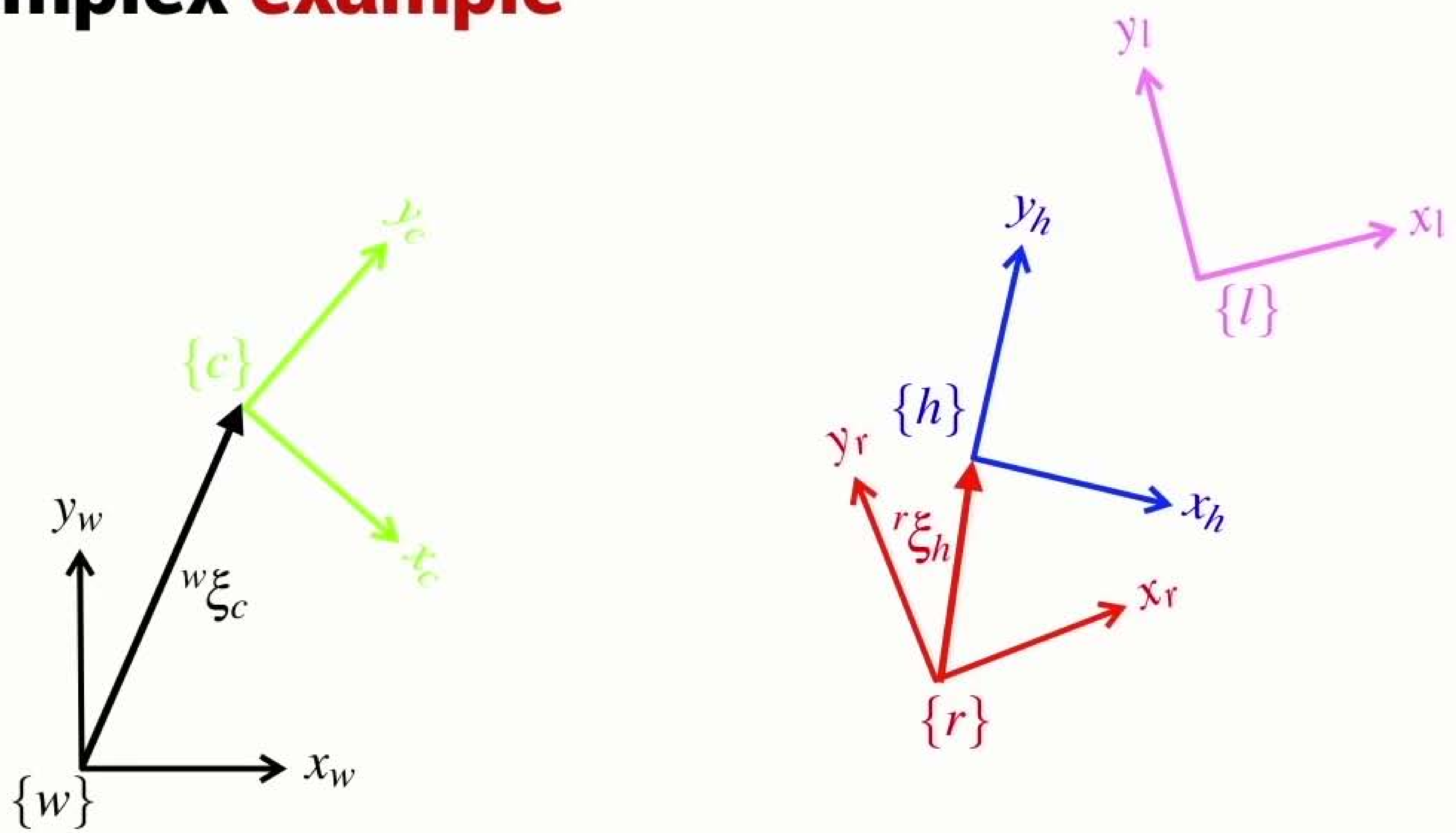
Complex **example**



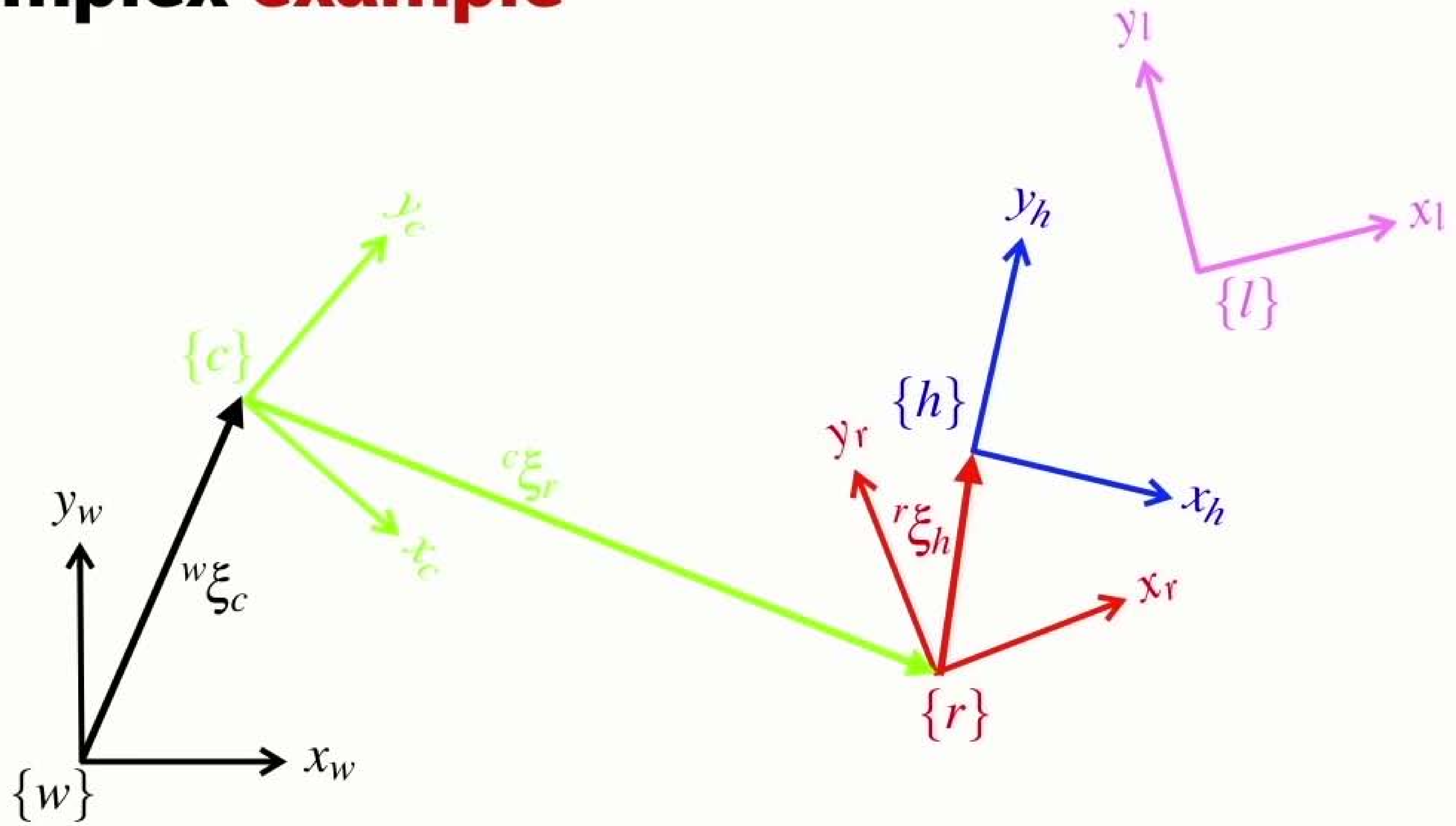
Complex example



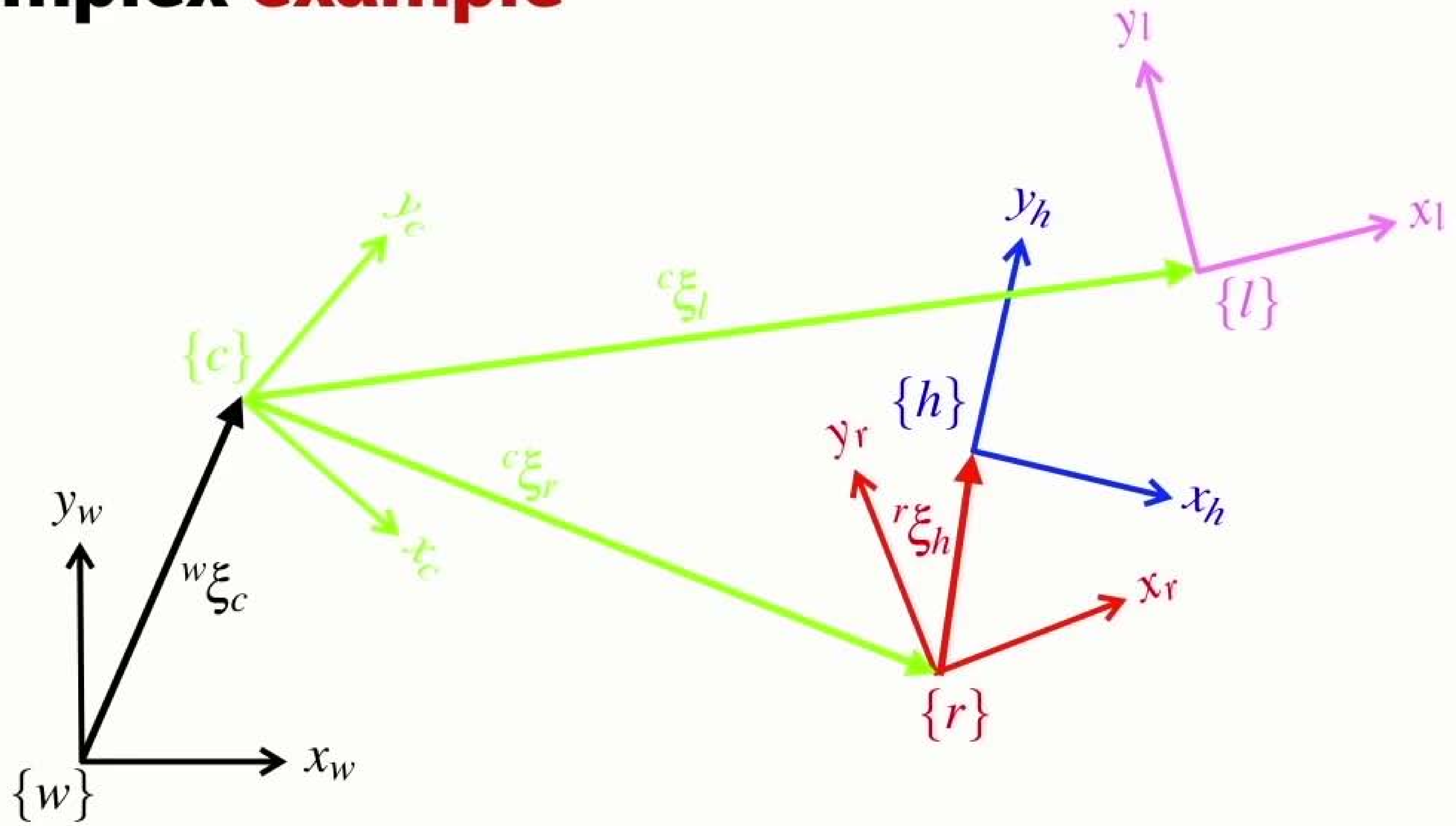
Complex example



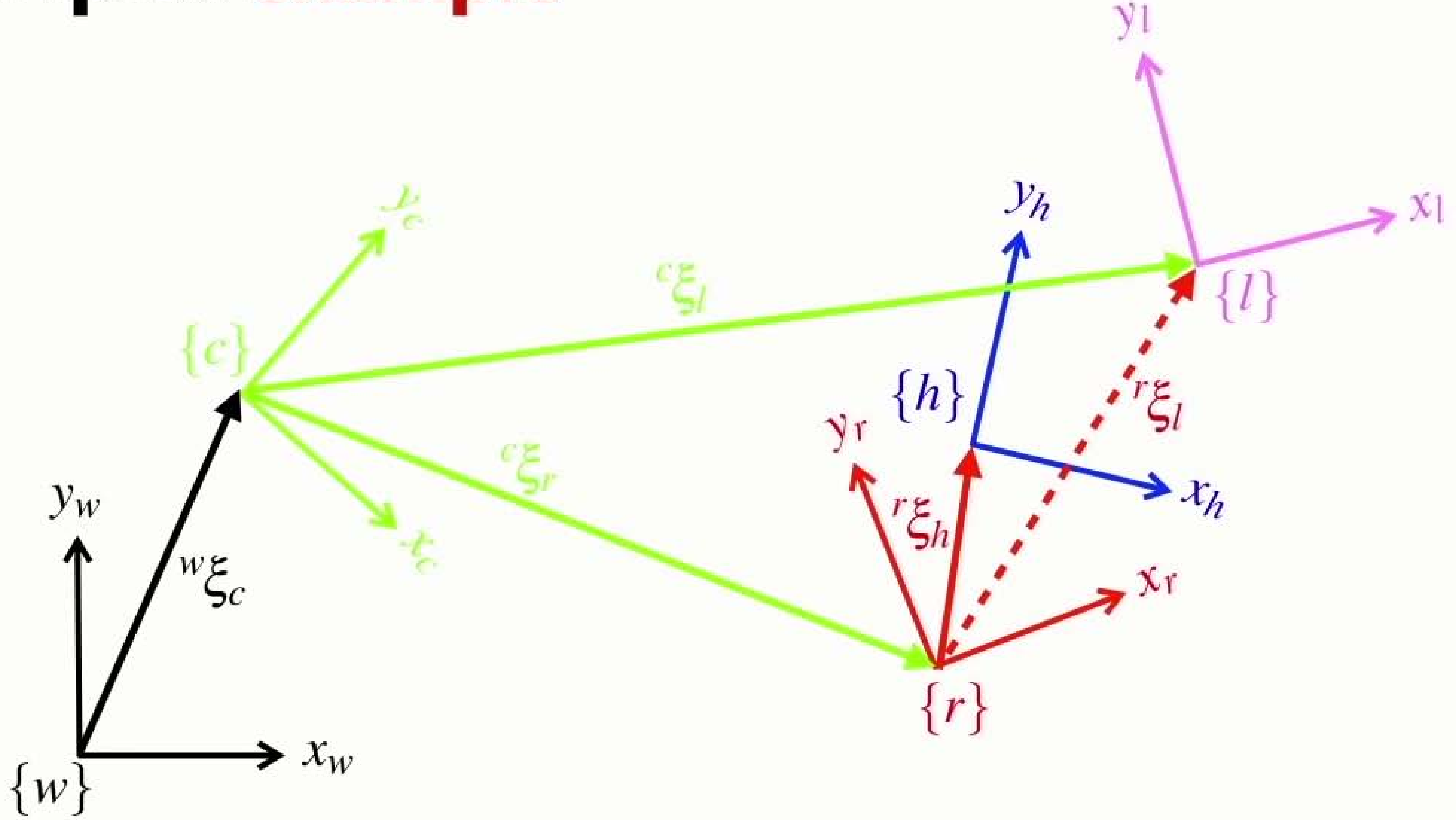
Complex example



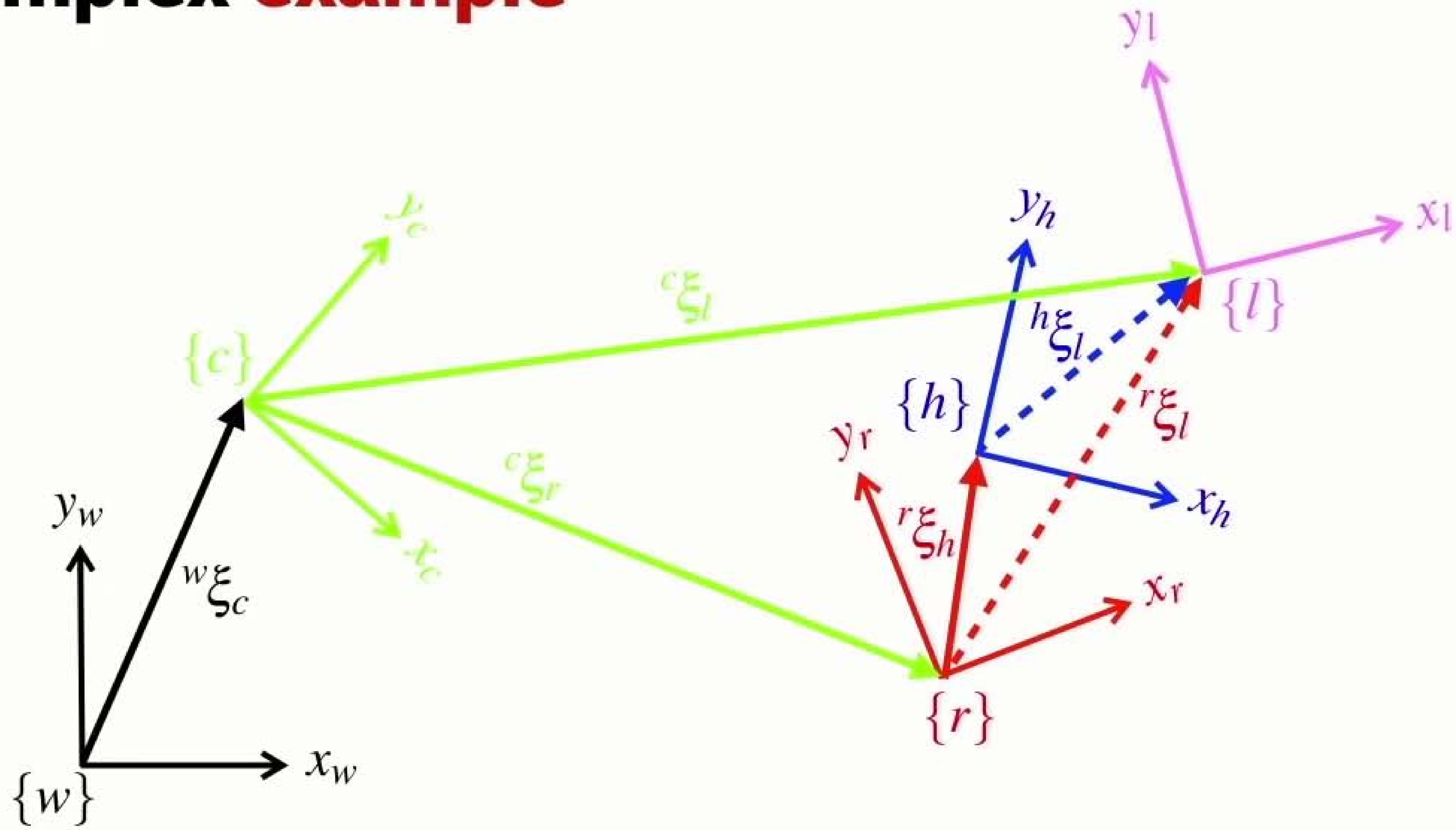
Complex example



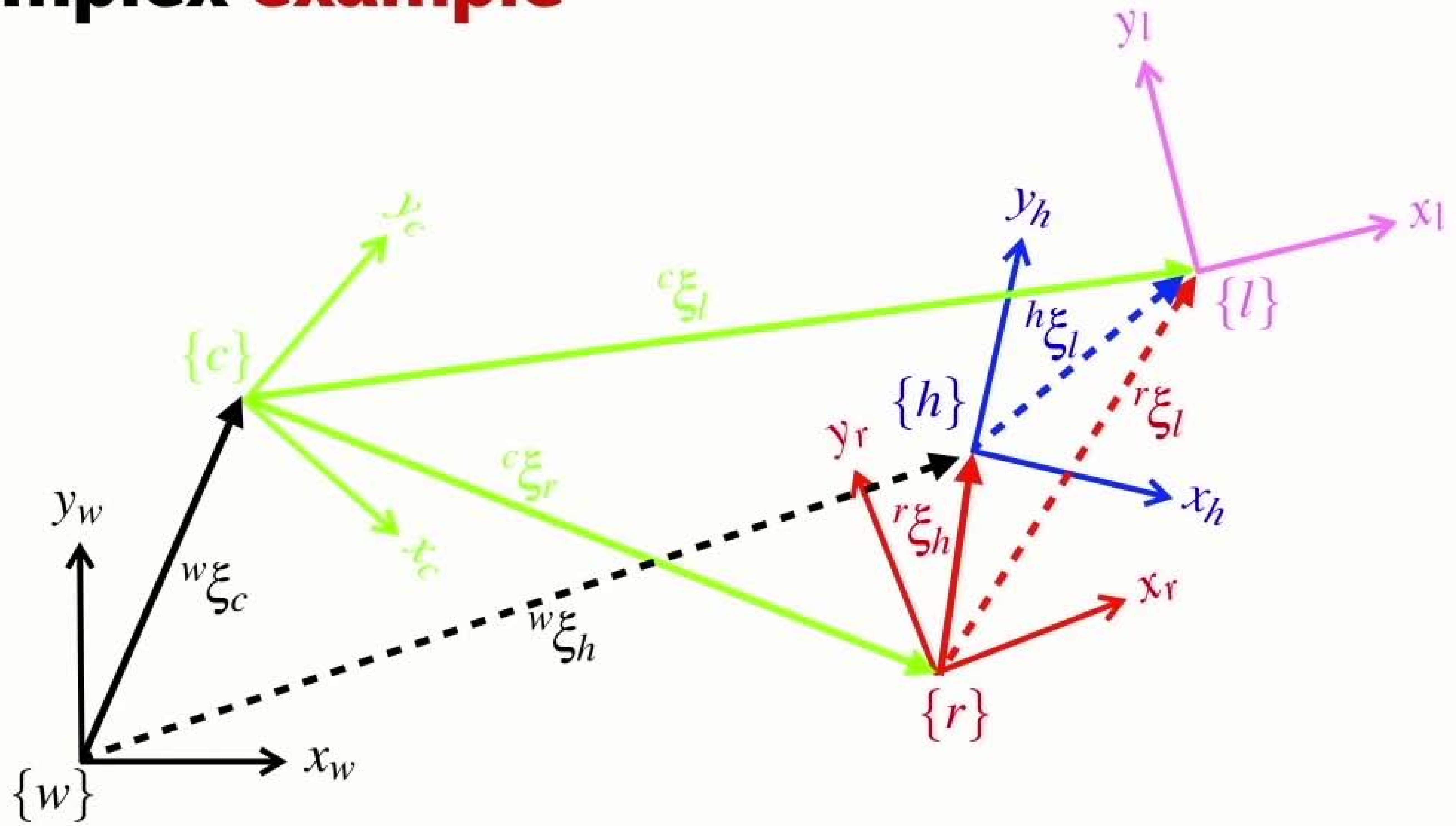
Complex **example**



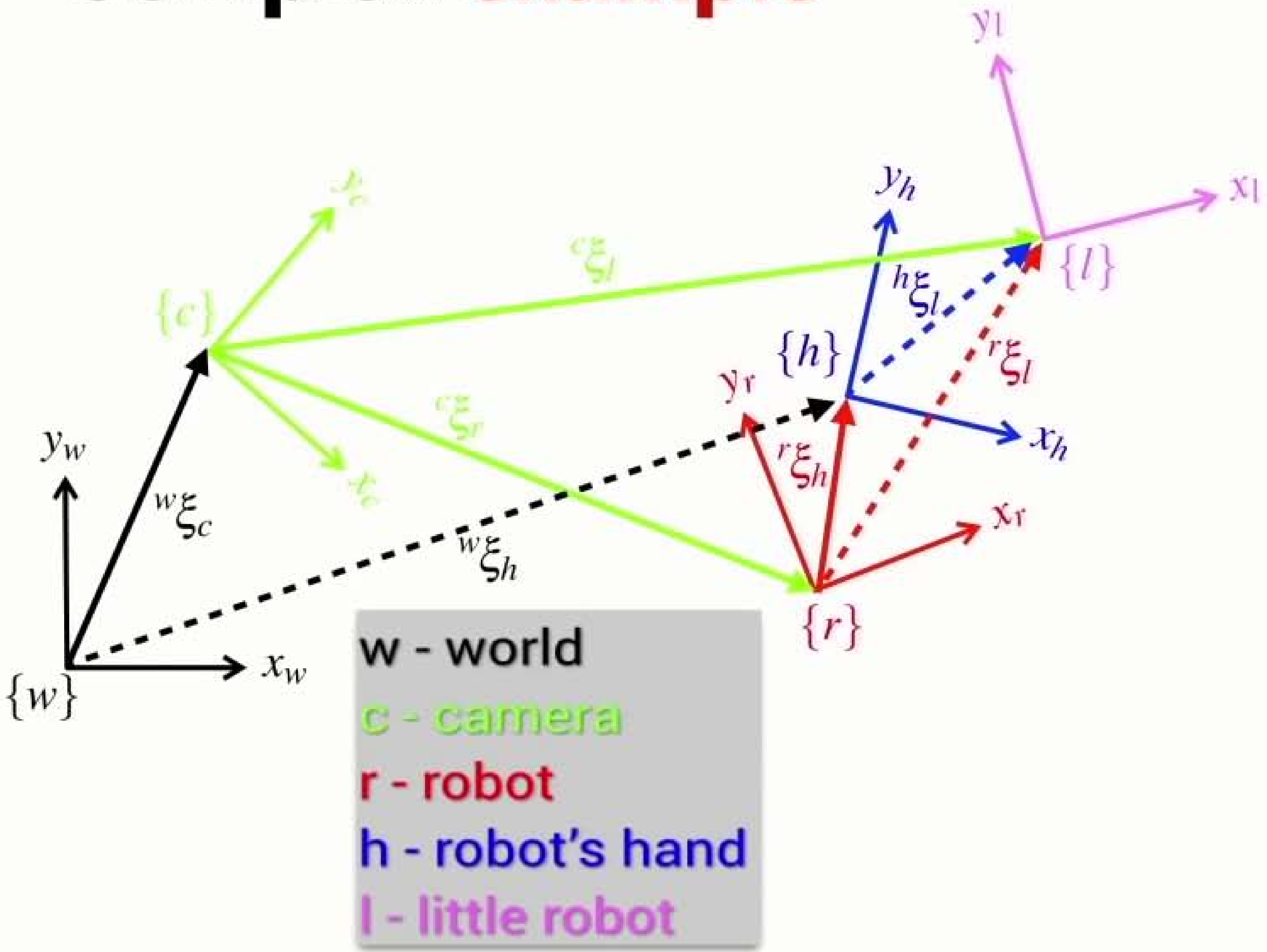
Complex example



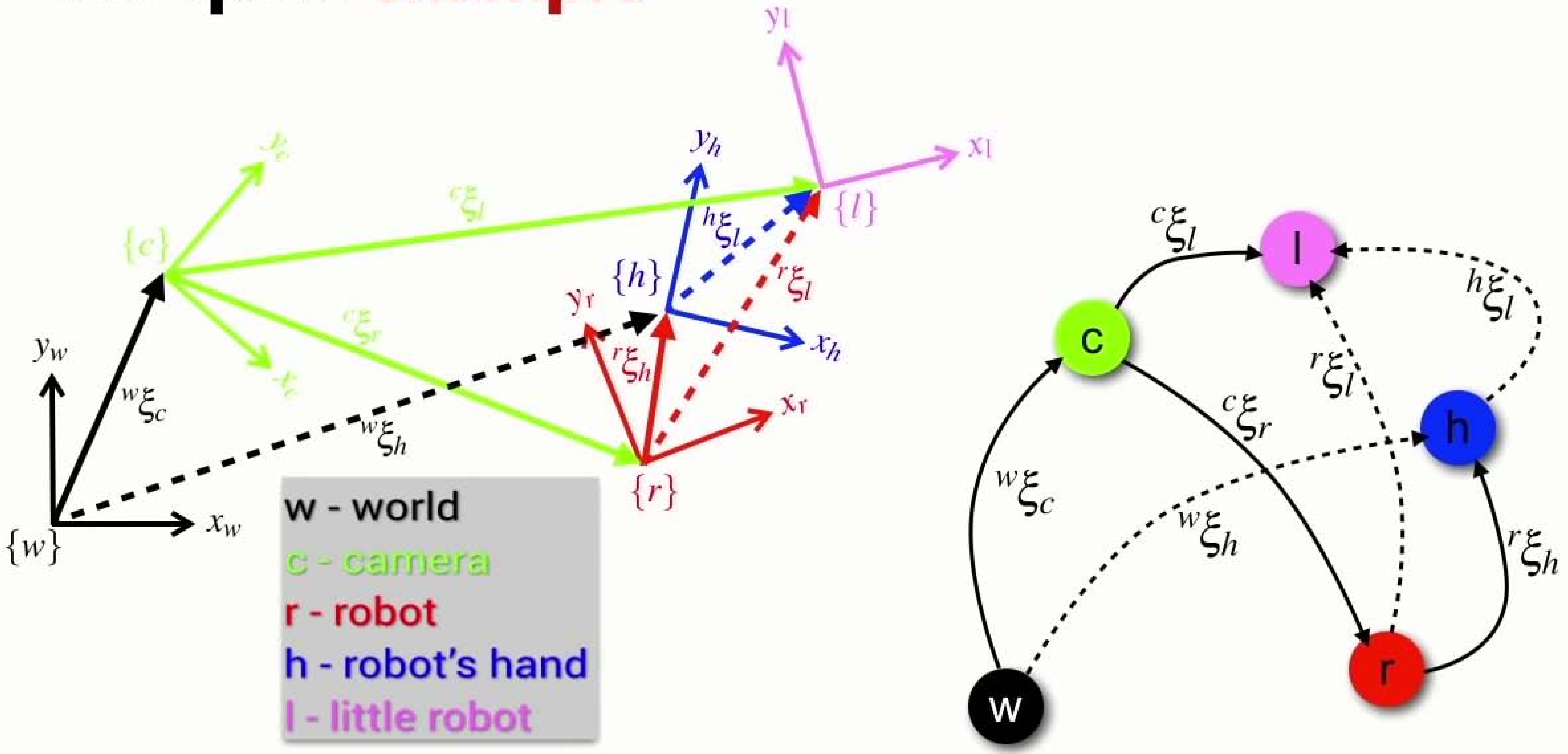
Complex example



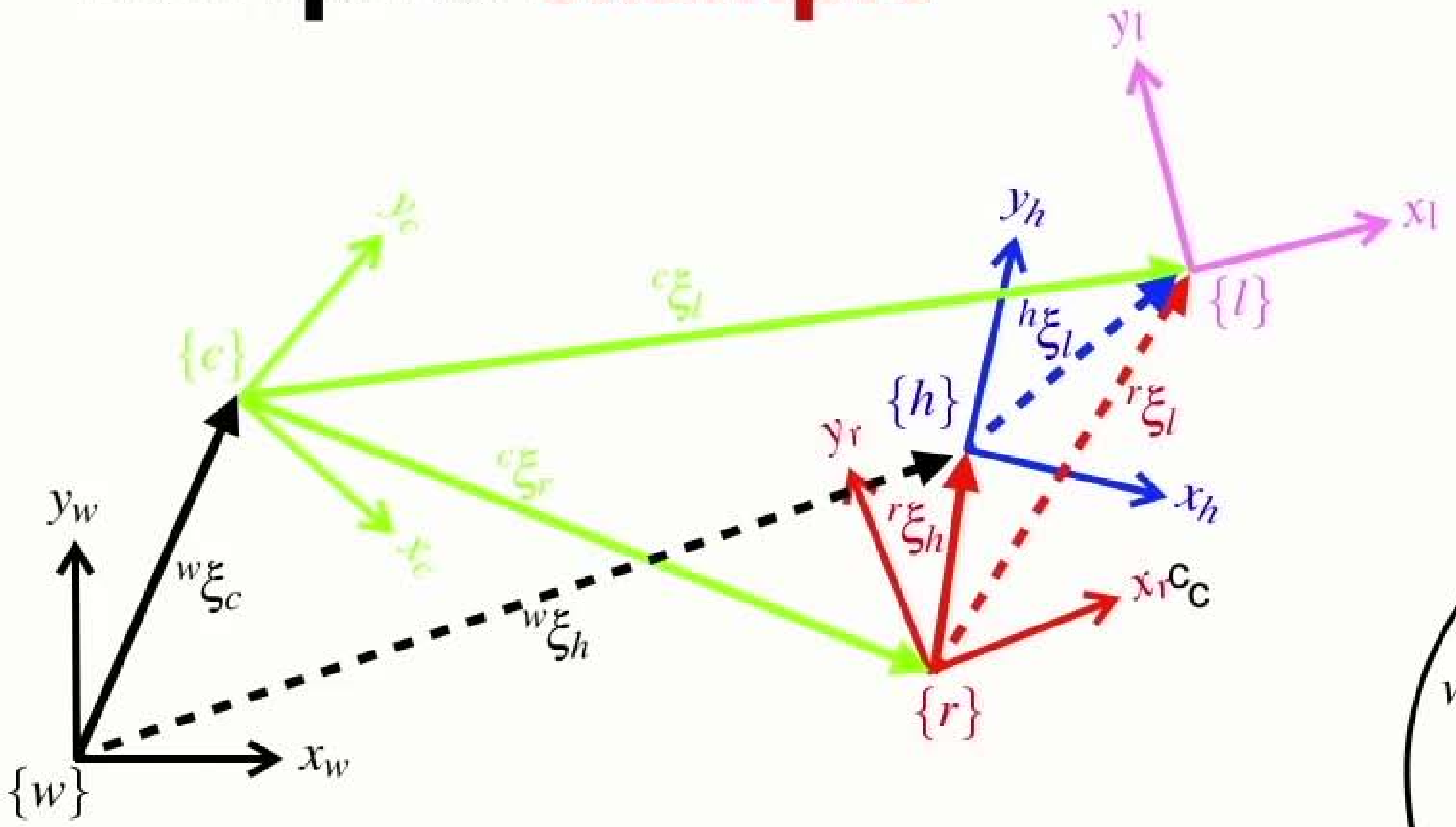
Complex example

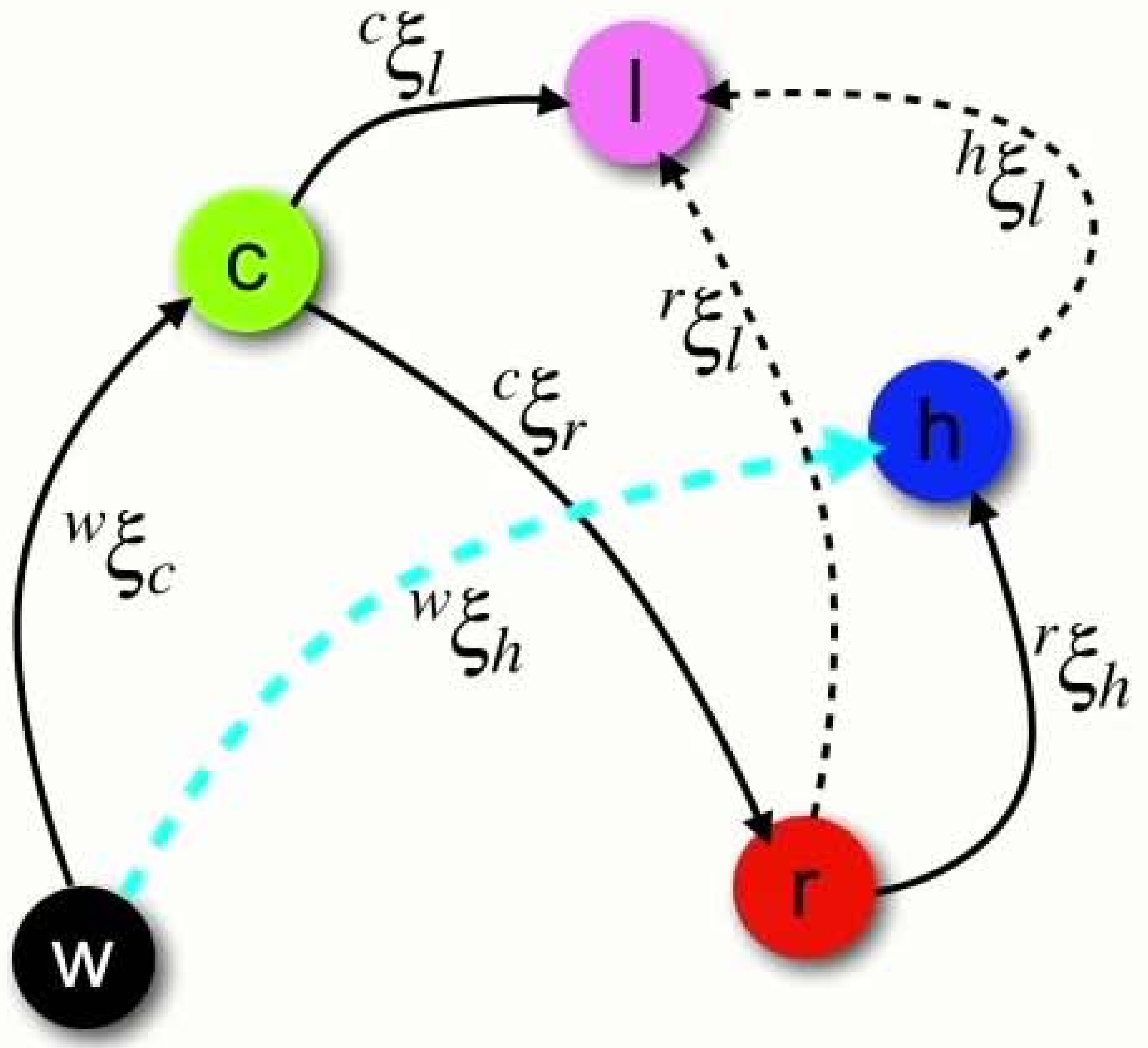


Complex example

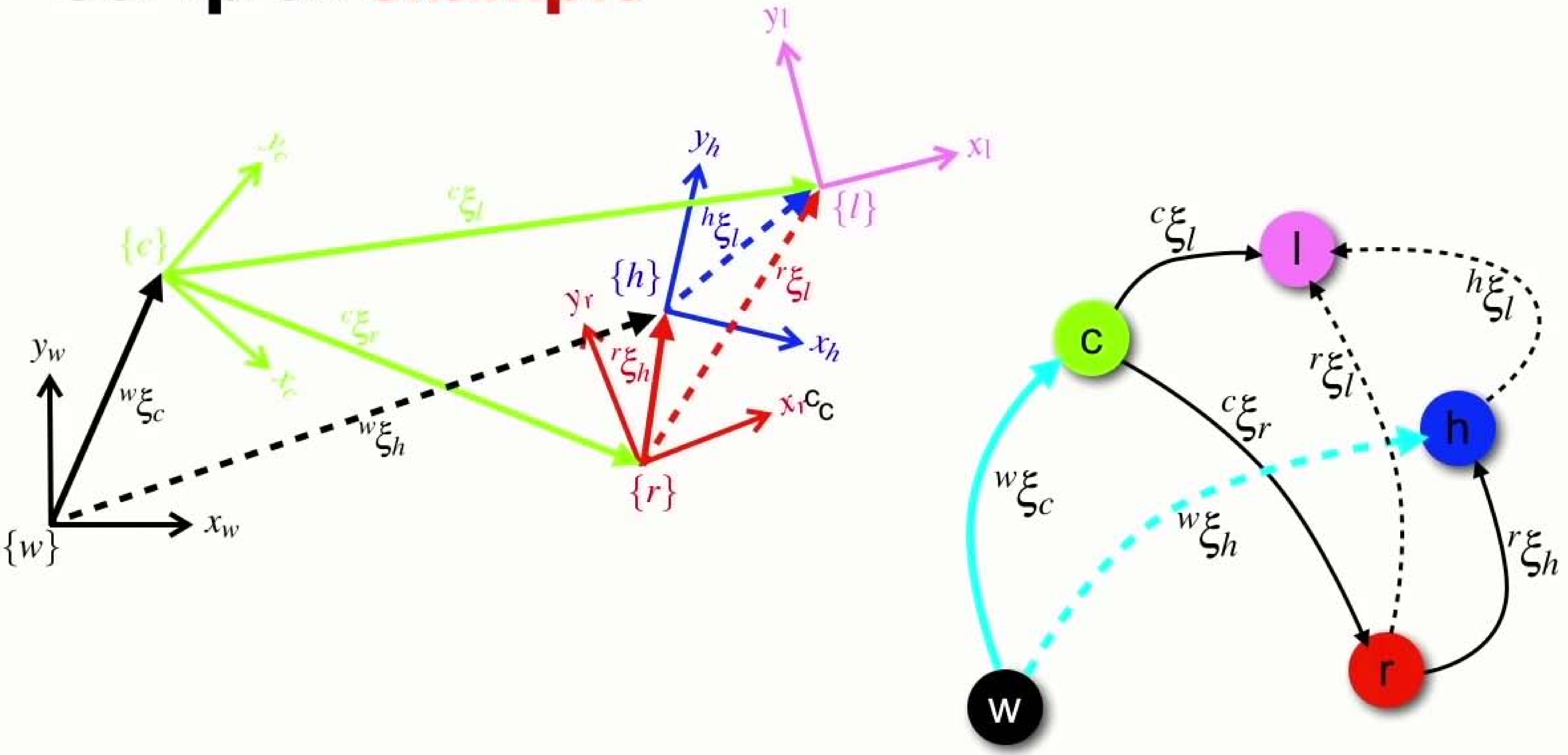


Complex example

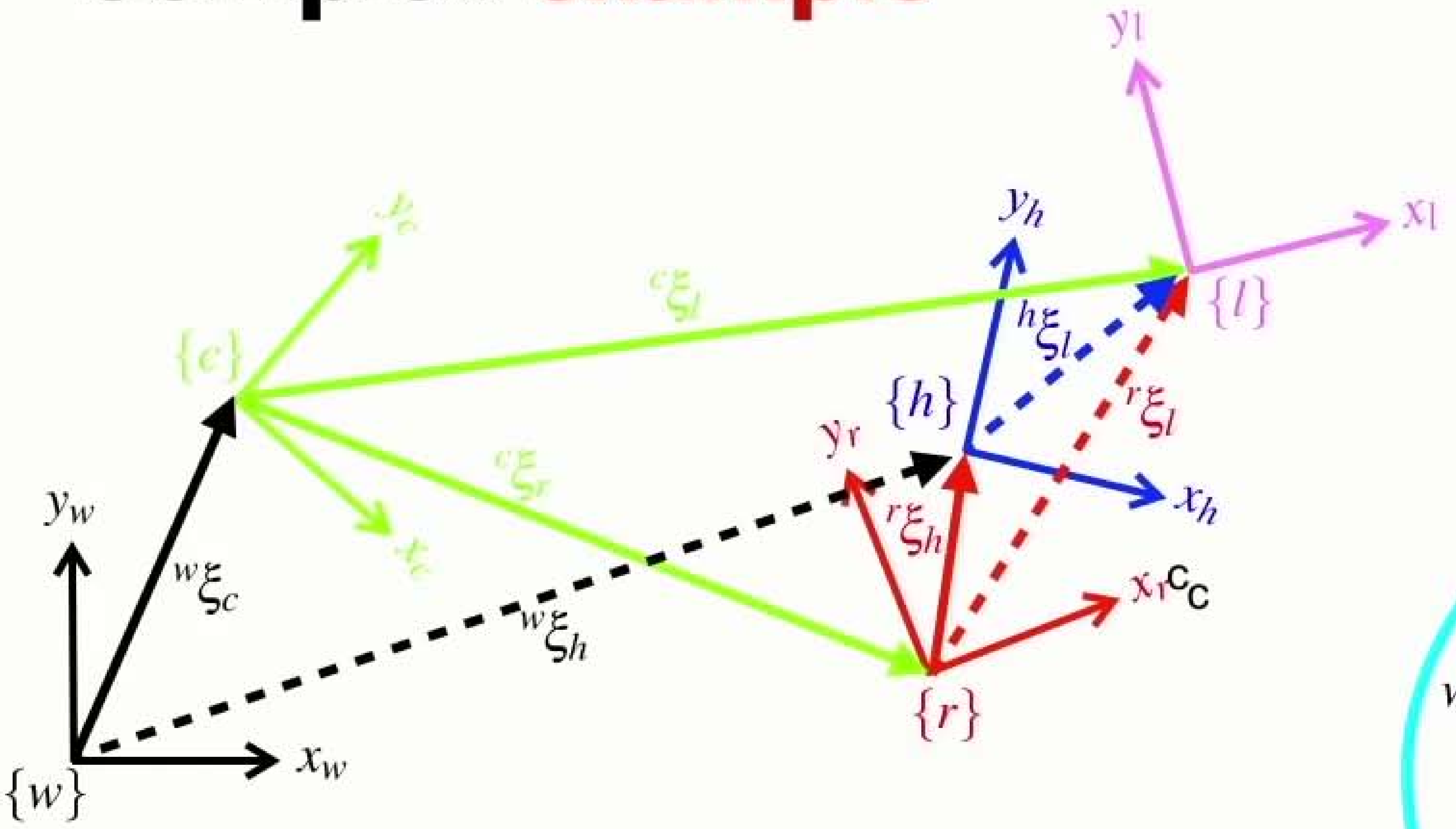


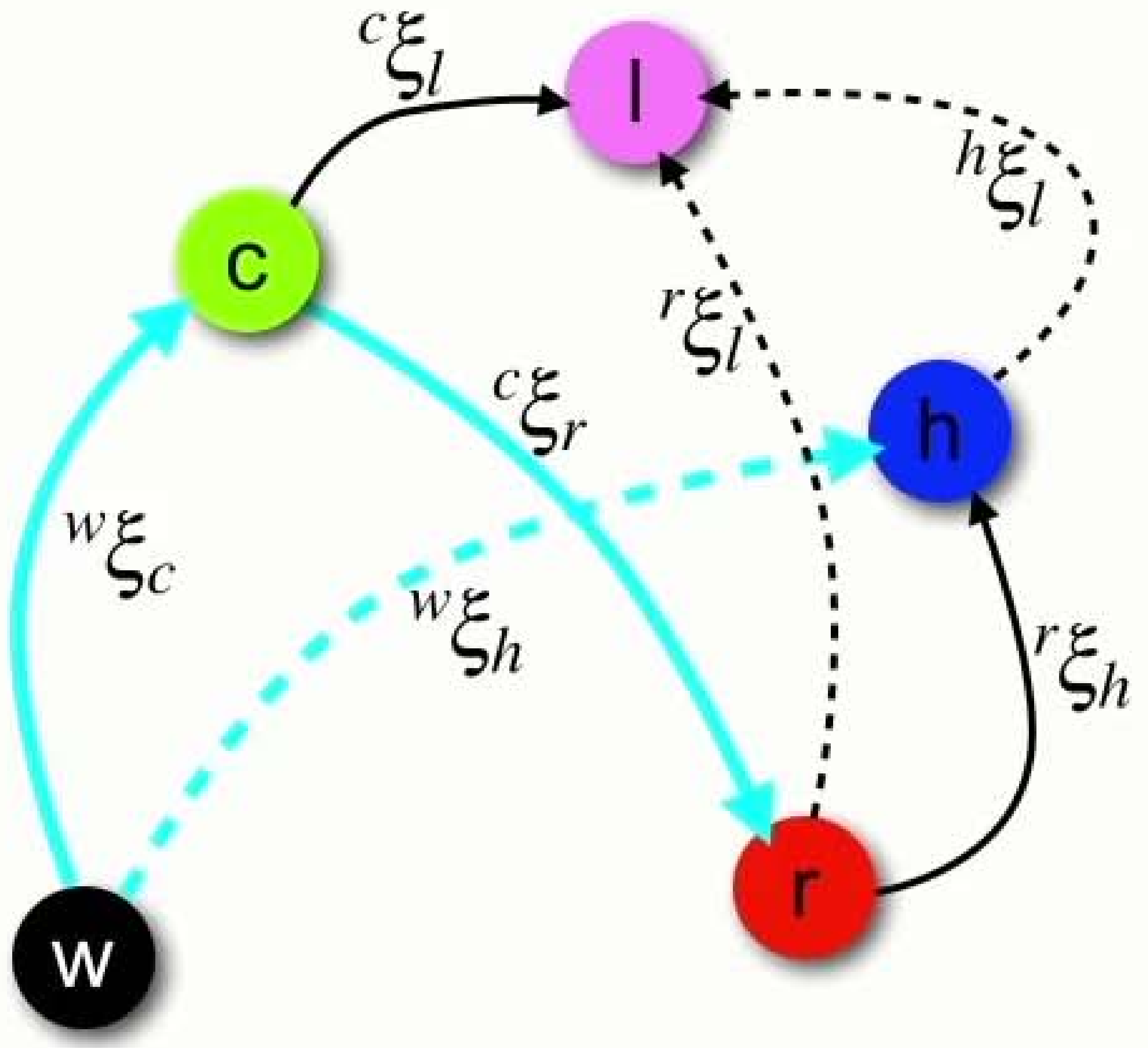


Complex example

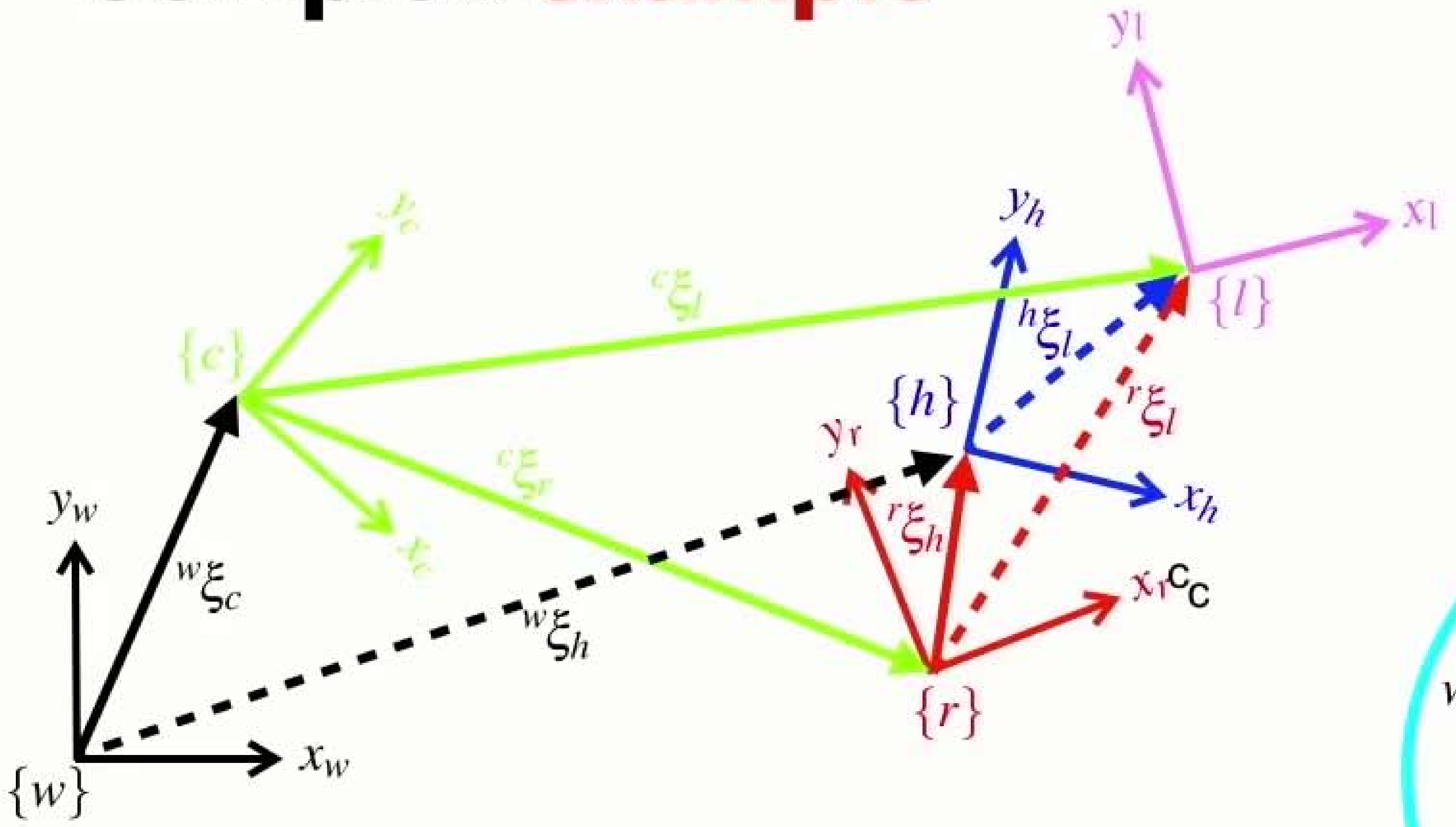


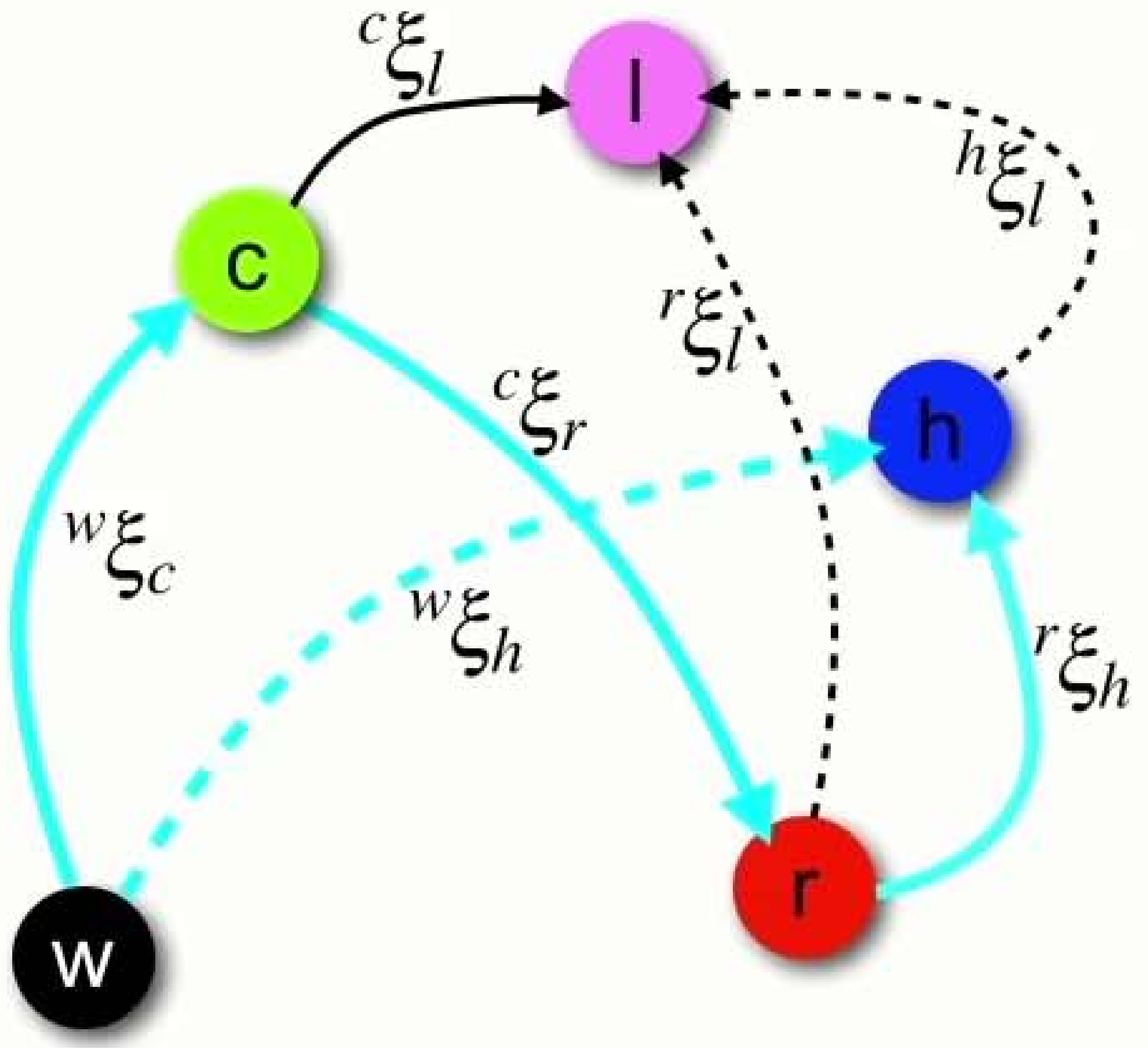
Complex example



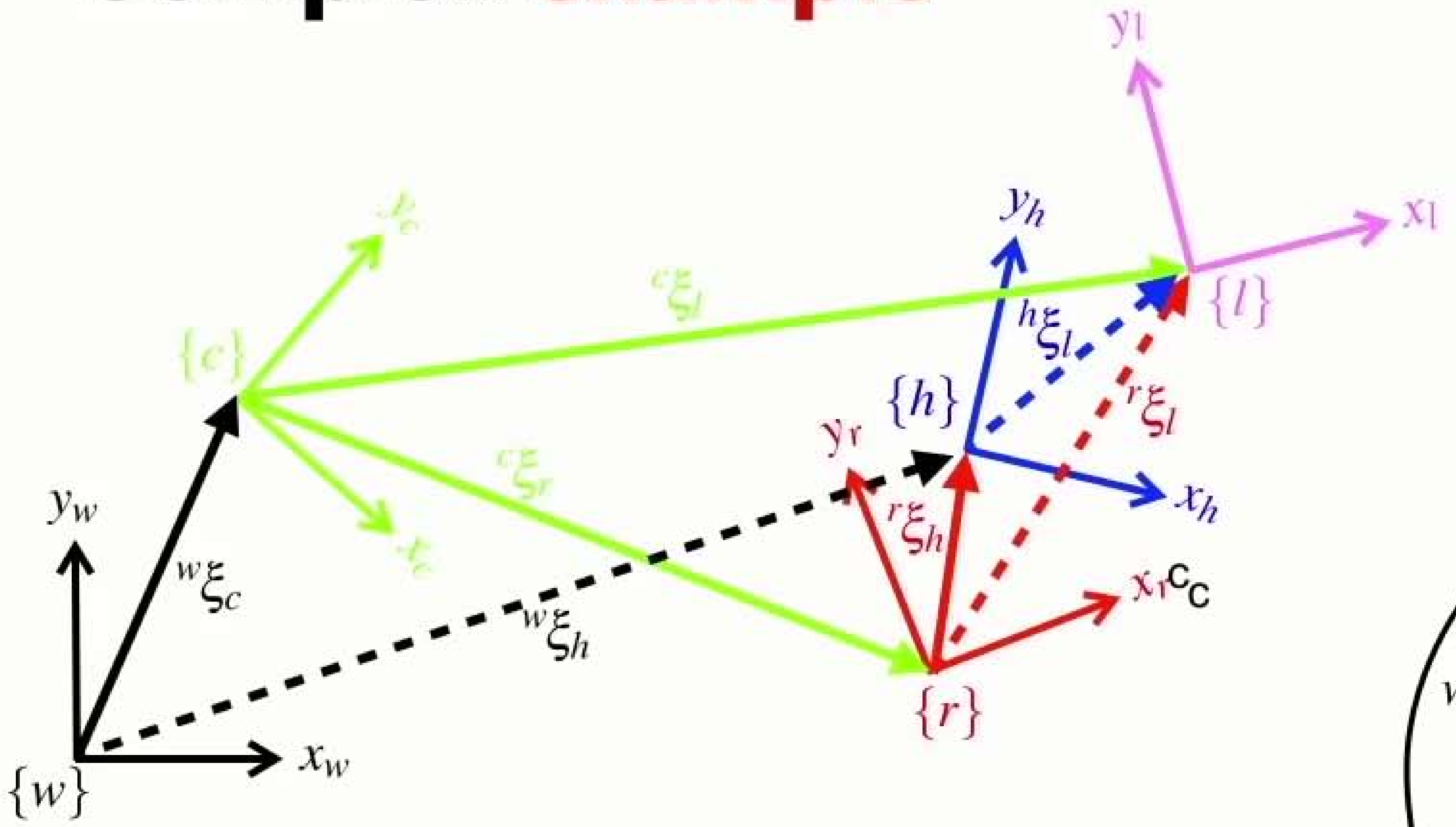


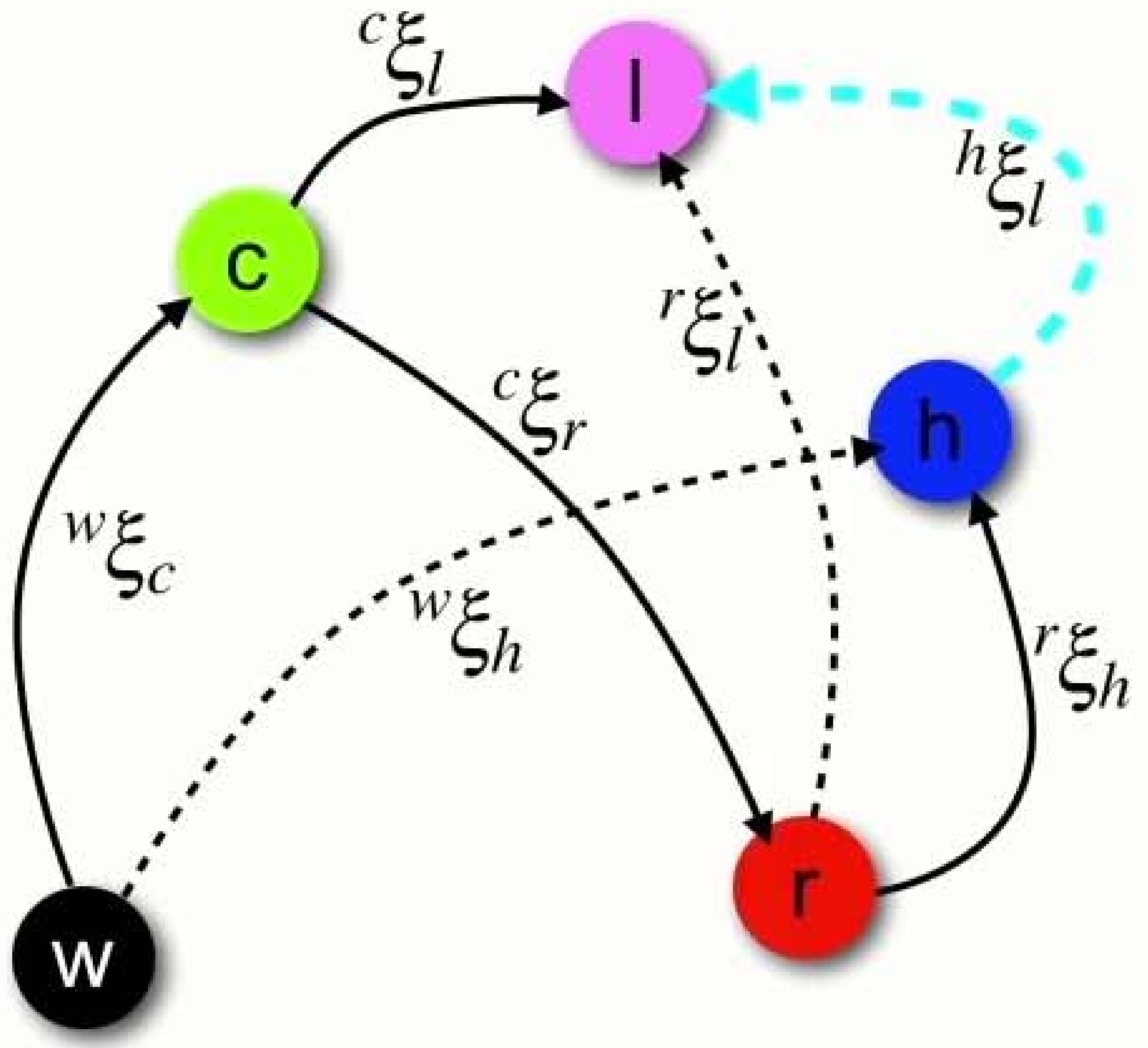
Complex example



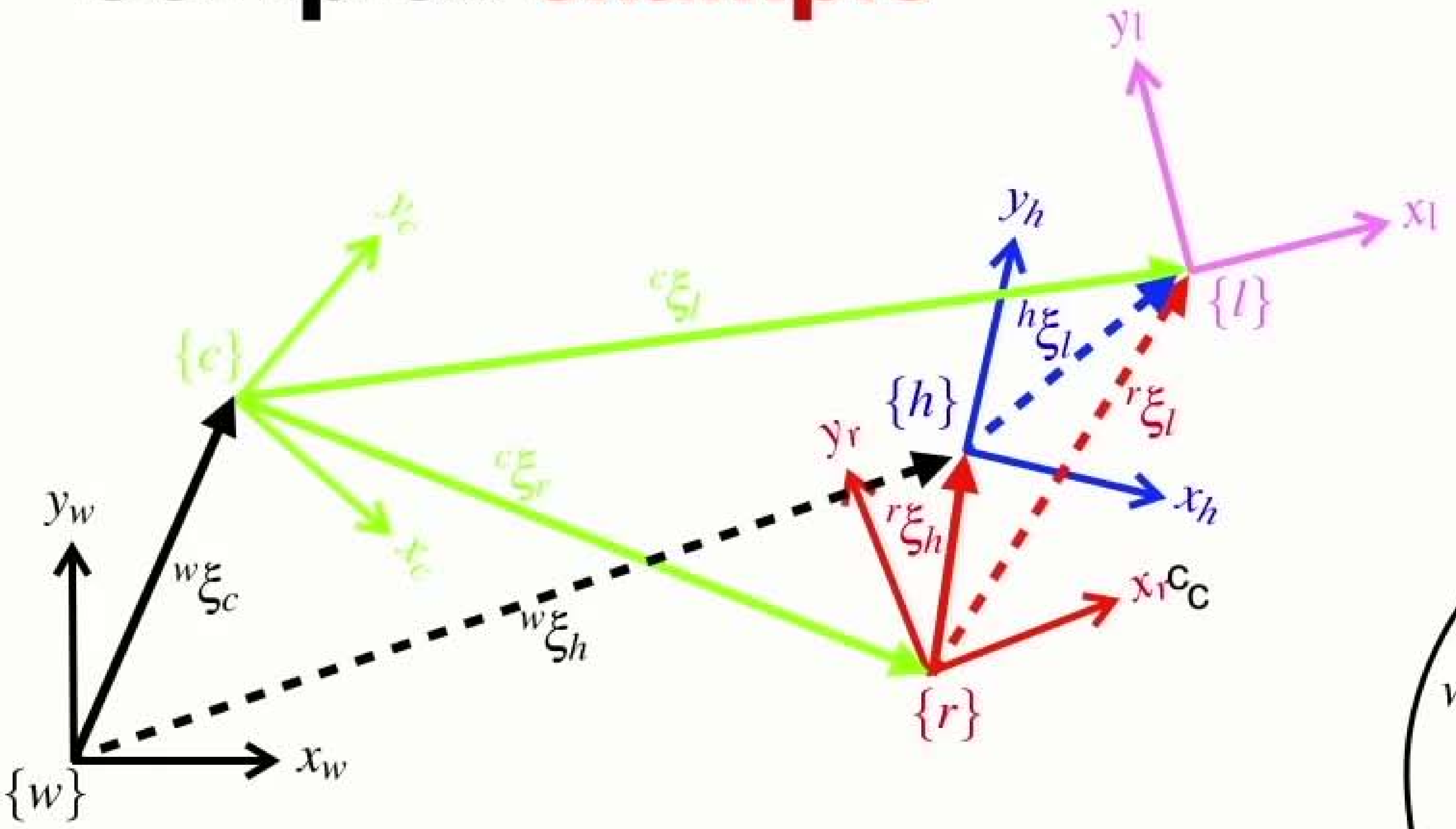


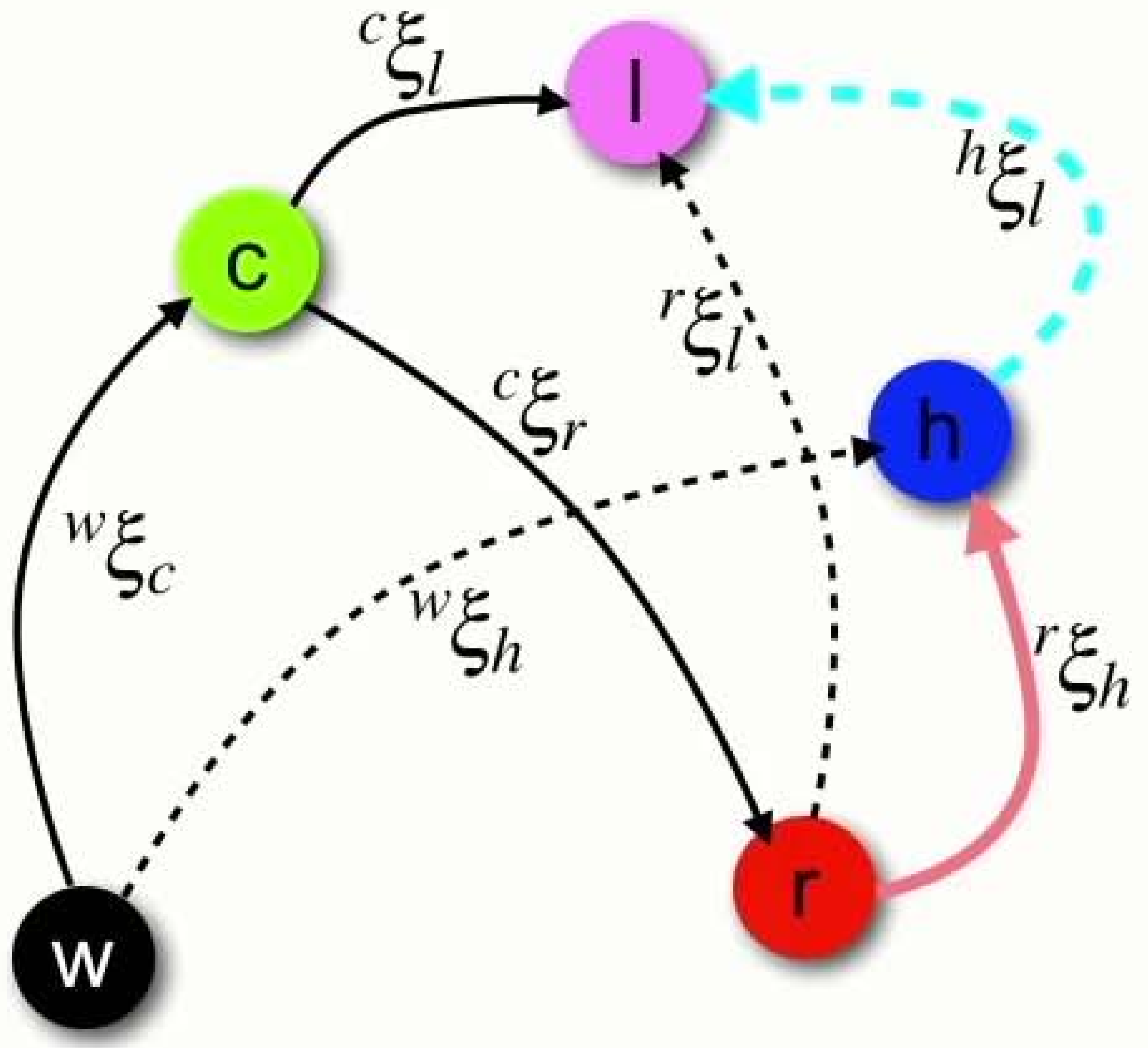
Complex example



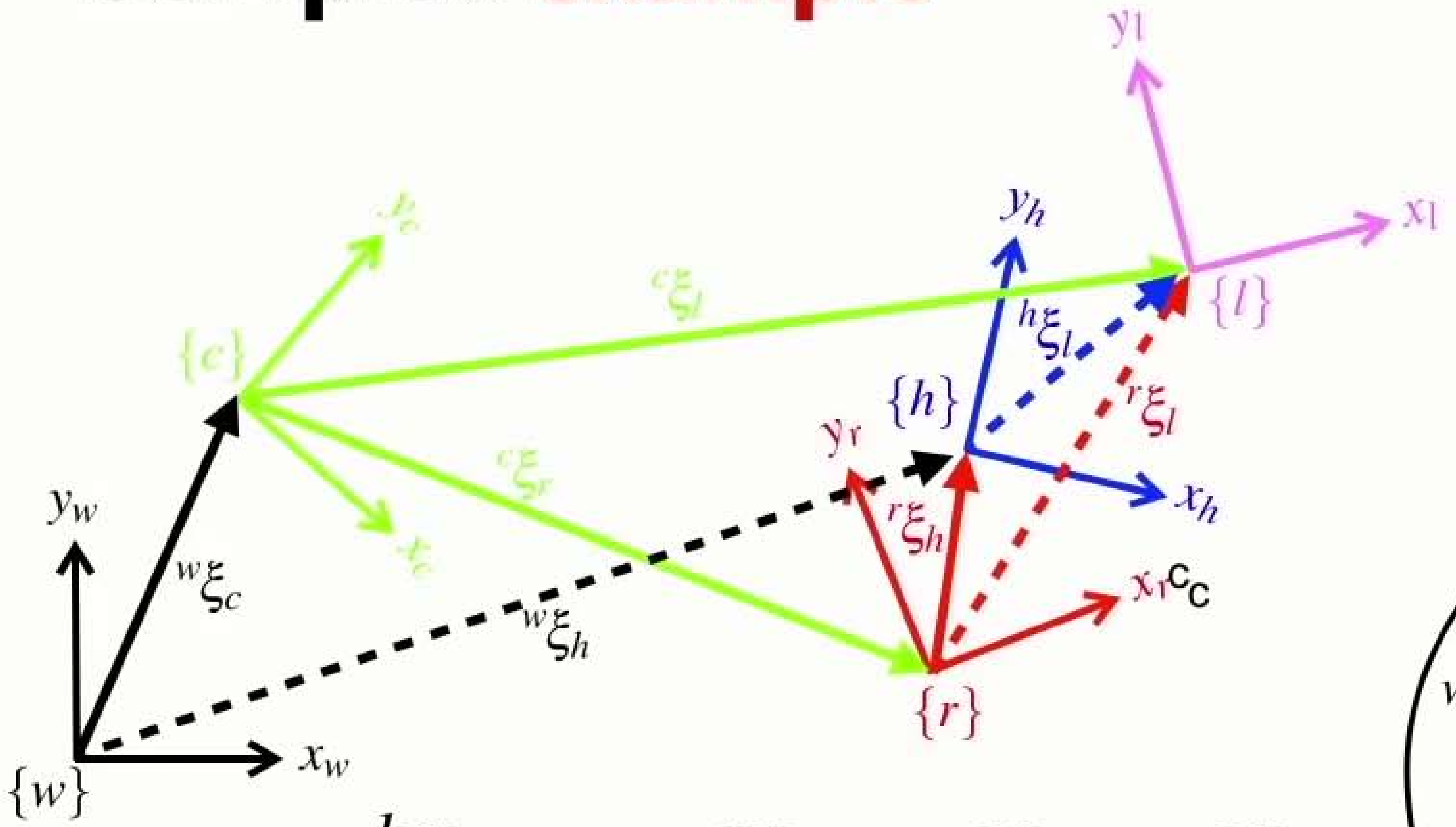


Complex example



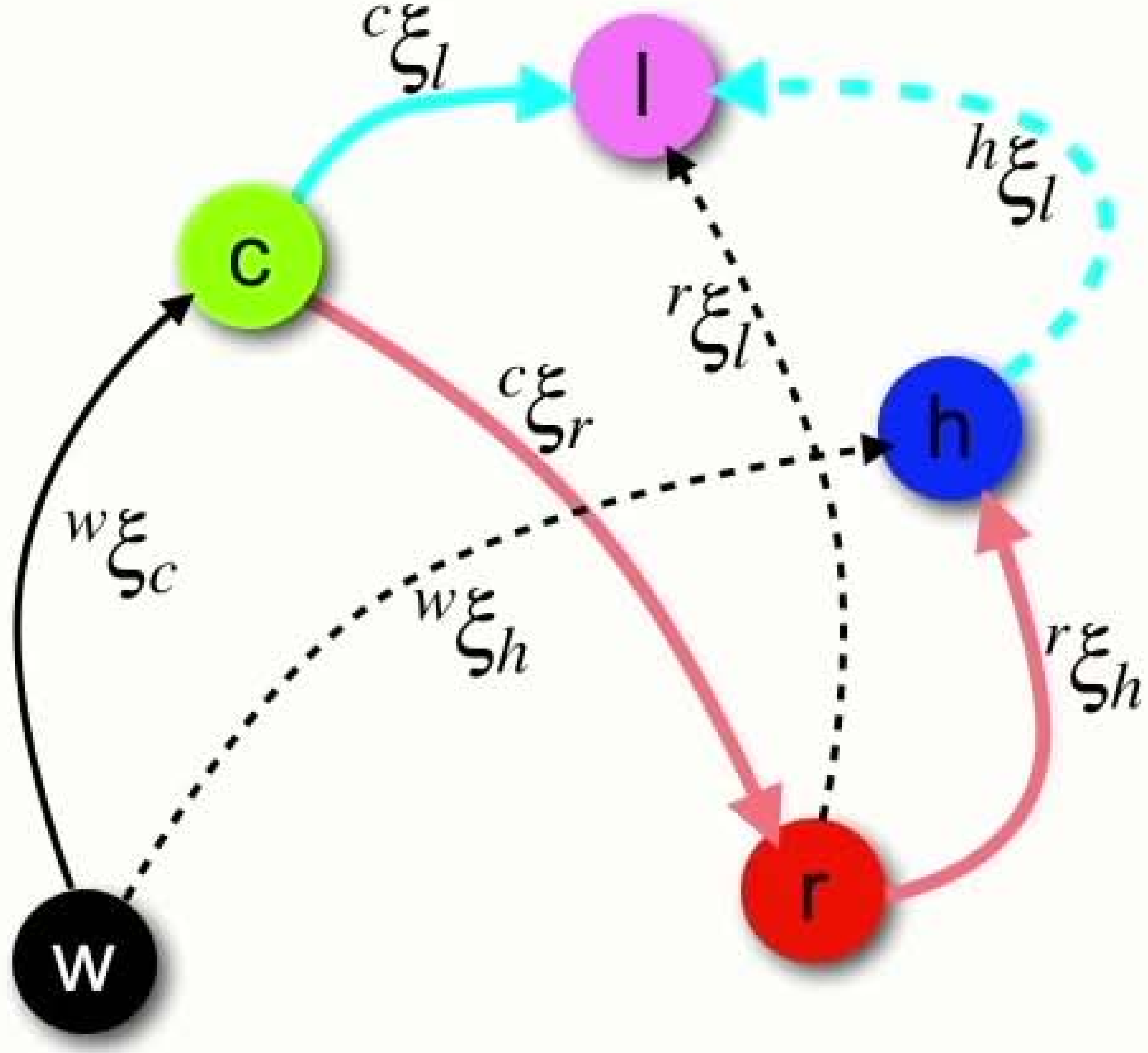


Complex example

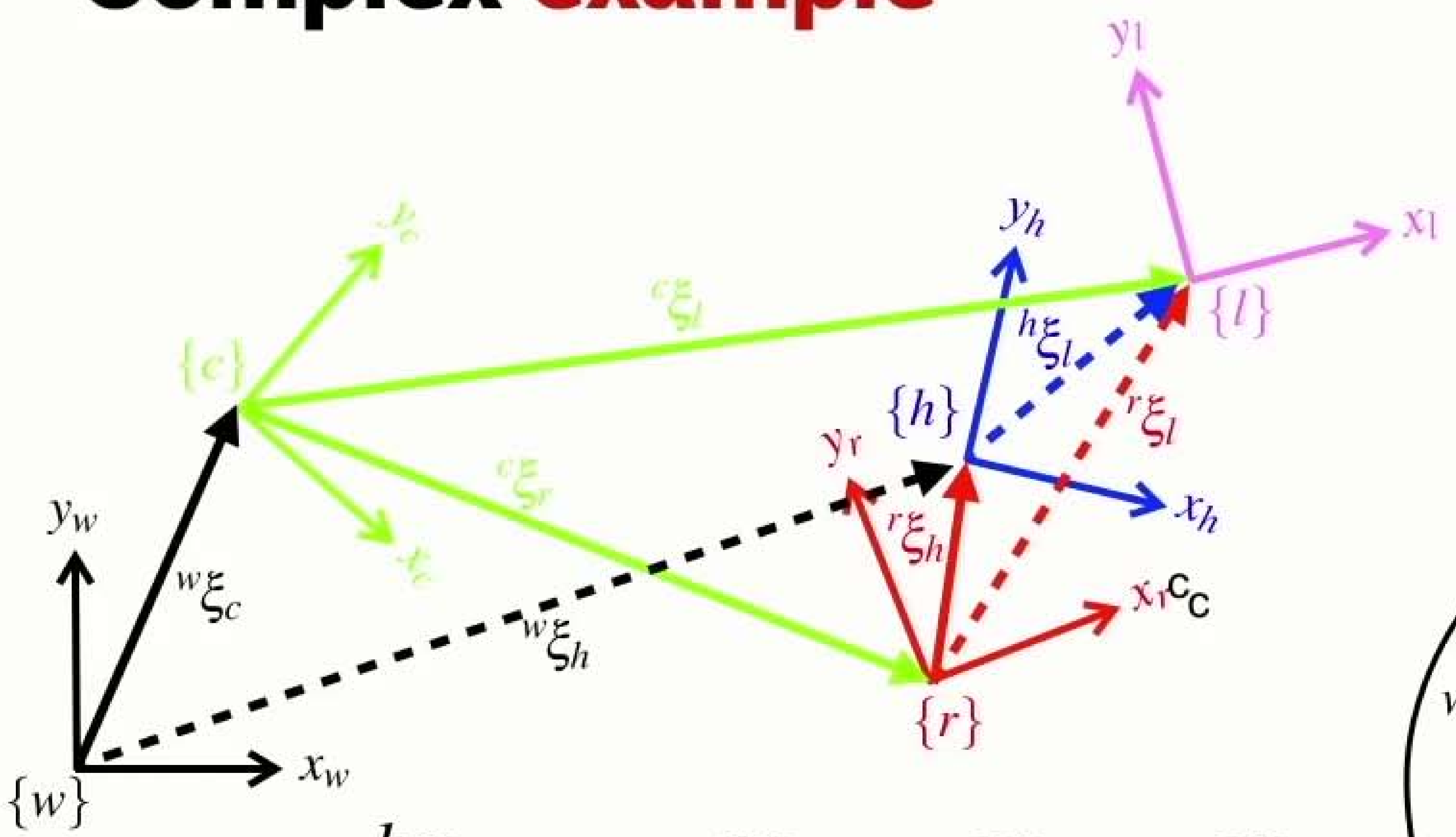


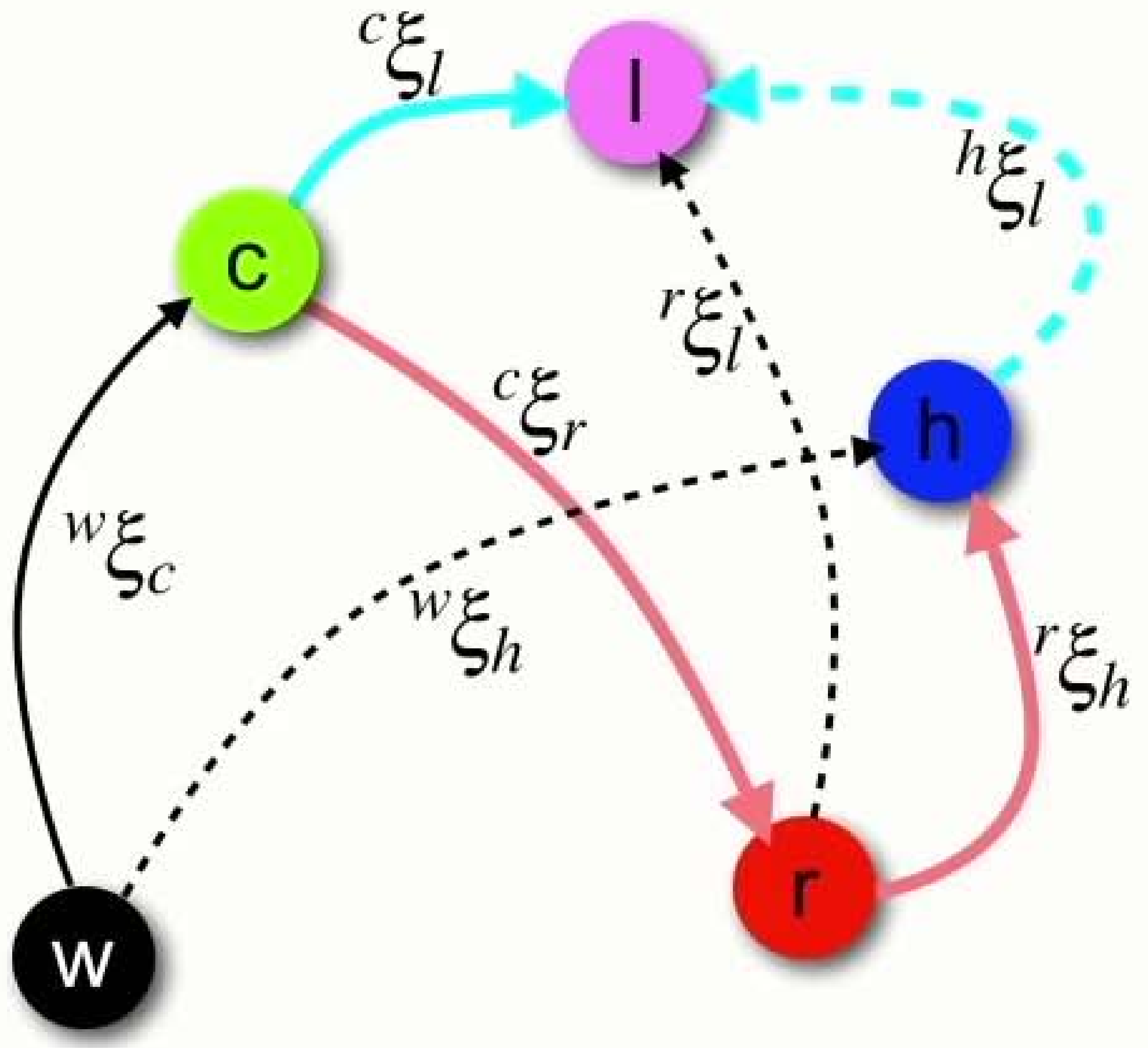
$$ {}^h\xi_l = \ominus\, {}^r\xi_h \ominus {}^c\xi_r \oplus {}^c\xi_l $$

reverse direction



Complex **example**





$$^h\xi_l = \ominus\, {}^r\xi_h \ominus {}^c\xi_r \oplus {}^c\xi_l$$

reverse direction

Pose **algebra**

- There are just a few rules

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

Pose **algebra**

- There are just a few rules

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

Pose algebra

- There are just a few rules

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

Pose **algebra**

- There are just a few rules

$$ {}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z} $$

Pose **algebra**

- There are just a few rules

$$ {}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z} $$

$$ \xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1 $$

Pose **algebra**

- There are just a few rules

$$ {}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z} $$

$$ \xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1 $$

$$ \ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X} $$

Pose algebra

- There are just a few rules

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X}$$

$$\xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0$$

Pose **algebra**

- There are just a few rules

$$ {}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z} $$

$$ \xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1 $$

$$ \ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X} $$

$$ \xi \ominus \xi = 0 \qquad \ominus \xi \oplus \xi = 0 $$

$$ \xi \ominus 0 = \xi \qquad \xi \oplus 0 = \xi $$

Pose **algebra**

- There are just a few rules

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

$$^{X}\boldsymbol{p} = {}^{X}\xi_{Y} \bullet {}^{Y}\boldsymbol{p}$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X}$$

$$\xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \qquad \xi \oplus 0 = \xi$$

Pose **algebra**

- There are just a few rules

$$ {}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z} \qquad {}^{X}\boldsymbol{p} = {}^{X}\xi_{Y} \bullet {}^{Y}\boldsymbol{p} $$

$$ \xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1 $$

$$ \ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X} $$

$$ \xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0 $$

$$ \xi \ominus 0 = \xi \qquad \xi \oplus 0 = \xi $$

Pose **algebra**

- There are just a few rules

$$^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z \qquad {}^X\boldsymbol{p} = {}^X\xi_Y \bullet {}^Y\boldsymbol{p}$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus\, {}^X\xi_Y = {}^Y\xi_X$$

$$\xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \qquad \xi \oplus 0 = \xi$$

Section 5 Developing a real representation of pose

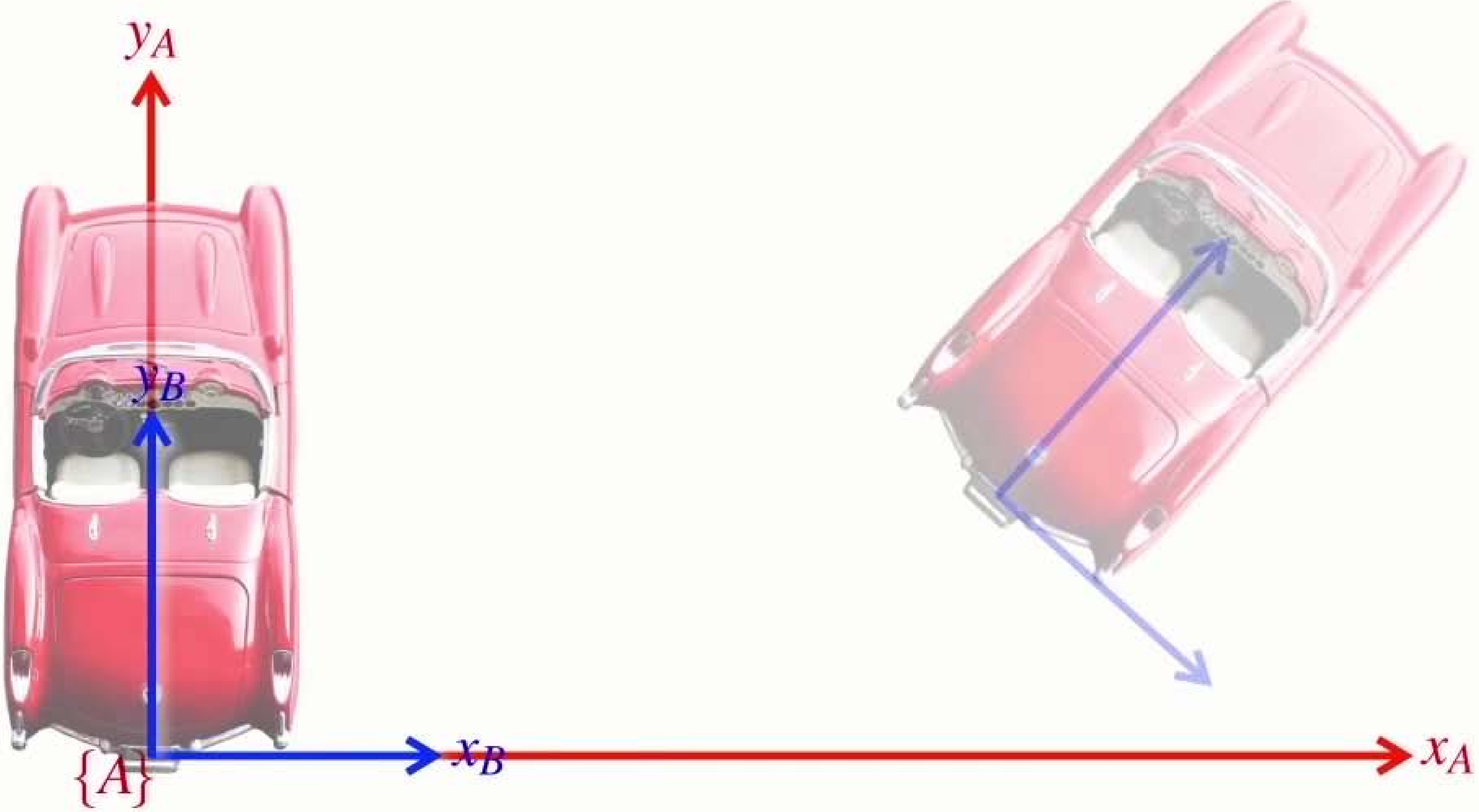
Pose

$$\xi$$

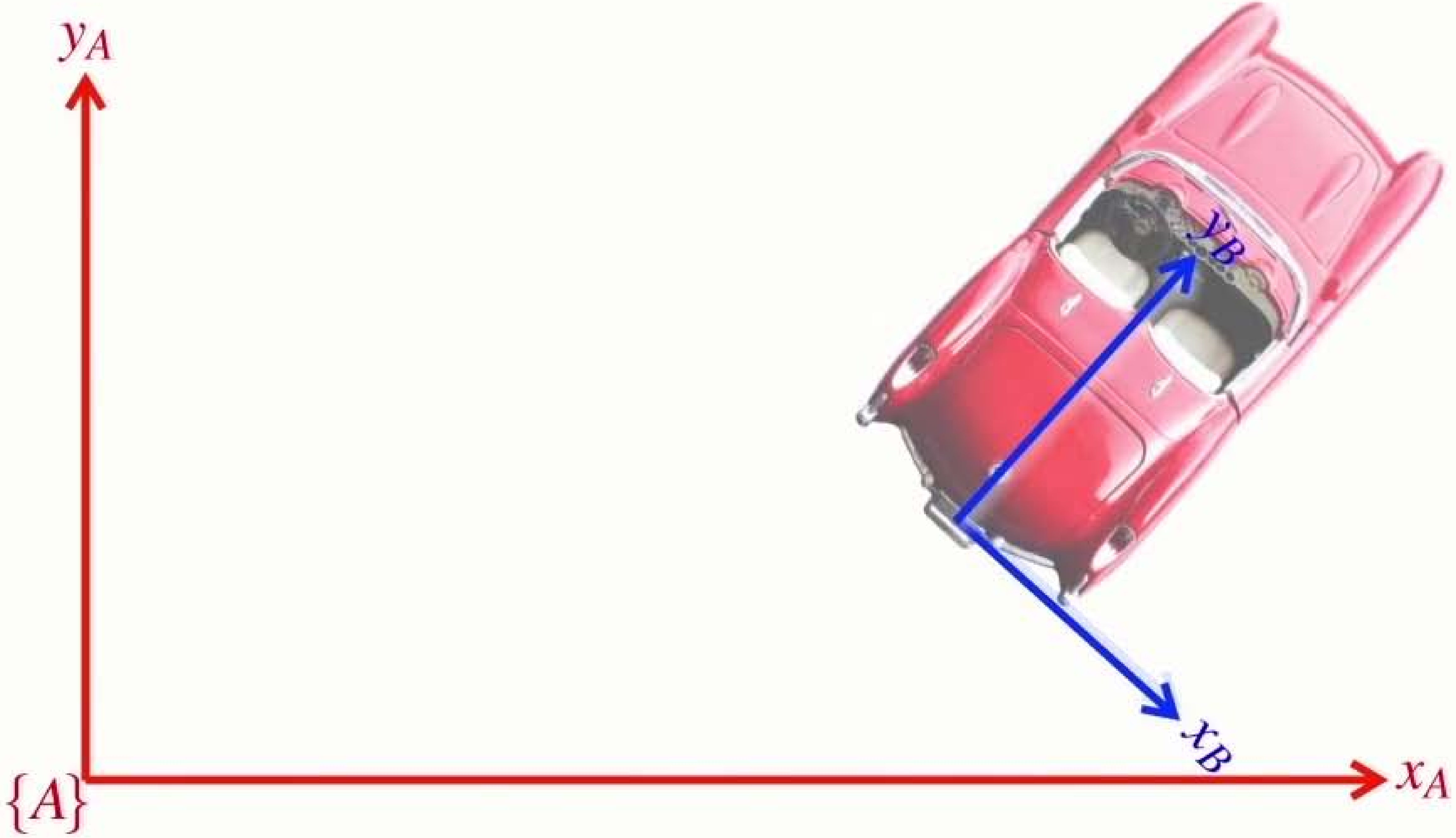
pronounced ksi

- It has 3 parameters: (x, y, θ)

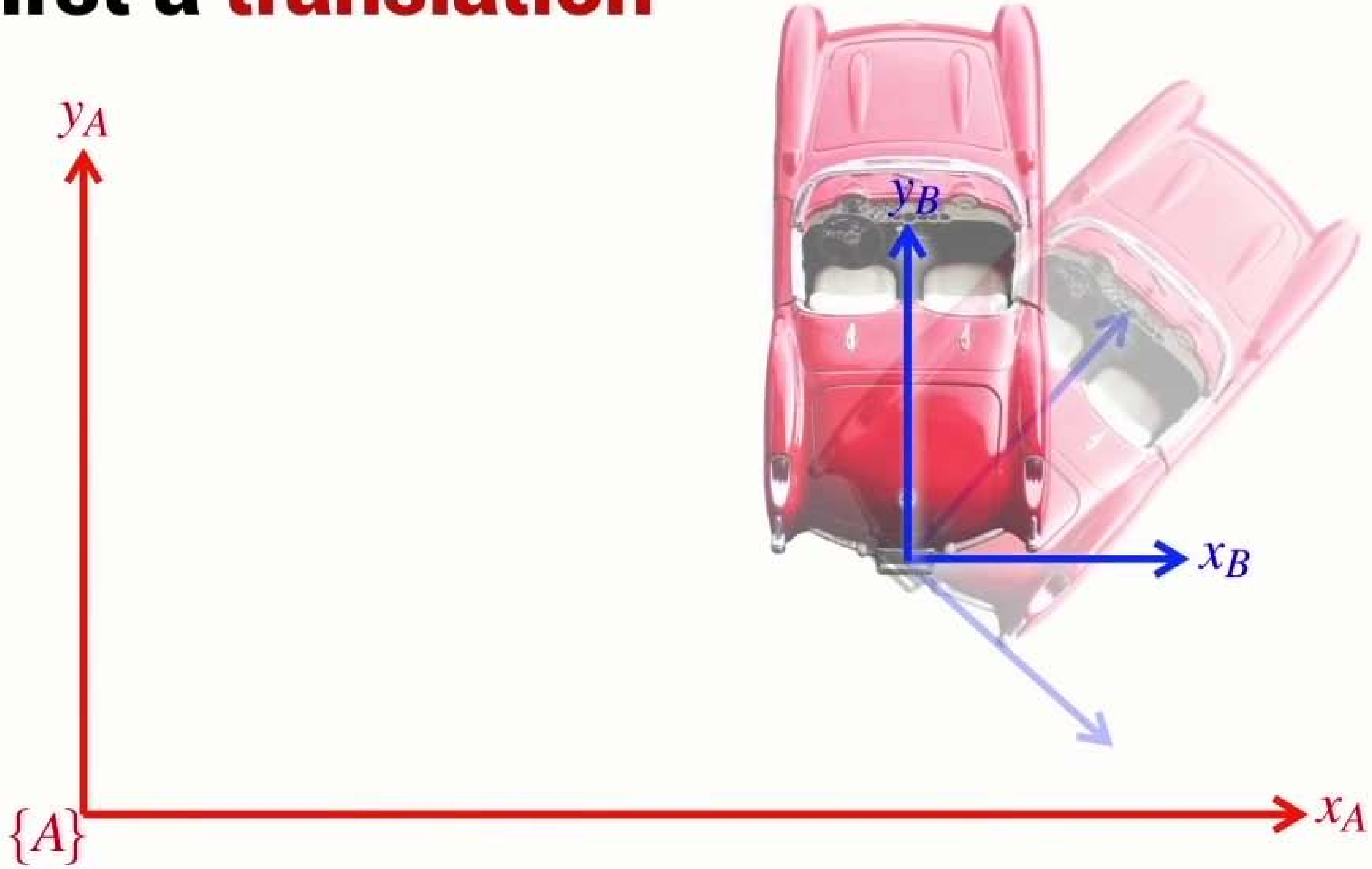
Separating **rotation** and **translation**



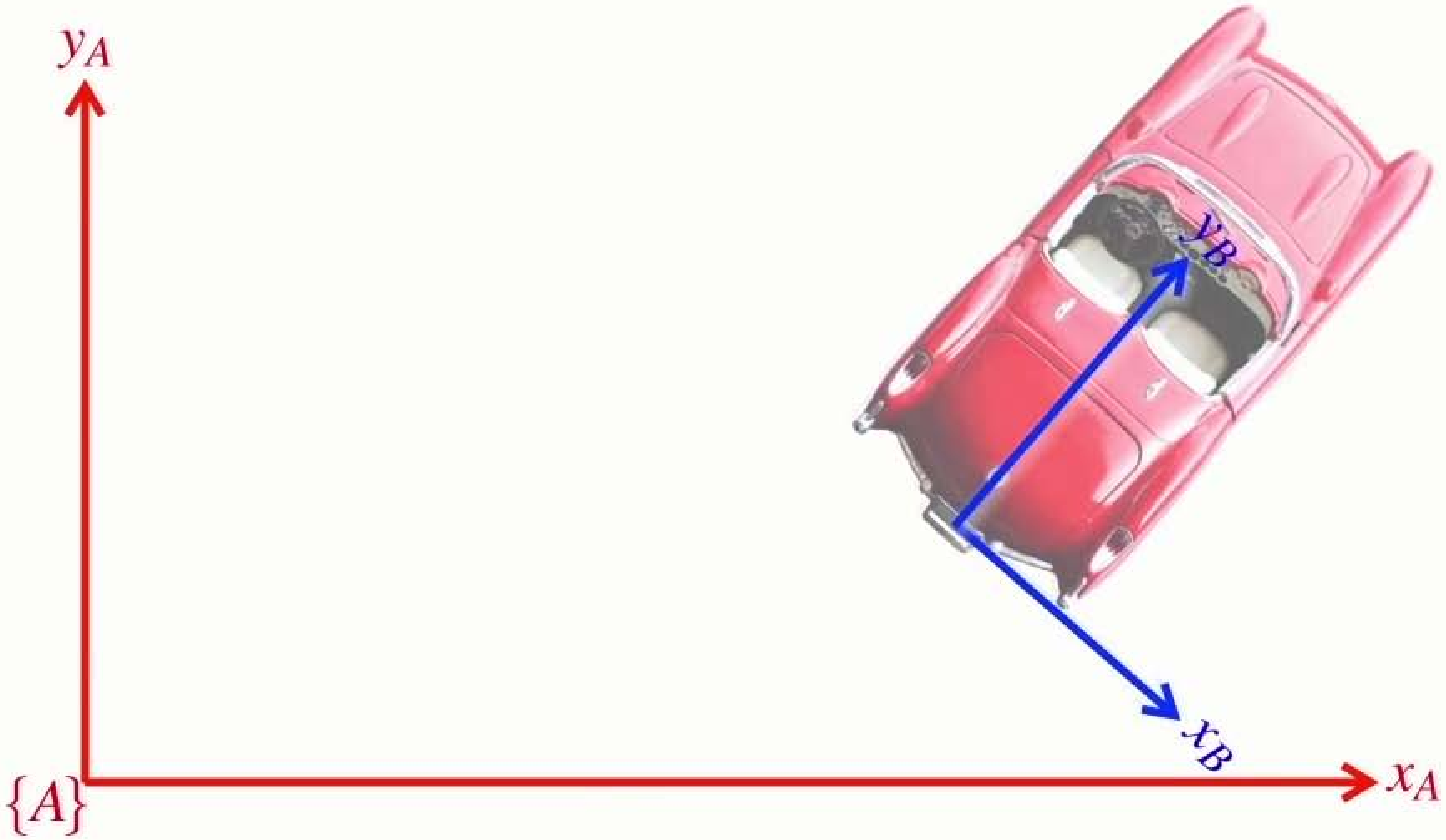
Separating **rotation** and **translation**



First a **translation**

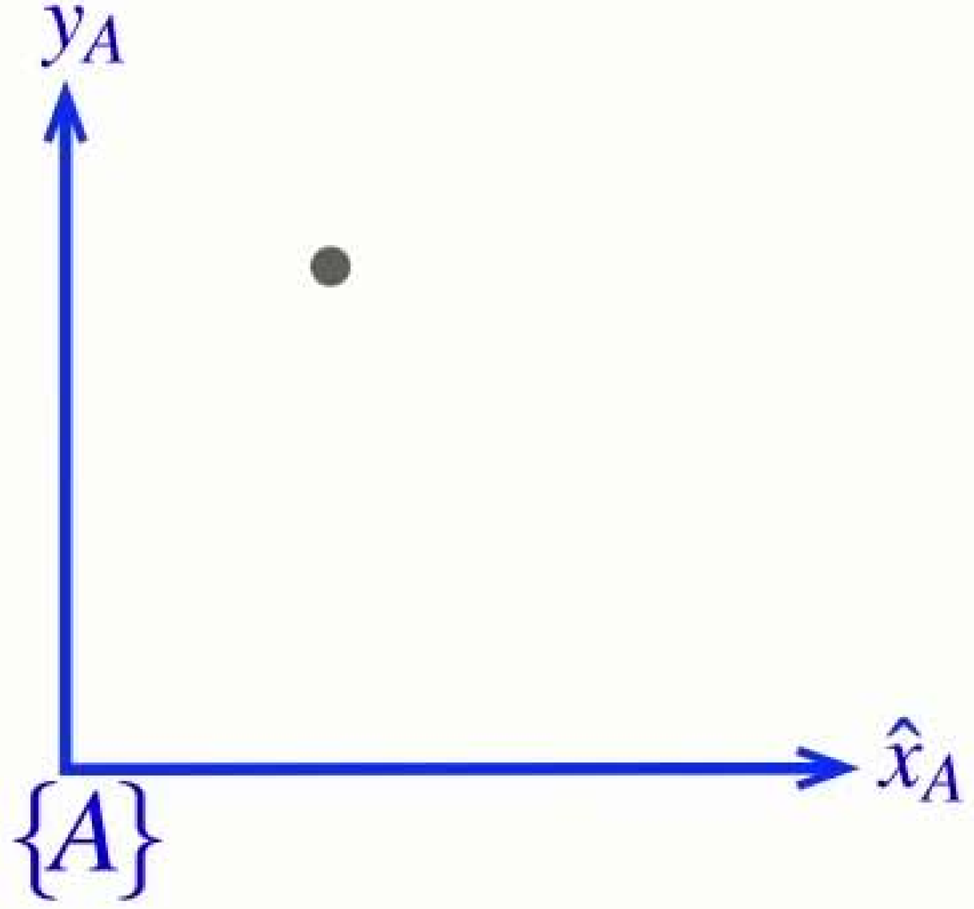


Then rotation

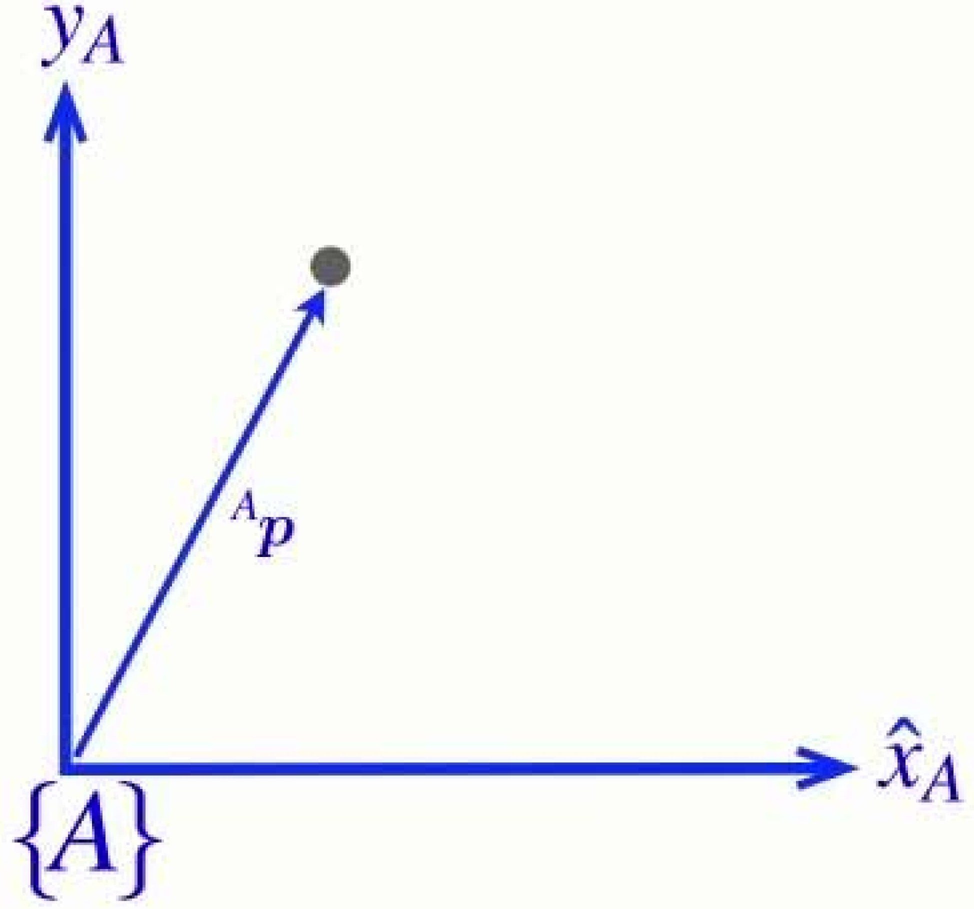


Section 6 Describing rotation

Understanding rotation

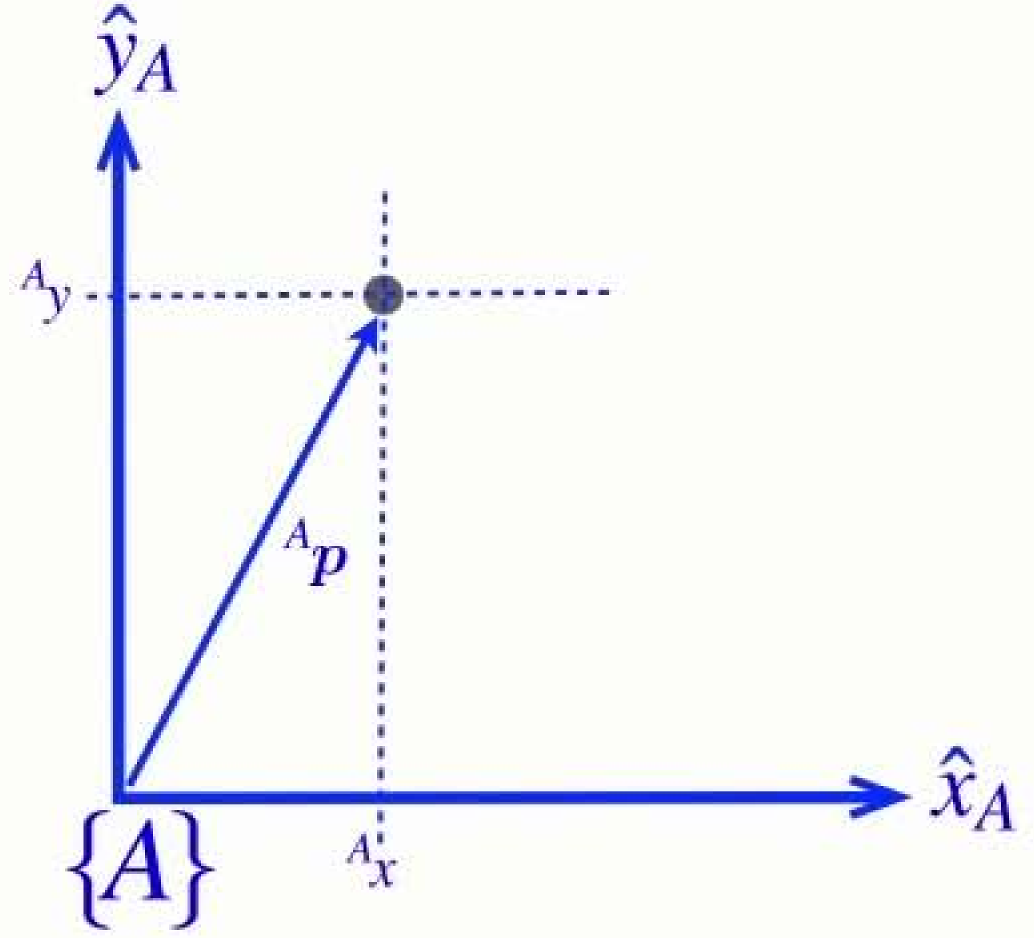


Understanding **rotation**



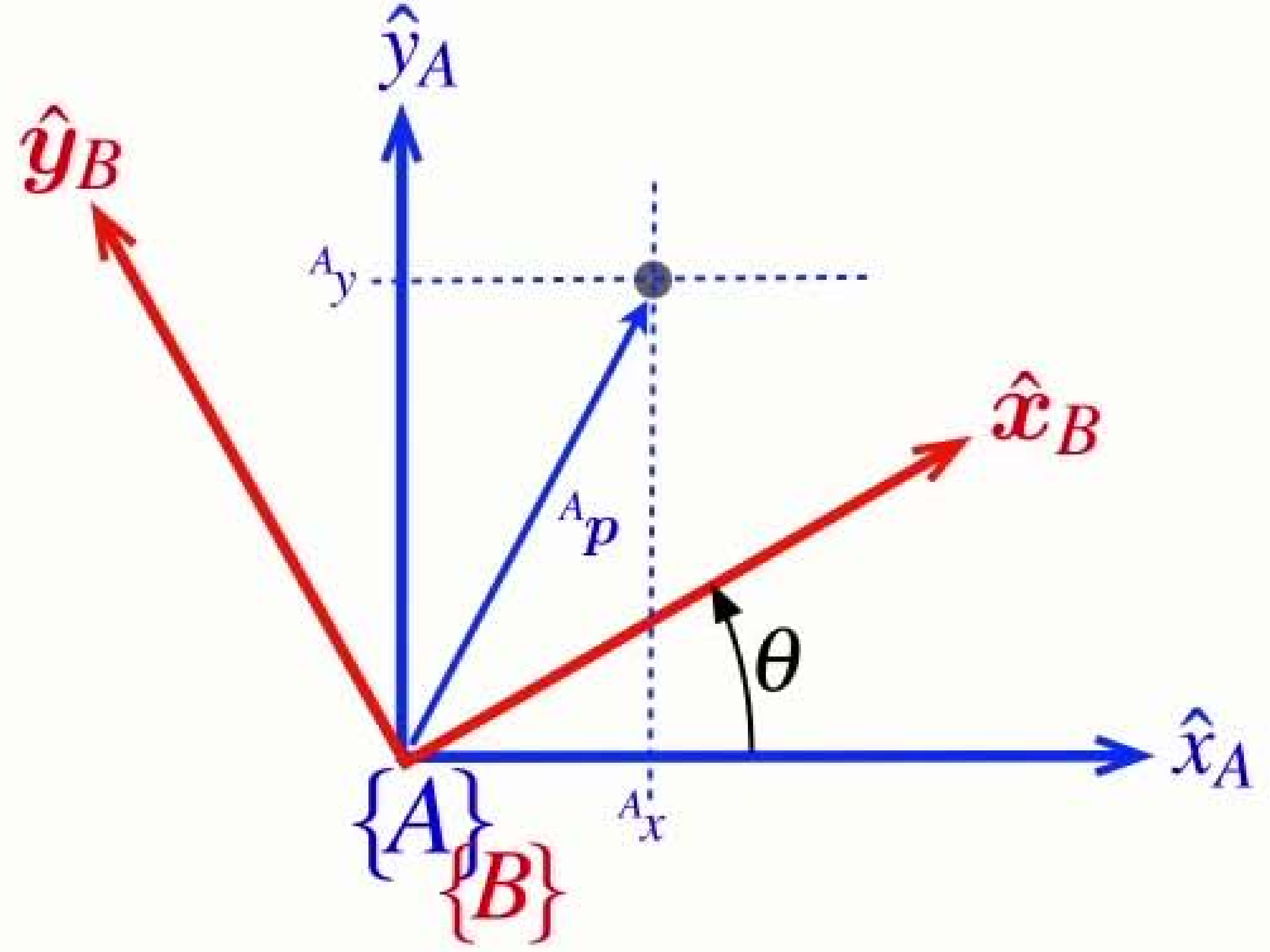
Understanding **rotation**

$${}^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$



Understanding **rotation**

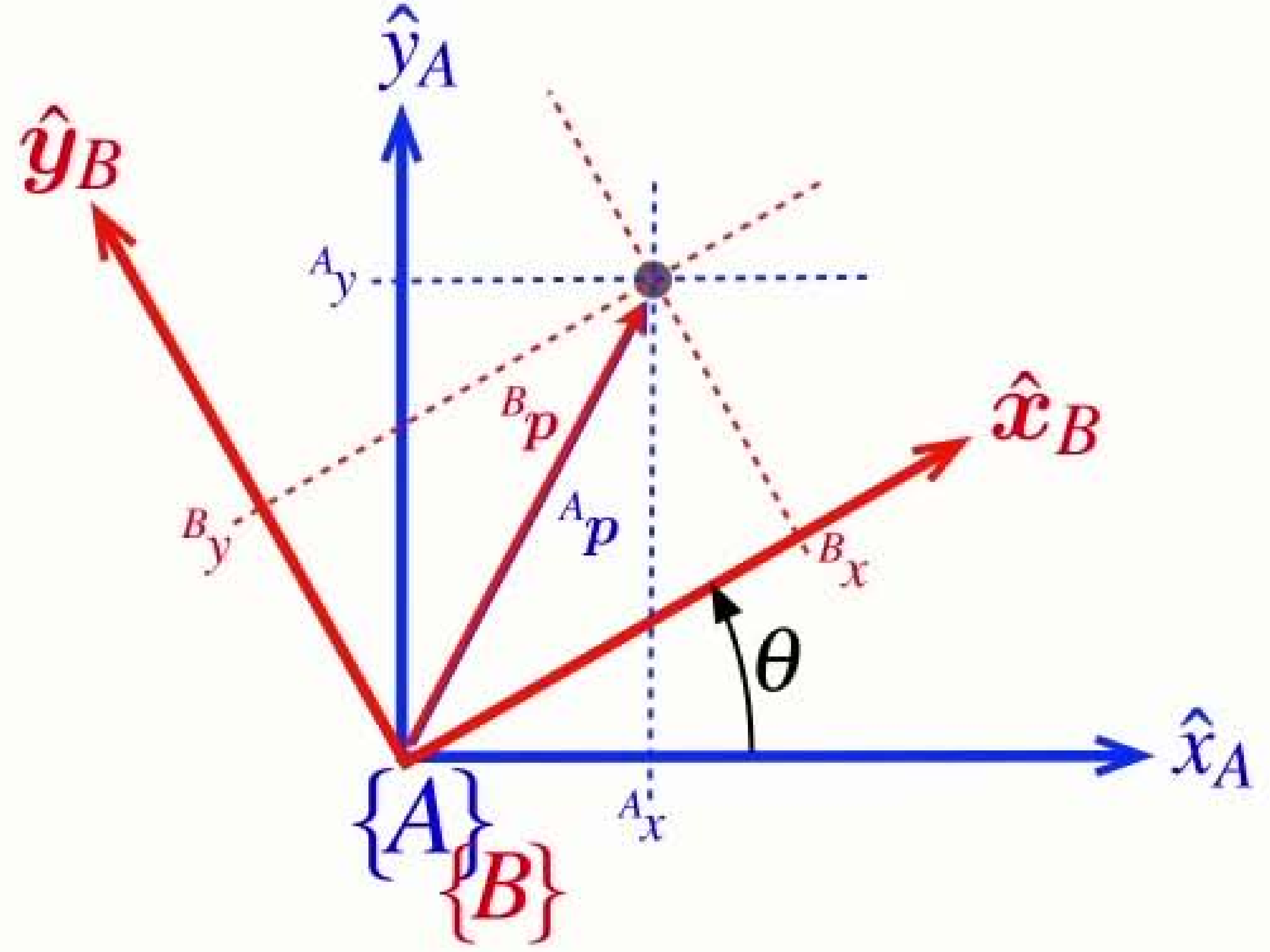
$$^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$



Understanding rotation

$$^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$

$$^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B$$

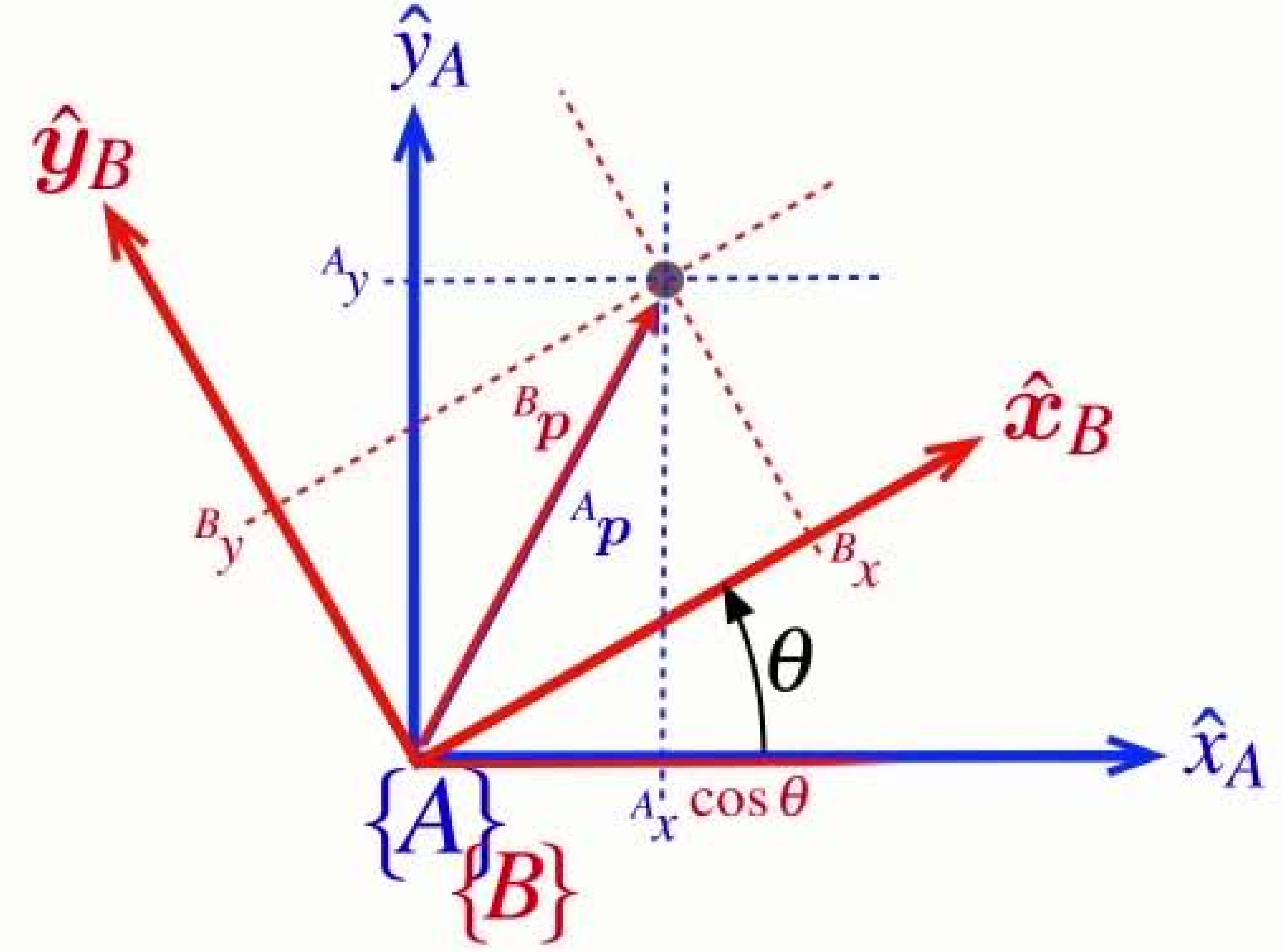


Understanding rotation

$${}^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$

$${}^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B$$

$$\hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A$$



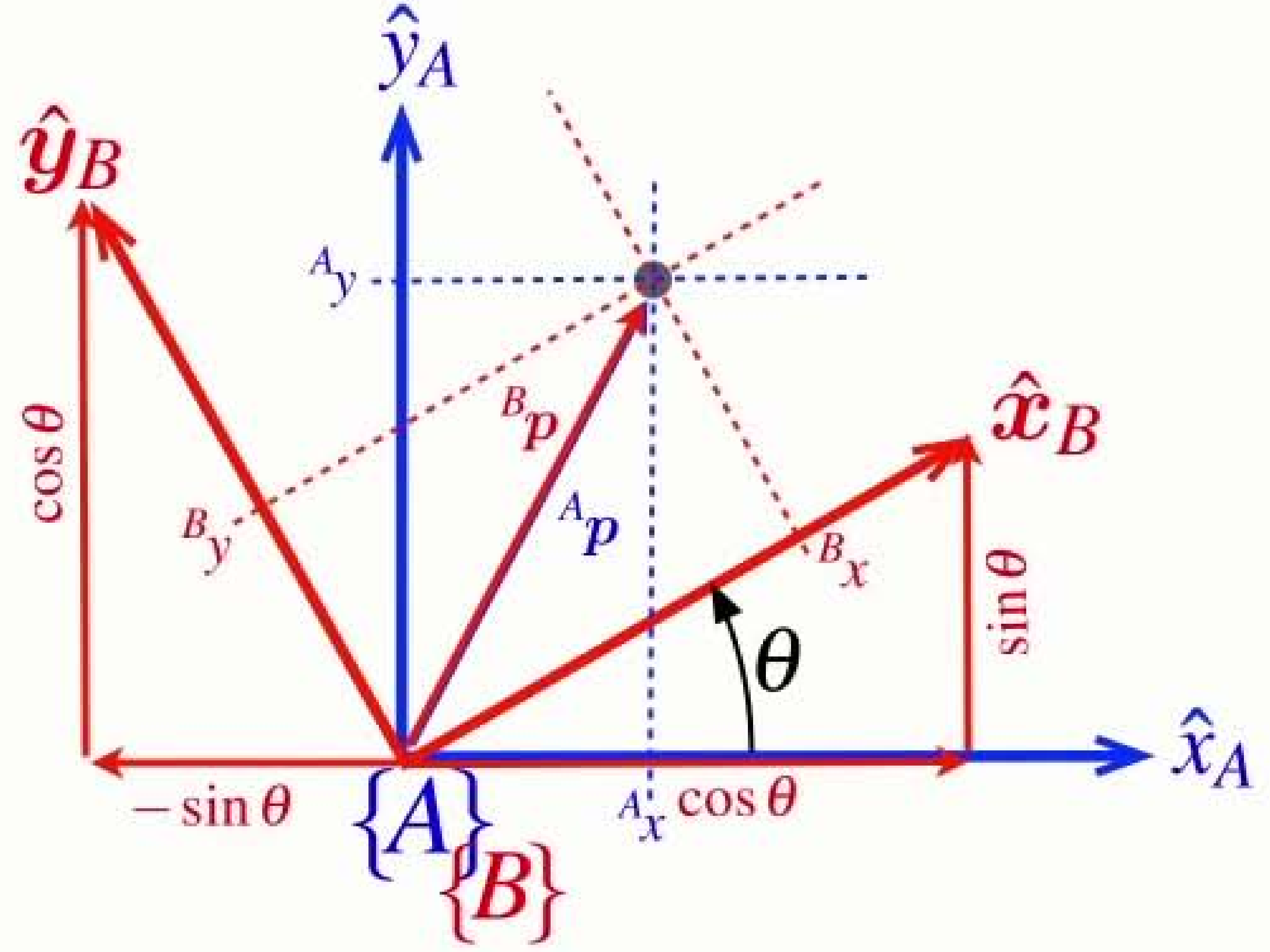
Understanding rotation

$$^{A}\boldsymbol{p} = {^{A}x}\,\hat{\boldsymbol{x}}_A + {^{A}y}\,\hat{\boldsymbol{y}}_A$$

$$^{B}\boldsymbol{p} = {^{B}x}\,\hat{\boldsymbol{x}}_B + {^{B}y}\,\hat{\boldsymbol{y}}_B$$

$$\hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A$$

$$\hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A$$



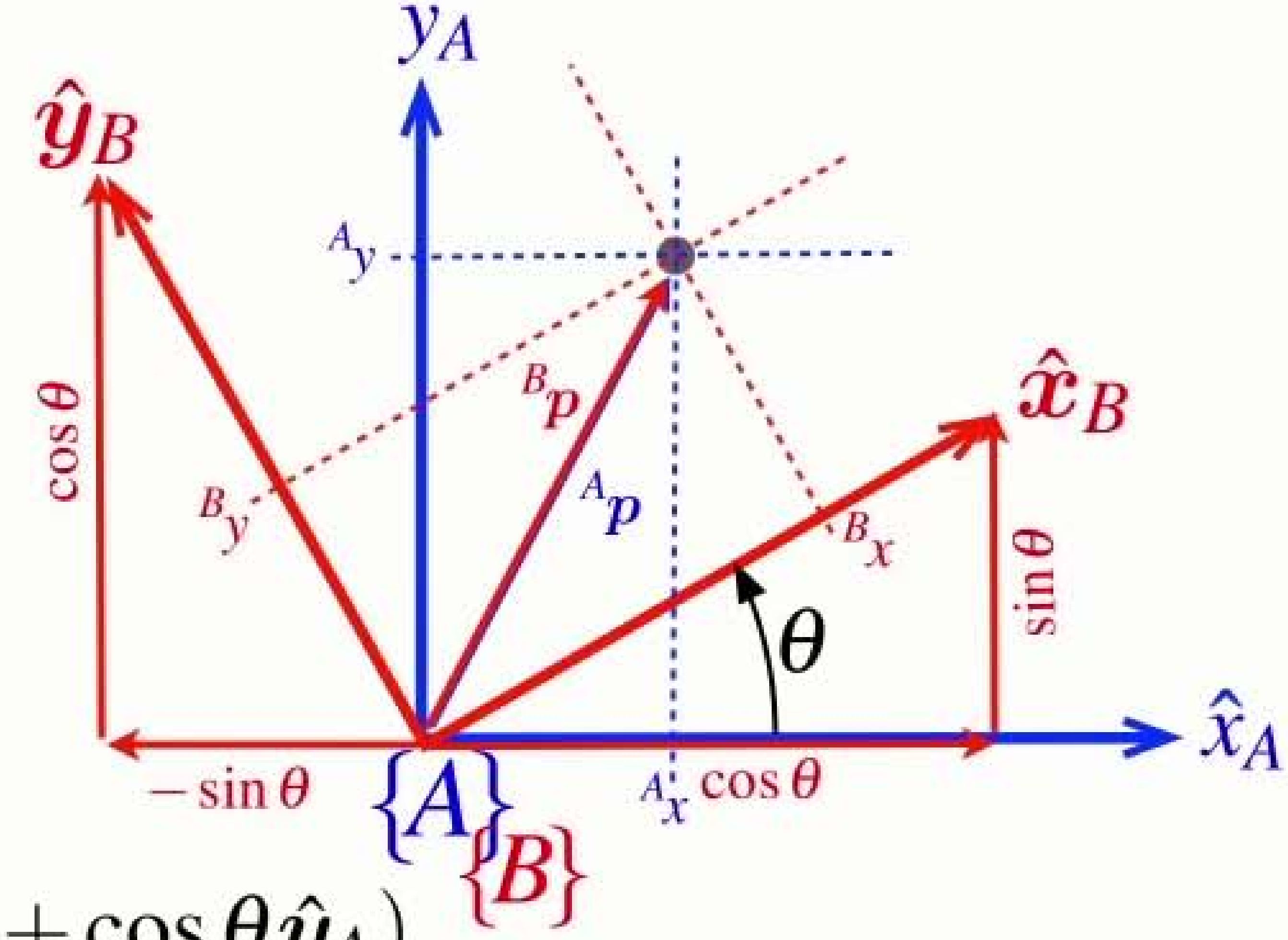
Understanding **rotation**

$$^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$

$$^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B$$

$$\hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A$$

$$\hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A$$



$$^{B}\boldsymbol{p} = {}^{B}x\left(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A\right) + {}^{B}y\left(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A\right)$$

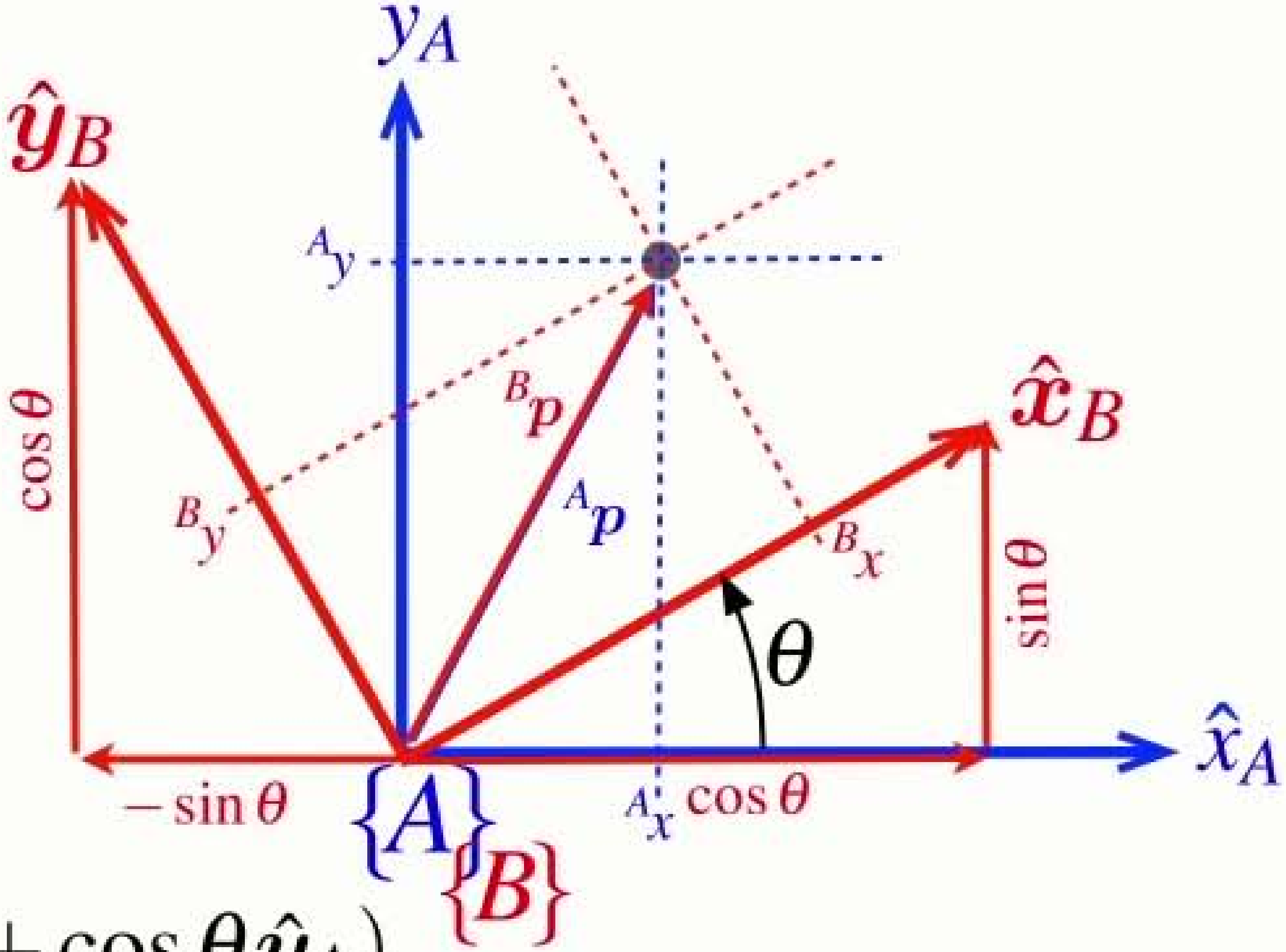
Understanding **rotation**

$$^{A}\boldsymbol{p} = {^{A}x}\,\hat{\boldsymbol{x}}_A + {^{A}y}\,\hat{\boldsymbol{y}}_A$$

$$^{B}\boldsymbol{p} = {^{B}x}\,\hat{\boldsymbol{x}}_B + {^{B}y}\,\hat{\boldsymbol{y}}_B$$

$$\hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A$$

$$\hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A$$



$$^{B}\boldsymbol{p} = {^{B}x}\left(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A\right) + {^{B}y}\left(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A\right)$$

$$^{B}\boldsymbol{p} = \left({^{B}x}\cos\theta - {^{B}y}\sin\theta\right)\hat{\boldsymbol{x}}_A + \left({^{B}x}\sin\theta + {^{B}y}\cos\theta\right)\hat{\boldsymbol{y}}_A$$

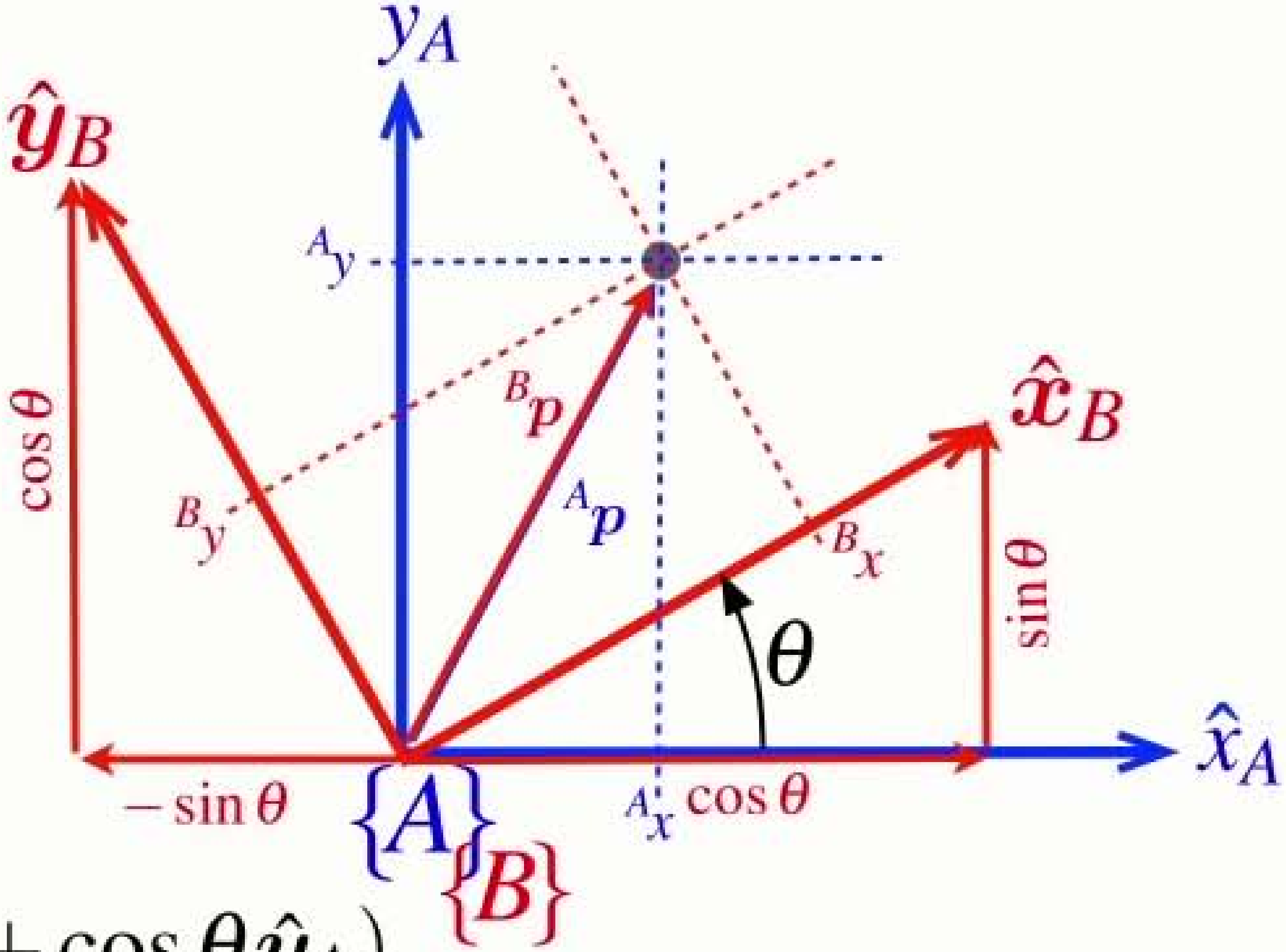
Understanding **rotation**

$$ {}^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A $$

$$ {}^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B $$

$$ \hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A $$

$$ \hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A $$



$$ {}^{B}\boldsymbol{p} = {}^{B}x\left(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A\right) + {}^{B}y\left(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A\right) $$

$$ {}^{B}\boldsymbol{p} = \left({}^{B}x\cos\theta - {}^{B}y\sin\theta\right)\hat{\boldsymbol{x}}_A + \left({}^{B}x\sin\theta + {}^{B}y\cos\theta\right)\hat{\boldsymbol{y}}_A $$

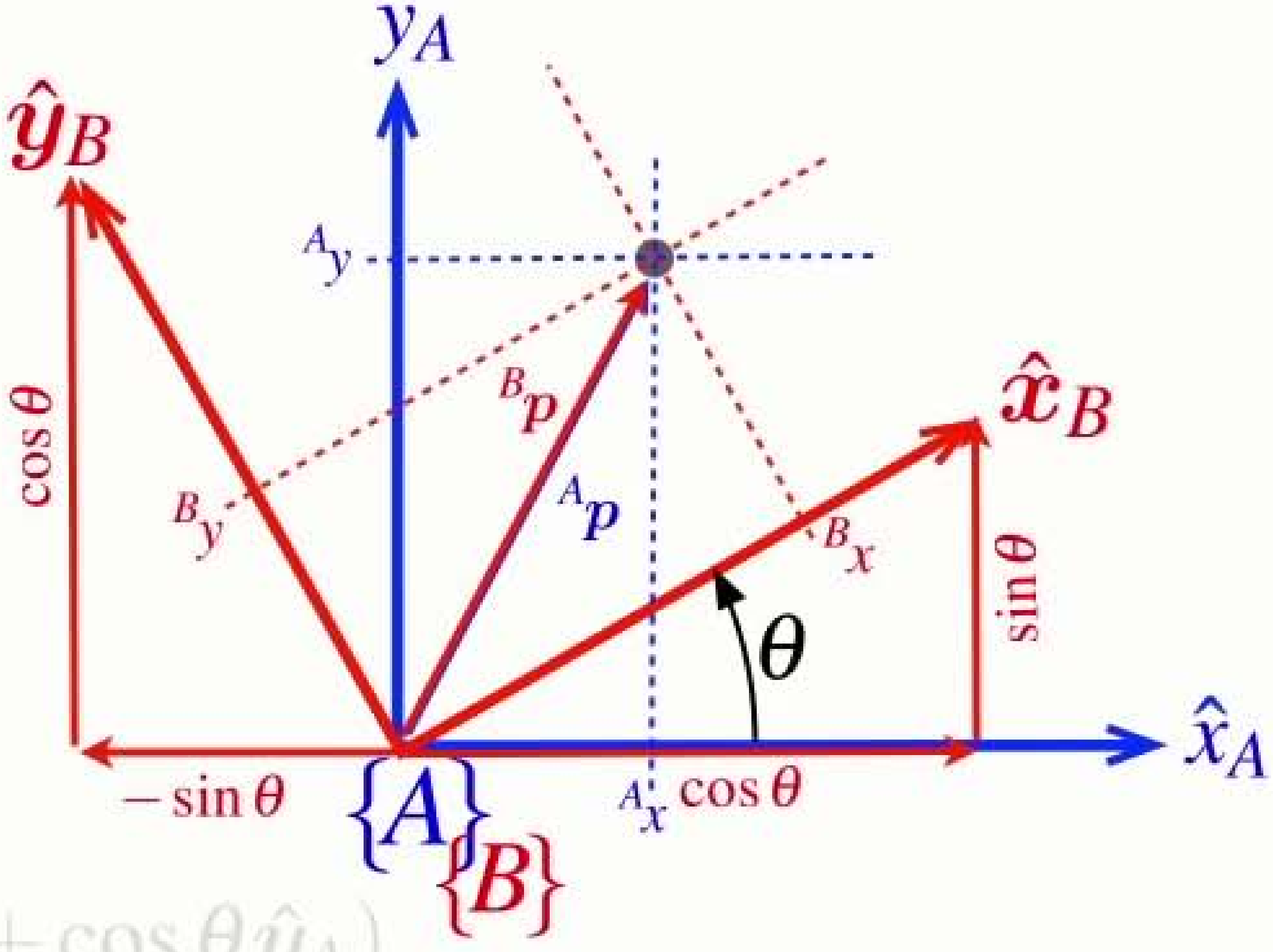
Understanding **rotation**

$$ {}^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A $$

$$ {}^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B $$

$$ \hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A $$

$$ \hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A $$



$$ {}^{B}\boldsymbol{p} = {}^{B}x\left(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A\right) + {}^{B}y\left(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A\right) $$

$$ {}^{B}\boldsymbol{p} = \left({}^{B}x\cos\theta - {}^{B}y\sin\theta\right)\hat{\boldsymbol{x}}_A + \left({}^{B}x\sin\theta + {}^{B}y\cos\theta\right)\hat{\boldsymbol{y}}_A $$

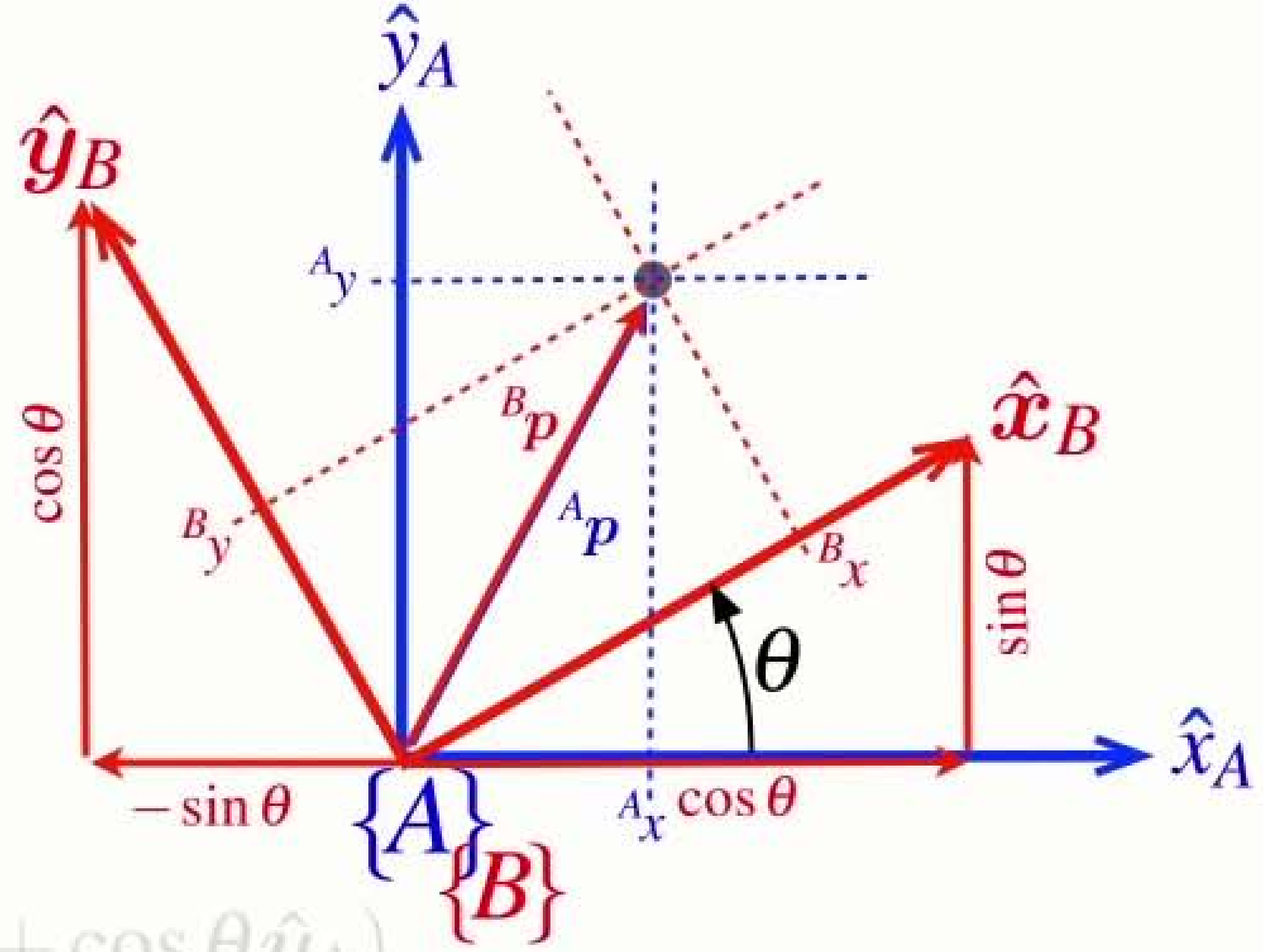
Understanding **rotation**

$$^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$

$$^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B$$

$$\hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A$$

$$\hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A$$



$$^{B}\boldsymbol{p} = {}^{B}x\left(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A\right) + {}^{B}y\left(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A\right)$$

$$^{B}\boldsymbol{p} = \left({}^{B}x\cos\theta - {}^{B}y\sin\theta\right)\hat{\boldsymbol{x}}_A + \left({}^{B}x\sin\theta + {}^{B}y\cos\theta\right)\hat{\boldsymbol{y}}_A$$

$$^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta$$

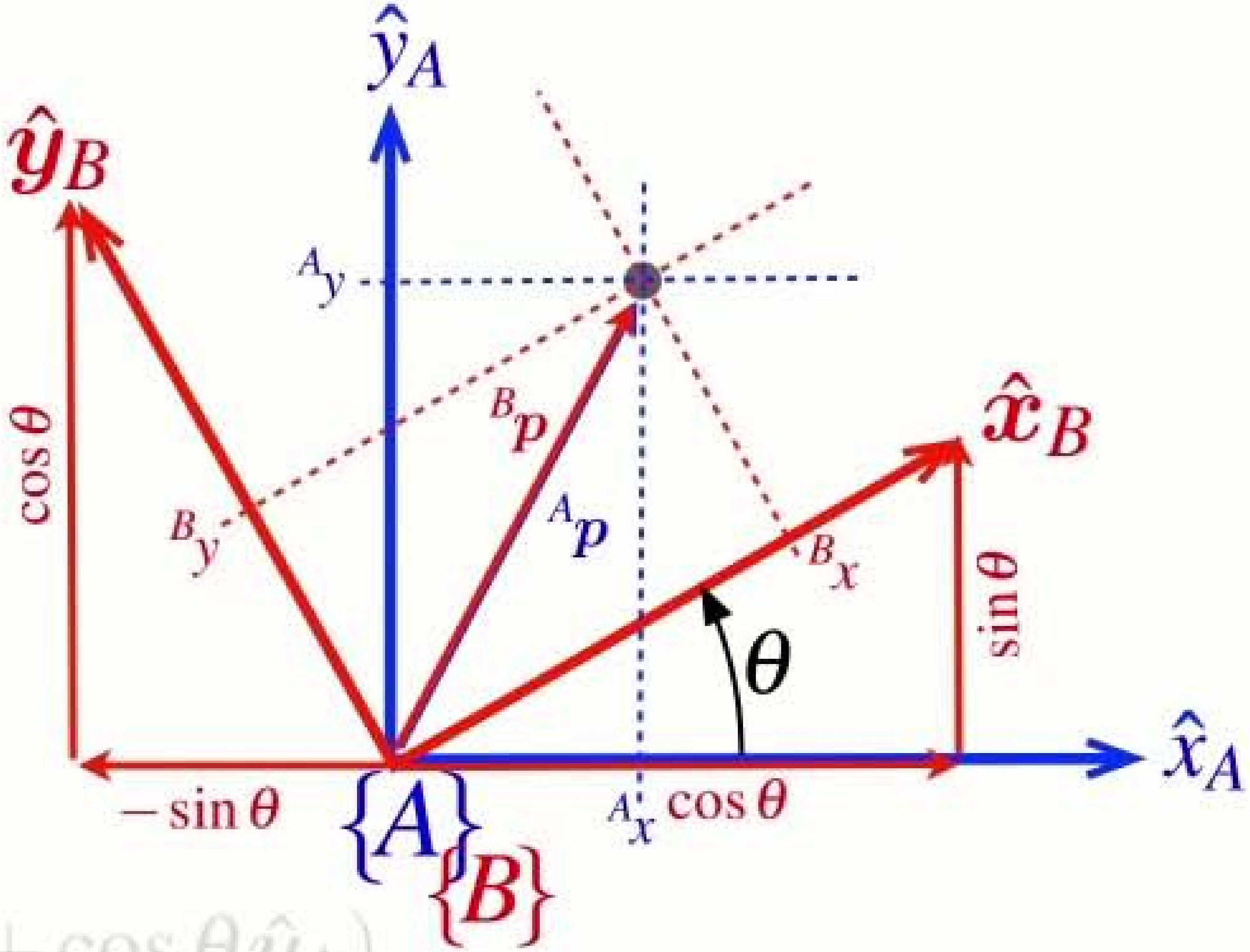
Understanding **rotation**

$${}^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A$$

$${}^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B$$

$$\hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A$$

$$\hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A$$



$${}^{B}\boldsymbol{p} = {}^{B}x\left(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A\right) + {}^{B}y\left(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A\right)$$

$${}^{B}\boldsymbol{p} = \left({}^{B}x\cos\theta - {}^{B}y\sin\theta\right)\hat{\boldsymbol{x}}_A + \left({}^{B}x\sin\theta + {}^{B}y\cos\theta\right)\hat{\boldsymbol{y}}_A$$

$${}^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta$$

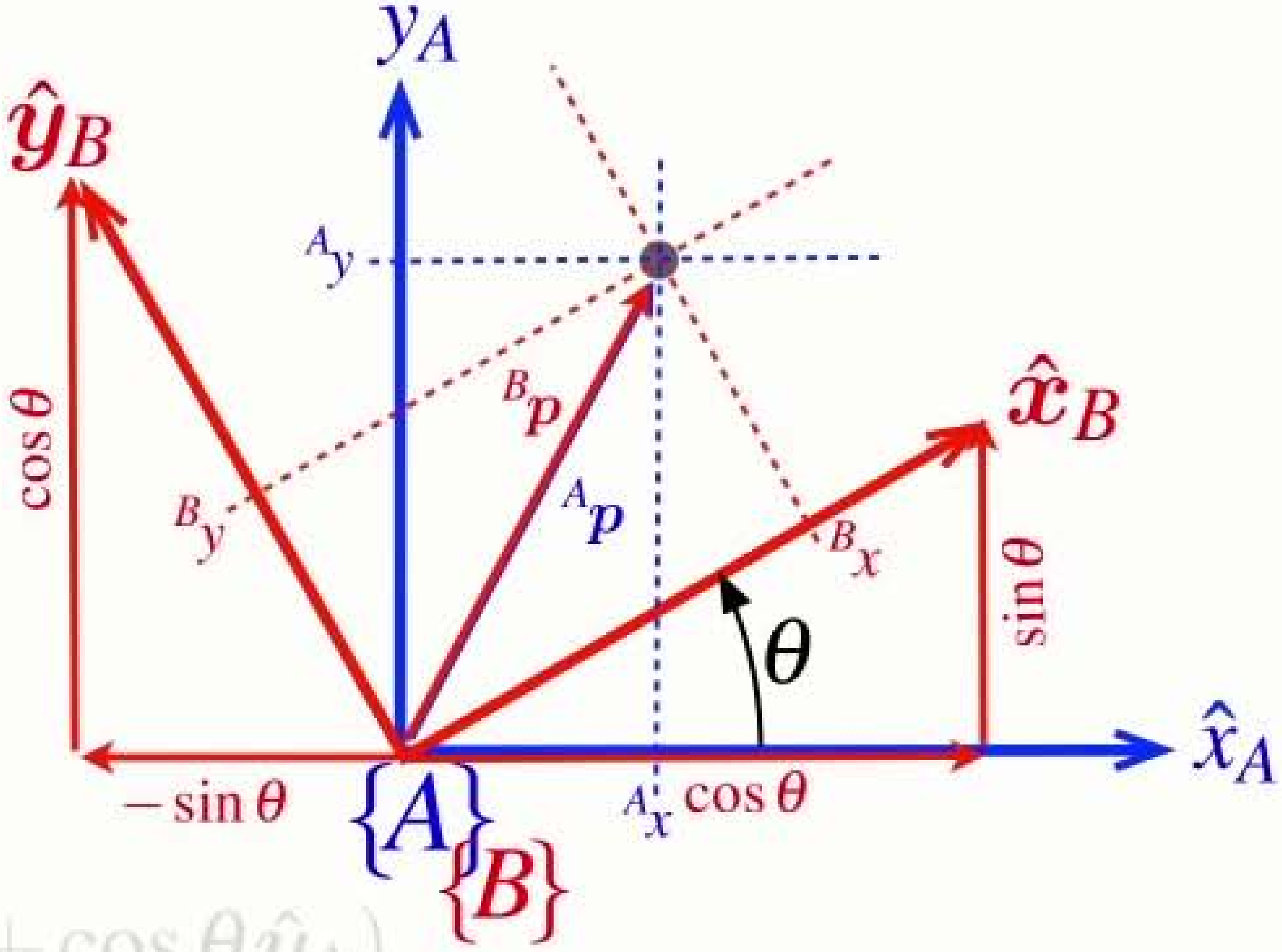
Understanding **rotation**

$$ {}^{A}\boldsymbol{p} = {}^{A}x\,\hat{\boldsymbol{x}}_A + {}^{A}y\,\hat{\boldsymbol{y}}_A $$

$$ {}^{B}\boldsymbol{p} = {}^{B}x\,\hat{\boldsymbol{x}}_B + {}^{B}y\,\hat{\boldsymbol{y}}_B $$

$$ \hat{\boldsymbol{x}}_B = \cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A $$

$$ \hat{\boldsymbol{y}}_B = -\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A $$



$$ {}^{B}\boldsymbol{p} = {}^{B}x\,(\cos\theta\,\hat{\boldsymbol{x}}_A + \sin\theta\,\hat{\boldsymbol{y}}_A) + {}^{B}y\,(-\sin\theta\,\hat{\boldsymbol{x}}_A + \cos\theta\,\hat{\boldsymbol{y}}_A) $$

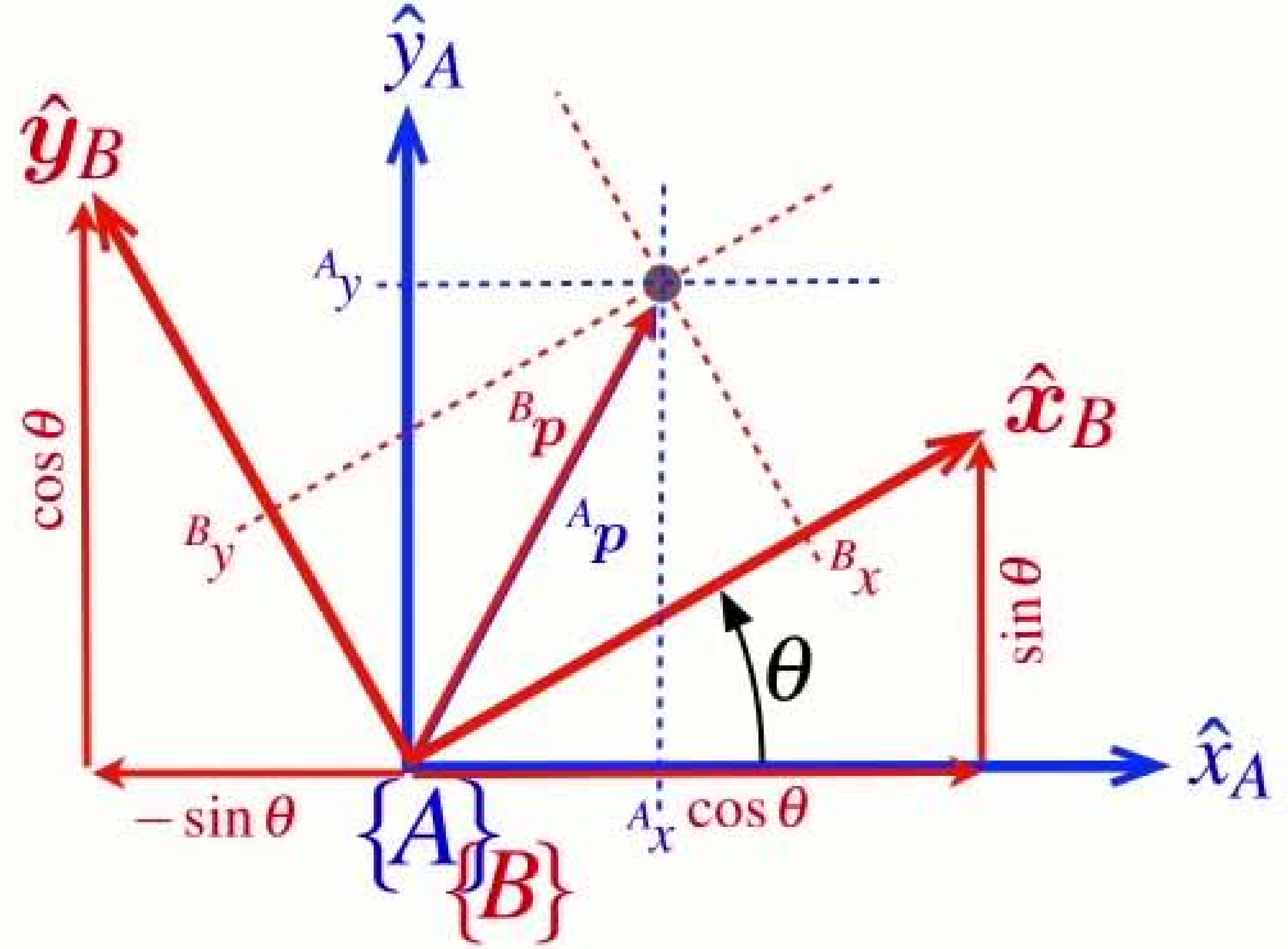
$$ {}^{B}\boldsymbol{p} = \left({}^{B}x\cos\theta - {}^{B}y\sin\theta\right)\hat{\boldsymbol{x}}_A + \left({}^{B}x\sin\theta + {}^{B}y\cos\theta\right)\hat{\boldsymbol{y}}_A $$

$$ {}^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta \qquad {}^{A}y = {}^{B}x\sin\theta + {}^{B}y\cos\theta $$

Consider rotation first

$$^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta$$

$$^{A}y = {}^{B}x\sin\theta + {}^{B}y\cos\theta$$

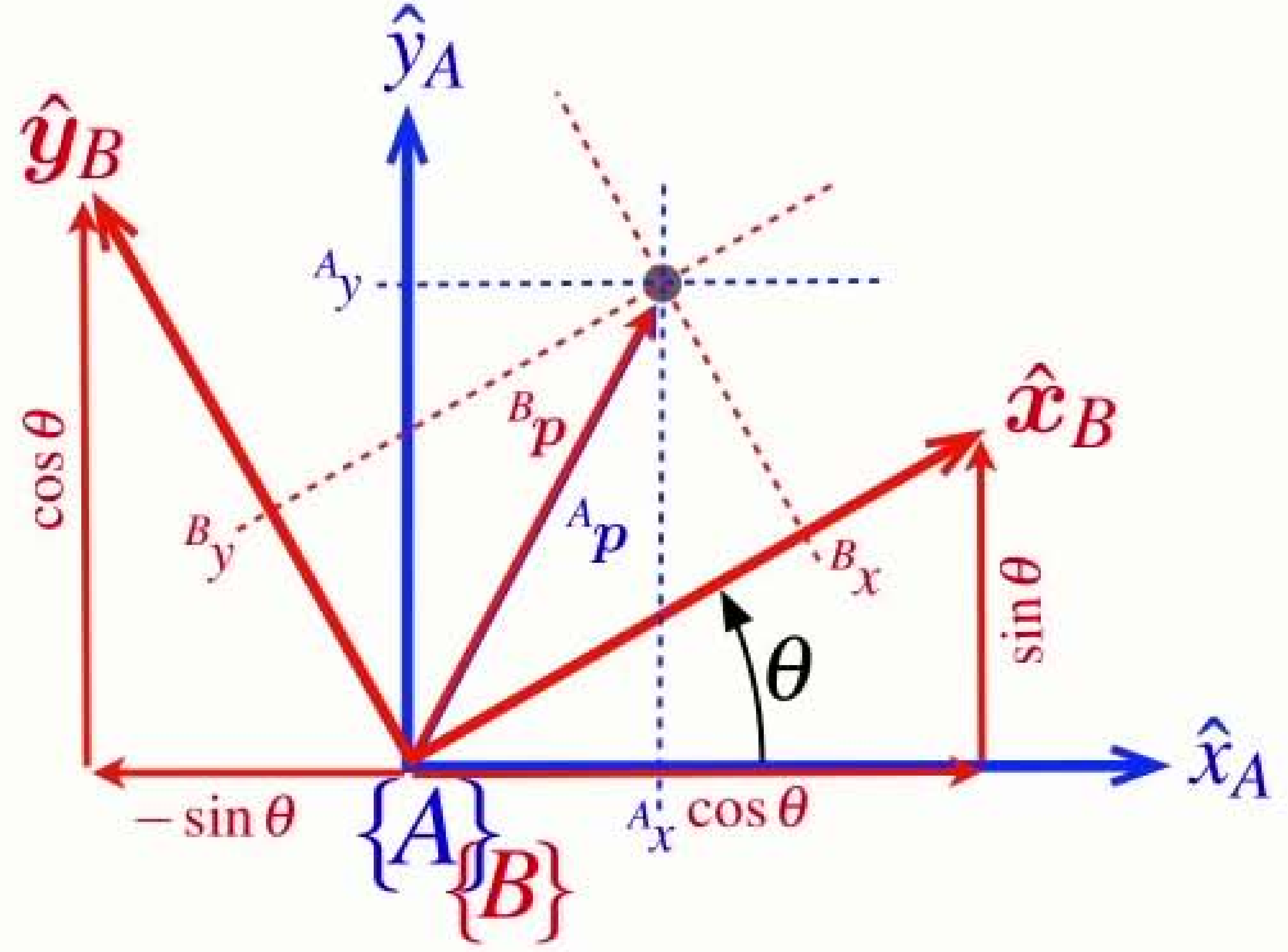


Consider **rotation first**

$$^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta$$

$$^{A}y = {}^{B}x\sin\theta + {}^{B}y\cos\theta$$

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \end{pmatrix}$$



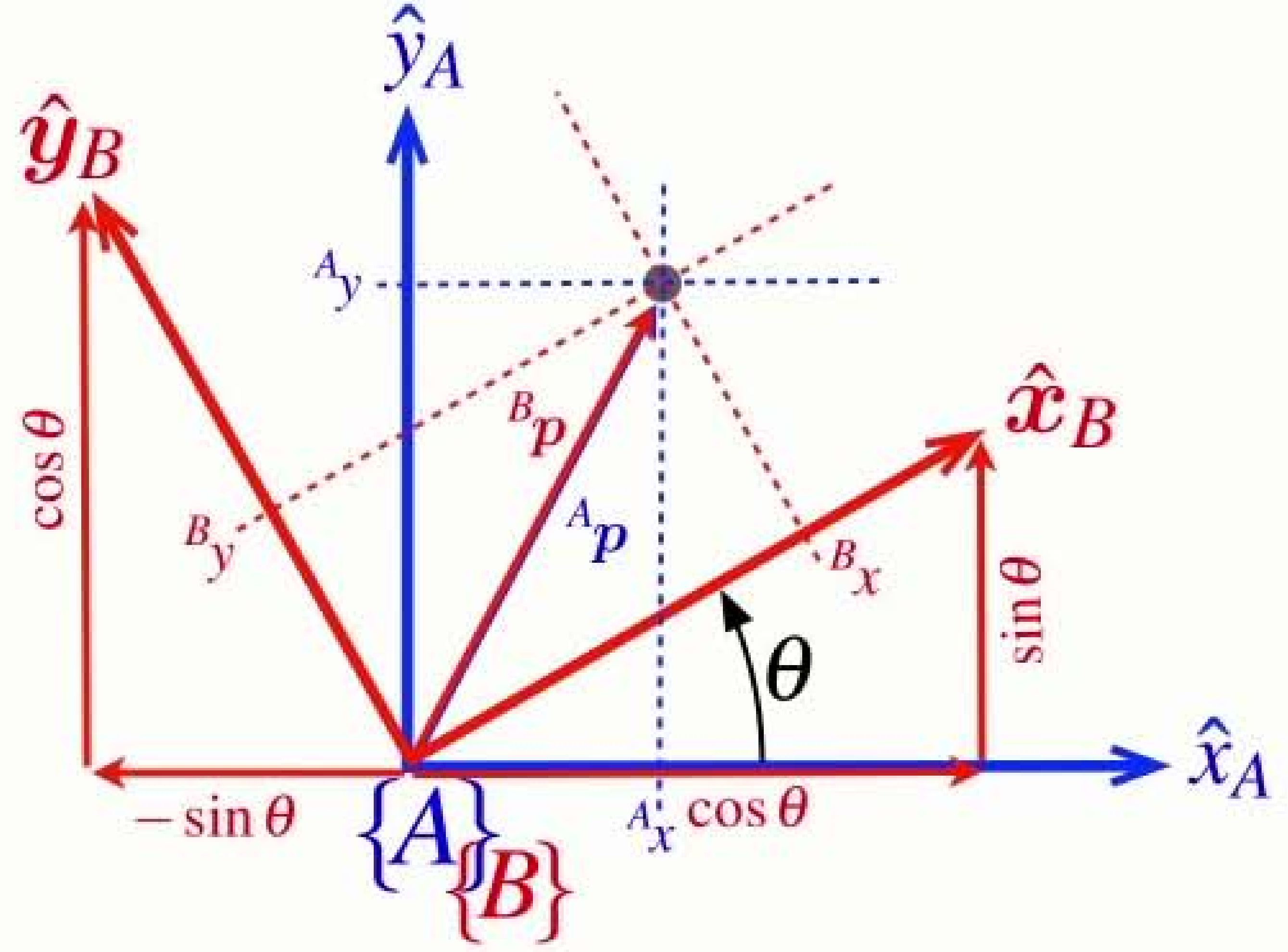
Consider rotation first

$$^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta$$

$$^{A}y = {}^{B}x\sin\theta + {}^{B}y\cos\theta$$

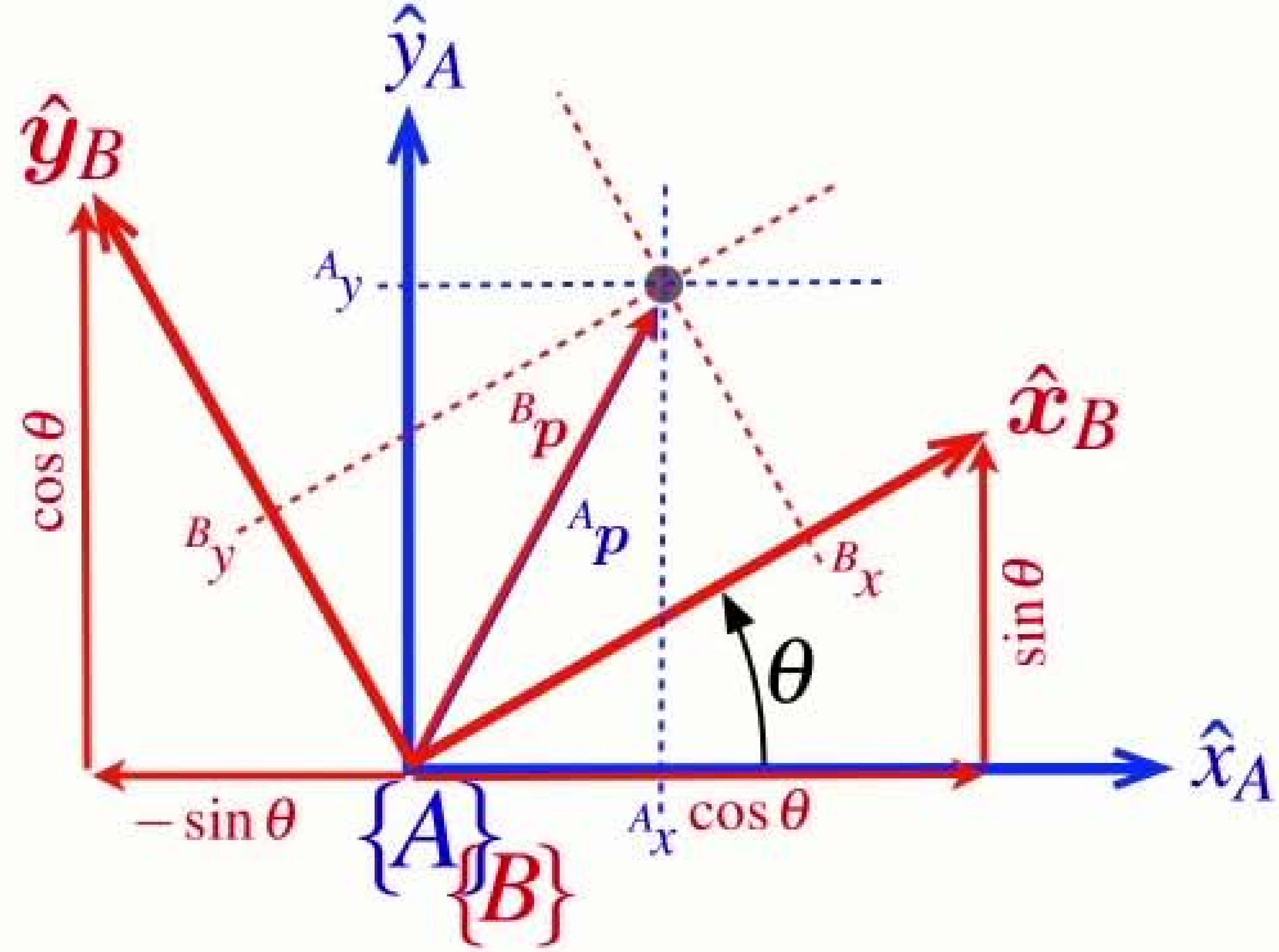
$$\begin{pmatrix} ^{A}x \\ ^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} ^{B}x \\ ^{B}y \end{pmatrix}$$

$$^{A}\boldsymbol{p} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} {}^{B}\boldsymbol{p}$$



Consider rotation first

$$^{A}x = {}^{B}x\cos\theta - {}^{B}y\sin\theta$$
$$^{A}y = {}^{B}x\sin\theta + {}^{B}y\cos\theta$$



$$\begin{pmatrix} ^{A}x \\ ^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} ^{B}x \\ ^{B}y \end{pmatrix}$$

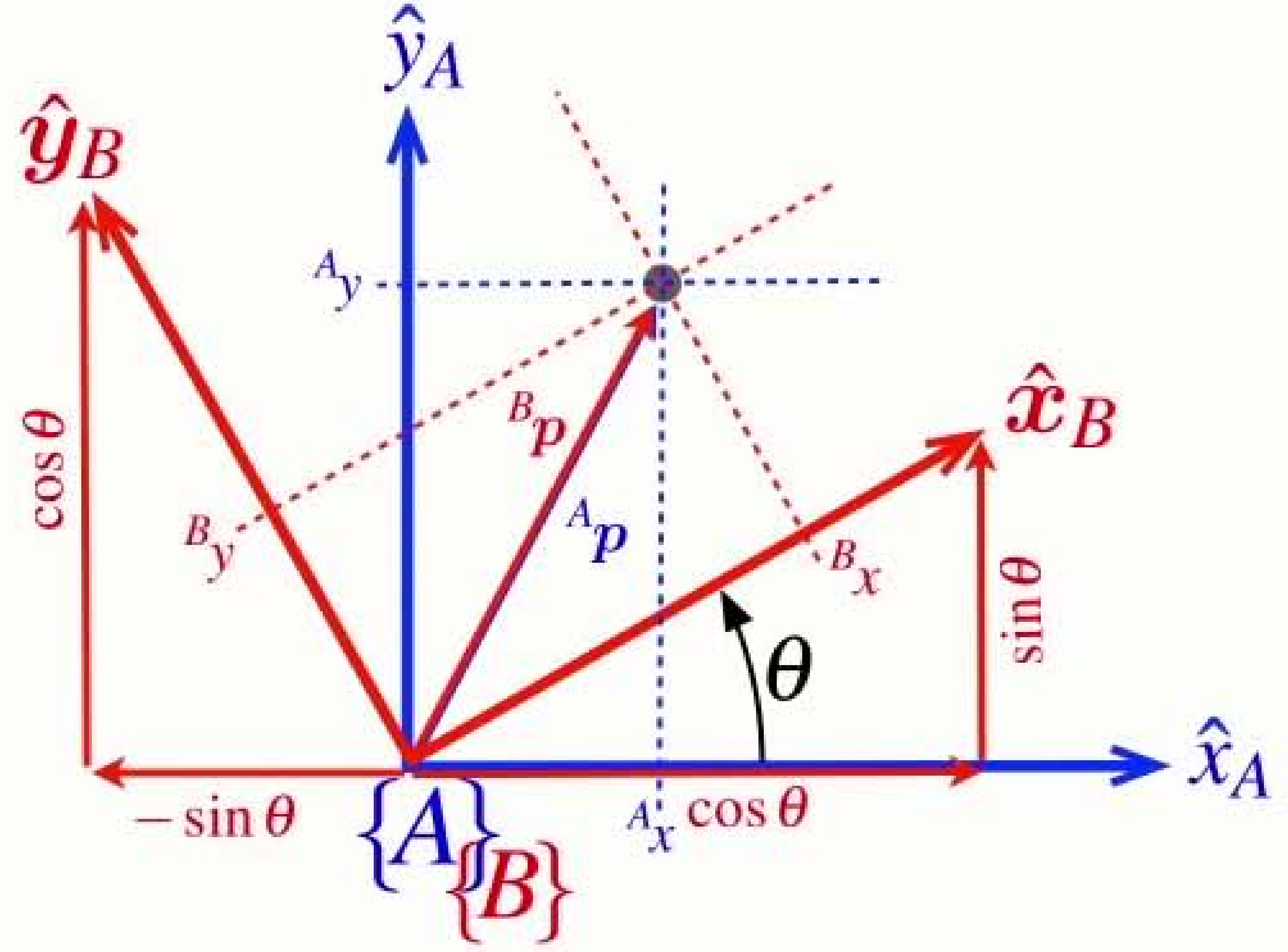
$$^{A}\boldsymbol{p} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} {}^{B}\boldsymbol{p}$$

$$^{A}\boldsymbol{p} = \mathbf{R}\,{}^{B}\boldsymbol{p} \qquad \mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Consider rotation first

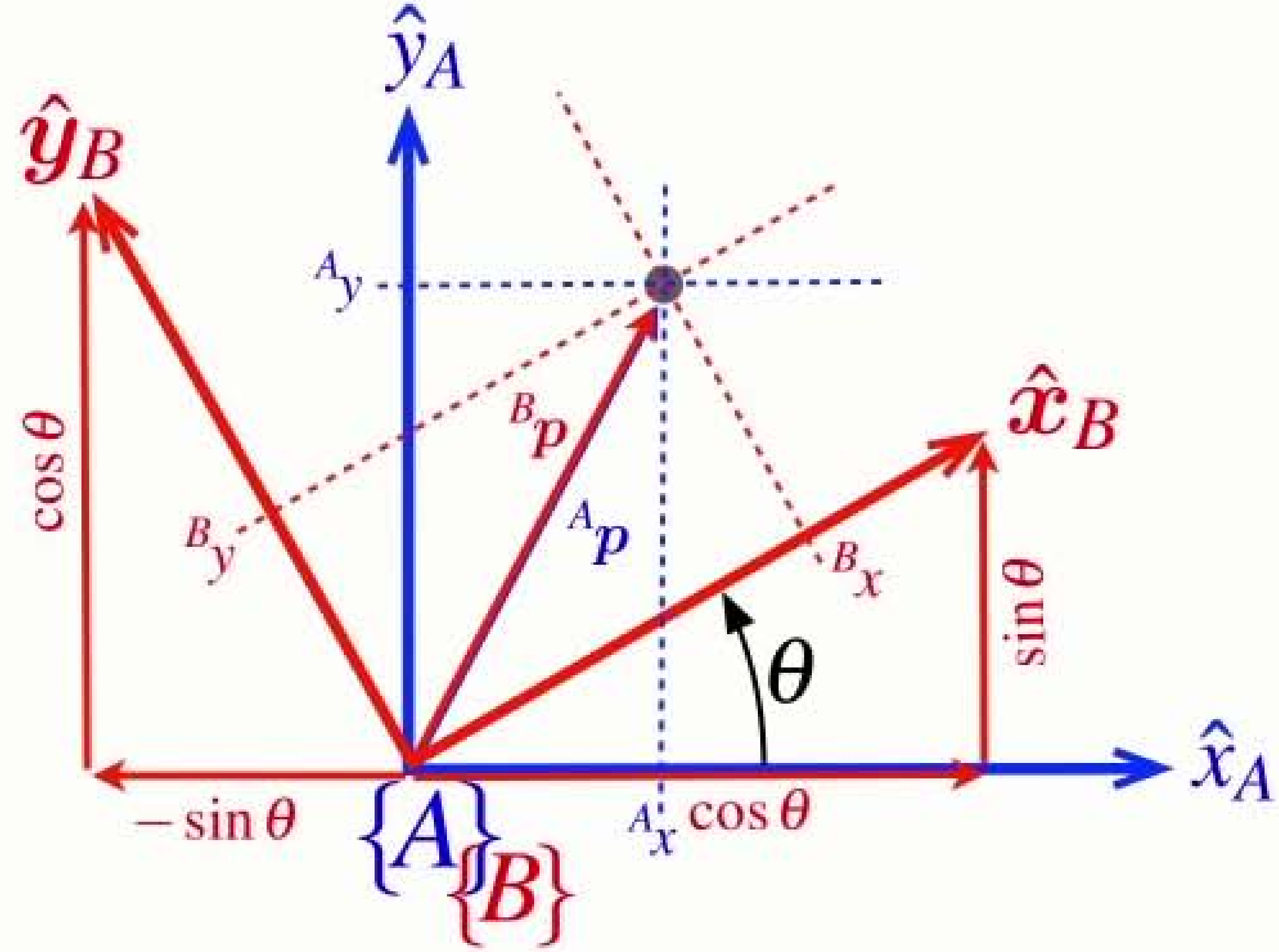
$$ {}^{A}\boldsymbol{p} = \mathbf{R}\,{}^{B}\boldsymbol{p} \qquad \mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$

$$ {}^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p} \qquad {}^{A}\mathbf{R}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$



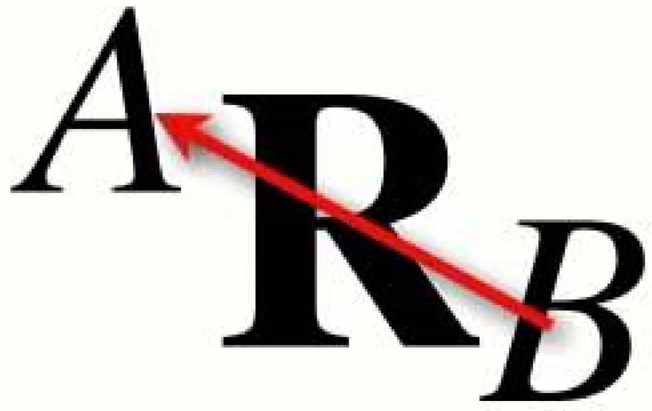
Consider rotation first

$$^{A}\boldsymbol{p} = \mathbf{R}\,^{B}\boldsymbol{p} \qquad \mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$



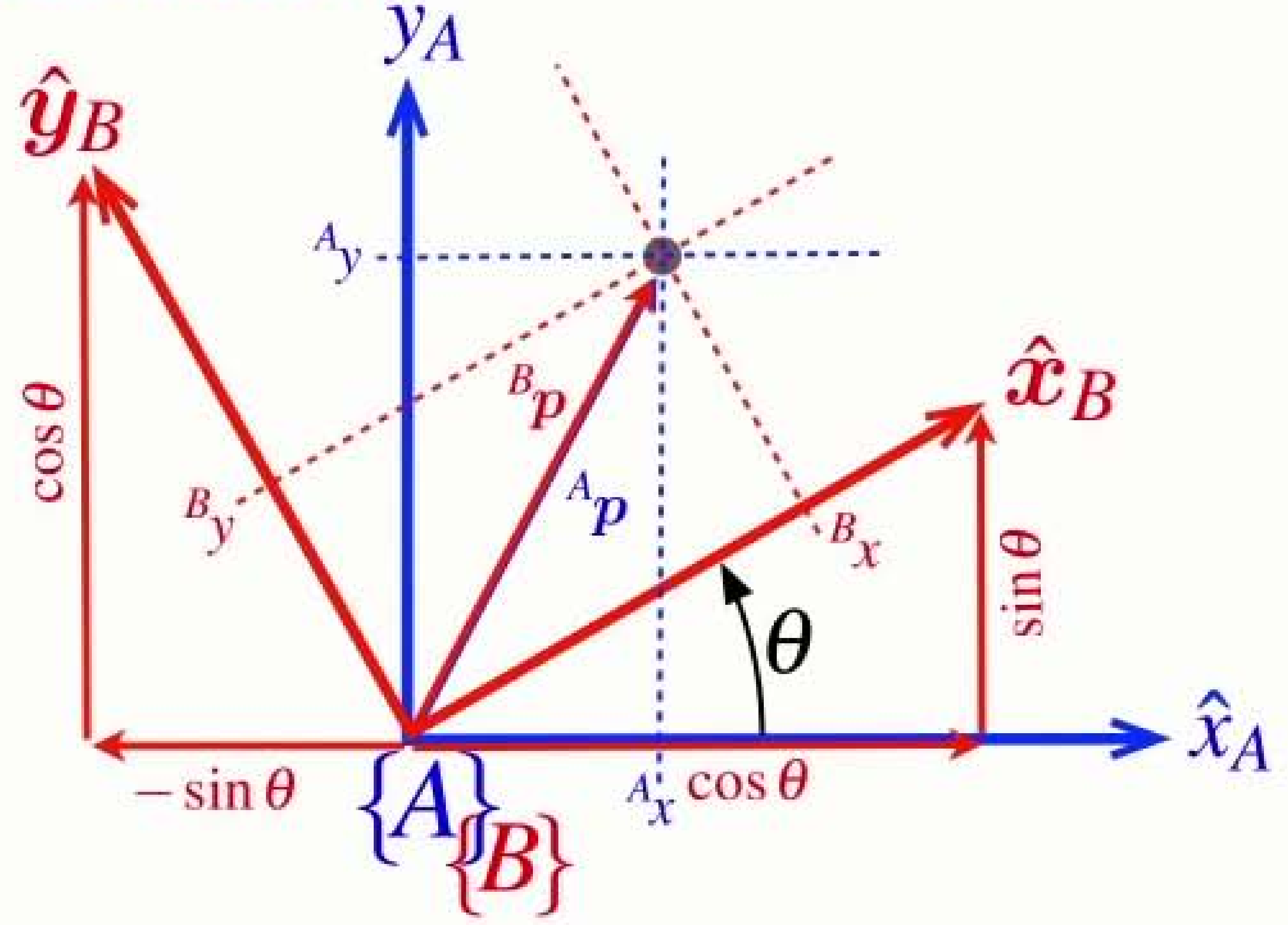
$$^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\,^{B}\boldsymbol{p} \qquad {}^{A}\mathbf{R}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

- **R** is a rotation matrix that:
  - ➥ rotates vectors from frame {B} to frame {A}



Properties of the rotation matix

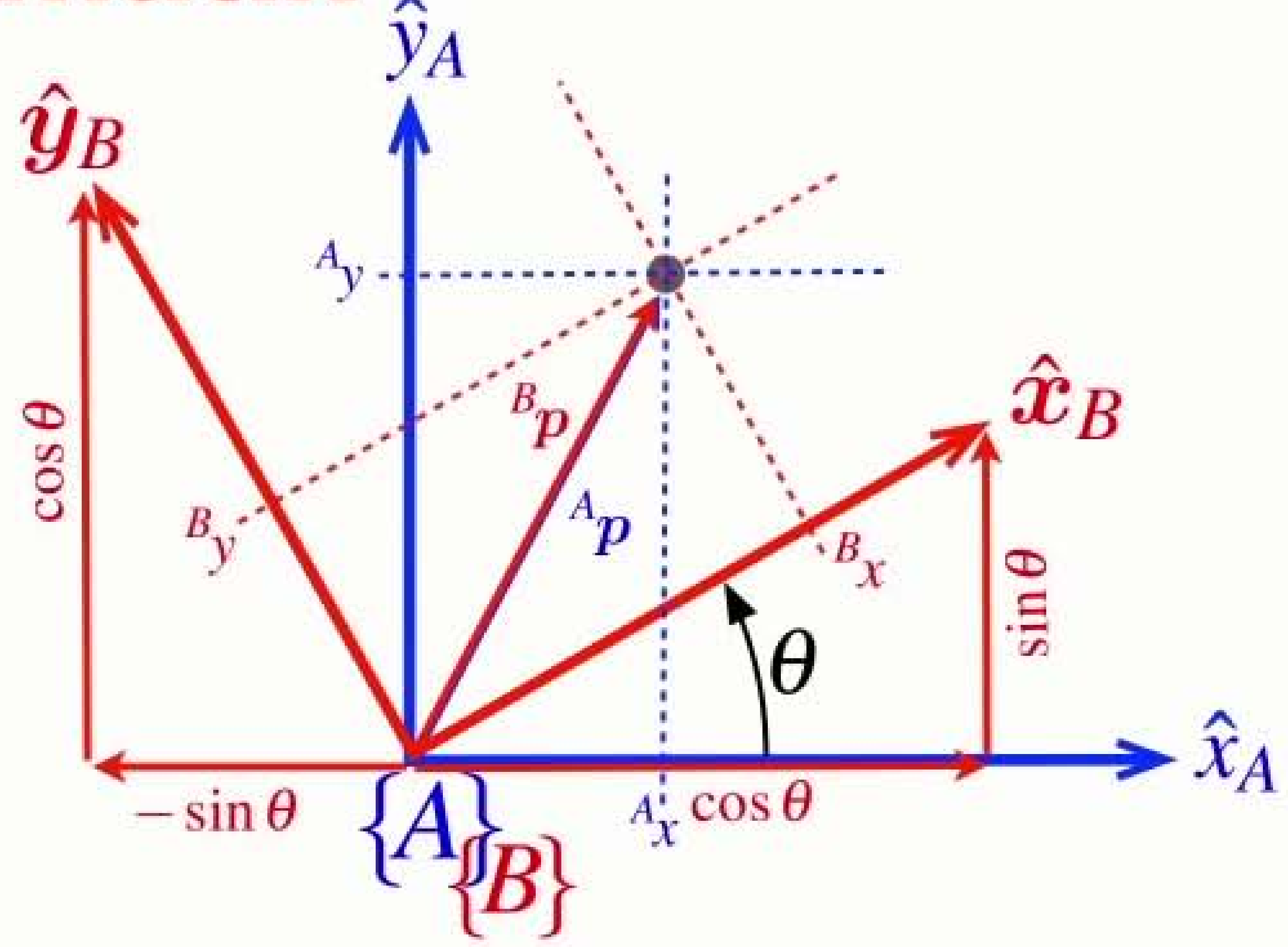
$$ {}^{A}\mathbf{R}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$



Properties of the rotation matix

$$
{}^{A}\mathbf{R}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}
$$

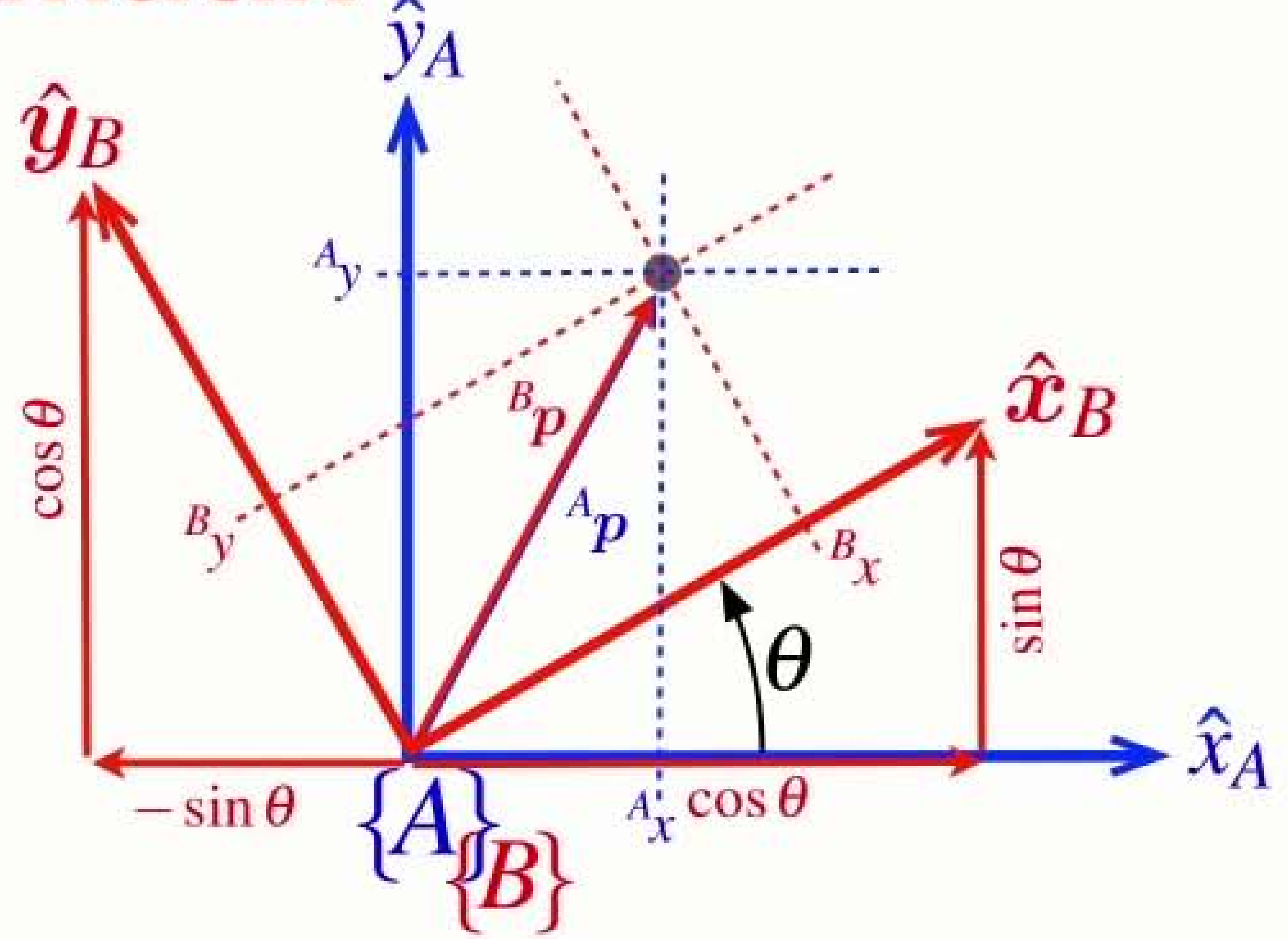
- An orthogonal (orthonormal) matrix



Properties of the rotation matix

$$ {}^{A}\mathbf{R}_B = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$

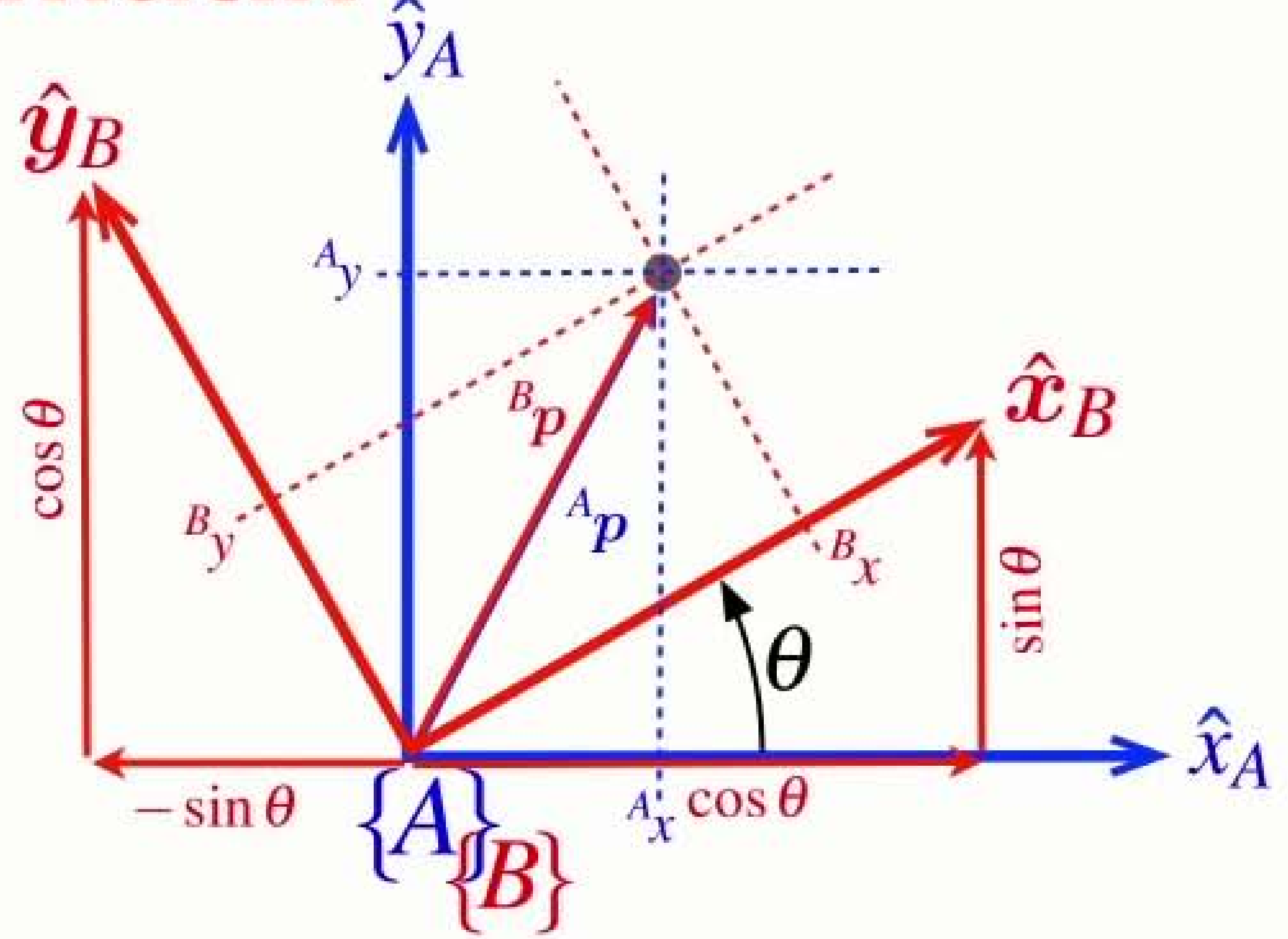
- An orthogonal (orthonormal) matrix
  - ➥ Each column is a unit length vector



Properties of the rotation matix

$$^{A}\mathbf{R}_B = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

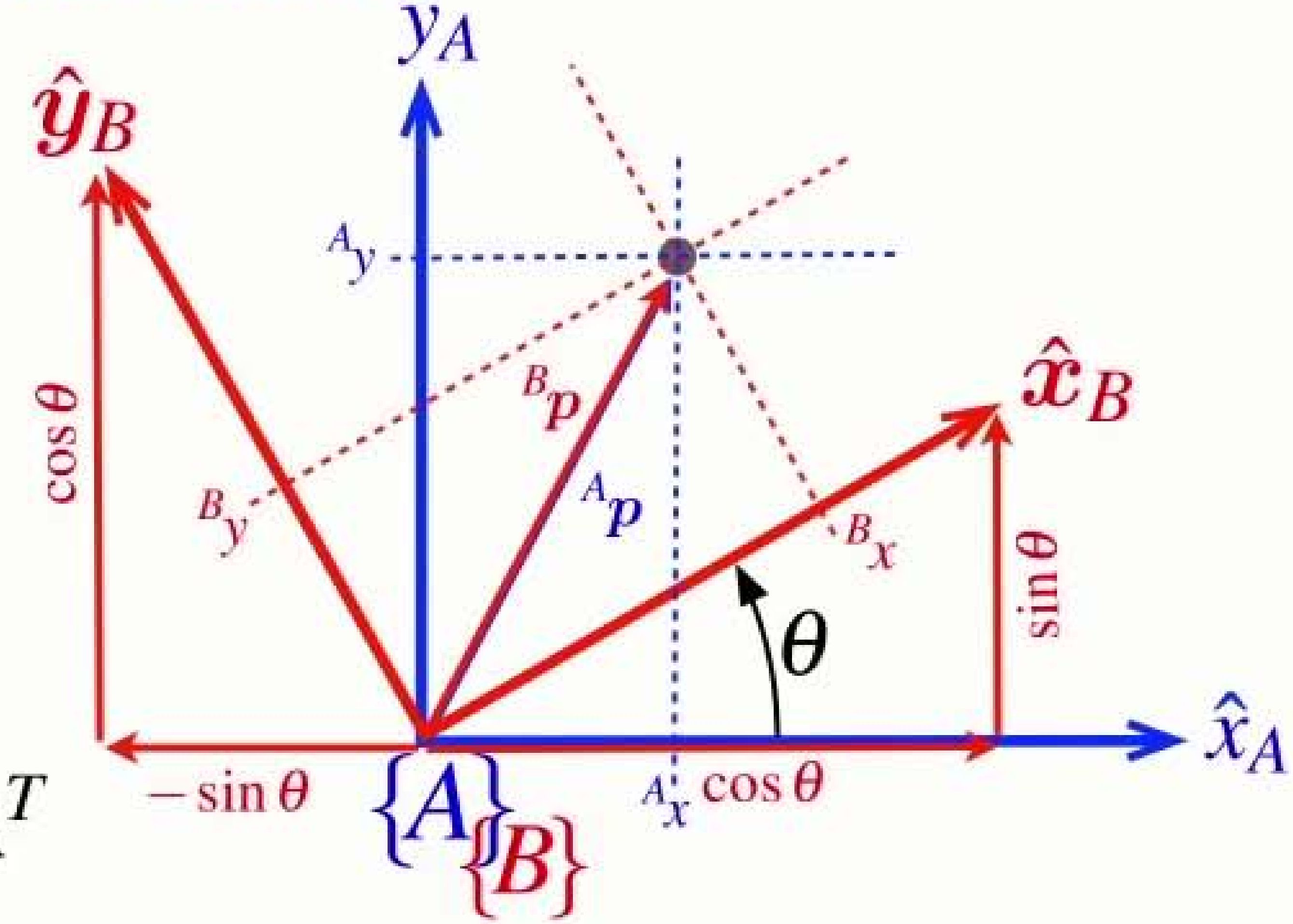
- An orthogonal (orthonormal) matrix
  - ➥ Each column is a unit length vector
  - ➥ Each column is orthogonal to all other columns



Properties of the rotation matix

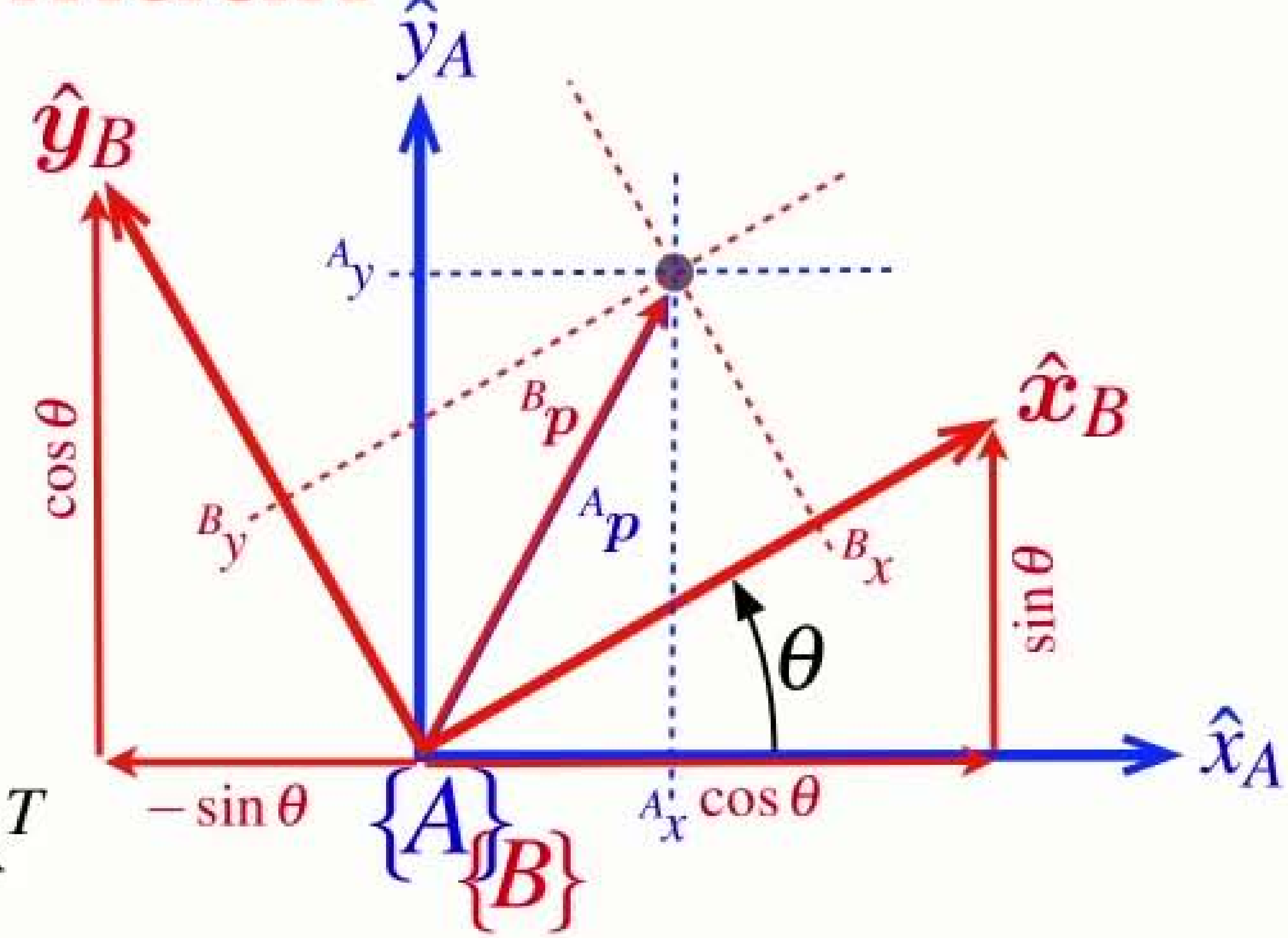
$$ {}^{A}\mathbf{R}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$

- An orthogonal (orthonormal) matrix
  - ➥ Each column is a unit length vector
  - ➥ Each column is orthogonal to all other columns
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^{T}$



Properties of the rotation matix

$$ {}^{A}\mathbf{R}_B = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$



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  - ➡ Each column is a unit length vector
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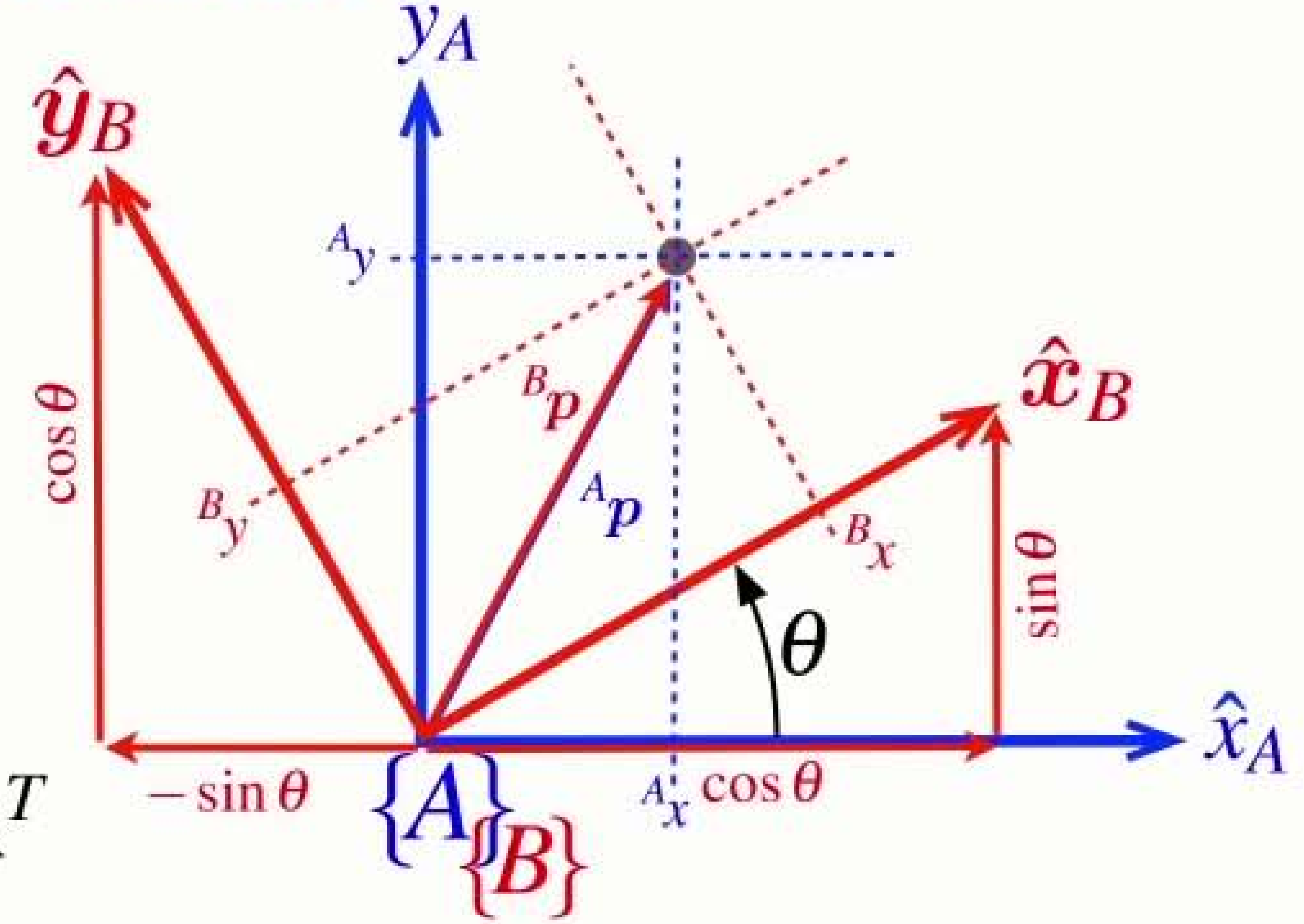
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^T$
- The determinant is +1 $det(\mathbf{R}) = 1$
  - ➡ the length of a vector is unchanged by rotation

Properties of the rotation matix

$$^{A}\mathbf{R}_B = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$



- An orthogonal (orthonormal) matrix
  - ➡ Each column is a unit length vector
  - ➡ Each column is orthogonal to all other columns
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^T$
- The determinant is +1 $det(\mathbf{R}) = 1$
  - ➡ the length of a vector is unchanged by rotation
- Rotation matrices belong to the Special Orthogonal group of dimension 2 $\mathbf{R} \in SO(2)$

Inverse rotation

$$^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

Inverse rotation

$$ {}^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\, {}^{B}\boldsymbol{p} $$

$$ {}^{B}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}^{-1}\, {}^{A}\boldsymbol{p} $$

Inverse rotation

- To rotate a vector from frame {A} to frame {B} we use the inverse rotation matrix
  - ➥ The inverse is simply the transpose

$$ {}^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\, {}^{B}\boldsymbol{p} $$

$$ {}^{B}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}^{-1}\, {}^{A}\boldsymbol{p} $$

Inverse rotation

$$^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\, {}^{B}\boldsymbol{p}$$

- To rotate a vector from frame {A} to frame {B} we use the inverse rotation matrix
  - ➥ The inverse is simply the transpose

$$^{B}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}^{-1}\, {}^{A}\boldsymbol{p}$$

$$^{B}\boldsymbol{p} = {}^{B}\mathbf{R}_{A}\, {}^{A}\boldsymbol{p}$$

Inverse rotation

$${}^{A}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

- To rotate a vector from frame {A} to frame {B} we use the inverse rotation matrix
  - ➥ The inverse is simply the transpose

$${}^{B}\boldsymbol{p} = {}^{A}\mathbf{R}_{B}^{-1}\,{}^{A}\boldsymbol{p}$$

$${}^{B}\boldsymbol{p} = {}^{B}\mathbf{R}_{A}\,{}^{A}\boldsymbol{p}$$

$${}^{B}\mathbf{R}_{A} = {}^{A}\mathbf{R}_{B}^{-1}$$

Table 2.1. Summary of the various concrete representations of pose ξ introduced in this chapter

Representation	⊕	⊖	transl.	rotn.	dim	MATLAB
$(x, y, \theta) \in \mathbb{R}^2 \times \mathbb{S}$			✓	✓	2D	
$\boldsymbol{T} \in SE(2)$	$\boldsymbol{T}_1\boldsymbol{T}_2$	$\boldsymbol{T}^{-1}$	✓	×	2D	`se2(x, y)`
$\boldsymbol{R} \in SO(2)$	$\boldsymbol{R}_1\boldsymbol{R}_2$	$\boldsymbol{R}^T$	×	✓	2D	`se2(0, 0, th)`
$\boldsymbol{T} \in SE(2)$	$\boldsymbol{T}_1\boldsymbol{T}_2$	$\boldsymbol{T}^{-1}$	✓	✓	2D	`se2(x, y, th)`
$(x, y, z, \Gamma) \in \mathbb{R}^3 \times \mathbb{S}^3$			✓	✓	3D	
$\boldsymbol{R} \in SO(3)$	$\boldsymbol{R}_1\boldsymbol{R}_2$	$\boldsymbol{R}^T$	×	✓	3D	`rotx,roty,...`
$\Gamma \in \mathbb{S}^3$	×			✓	3D	`tr2eul,eul2tr`
$\Gamma \in \mathbb{S}^3$	×			✓	3D	`tr2rpy1,rpy2tr`
$\boldsymbol{T} \in SE(3)$	$\boldsymbol{T}_1\boldsymbol{T}_2$	$\boldsymbol{T}^{-1}$	✓	✓	3D	`transl(x,y,z)`
$\mathring{q} \in \mathbb{Q}$	$\mathring{q}_1\mathring{q}_2$	$\mathring{q}^{-1}$	×	✓	3D	`quaternion`

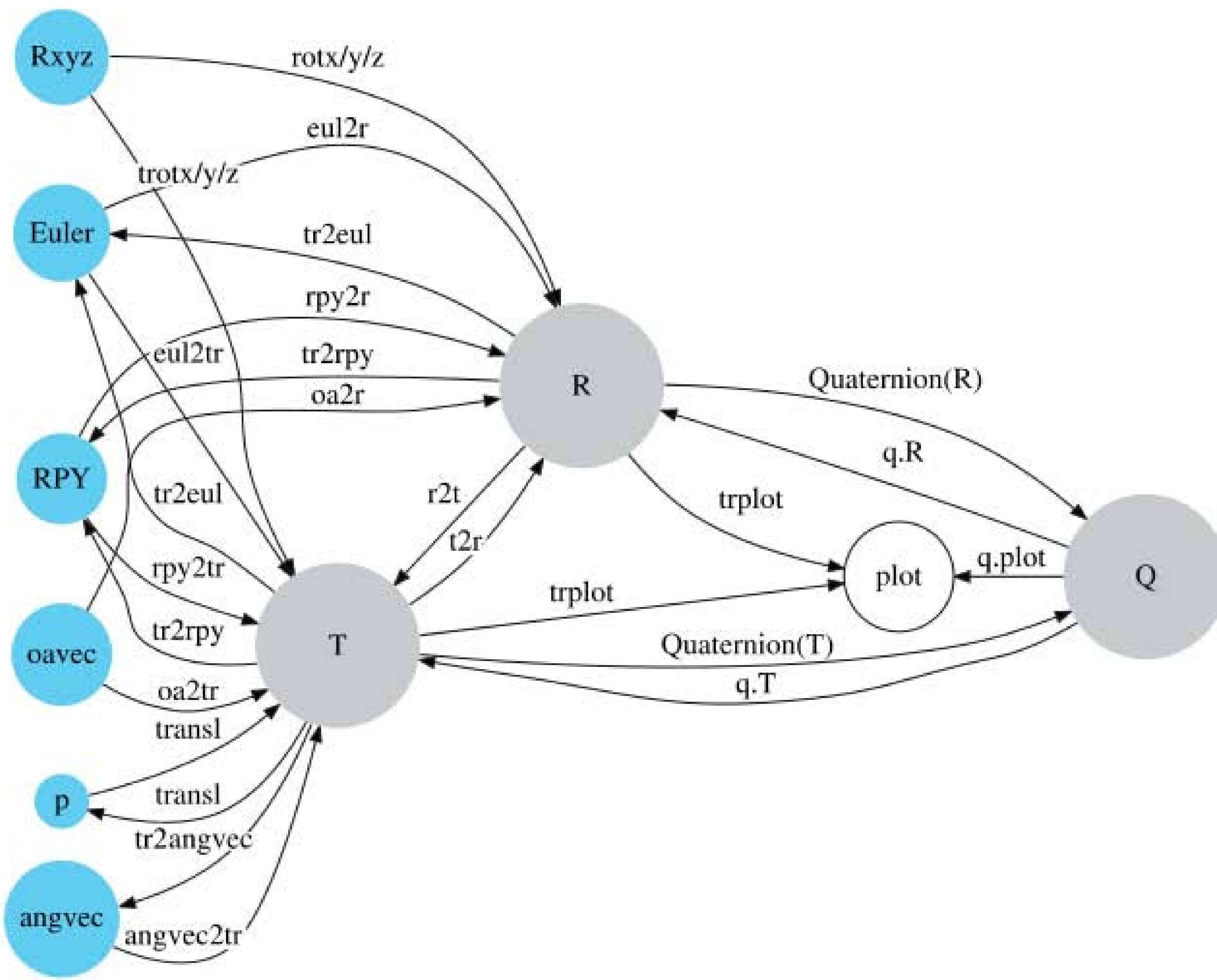


Fig. 2.15. Conversion between rotational representations

Live Editor - /media/vmrguser/hdd/school/teaching/_robotics/lecs/lec01-2DPos/code/rvc1_ch2.mlx *

rvc1_ch2.mlx *

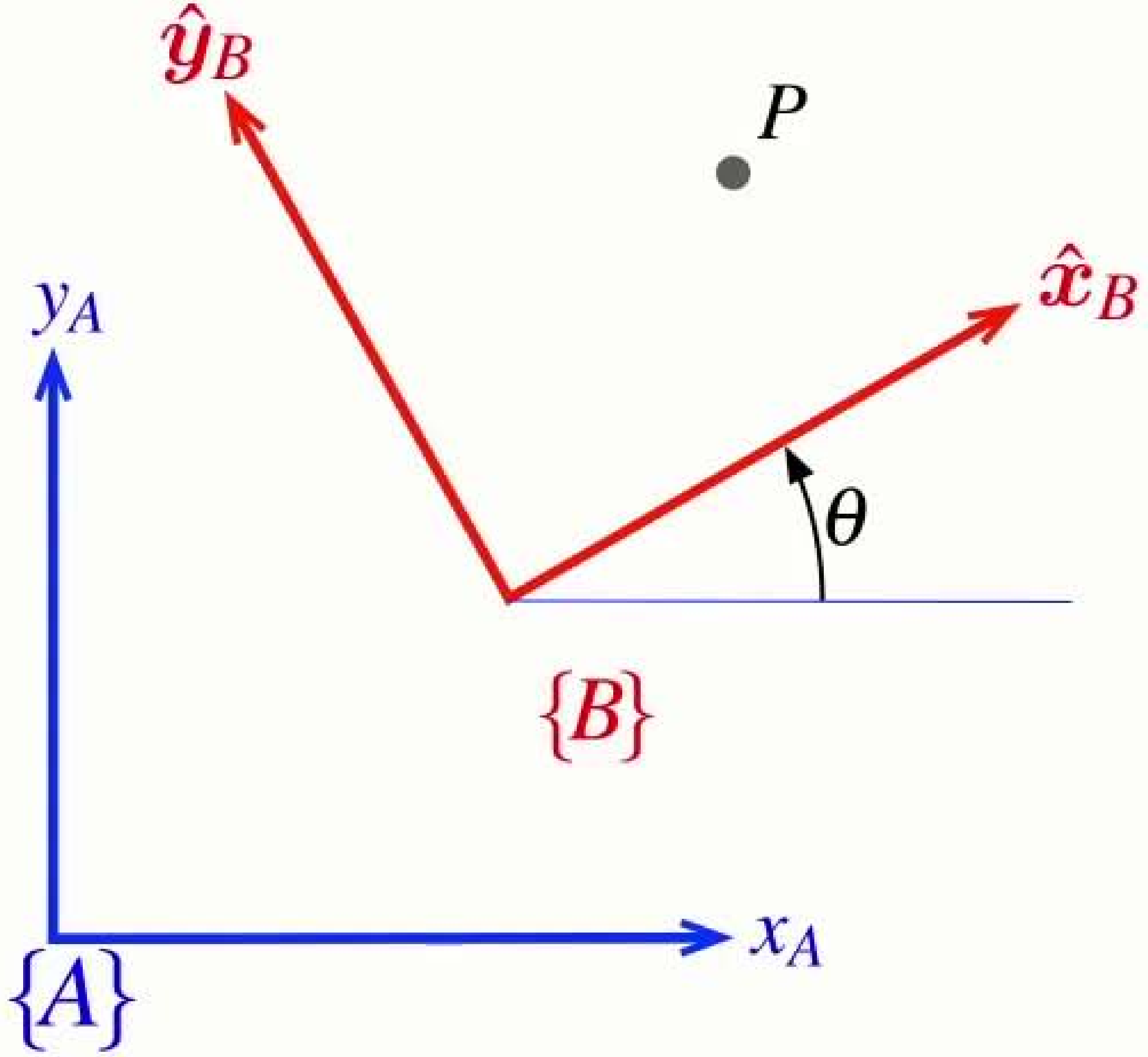
RVC 1, Ch 2

```
clear variables
close all
clc
clear livescript
```

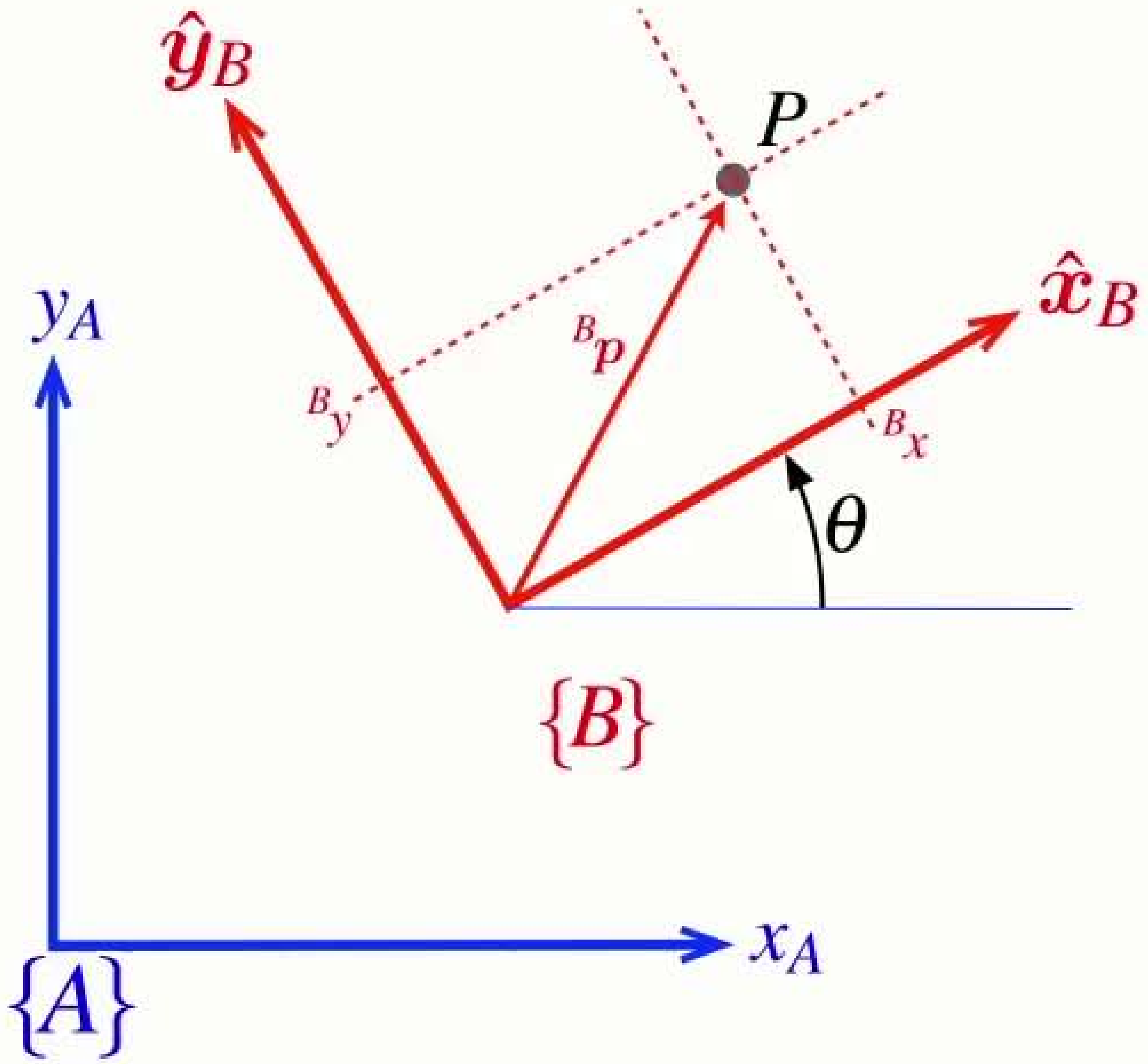
2.6 2D rotations

```
% Create a rotation matrix
R = rot2(0)

% Rotate by 0.2 radians
R=rot2(0.2)

% Rotate by 30 degrees
R=rot2(30,'deg')

% Rotation matrices are
% 1) orthnormal: columns are orthogonal to each other and unit length
% 2) full rank: independent/invertible
% 3) symmetrical: the inverse is equal to the transpose
c1=R(:,1)
c2=R(:,2)

norm(c1),norm(c2)
dot(c1,c2)

% Full rank
det(R)
```

Section 7 Describing rotation and translation

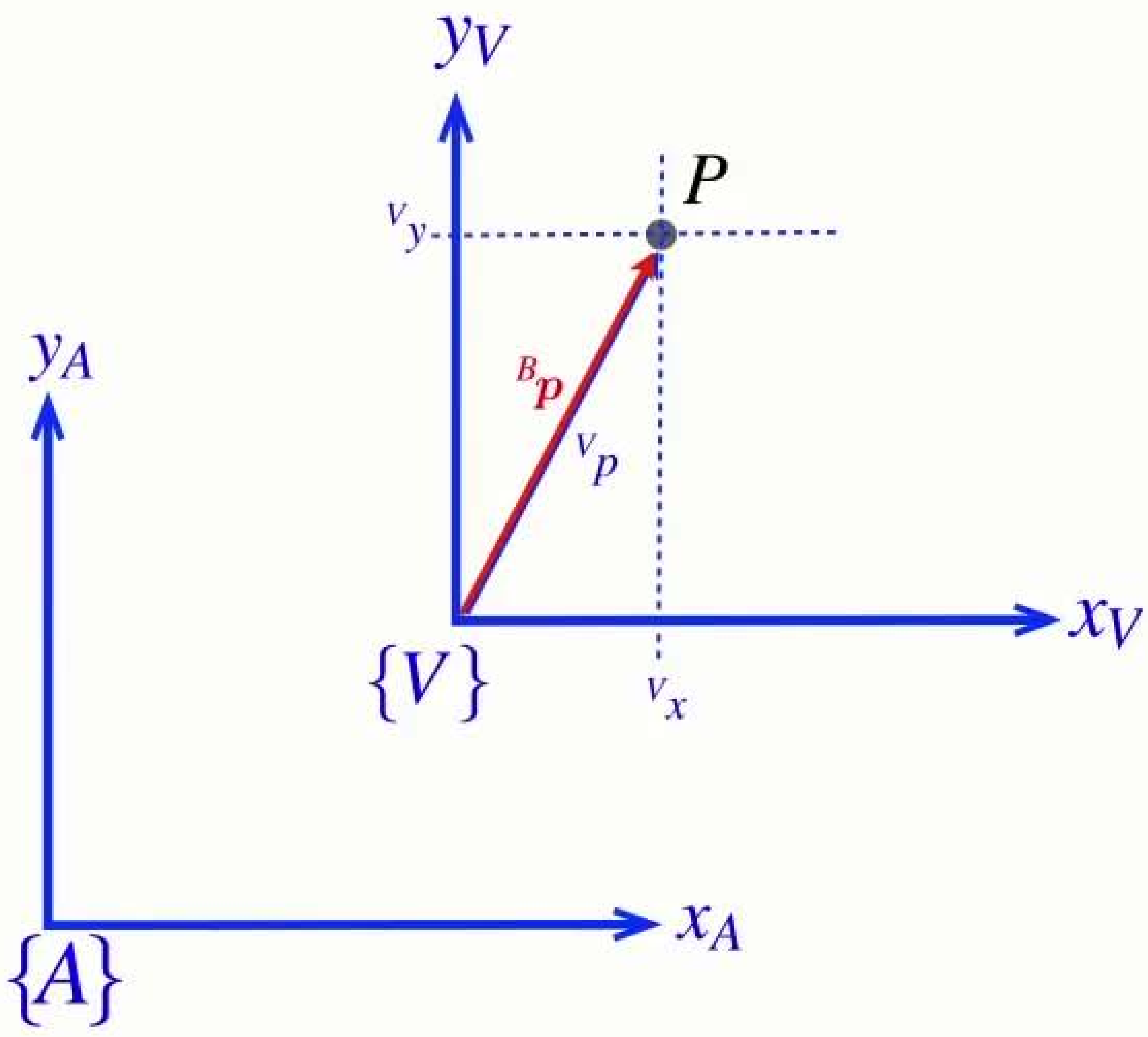


$\hat{y}_B$
P
$\hat{x}_B$
y_A
θ
$\{B\}$
x_A
$\{A\}$

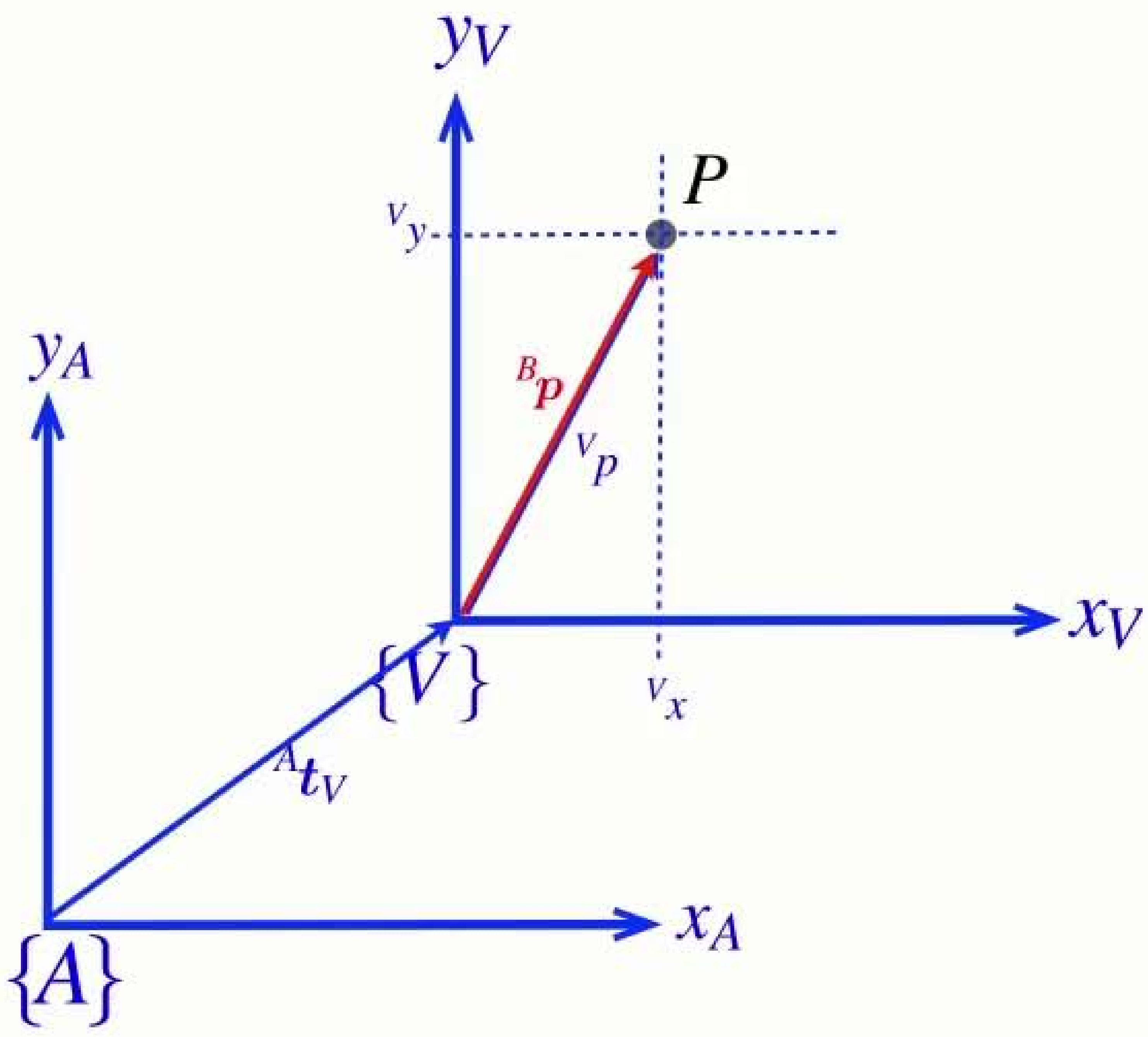


$\hat{y}_B$
P
$\hat{x}_B$
y_A
${}^B\mathbf{p}$
${}^By$
${}^Bx$
θ
$\{B\}$
x_A
$\{A\}$

$$ {}^{V}\boldsymbol{p} = {}^{V}\mathbf{R}_B \, {}^{B}\boldsymbol{p} $$

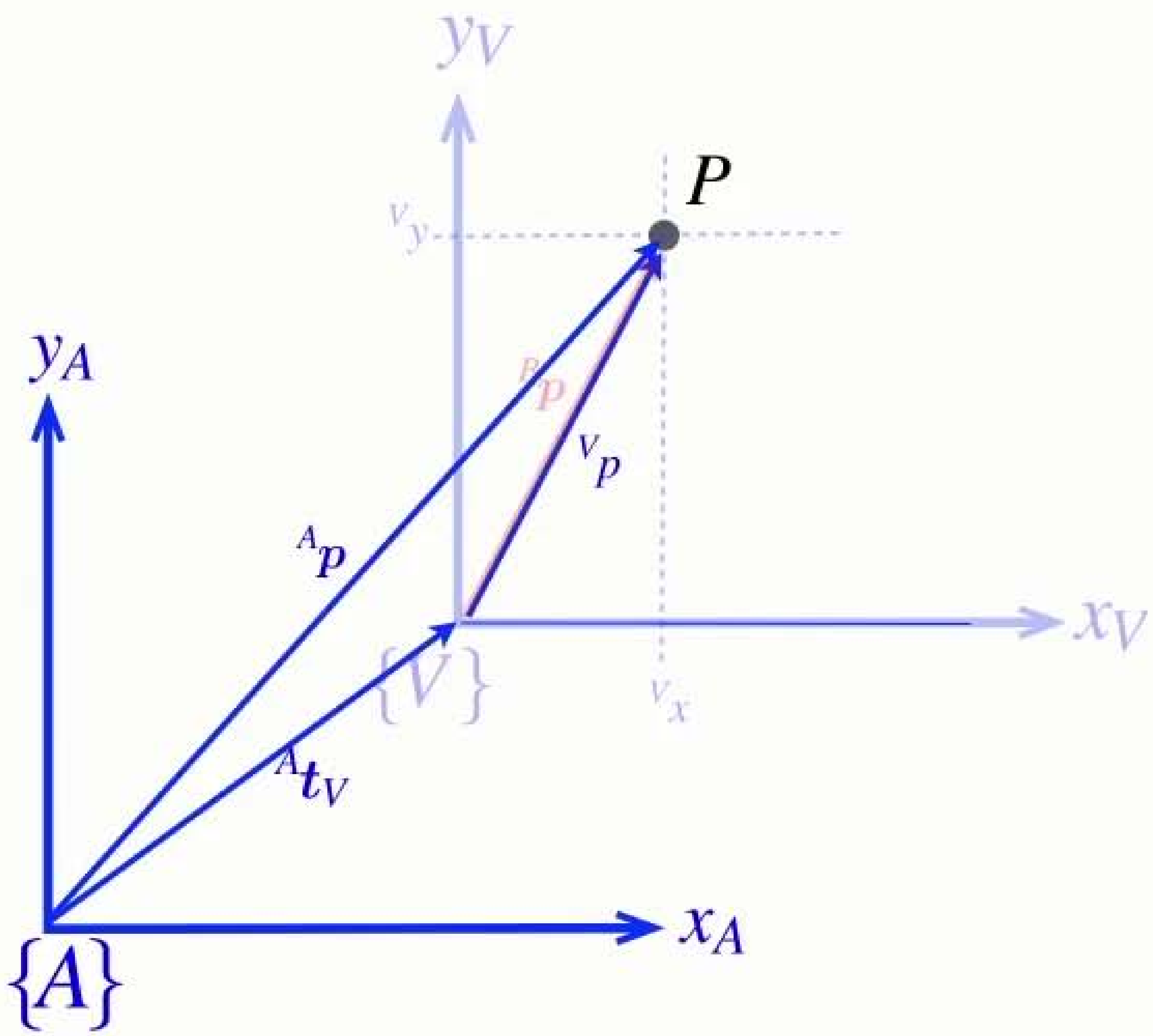


$$ {}^{V}\boldsymbol{p} = {}^{V}\mathbf{R}_{B}\, {}^{B}\boldsymbol{p} $$



$$^{V}\boldsymbol{p} = {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\boldsymbol{p} \qquad \{V\} \parallel \{A\}$$

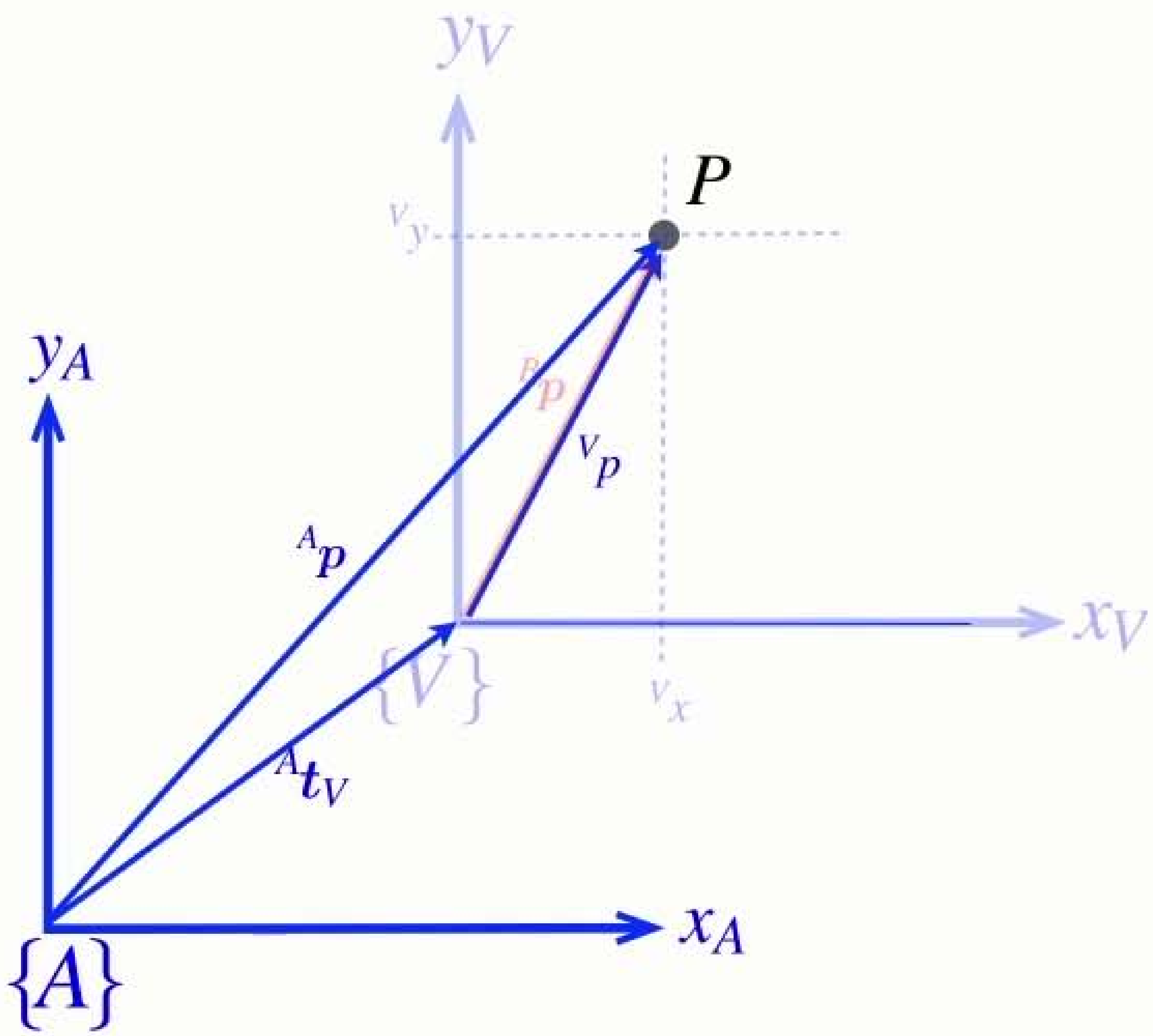


- Only add vectors in the same (or parallel) reference frames

$$^{V}\boldsymbol{p} = {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\boldsymbol{p} \qquad \{V\} \parallel \{A\}$$

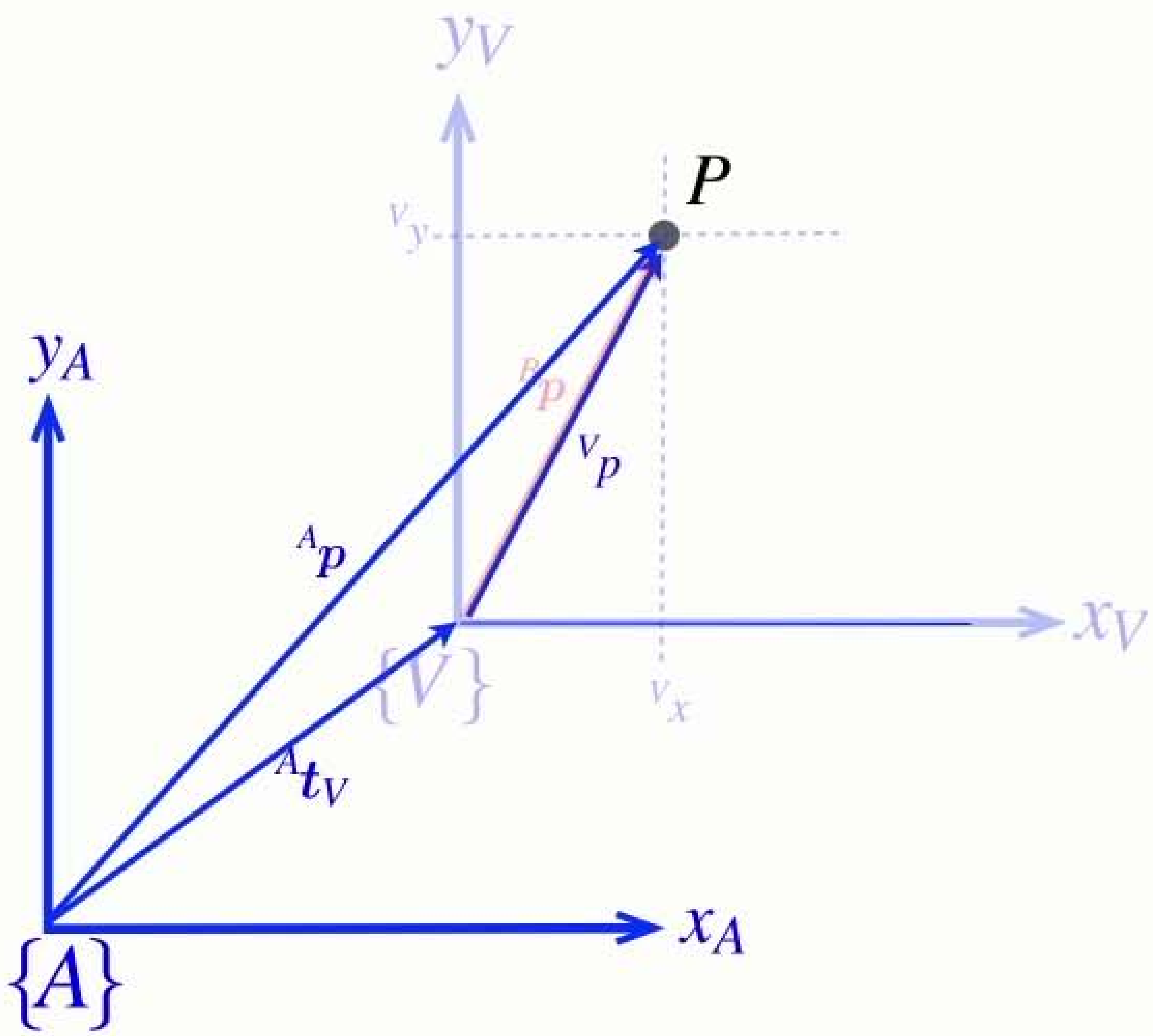
$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$



- Only add vectors in the same (or parallel) reference frames

$$ {}^{V}\boldsymbol{p} = {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p} $$

$$ {}^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\boldsymbol{p} \qquad \{V\} \parallel \{A\} $$



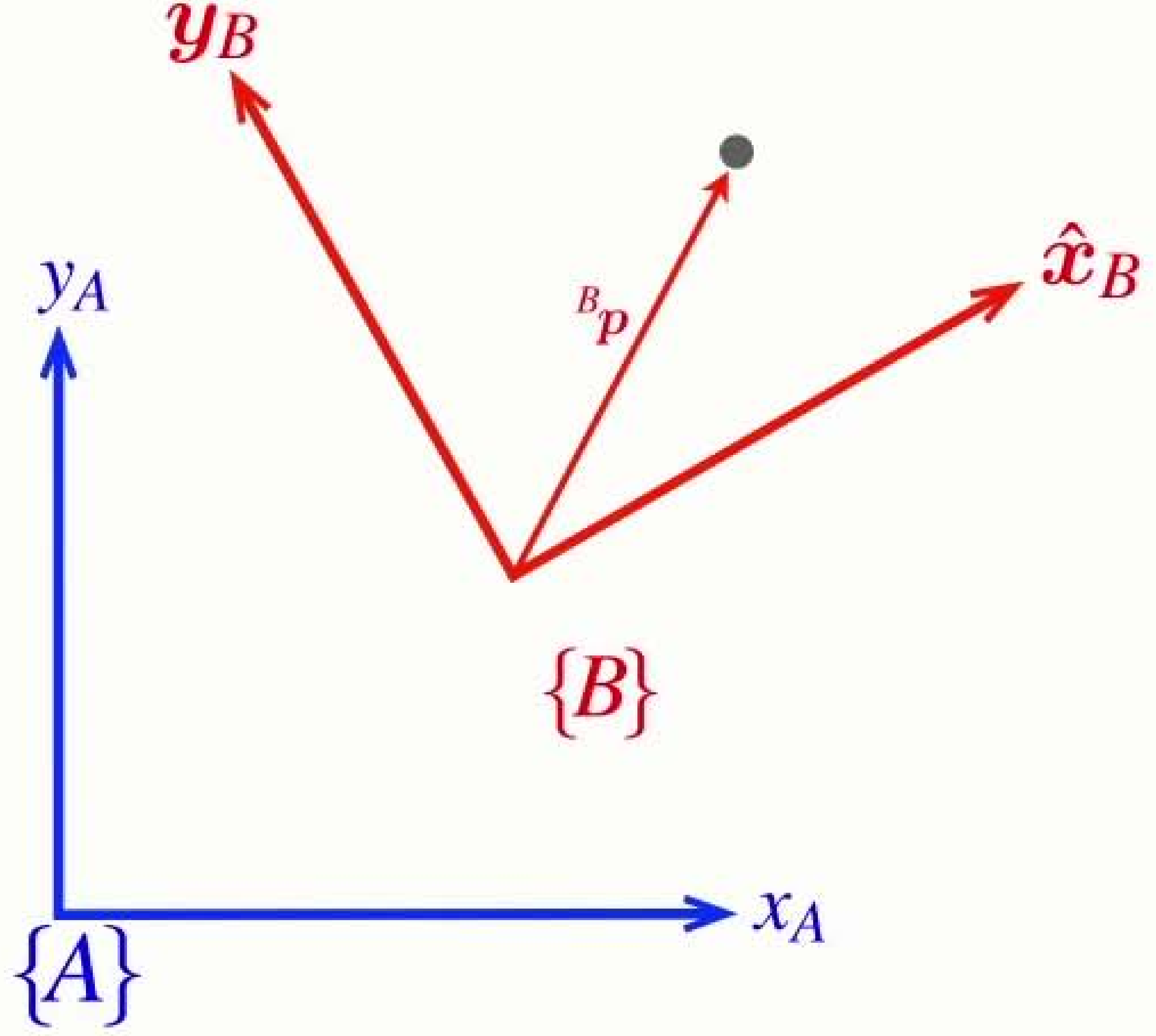
$$ {}^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p} $$

$$ {}^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{B} + {}^{A}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p} $$

- Only add vectors in the same (or parallel) reference frames

Homogeneous form

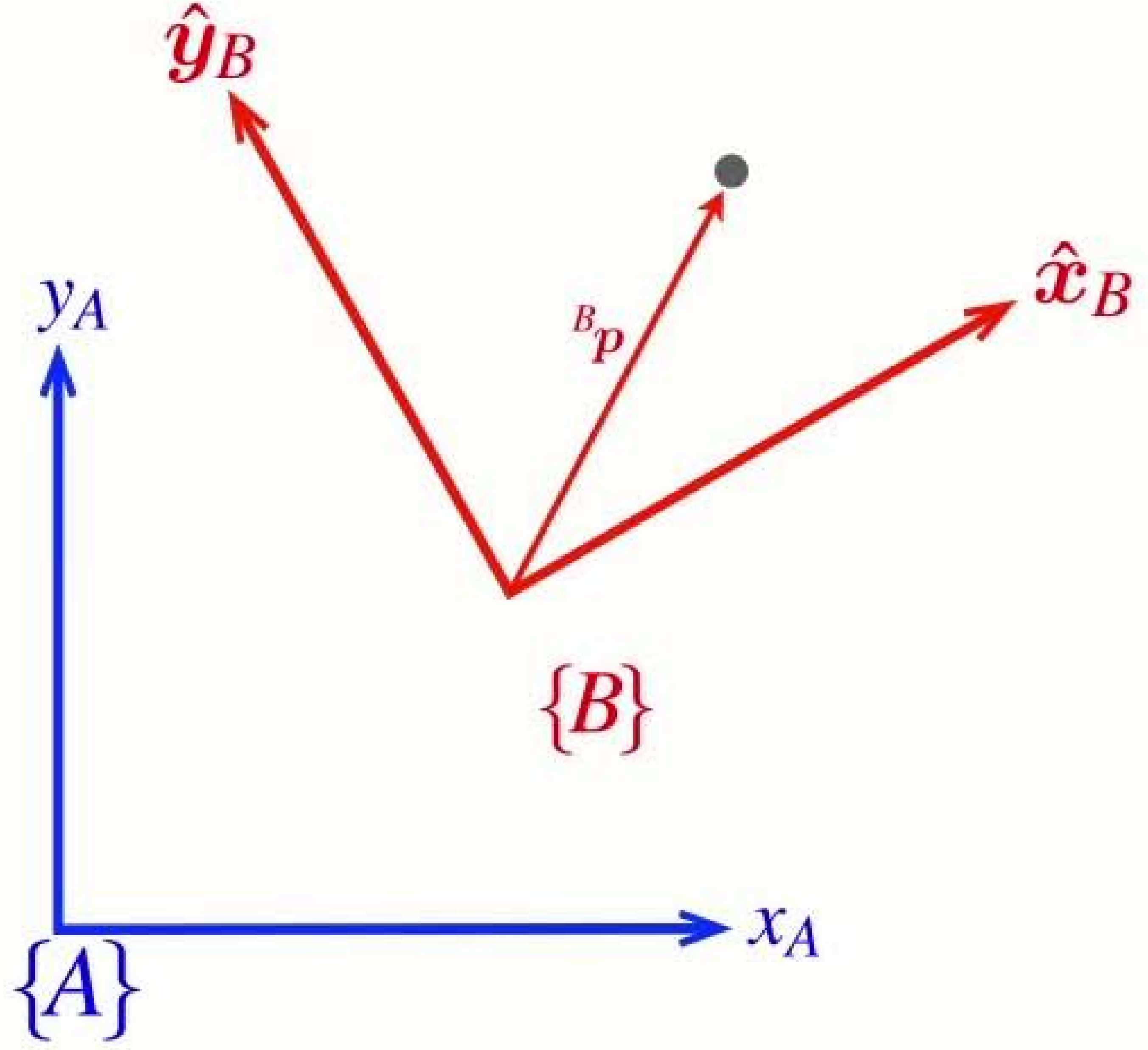
$${}^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$



Homogeneous **form**

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_V + {}^{V}\mathbf{R}_B\,{}^{B}\boldsymbol{p}$$

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \end{pmatrix} + \begin{pmatrix} {}^{A}x_B \\ {}^{A}y_B \end{pmatrix}$$



Homogeneous form

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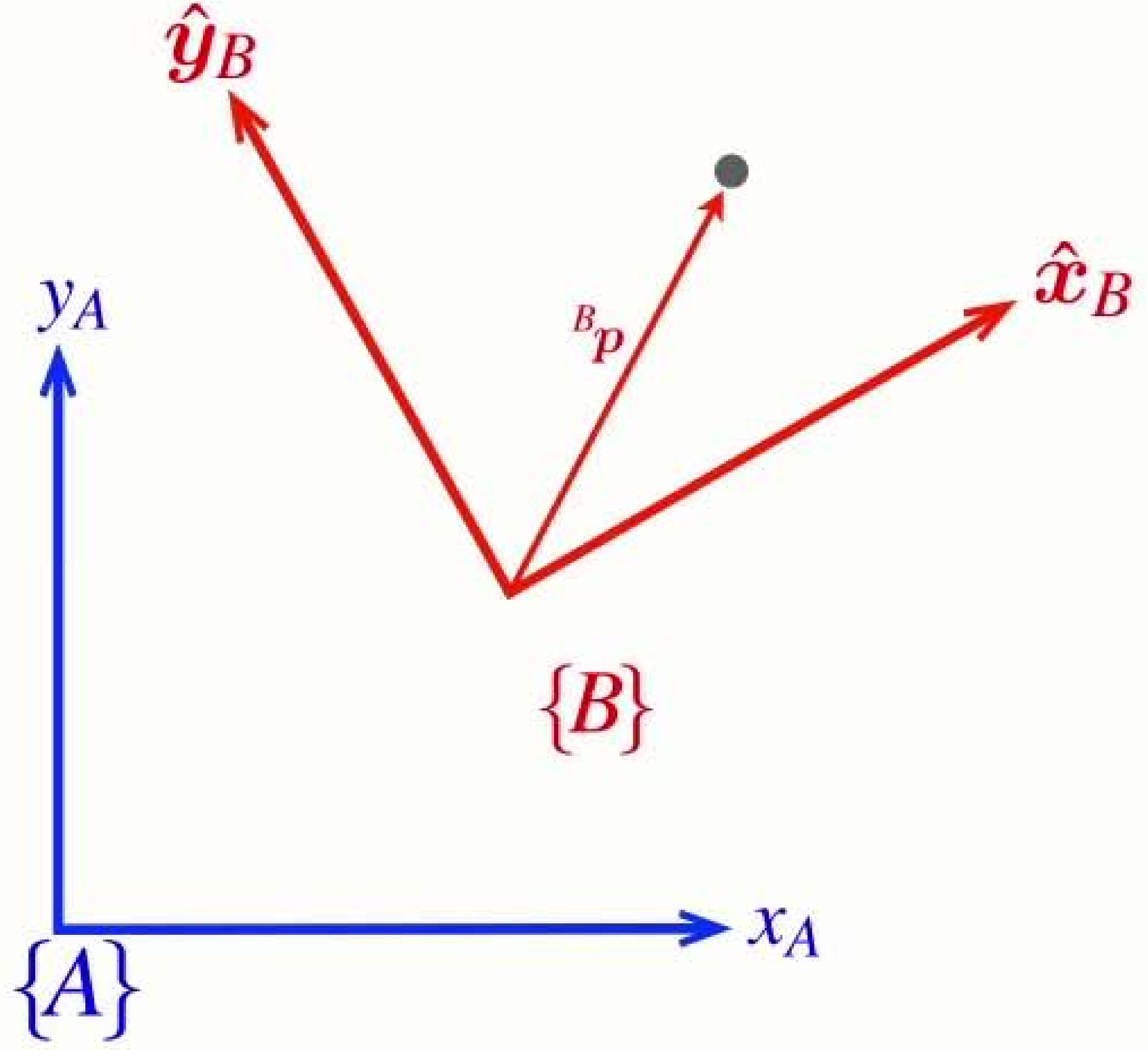
$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_{B} \\ \sin\theta & \cos\theta & {}^{A}y_{B} \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix}$$



Homogeneous **form**

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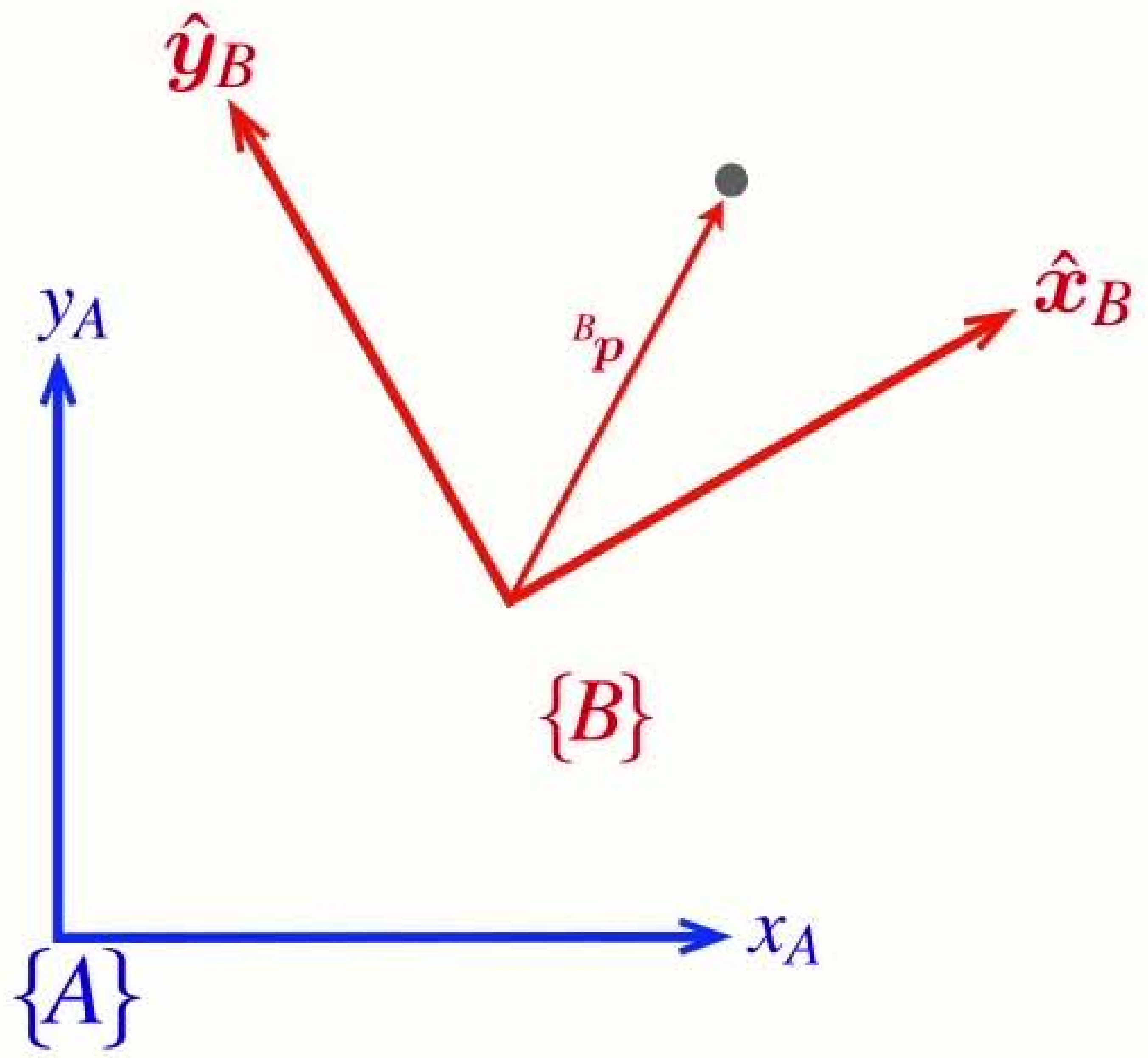
$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_B \\ \sin\theta & \cos\theta & {}^{A}y_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix}$$



Homogeneous form

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$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_B \\ \sin\theta & \cos\theta & {}^{A}y_B \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix}$$

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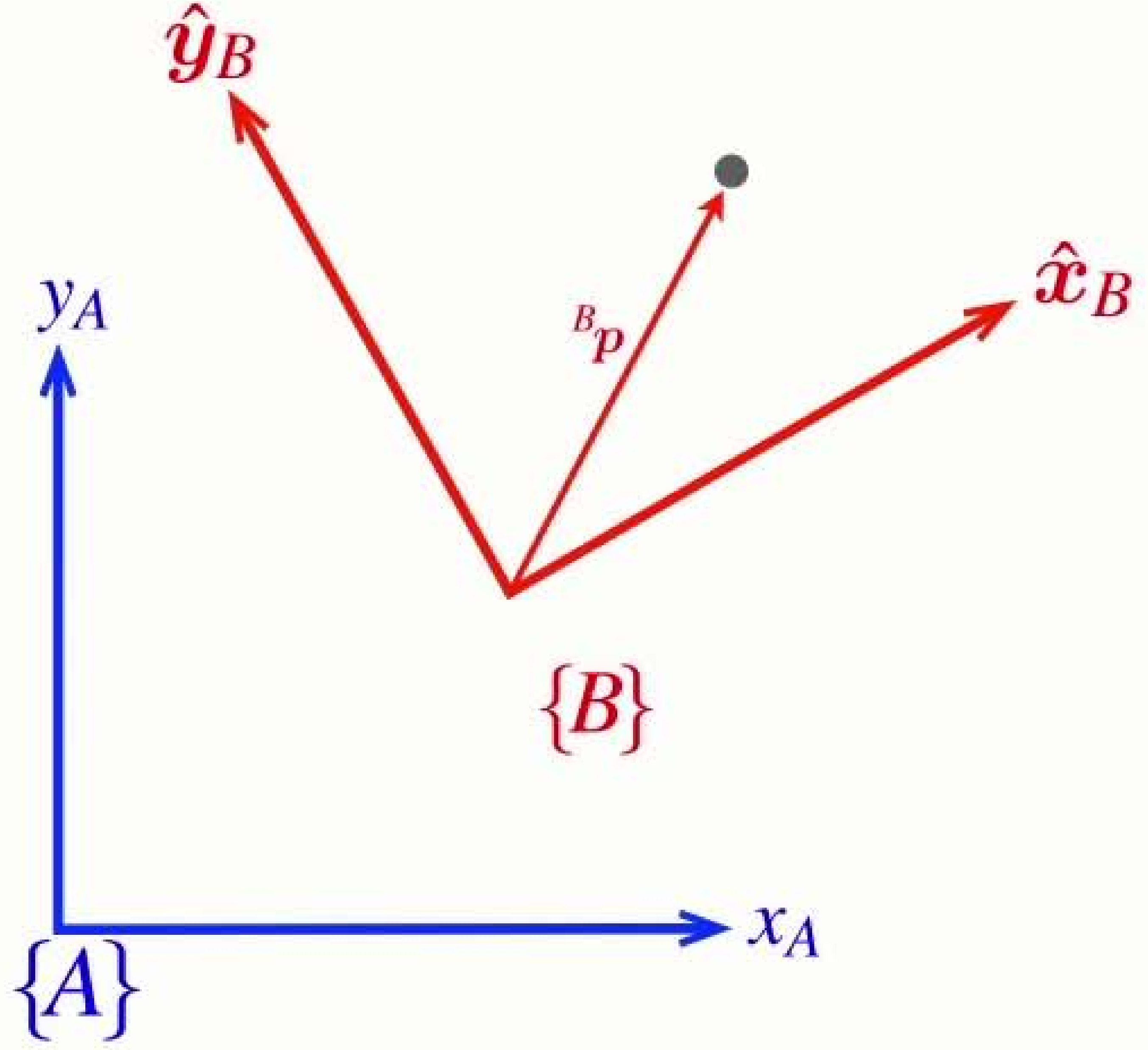


$${}^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} {}^{A}\mathbf{R}_B & {}^{A}\boldsymbol{t}_B \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$

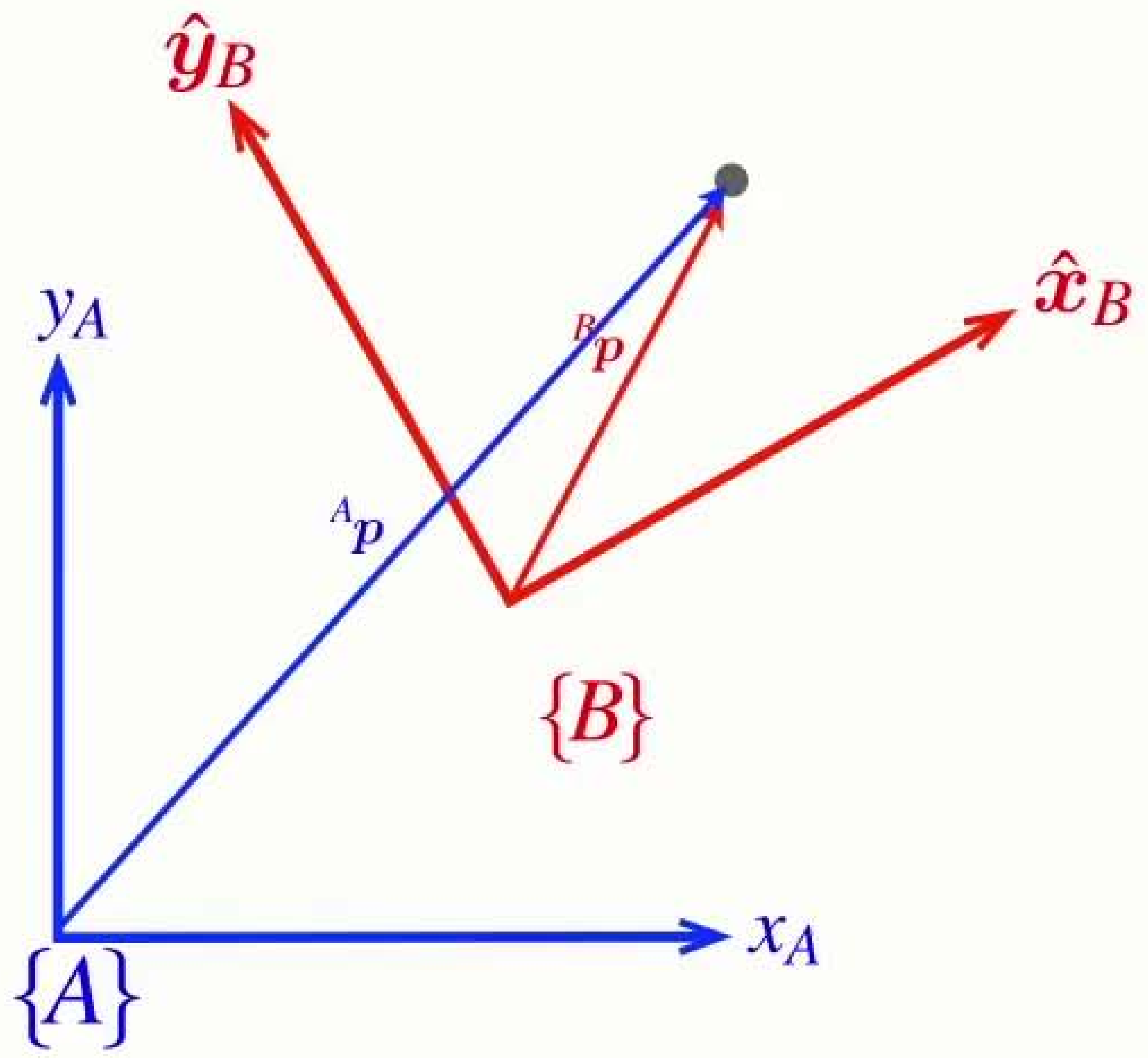
Homogeneous **form**

$$ {}^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_V + {}^{V}\mathbf{R}_B\, {}^{B}\boldsymbol{p} $$

$$ \begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \end{pmatrix} + \begin{pmatrix} {}^{A}x_B \\ {}^{A}y_B \end{pmatrix} $$

$$ \begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_B \\ \sin\theta & \cos\theta & {}^{A}y_B \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix} $$

$$ \begin{pmatrix} {}^{A}x \\ {}^{A}y \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_B \\ \sin\theta & \cos\theta & {}^{A}y_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix} $$



$$ {}^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} {}^{A}\mathbf{R}_B & {}^{A}\boldsymbol{t}_B \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}} $$

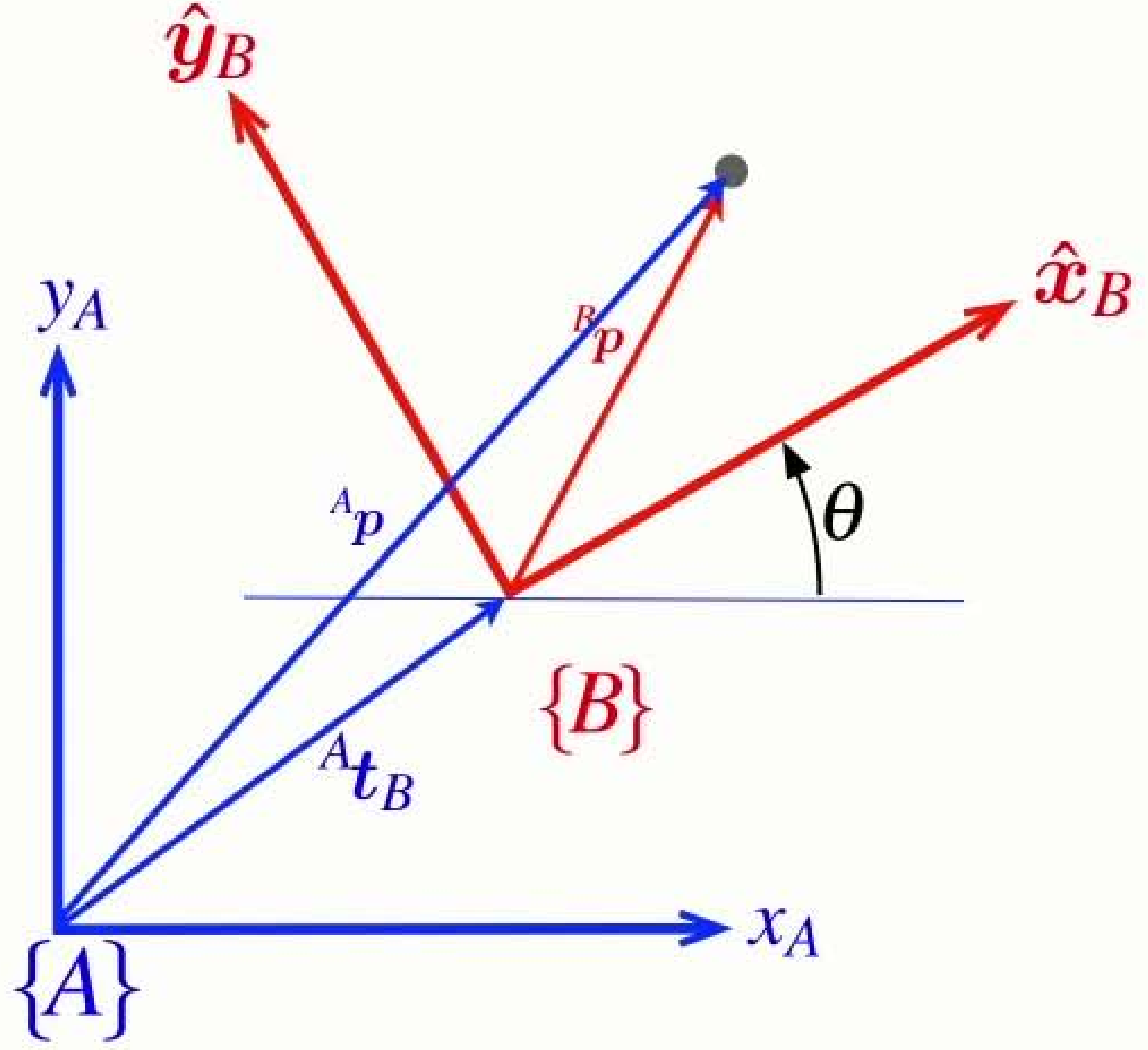
Homogeneous **form**

$$ {}^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_V + {}^{V}\mathbf{R}_B\, {}^{B}\boldsymbol{p} $$

$$ \begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \end{pmatrix} + \begin{pmatrix} {}^{A}x_B \\ {}^{A}y_B \end{pmatrix} $$

$$ \begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_B \\ \sin\theta & \cos\theta & {}^{A}y_B \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix} $$

$$ \begin{pmatrix} {}^{A}x \\ {}^{A}y \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & {}^{A}x_B \\ \sin\theta & \cos\theta & {}^{A}y_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix} $$



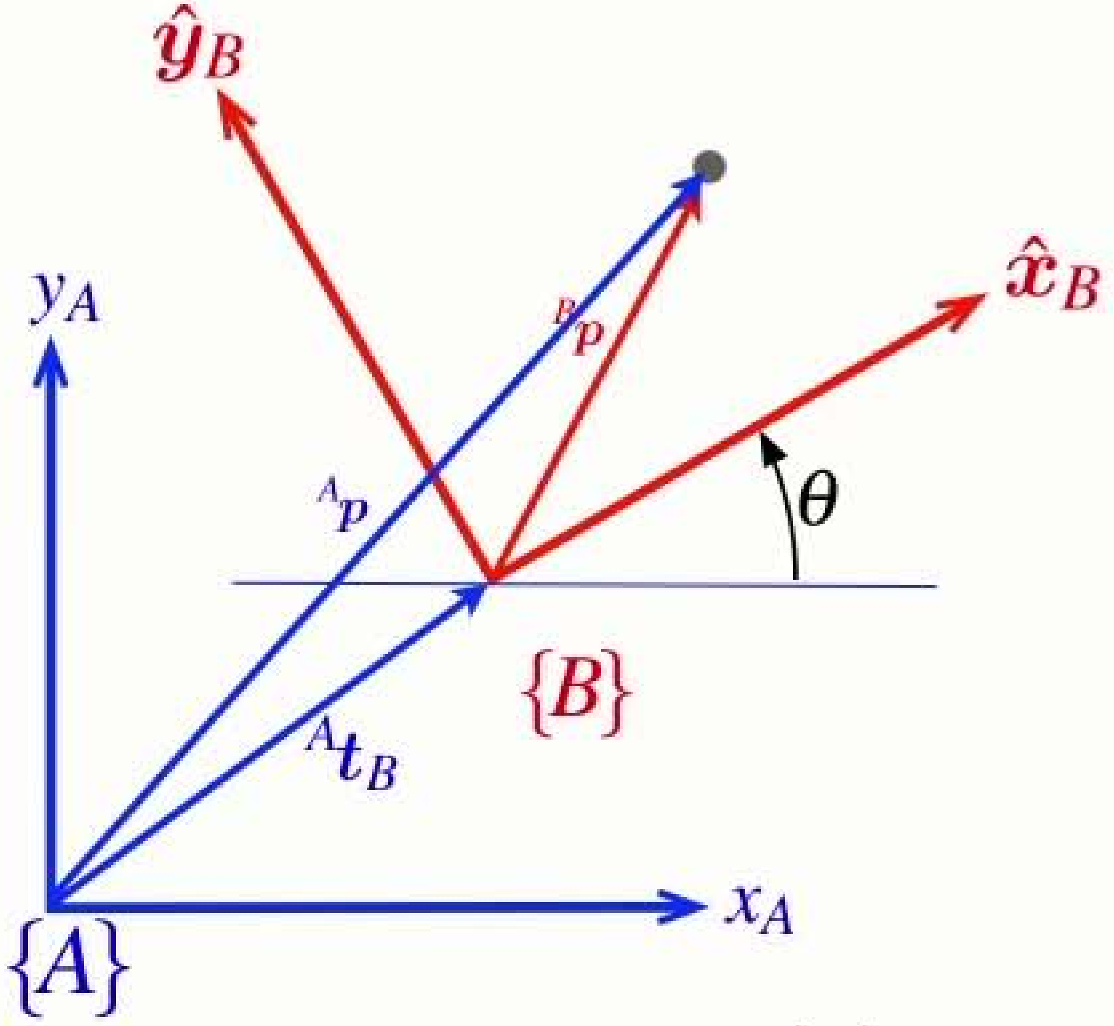
$$ {}^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} {}^{A}\mathbf{R}_B & {}^{A}\boldsymbol{t}_B \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}} $$

Properties of the **homogeneous transform**

homogeneous vector

homogeneous vector

$${}^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

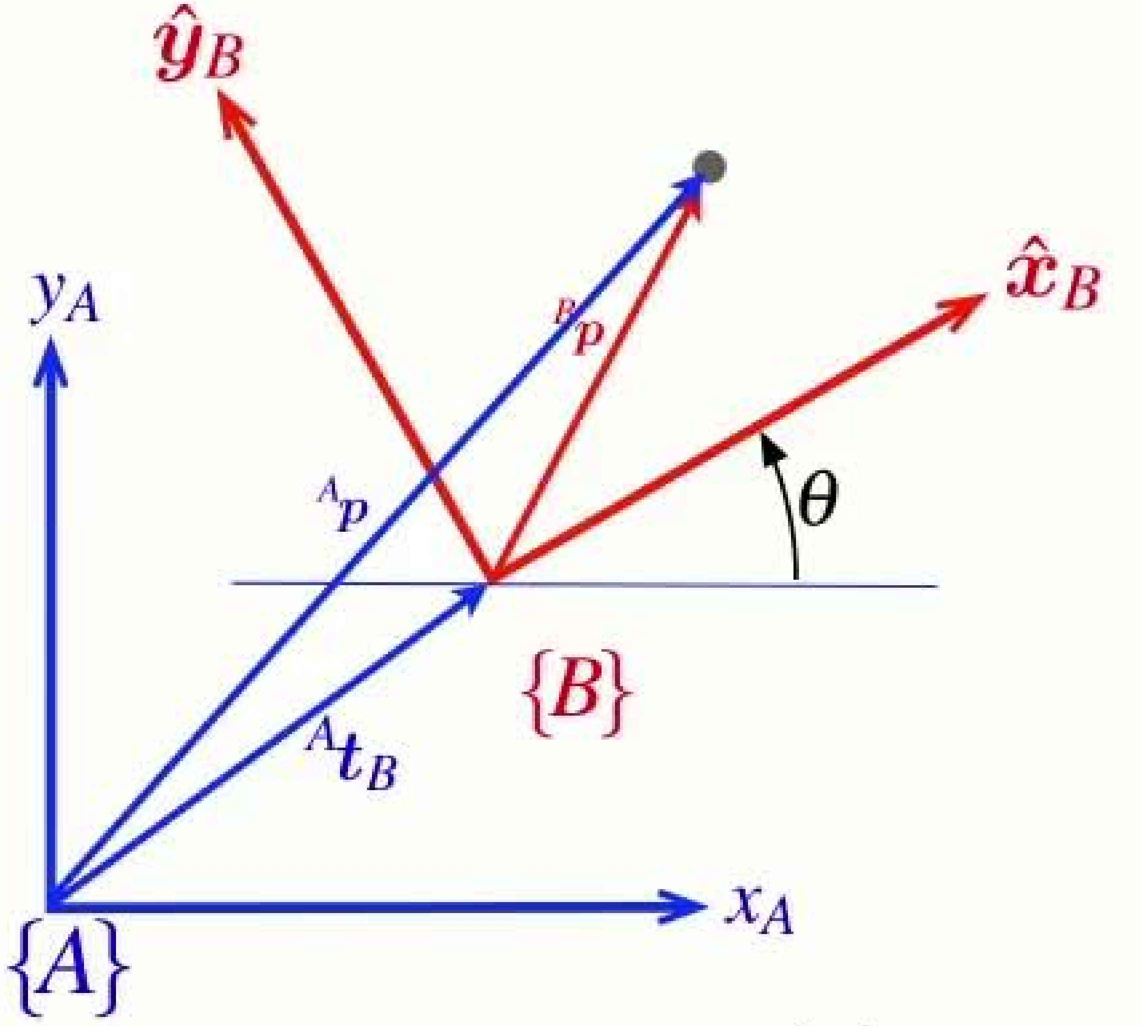
Properties of the homogeneous transform

homogeneous vector

homogeneous transform

homogeneous vector

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

Properties of the homogeneous transform

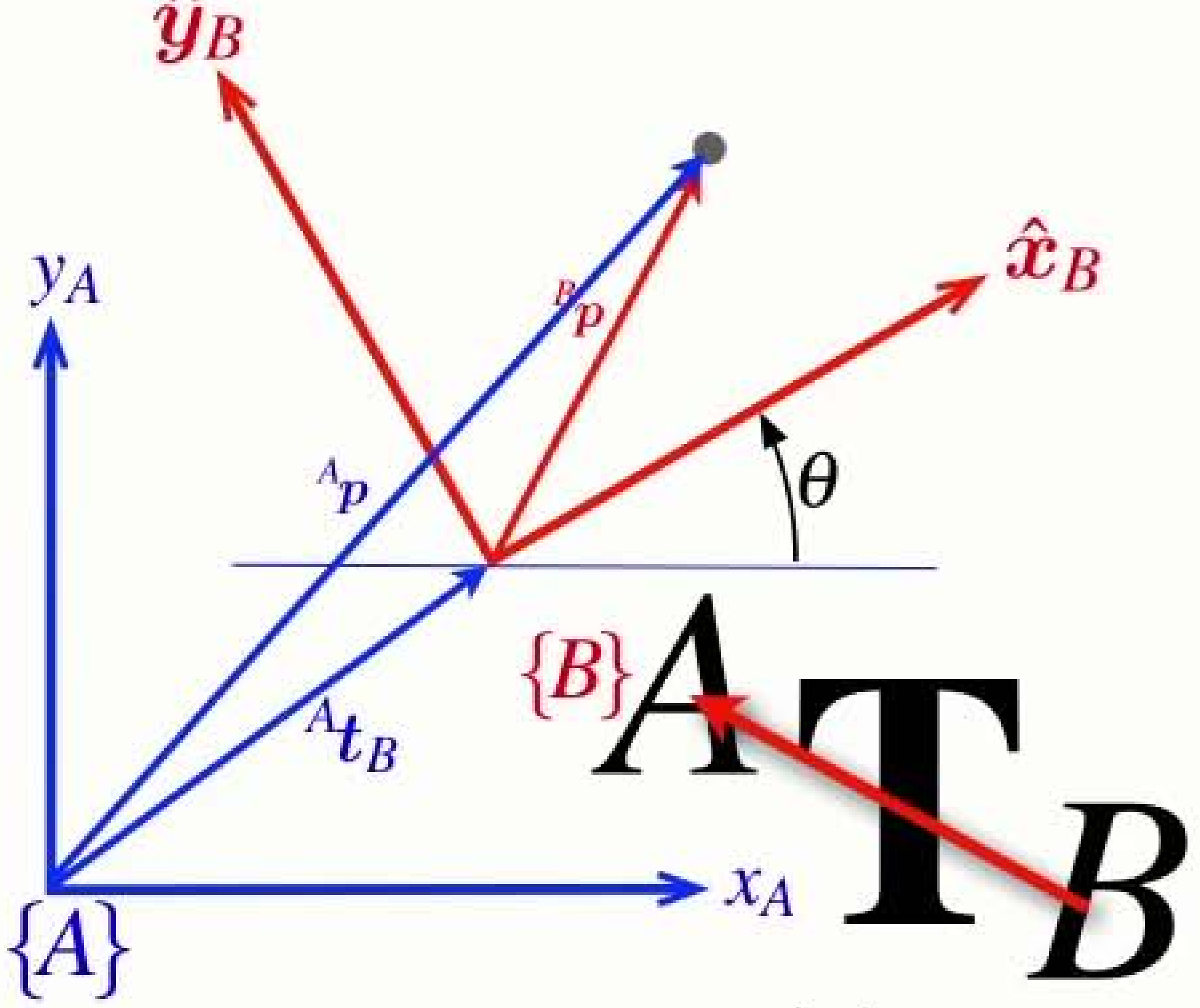
homogeneous vector

homogeneous transform

homogeneous vector

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$

$$^{A}\tilde{\boldsymbol{p}} = {}^{A}\mathbf{T}_{B}\,{}^{B}\tilde{\boldsymbol{p}}$$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

Properties of the homogeneous transform

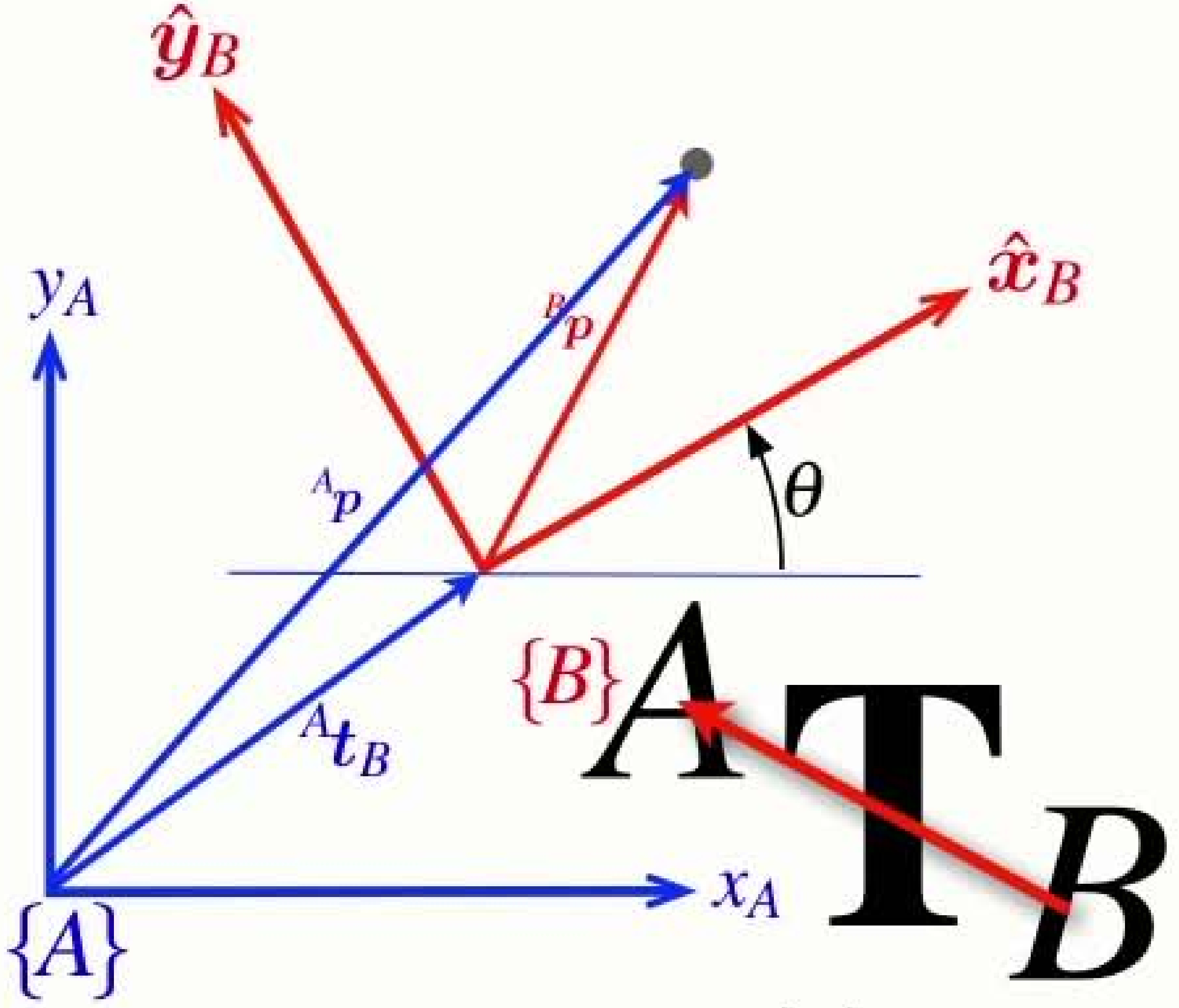
homogeneous vector

homogeneous transform

homogeneous vector

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$

$$^{A}\tilde{\boldsymbol{p}} = {}^{A}\mathbf{T}_{B}\,{}^{B}\tilde{\boldsymbol{p}} \qquad {}^{A}\mathbf{T}_{B} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix}$$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

Properties of the homogeneous transform

homogeneous vector

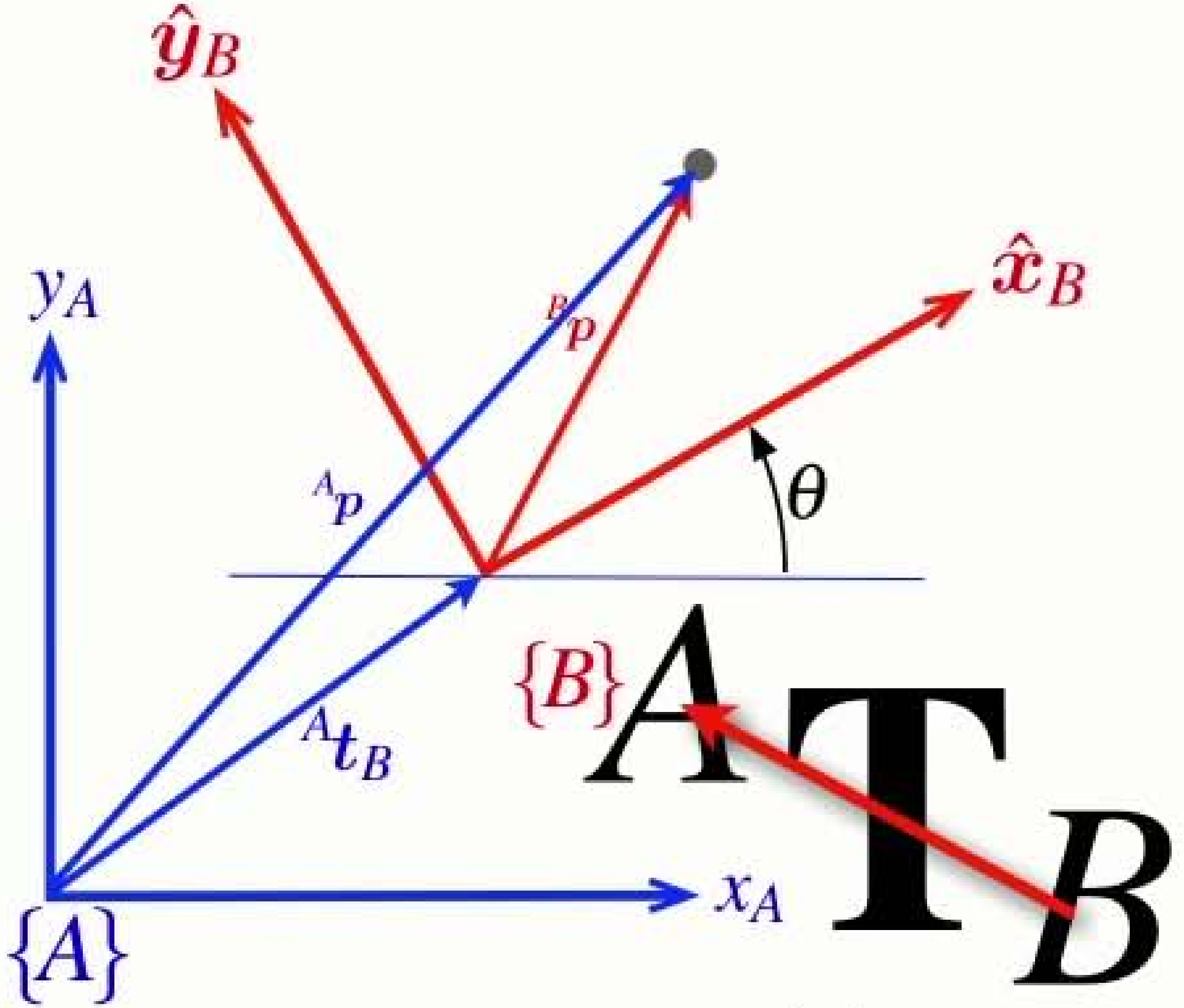
homogeneous transform

homogeneous vector

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix} {^{B}\tilde{\boldsymbol{p}}}$$

$$^{A}\tilde{\boldsymbol{p}} = {^{A}\mathbf{T}_{B}} {^{B}\tilde{\boldsymbol{p}}} \qquad ^{A}\mathbf{T}_{B} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix}$$

- Describes a relative pose as a 3x3 matrix



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

Properties of the **homogeneous transform**

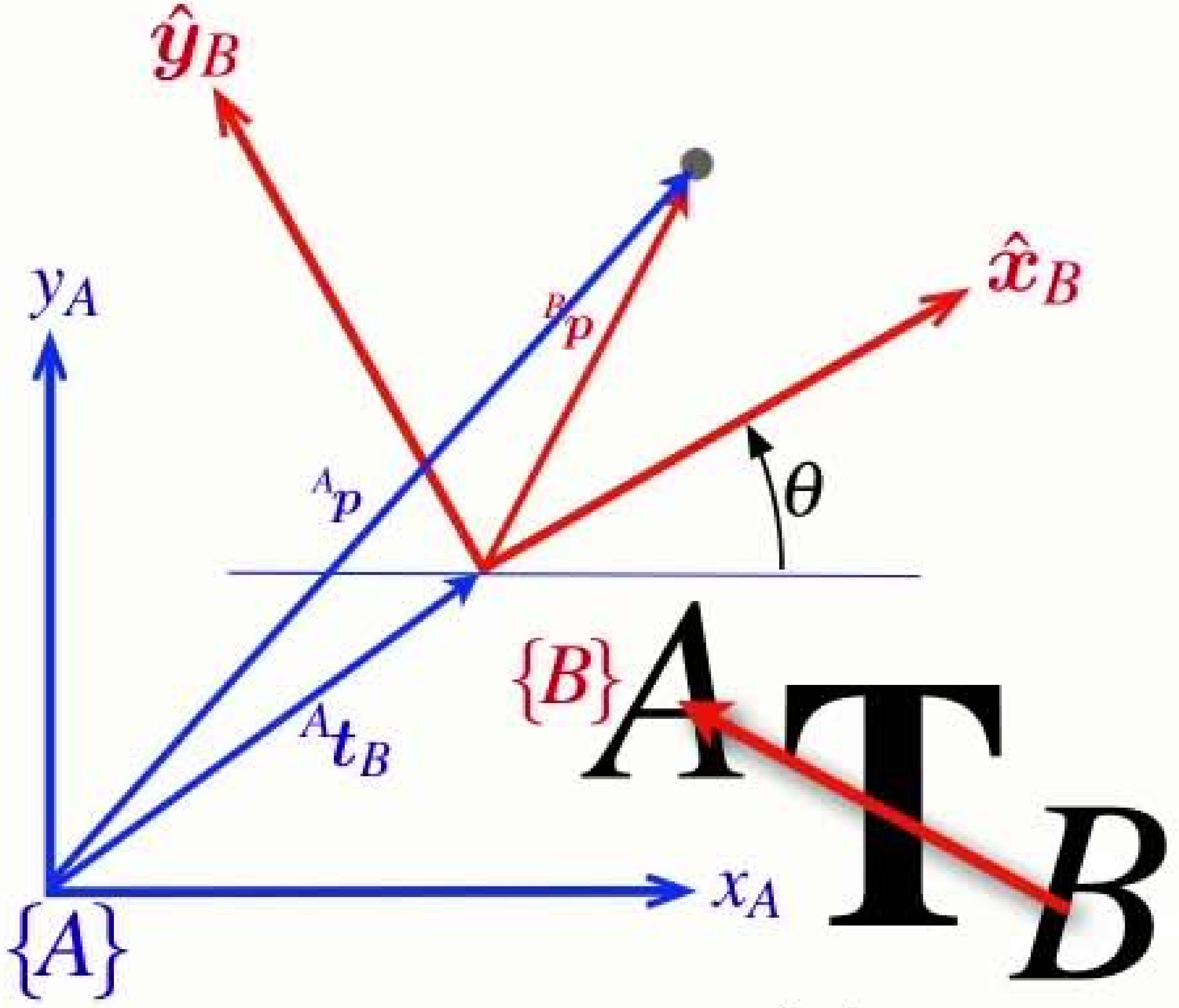
homogeneous vector

homogeneous transform

homogeneous vector

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} ^{A}\mathbf{R}_B & ^{A}\boldsymbol{t}_B \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$

$$^{A}\tilde{\boldsymbol{p}} = {}^{A}\mathbf{T}_B \, {}^{B}\tilde{\boldsymbol{p}} \qquad {}^{A}\mathbf{T}_B = \begin{pmatrix} ^{A}\mathbf{R}_B & ^{A}\boldsymbol{t}_B \\ 0,0 & 1 \end{pmatrix}$$

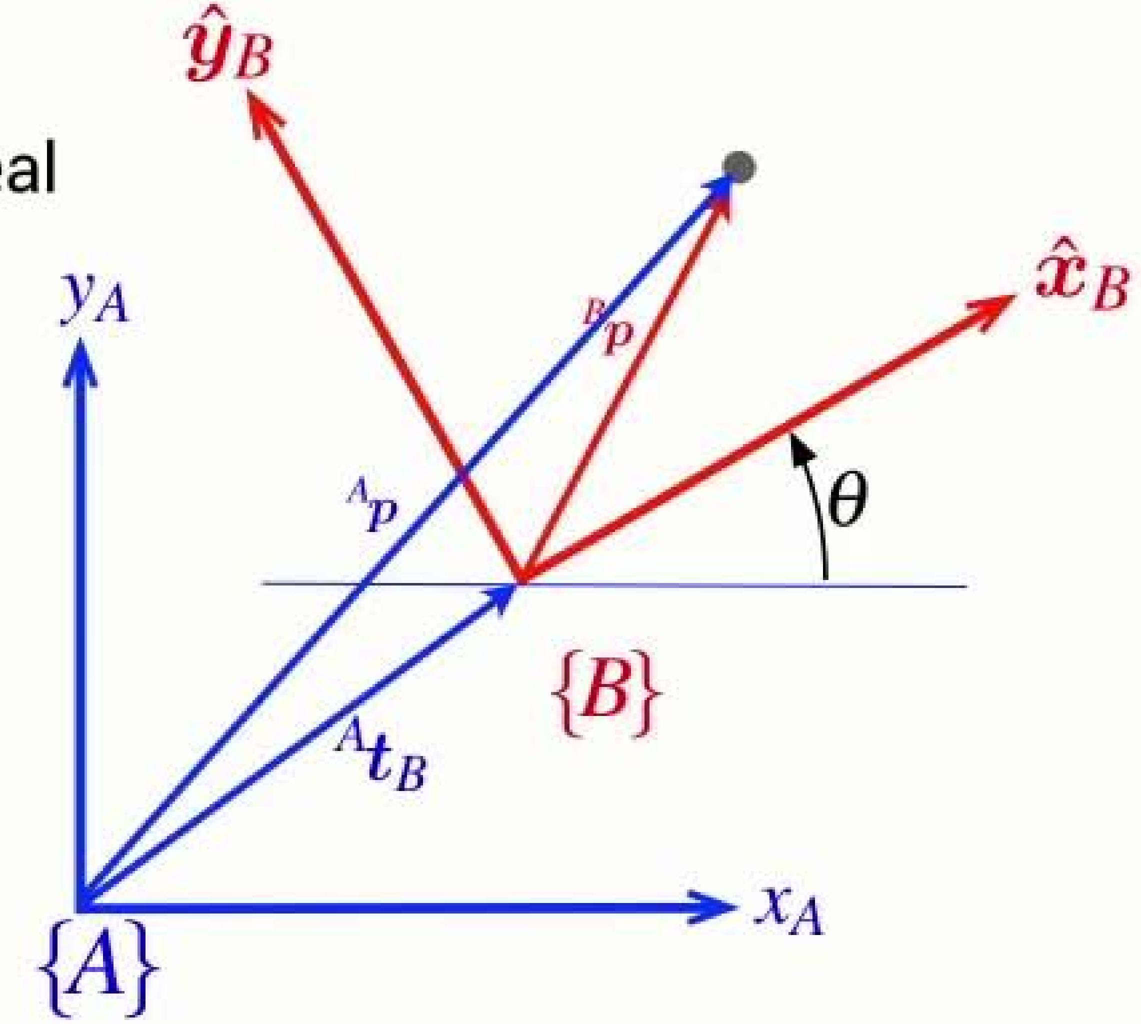


- Describes a relative pose as a 3x3 matrix
- Homogeneous transforms belong to the special Euclidean group of dimension 2 $\mathbf{T} \in SE(2)$

$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

From abstract to real

- Finally we can make these abstract symbols and concepts real
  - Pose is a matrix ${}^{A}\xi_{B} \sim {}^{A}\mathbf{T}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$

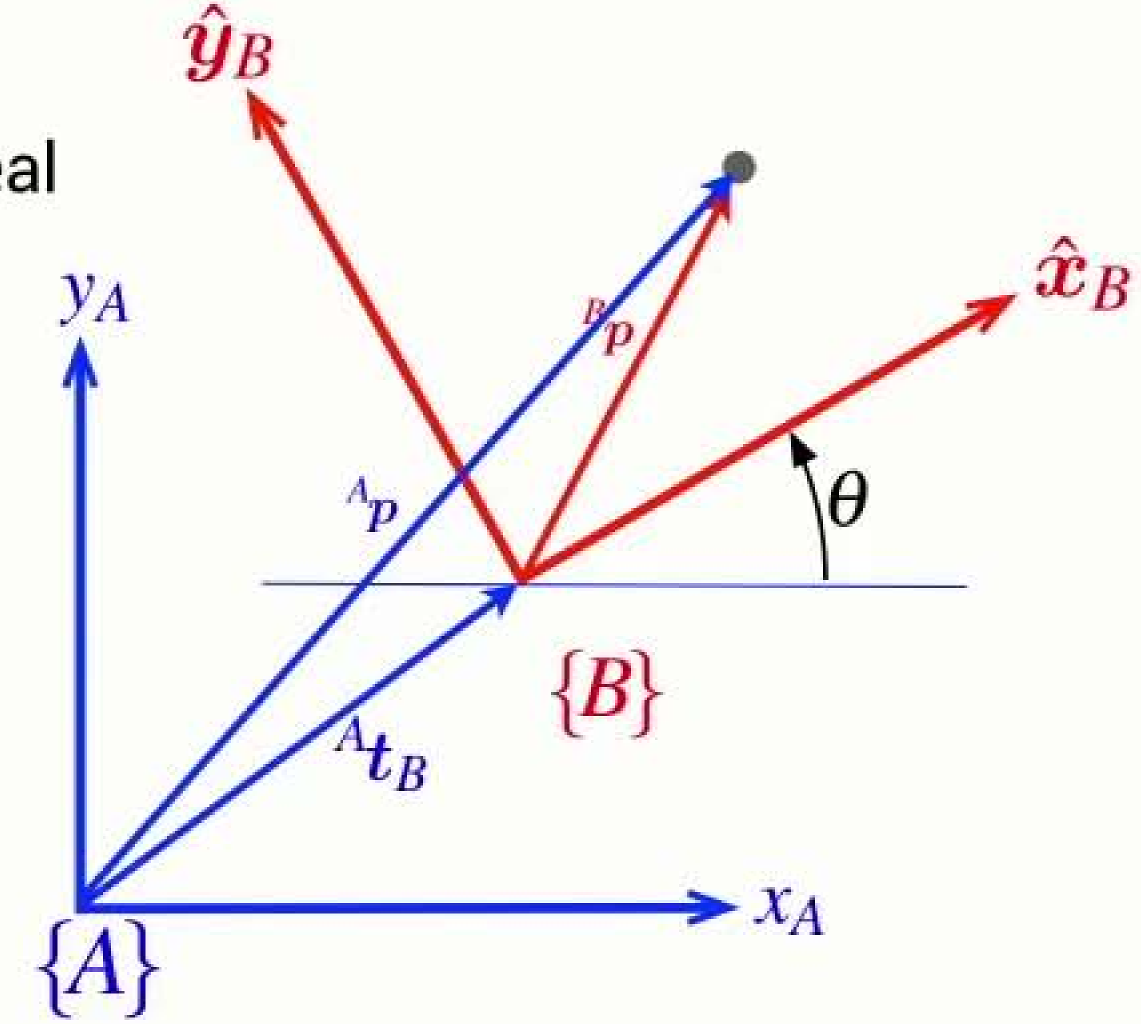


$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

From abstract to real

- Finally we can make these abstract symbols and concepts real
  - Pose is a matrix ${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$
  - Compounding (composition) is a matrix-matrix product

$${}^X\xi_Y \oplus {}^Y\xi_Z \mapsto {}^X\mathbf{T}_Y \, {}^Y\mathbf{T}_Z$$



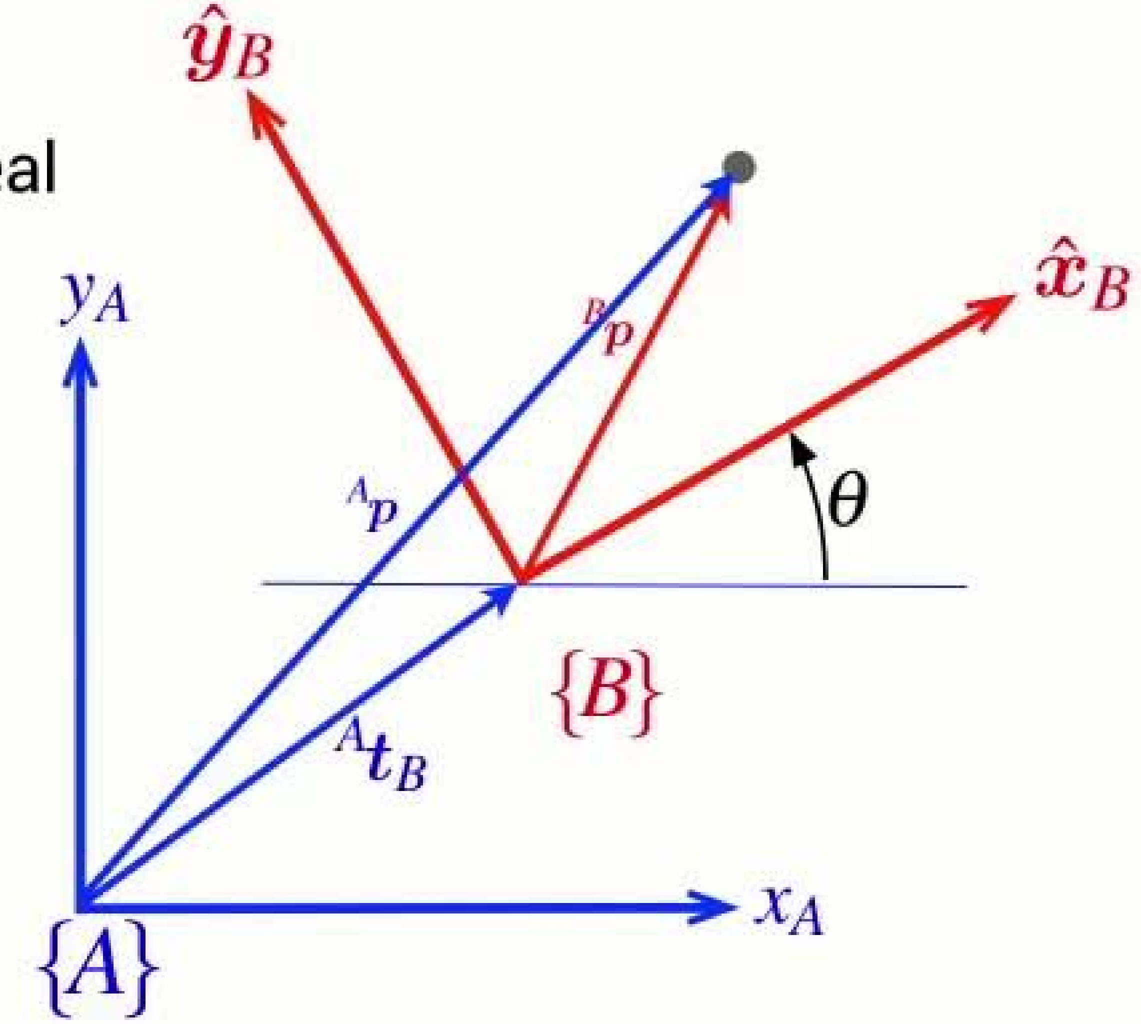
$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

From abstract to real

- Finally we can make these abstract symbols and concepts real
  - Pose is a matrix ${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$
  - Compounding (composition) is a matrix-matrix product

    ${}^X\xi_Y \oplus {}^Y\xi_Z \mapsto {}^X\mathbf{T}_Y\,{}^Y\mathbf{T}_Z$
  - Negation is a matrix inverse

    $\ominus\xi \mapsto \mathbf{T}^{-1}$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

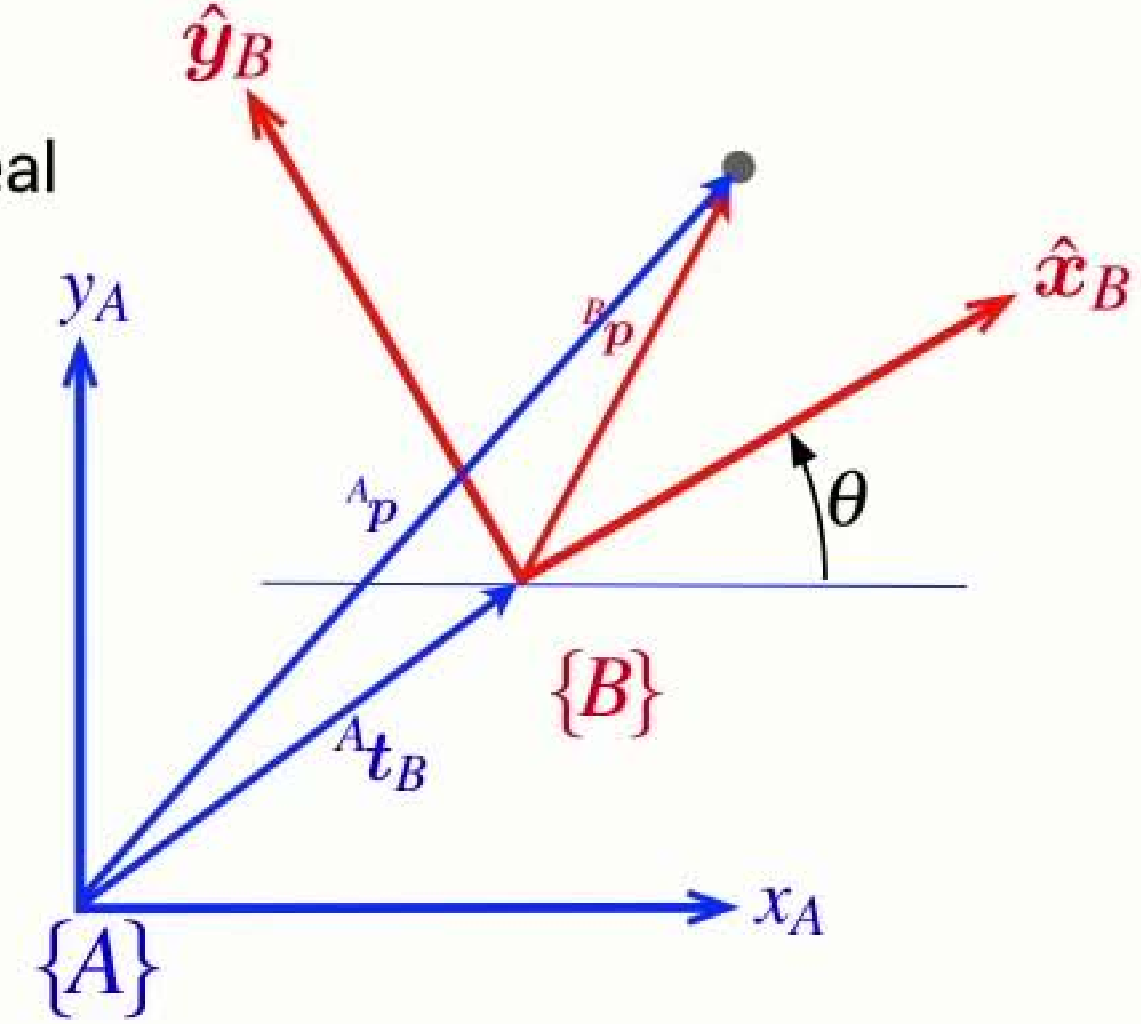
From abstract to real

- Finally we can make these abstract symbols and concepts real
  - Pose is a matrix ${}^{A}\xi_{B} \sim {}^{A}\mathbf{T}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$
  - Compounding (composition) is a matrix-matrix product

    $${}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} \mapsto {}^{X}\mathbf{T}_{Y}\,{}^{Y}\mathbf{T}_{Z}$$
  - Negation is a matrix inverse

    $$\ominus\xi \mapsto \mathbf{T}^{-1}$$
  - Vector transformation is a matrix-vector product

    $$\xi \cdot \boldsymbol{p} \mapsto \mathbf{T}\tilde{\boldsymbol{p}}$$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

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Current Folder

Command Window

```
>> transl2(1,2)

ans =

     1     0     1
     0     1     2
     0     0     1

>>
```

Workspace

Name ▲	Value	Class
ans	[1,0,1;0,1,2;0,0,1]	double

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Command Window

```
>> transl2(1,2)

ans =

     1     0     1
     0     1     2
     0     0     1

>> rot2(30, 'deg')

ans =

    0.8660   -0.5000
    0.5000    0.8660

>>
```

Workspace

Name ▲	Value	Class
ans	[0.8660,-0.5000;0.5000,...	double

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Command Window

```
>> transl2(1,2)

ans =

     1     0     1
     0     1     2
     0     0     1

>> rot2(30, 'deg')

ans =

    0.8660   -0.5000
    0.5000    0.8660

>> trot2(30, 'deg')

ans =

    0.8660   -0.5000         0
    0.5000    0.8660         0
         0         0    1.0000

>>
```

Current Folder

Workspace

Name ▲	Value	Class
ans	[0.8660,-0.5000,0;0.500...	double

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Current Folder

Command Window

```
>> transl2(1,2)

ans =

     1     0     1
     0     1     2
     0     0     1

>> rot2(30, 'deg')

ans =

    0.8660   -0.5000
    0.5000    0.8660

>> trot2(30, 'deg')

ans =

    0.8660   -0.5000         0
    0.5000    0.8660         0
         0         0    1.0000

>> transl2(1,2) * trot2(30, 'deg')

ans =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>>
```

Workspace

Name ▲	Value	Class
ans	[0.8660,-0.5000,1;0.500...	double

Command Window

```
>> transl2(1,2)

ans =

     1     0     1
     0     1     2
     0     0     1

>> rot2(30, 'deg')

ans =

    0.8660   -0.5000
    0.5000    0.8660

>> trot2(30, 'deg')

ans =

    0.8660   -0.5000         0
    0.5000    0.8660         0
         0         0    1.0000

>> transl2(1,2) * trot2(30, 'deg')

ans =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> se2(1, 2, 30, 'deg')

ans =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

fx >>
```

Current Folder

Workspace

Name	Value	Class
ans	[0.8660,-0.5000,1;0.500...	double

Command Window

```
    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> se2(1, 2, 30, 'deg')

ans =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> axis([0 5 0 5])
>> axis square
>>
```

Workspace

Name ▲	Value	Class
ans	[0.8660,-0.5000,1;0.500...	double

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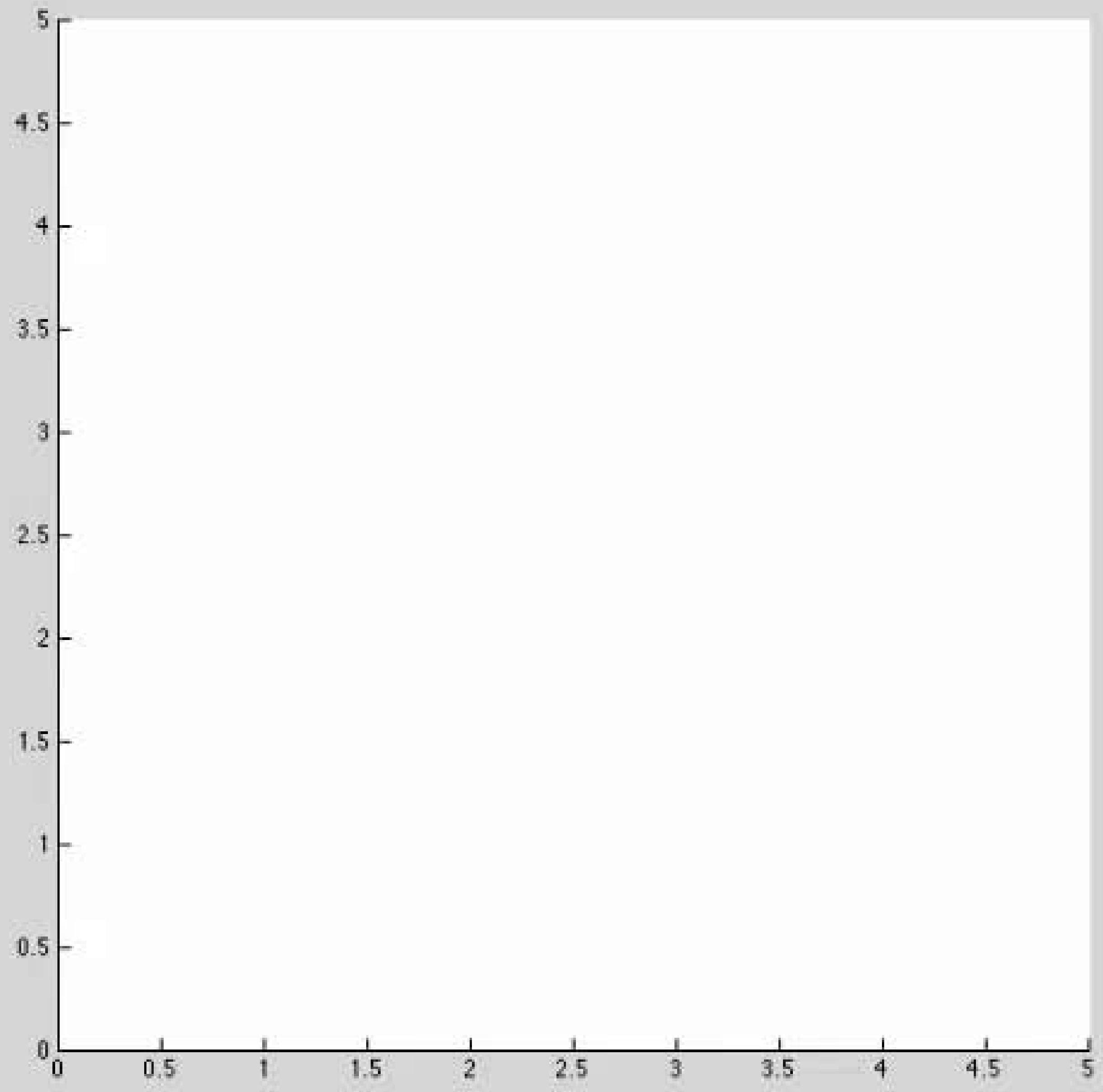
Current Folder

Figures

Figure 1



Command Window

```
    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> axis([0 5 0 5])
>> axis square
>> hold on
>> T1 = se2(1, 2, 30, 'deg')

T1 =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> 
```

Workspace

Name ▲	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

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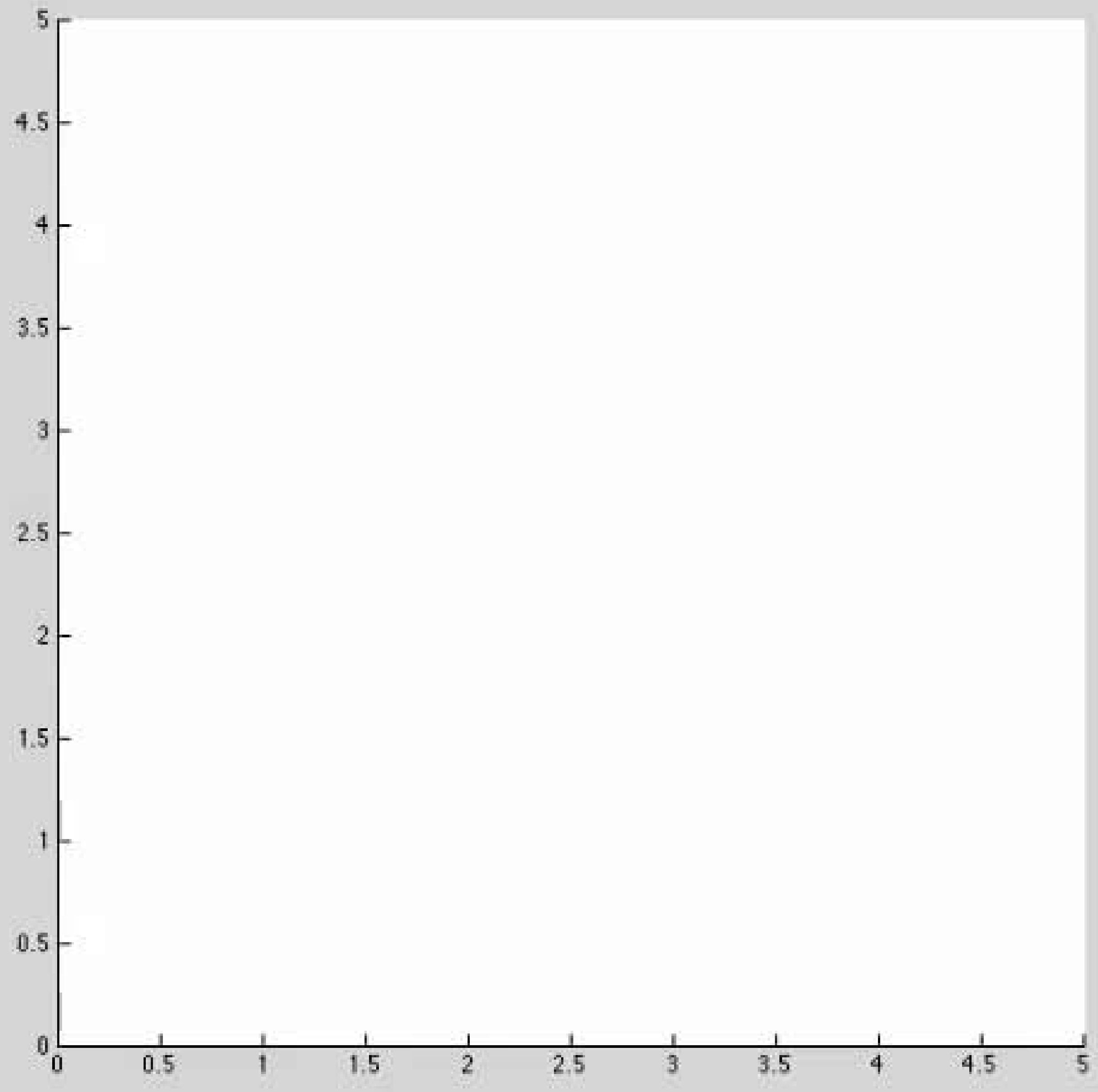
Current Folder

Figures

Figure 1



Command Window

```
    0.5000    0.8660    2.0000
         0         0    1.0000

>> axis([0 5 0 5])
>> axis square
>> hold on
>> T1 = se2(1, 2, 30, 'deg')

T1 =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> trplot2(T1, 'frame', '1', 'color', 'b')
>> 
```

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

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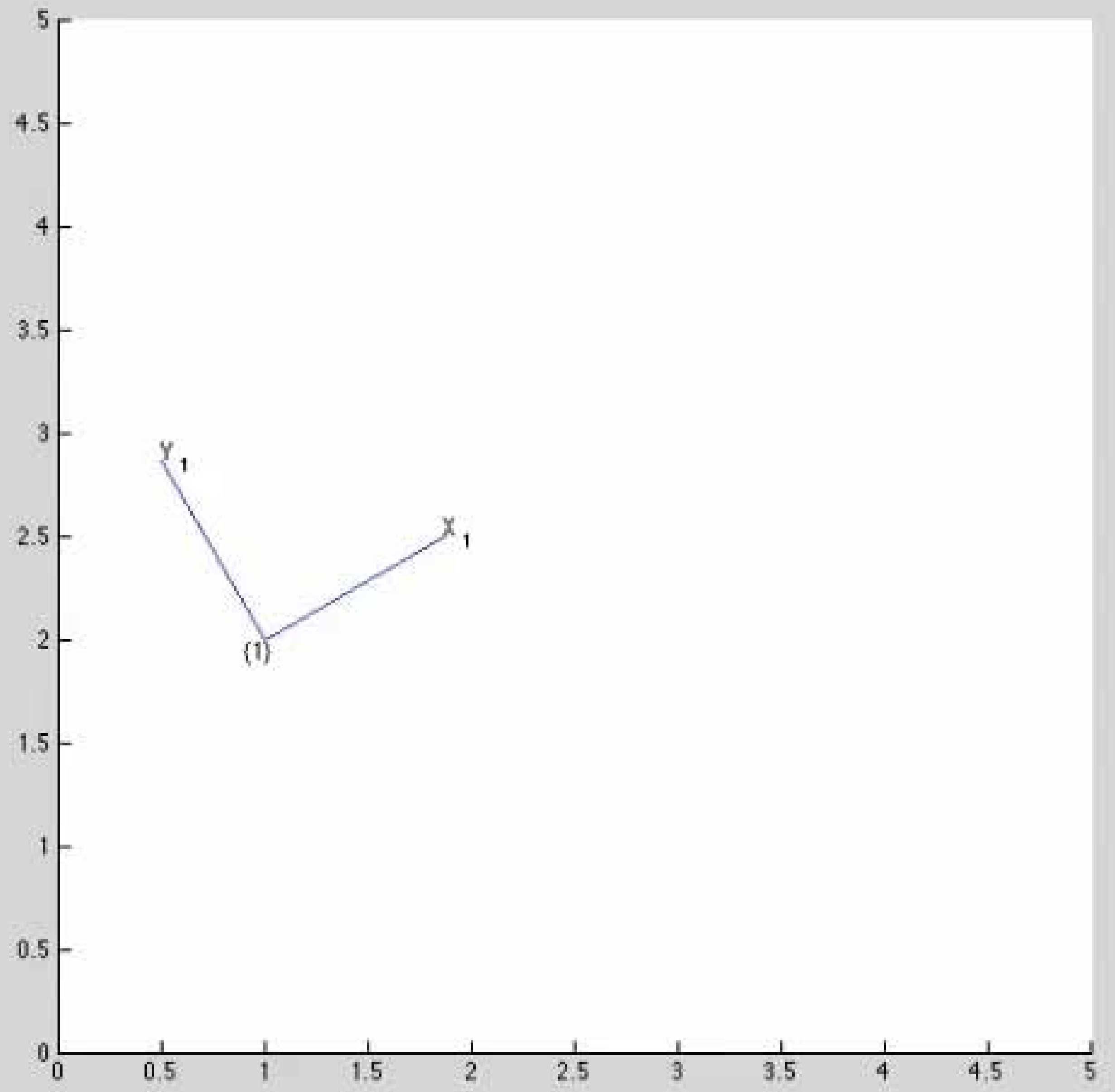
Figures

Figure 1



Command Window

```
T1 =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> trplot2(T1, 'frame', '1', 'color', 'b')
>> T2 = se2(2, 1, 0)

T2 =

     1     0     2
     0     1     1
     0     0     1

>>
```

Workspace

Name ▲	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
ans	[0.8660,-0.5000,1;0.500...	double

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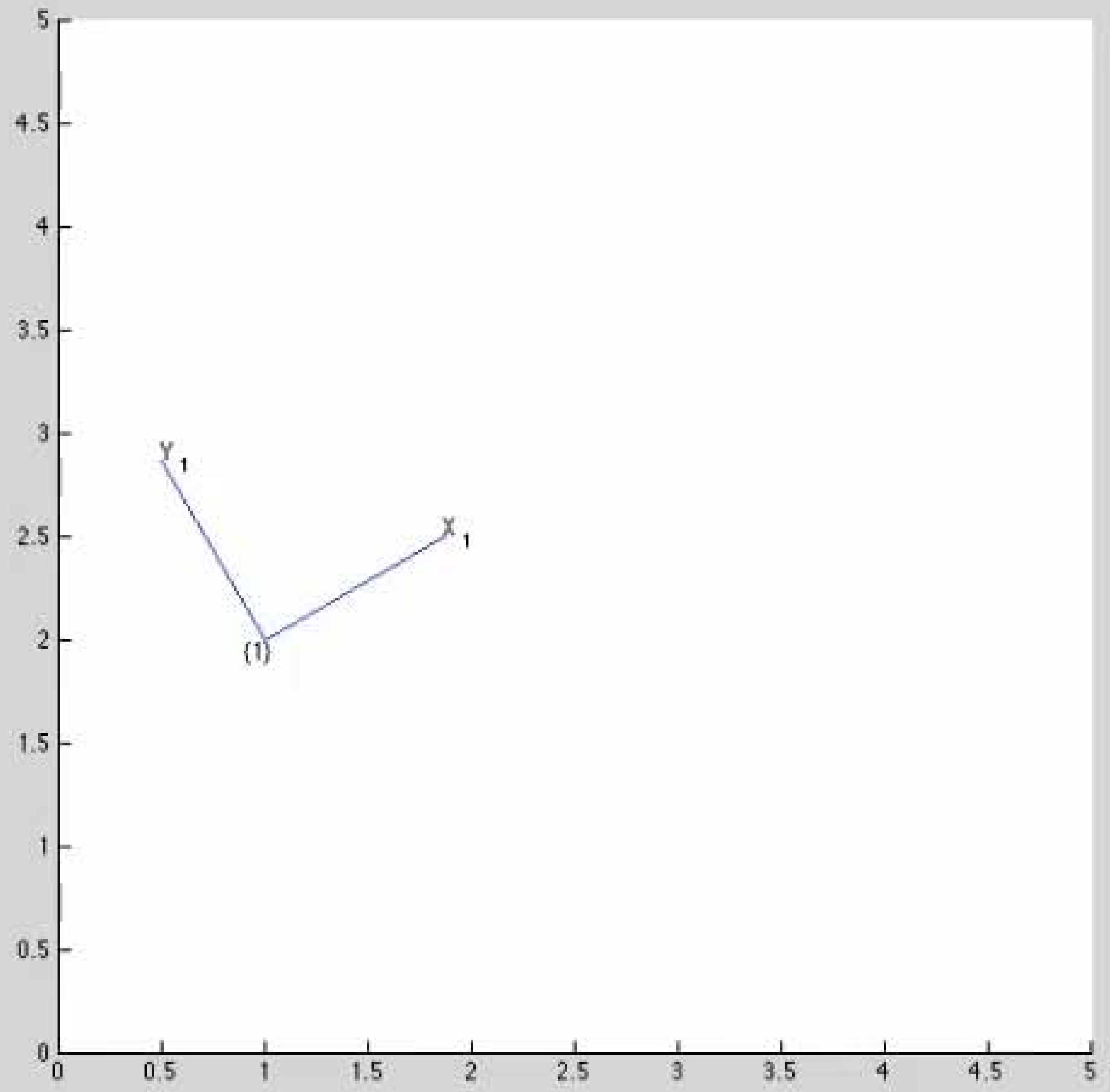
Current Folder

Figures

Figure 1



Command Window

```
    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000

>> trplot2(T1, 'frame', '1', 'color', 'b')
>> T2 = se2(2, 1, 0)

T2 =

     1     0     2
     0     1     1
     0     0     1

>> trplot2(T2, 'frame', '2', 'color', 'r')
>> 
```

Workspace

Name ▲	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
ans	[0.8660,-0.5000,1;0.500...	double

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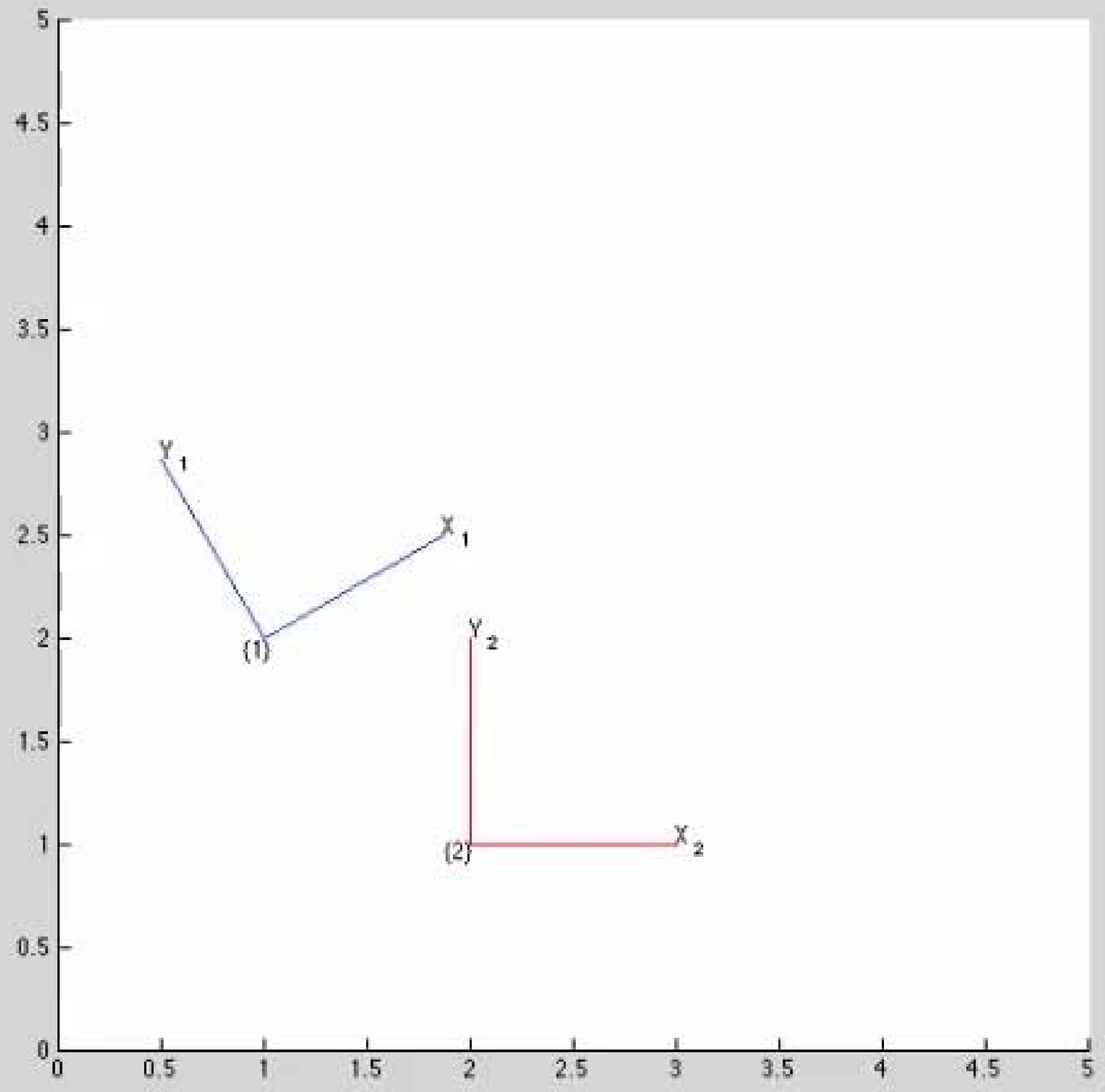
Current Folder

Figures

Figure 1



Command Window

```
T2 =

     1     0     2
     0     1     1
     0     0     1

>> trplot2(T2, 'frame', '2', 'color', 'r')
>> T3 = T1 * T2

T3 =

    0.8660   -0.5000    2.2321
    0.5000    0.8660    3.8660
         0         0    1.0000

>>
```

Workspace

Name ▲	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
ans	[0.8660,-0.5000,1;0.500...	double

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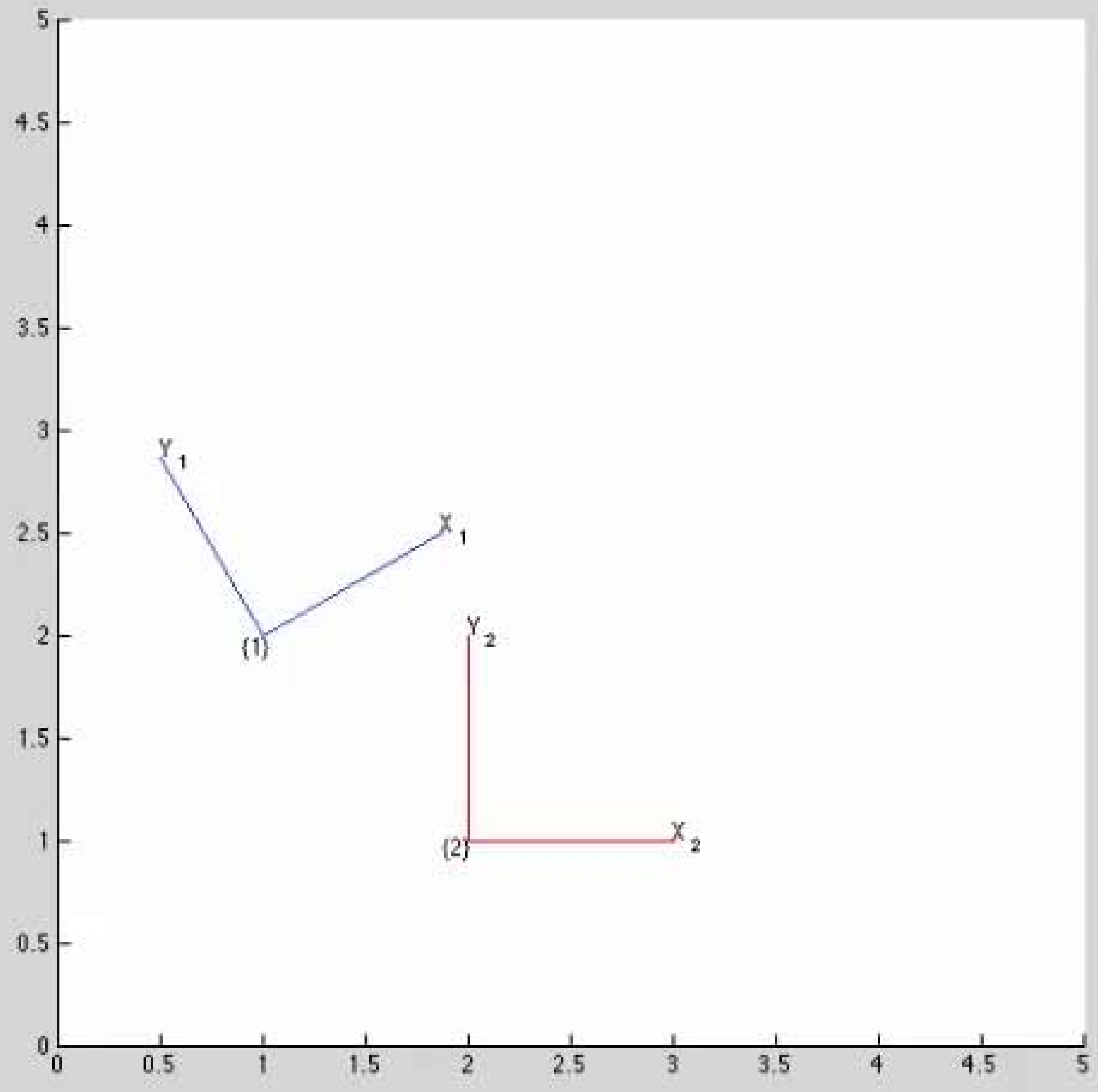
Figures

Figure 1



Command Window

```
     1     0     2
     0     1     1
     0     0     1

>> trplot2(T2, 'frame', '2', 'color', 'r')
>> T3 = T1 * T2

T3 =

    0.8660   -0.5000    2.2321
    0.5000    0.8660    3.8660
         0         0    1.0000

>> trplot2(T3, 'frame', '3', 'color', 'g')
>> 
```

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
ans	[0.8660,-0.5000,1;0.500...	double

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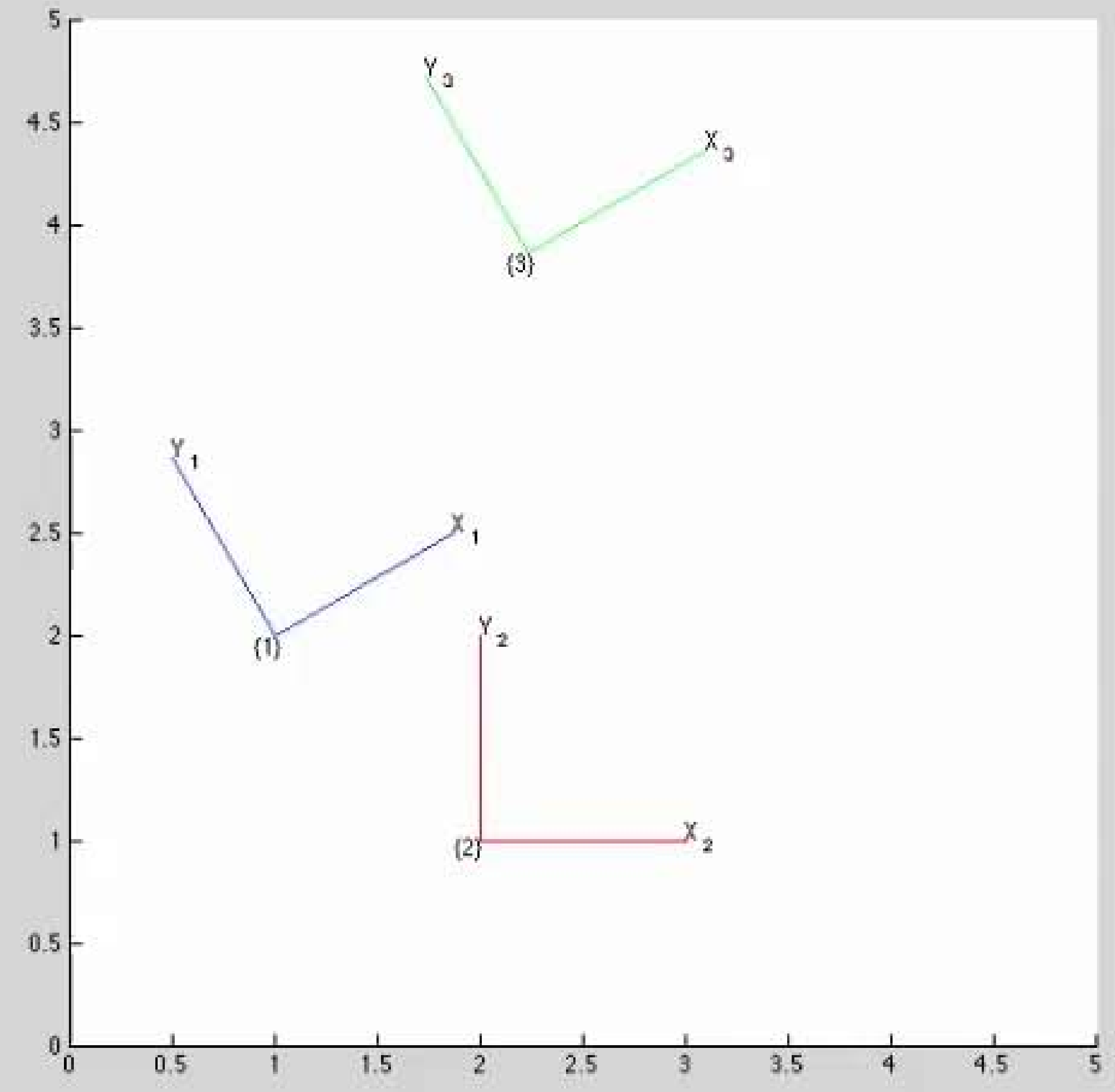
Current Folder

Figures

Figure 1



Command Window

```
    0.8660   -0.5000    2.2321
    0.5000    0.8660    3.8660
         0         0    1.0000

>> trplot2(T3, 'frame', '3', 'color', 'g')
>> T4 = T2 * T1

T4 =

    0.8660   -0.5000    3.0000
    0.5000    0.8660    3.0000
         0         0    1.0000

>> trplot2(T4, 'frame', '4', 'color', 'c')
>>
```

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
T4	[0.8660,-0.5000,3;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

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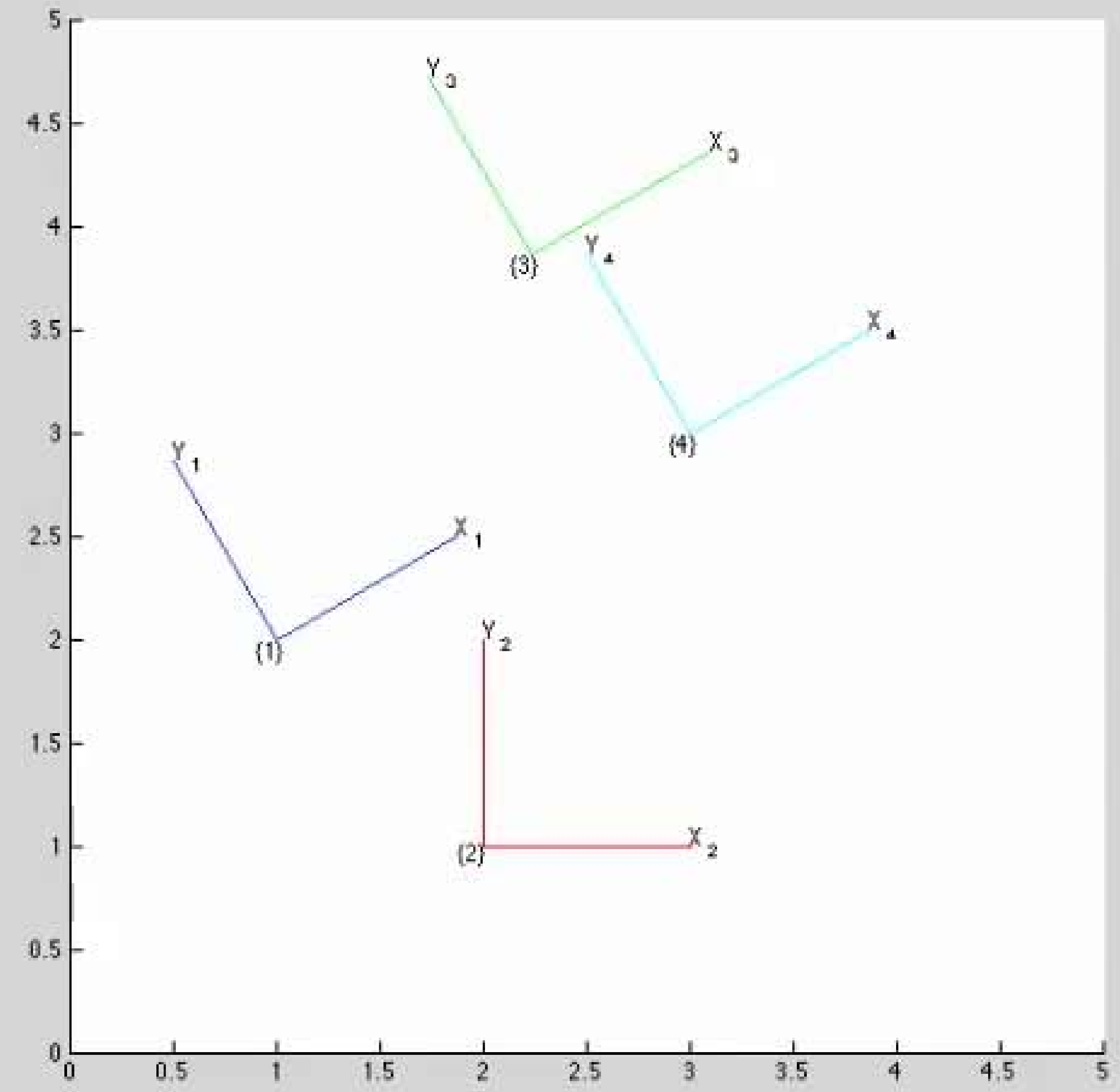
Figures

Figure 1



Command Window

```
T4 =

    0.8660   -0.5000    3.0000
    0.5000    0.8660    3.0000
         0         0    1.0000

>> trplot2(T4, 'frame', '4', 'color', 'c')
>> P = [ 3 2]'

P =

     3
     2

>> plot_point(P, '*')
>>
```

Workspace

Name	Value	Class
P	[3;2]	double
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
T4	[0.8660,-0.5000,3;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

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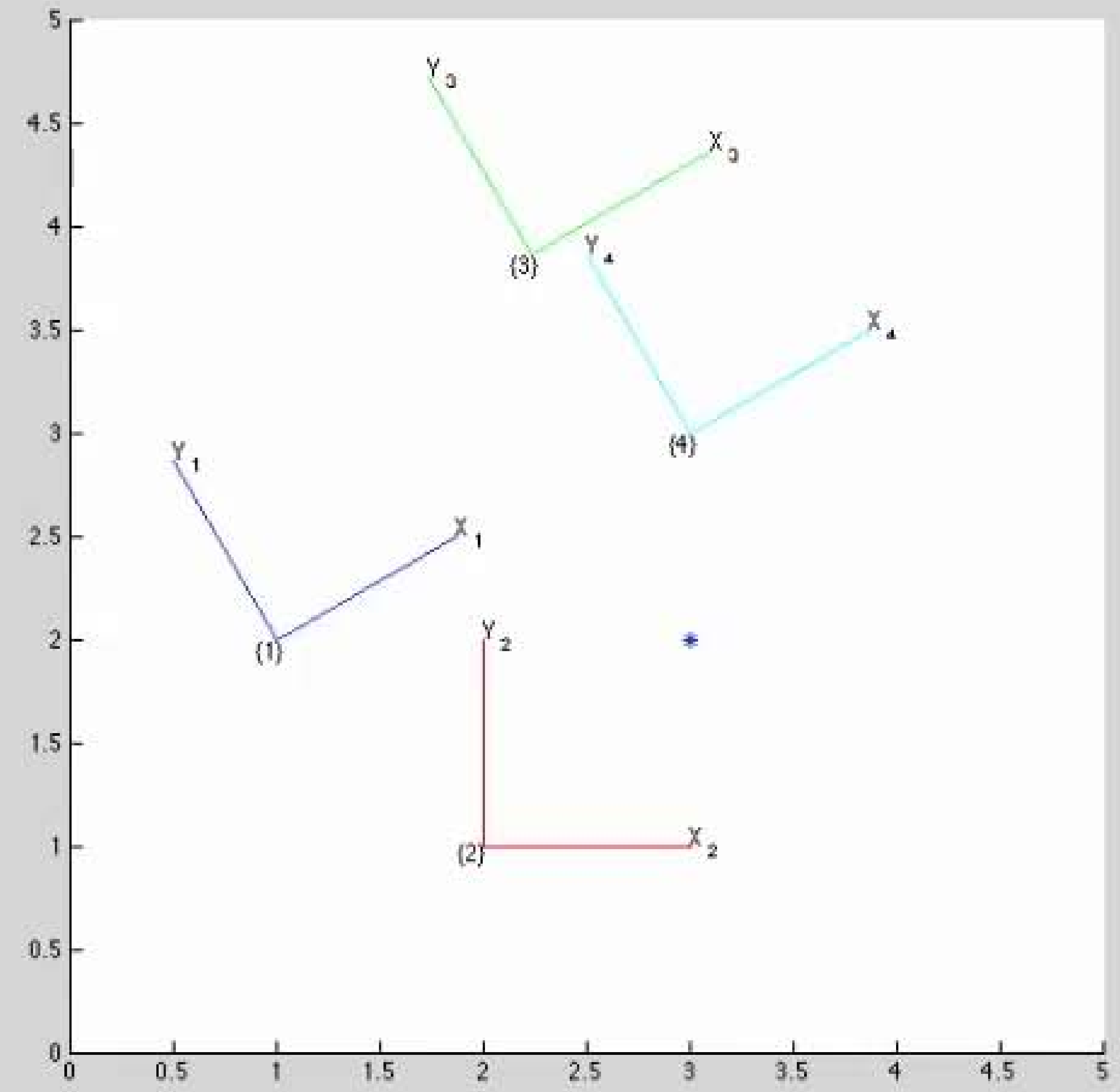
Figures

Figure 1



Command Window

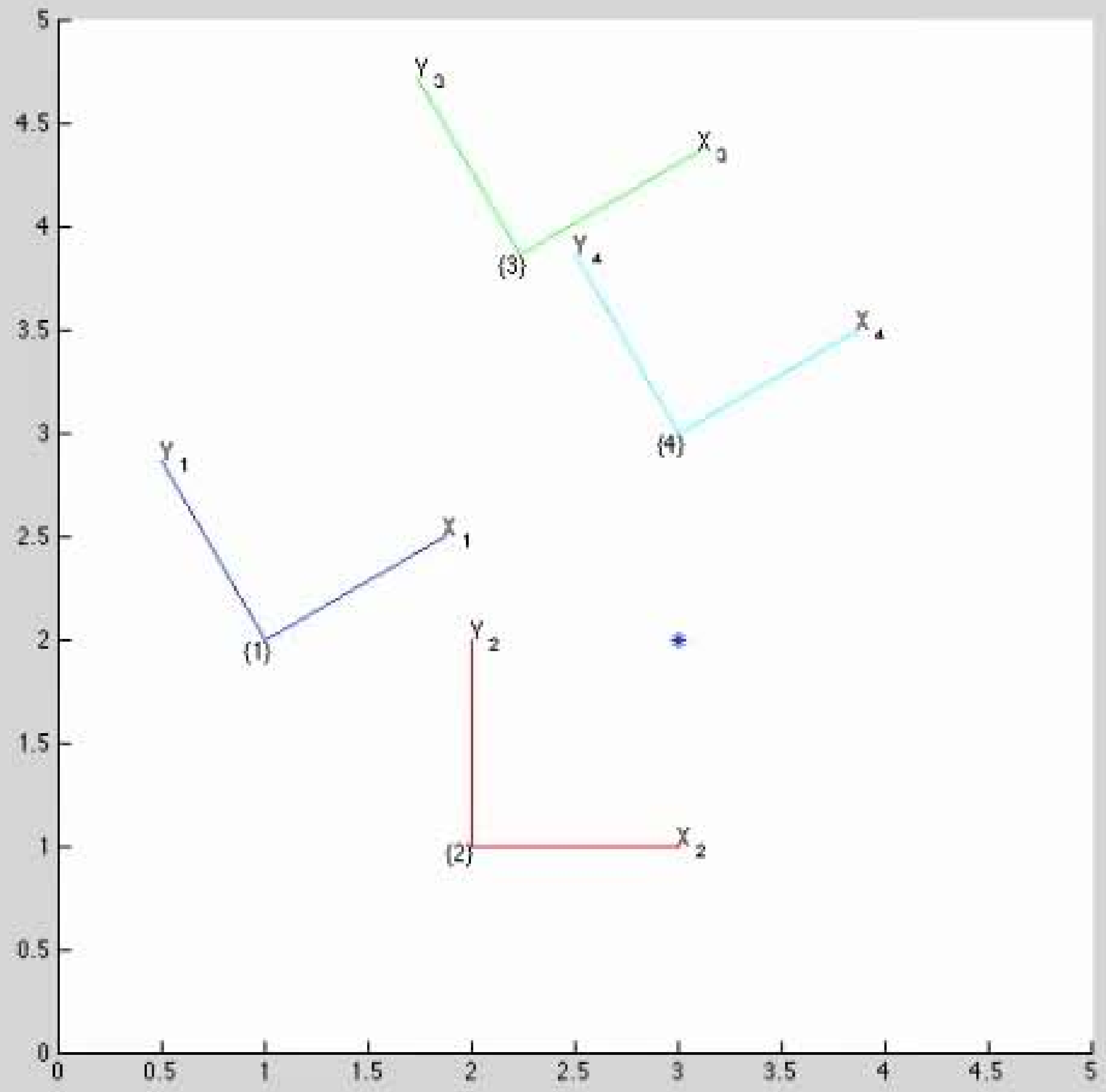
```
P =

     3
     2

>> plot_point(P, '*')
>> P1 = inv(T1) * [P ; 1]

P1 =

    1.7321
   -1.0000
    1.0000

>>
```

Workspace

Name ▲	Value	Class
P	[3;2]	double
P1	[1.7321;-1.0000;1]	double
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
T4	[0.8660,-0.5000,3;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

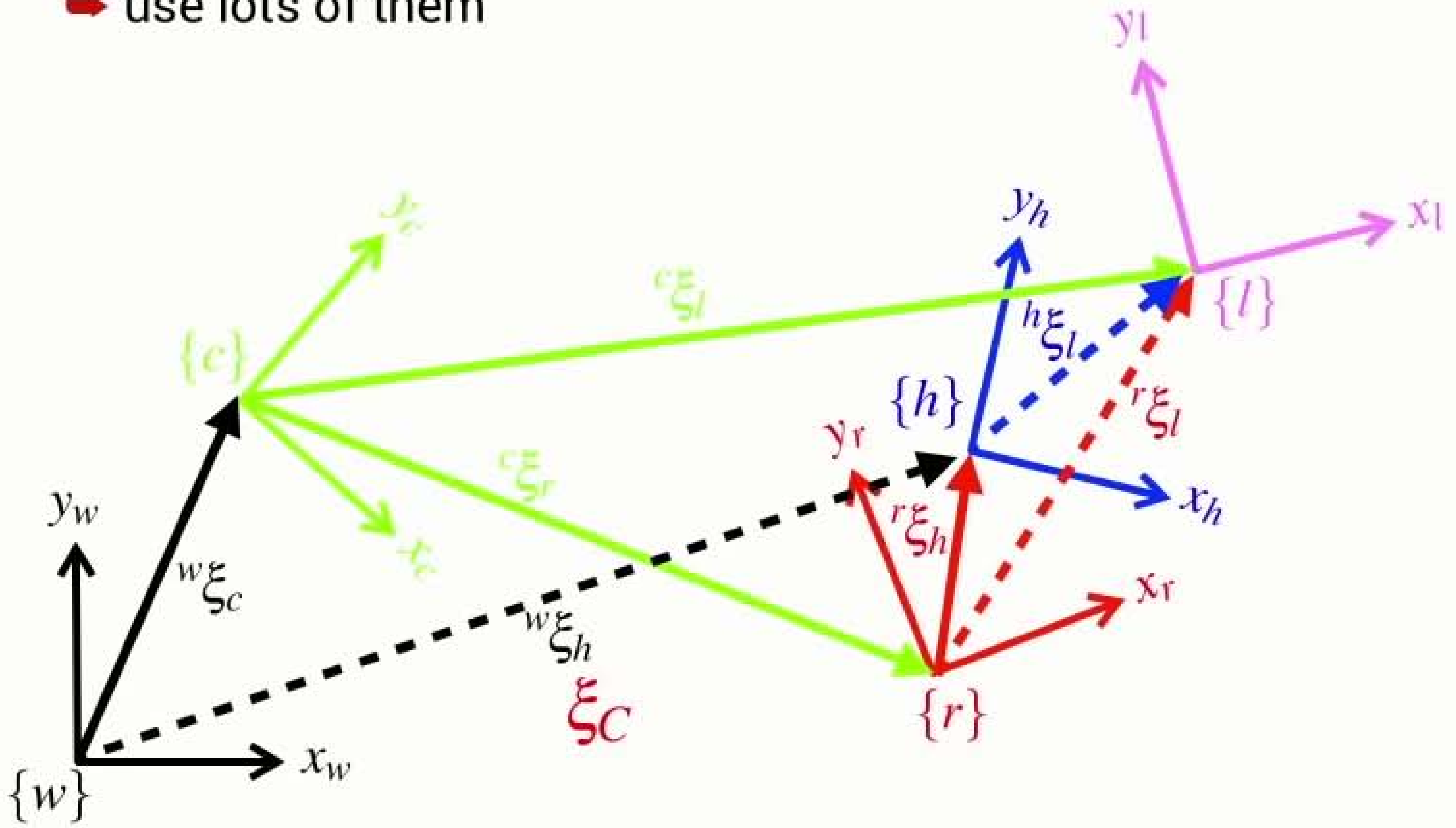
Figure 1



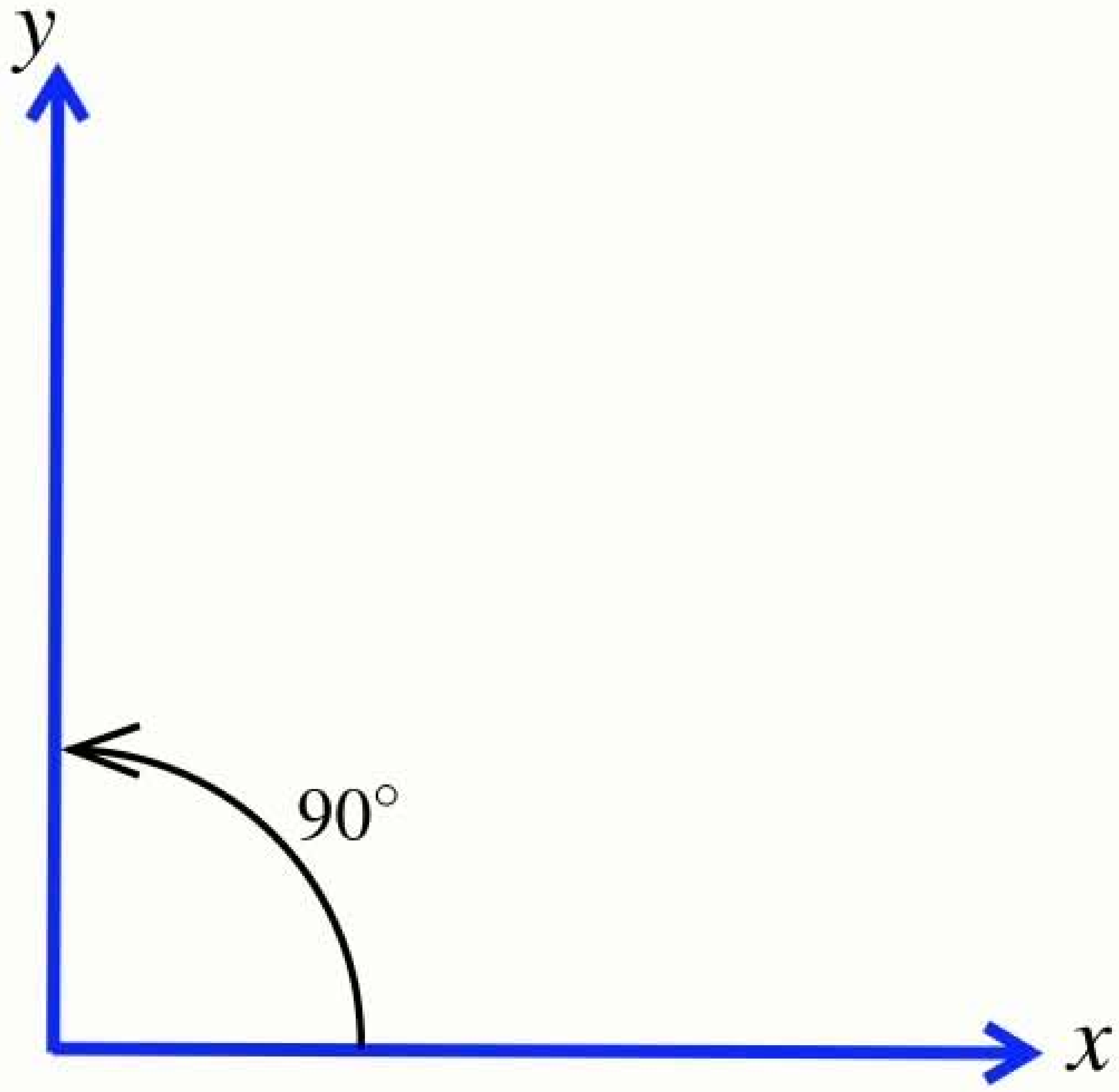
Section 8 Summary

Coordinate **frames**

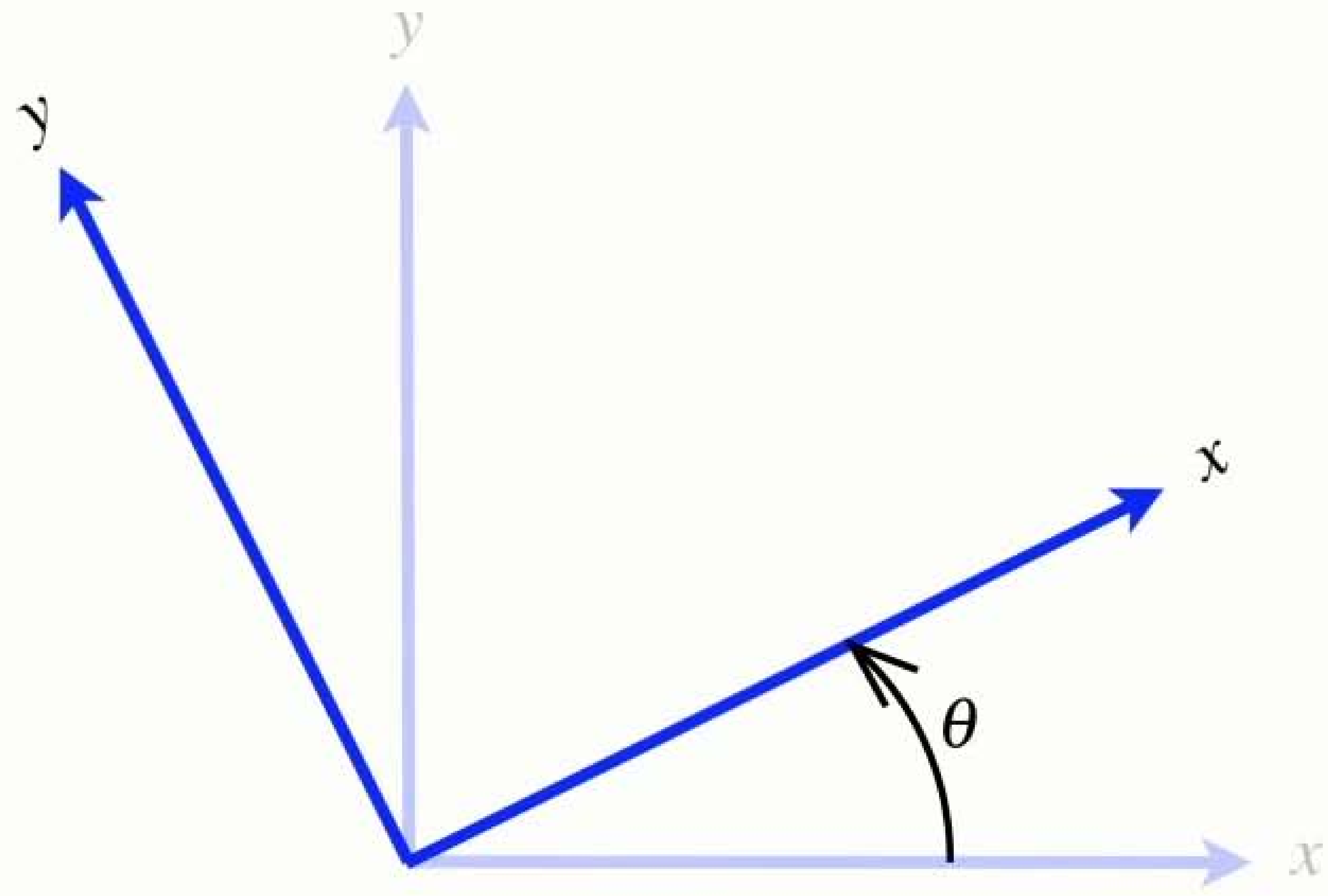
- Coordinate frames are your friend
  - ➥ they describe the pose of objects in the world
  - ➥ use lots of them



Right-handed coordinate frame



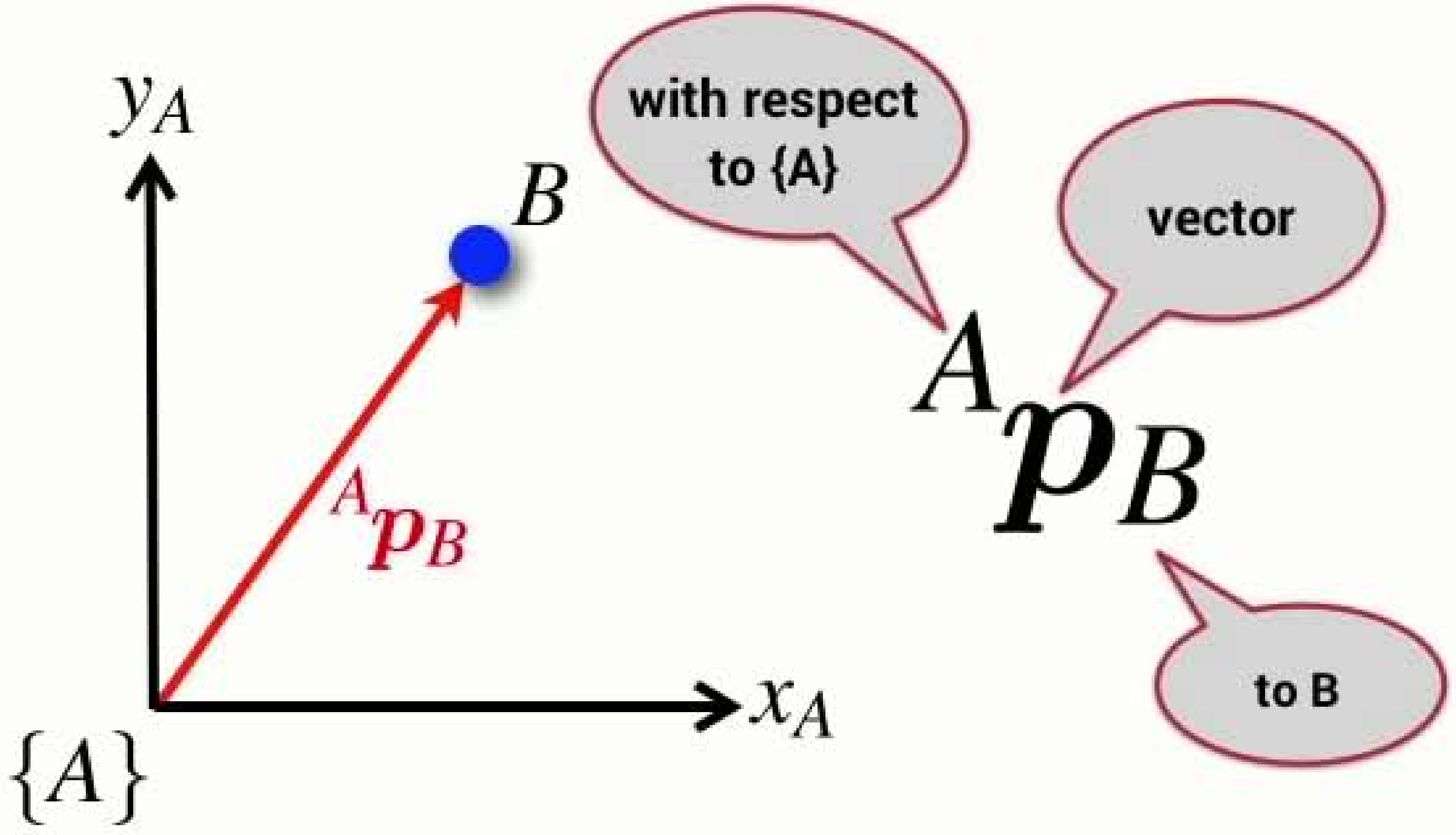
Coordinate frame rotation convention



- Angles increase positively in the anti-clockwise direction

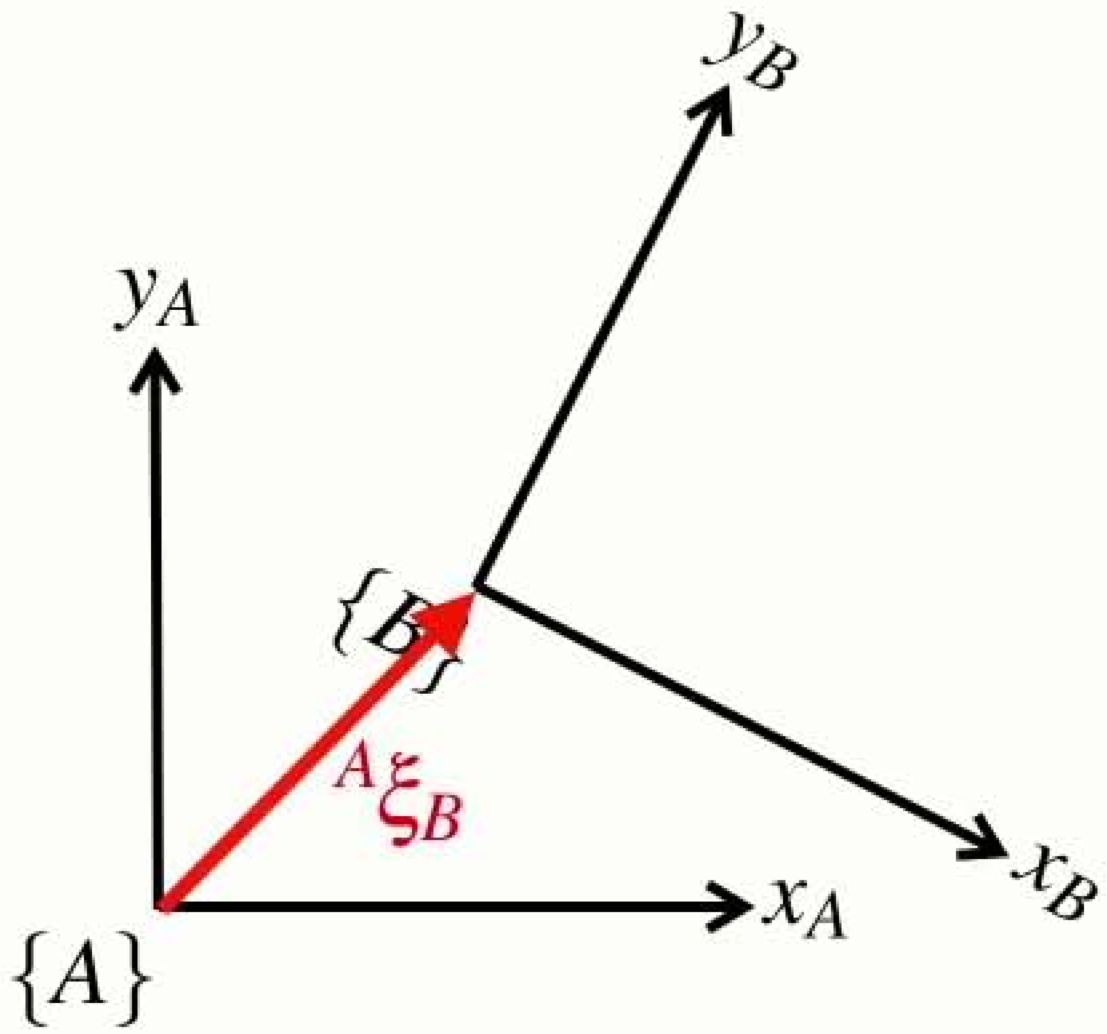
Points & vectors

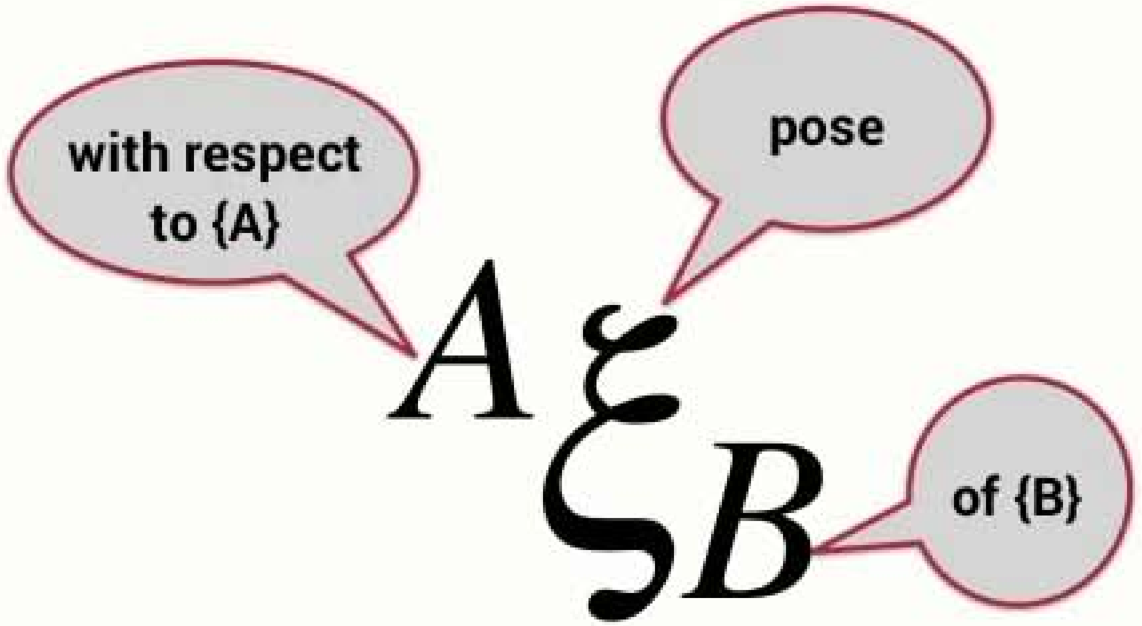
- A point is described by a vector with respect to a coordinate frame



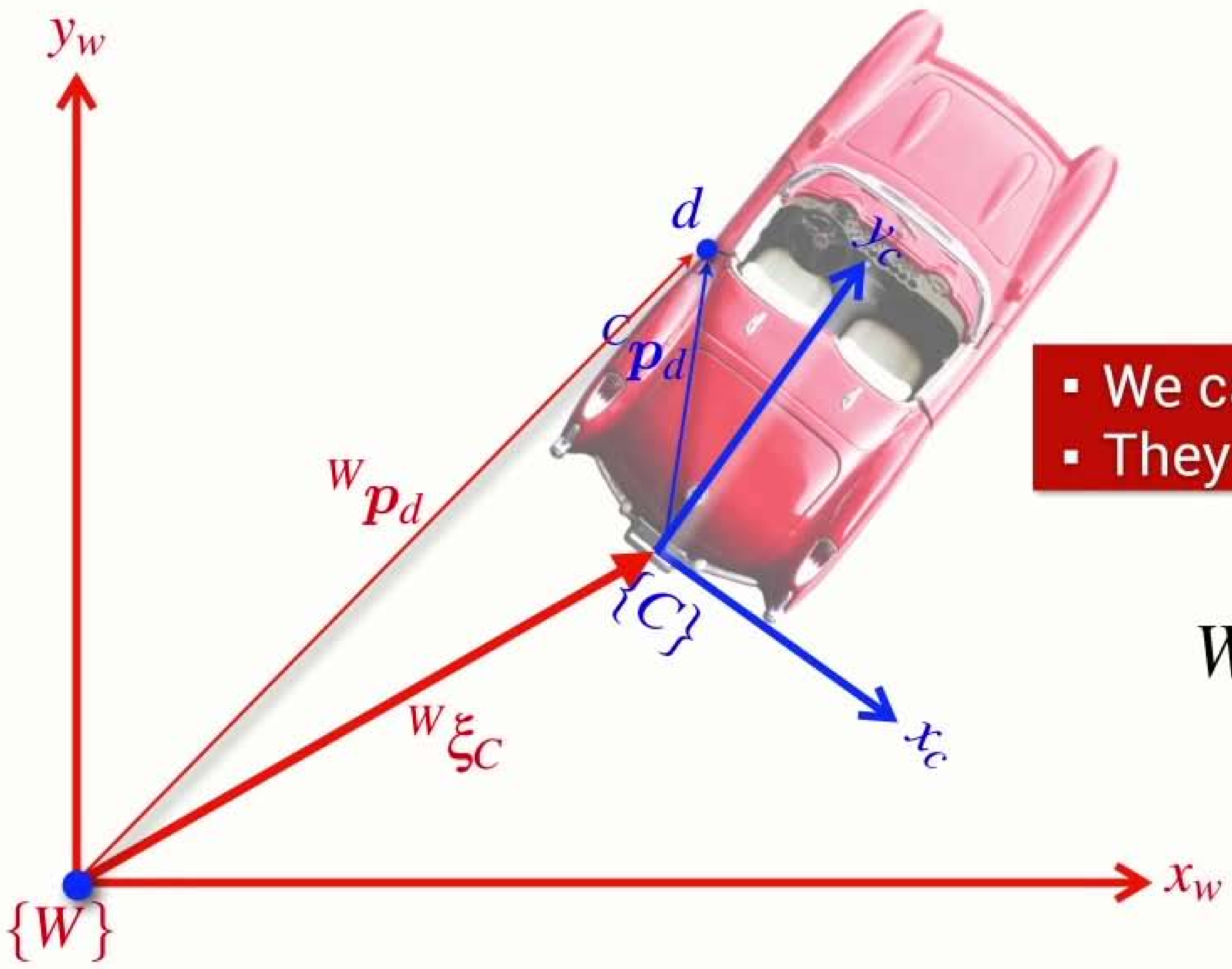
Pose

- The pose of one coordinate frame with respect to another is a relative pose
  - it has three parameters (x, y, θ)
  - we can consider it as a simple motion comprising translation and rotation





pronounced ksi

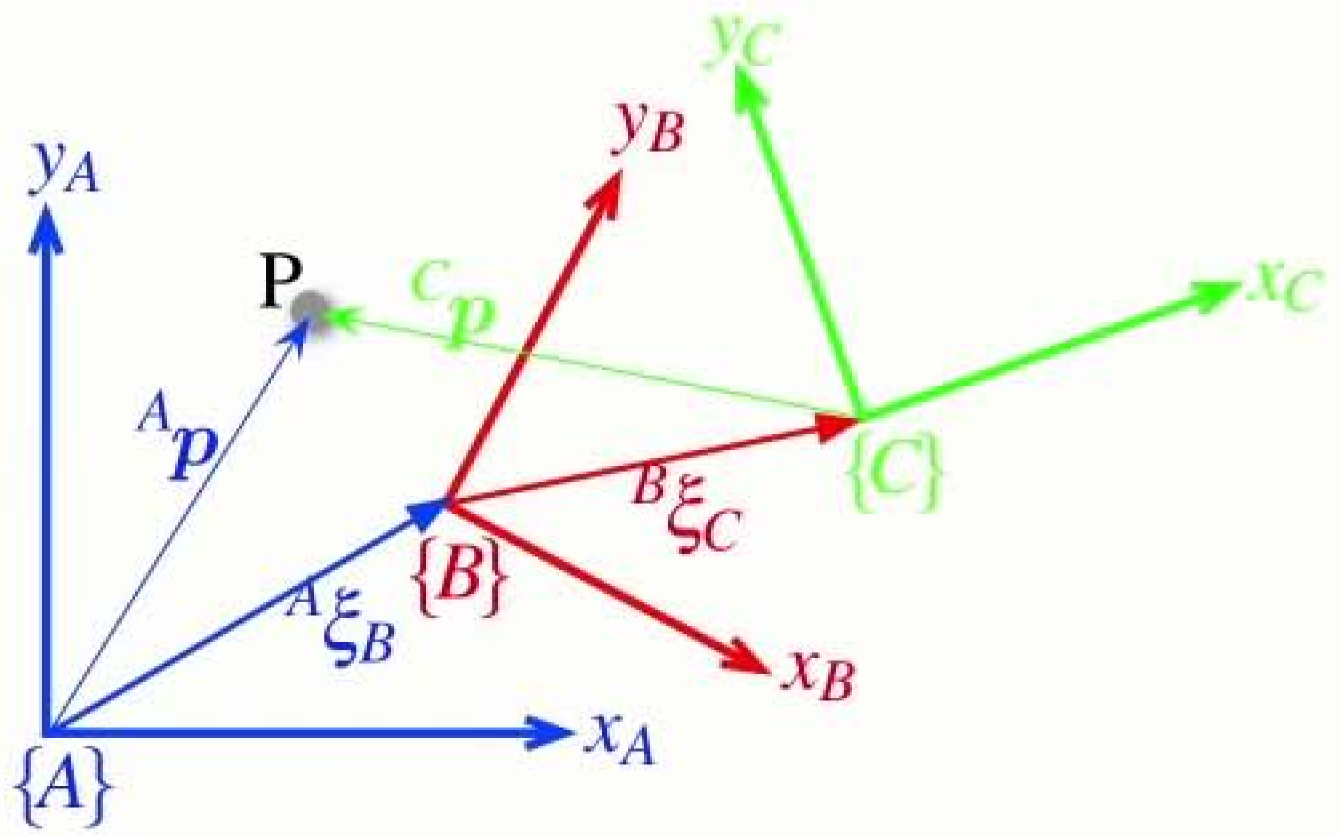


- We can't add **Poses** and **Vectors**
- They are different things

$$ {}^{W}\boldsymbol{p}_d = {}^{W}\xi_C \cdot {}^{C}\boldsymbol{p}_d $$

transform
vector from one frame
to another

Composition of **poses**



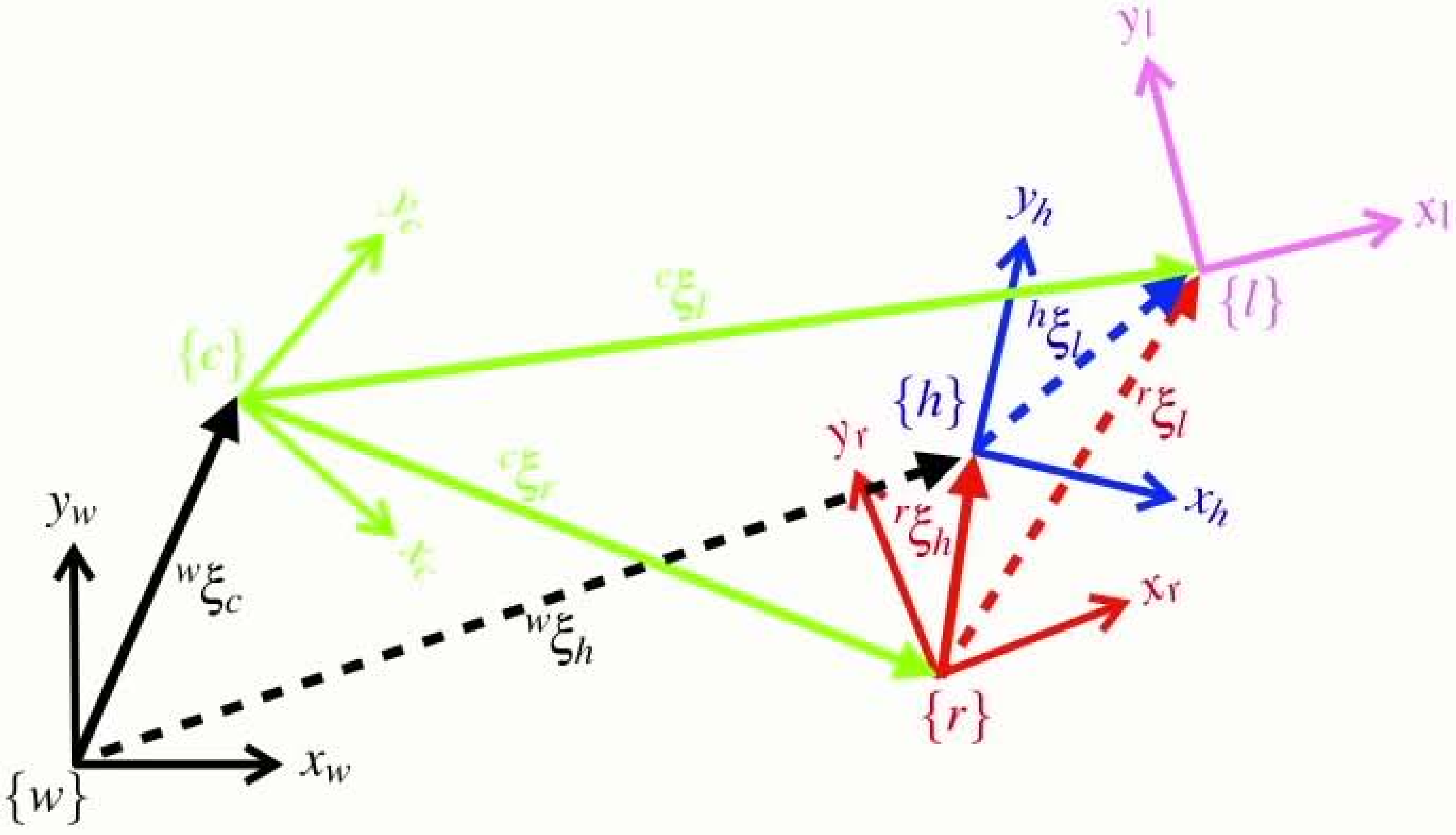
composition

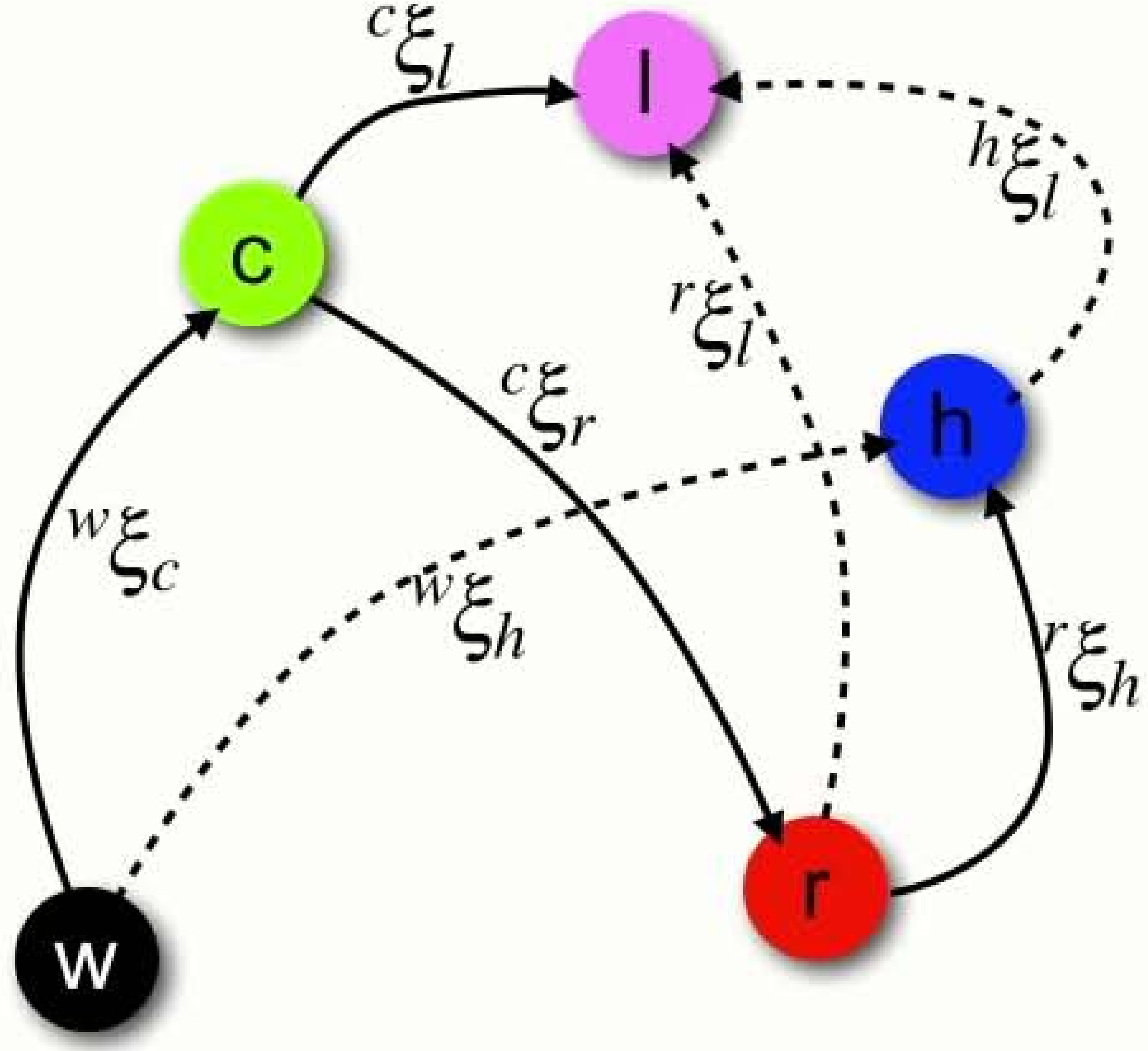
$$^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

- Relative poses can be **compounded** or **composed**
- We can extend this approach indefinitely...

Pose graphs

- We can express complex problems as pose graphs





$${}^{w}\xi_h = {}^{w}\xi_c \oplus {}^{c}\xi_r \oplus {}^{r}\xi_h$$

$${}^{h}\xi_l = \ominus\, {}^{r}\xi_h \ominus {}^{c}\xi_r \oplus {}^{c}\xi_l$$

Spatial algebra

- We can use a spatial algebra to solve for spatial relationships

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

$$^{X}\boldsymbol{p} = {}^{X}\xi_{Y} \bullet {}^{Y}\boldsymbol{p}$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\xi \ominus \xi = 0$$

$$\xi \oplus 0 = \xi$$

$$\ominus \xi \oplus \xi = 0$$

$$\ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X}$$

$$\xi \ominus 0 = \xi$$

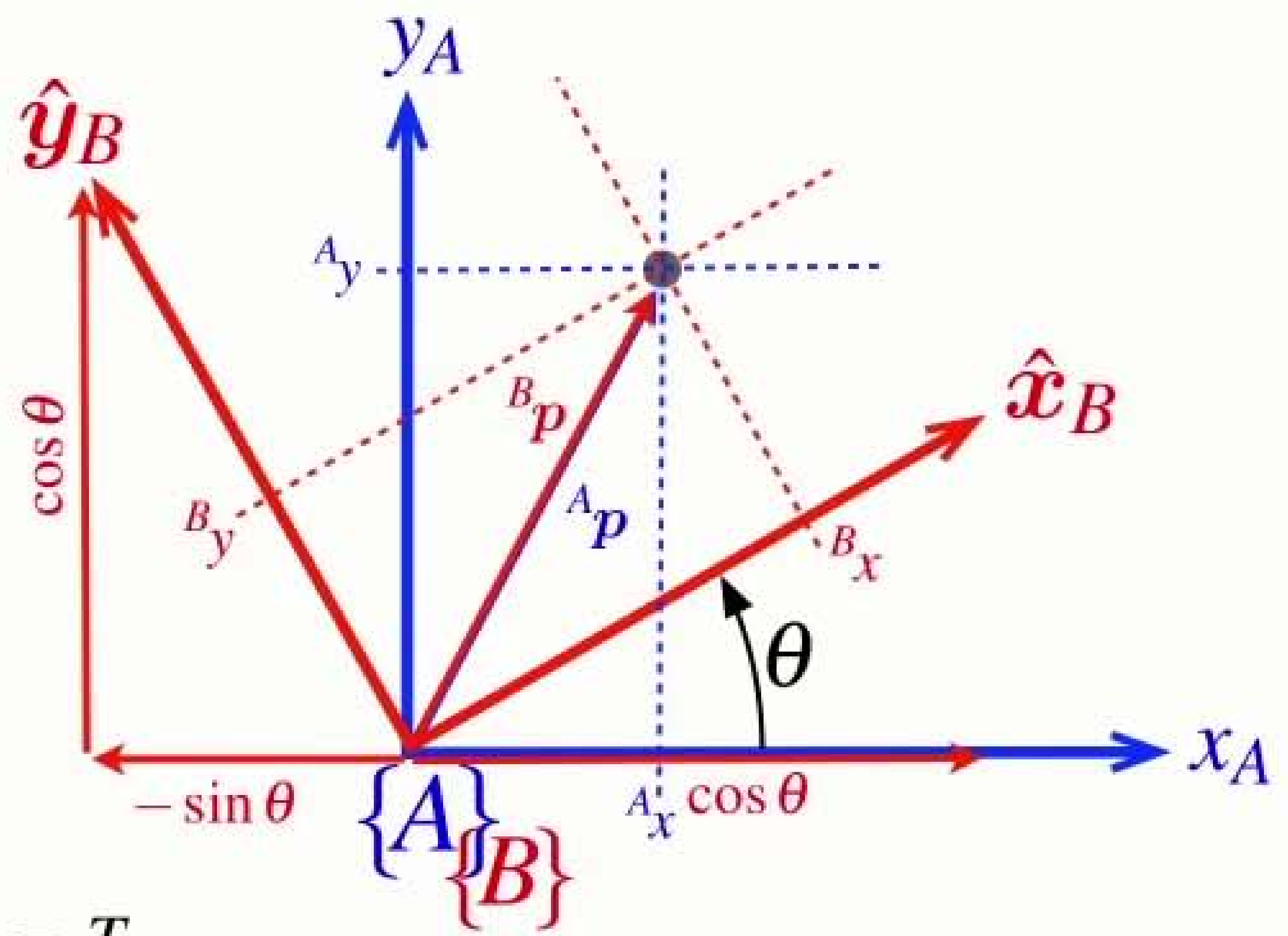
Rotation matrix

- We can rotate a vector from one frame to another

$$
{}^{A}\mathbf{R}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}
$$

$$
{}^{B}\mathbf{R}_{A} = {}^{A}\mathbf{R}_{B}^{-1}
$$

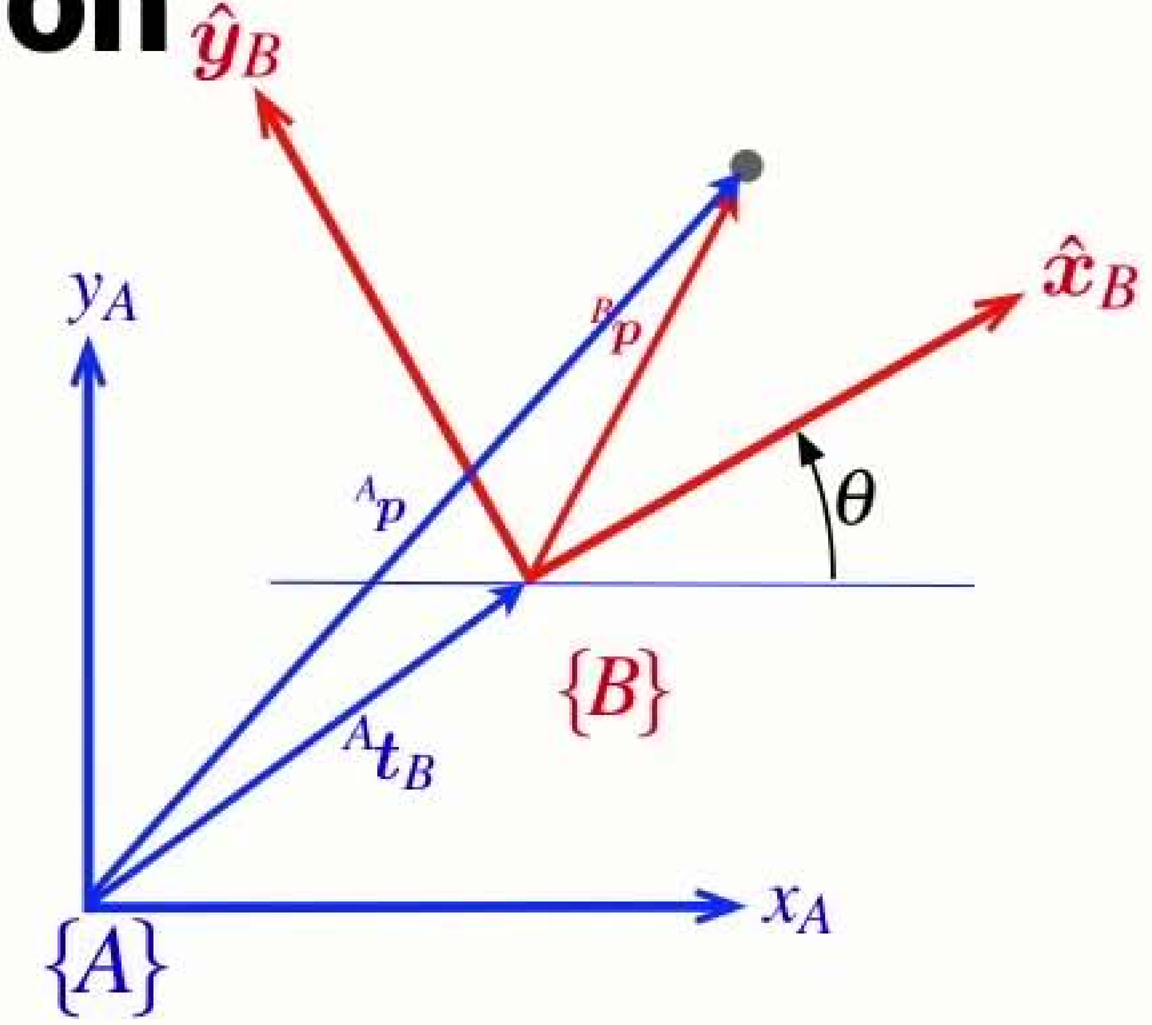
- Rotates vectors from frame {B} to frame {A}
- Is a function of the rotation angle from frame {A} to frame {B}
- An orthogonal (orthonormal) matrix
  - ➥ The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^{T}$
  - ➥ The determinant is +1
- Rotation matrices belong to the Special Orthogonal group of dimension 2 $\mathbf{R} \in SO(2)$



$$
{}^{A}\mathbf{R}_{B}
$$

Homogeneous transformation

- In concrete terms we can represent pose by 3x3 homogeneous transformation matrix $\mathbf{T} \in SE(2)$



$$^{A}\xi_B \sim {}^{A}\mathbf{T}_B = \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

$$^{X}\xi_Y \oplus {}^{Y}\xi_Z \mapsto {}^{X}\mathbf{T}_Y \, {}^{Y}\mathbf{T}_Z$$

$$\ominus\xi \mapsto \mathbf{T}^{-1}$$

$$^{A}\mathbf{T}_B \qquad \boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

$$\xi \cdot \boldsymbol{p} \mapsto \mathbf{T}\tilde{\boldsymbol{p}}$$